Configuration Summary

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DHW ANALYSIS - CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-10-17 19:45:40
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-- Data Preparation --

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

-- Model Training --

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 10
rollout_epoch_end...................... 60
rollout_loss_decay..................... 0.92
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 0.0001
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

-- Model Inference --

model_path............................. checkpoints_64x128/best_model.pt
input_data_path........................ data/processed_64x128/sst_final.nc
run_validation......................... True
run_test............................... True
forecast_lead_steps.................... 90
chain_stride_steps..................... 90
output_dir............................. results_64x128/
val_predictions_file................... predictions_val.nc
test_predictions_file.................. predictions_test.nc

-- Concat Chains --
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```
  predictions_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predictions_file_template..... predictions_{split}.nc
  std_params_path..... data/processed_64x128/std_params.nc
  trend_params_path..... data/processed_64x128/trend_params.nc
  output_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  actual_file_template..... sst_actual_{split}_continuous.nc
  predicted_file_template..... sst_predicted_{split}_chains.nc
  splits..... ['test', 'val']
  lon_correction_threshold..... 0.6

-- Calc DHW --

  mmm_path..... /home/272/xl5758/ct5km_climatology_v3.1.nc
  results_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predicted_file_template..... sst_predicted_{split}_chains.nc
  actual_file_template..... sst_actual_{split}_continuous.nc
  dhw_output_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_file_template..... dhw_products_{split}.nc
  run_test..... True
  run_validation..... True
  window_days..... 84
  hotspot_threshold..... 1

-- DHW Plots --

  dhw_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_pdf..... dhw_analysis_combined.pdf
  output_csv..... dhw_metrics_combined.csv
  spatial_lead_steps..... [1, 7, 14, 30, 60, 90]
  n_sample_chains..... 6
  plot_init_dates..... ['2016-05-20', '2016-10-15', '2017-02-10', '2017-07-09', '20
  dhw_vmax..... 20
  dhw_diff_vmax..... 8
  tropic_lat..... 30
  dhw_cmap..... ['#CCe6F0', '#441C94', '#6F47BF', '#815CCC', '#A383E6', '#ED
  error_cmap..... ['#053061', '#2166AC', '#92C5DE', '#F7F7F7', '#F4A582', '#D6
  rmse_cmap..... YlOrRd
  bias_cmap..... RdBu_r
  bleaching_thresholds..... [(0, 4, 'No Stress', '#27AE60'), (4, 8, 'Warning', '#F39C12'
  gbr_lat_bounds..... (-25.0, -10.0)
  gbr_lon_bounds..... (142.0, 157.0)
  reef_boxes..... [{'name': 'Great Barrier Reef', 'lat': (-25.0, -10.0), 'lon'
```

DHW Analysis - Summary

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DEGREE HEATING WEEKS (DHW) - COMBINED ANALYSIS
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METHODOLOGY  (NOAA Coral Reef Watch v3.1)
  SSTA      = SST - DC              (DC = daily climatology)
  HotSpot   = max(SST - MMM, 0)     (MMM = Maximum Monthly Mean)
  DHW_i     = (1/7) x sum(HotSpot_j) for j=i-83..i, HotSpot_j >= 1.0C

SLIDING WINDOW CONSTRUCTION  (always 84 days ending on valid date)
  Lead L:  (84-L) actual + L predicted SST days
  Lead 1:  83 actual +  1 predicted
  Lead 30: 54 actual + 30 predicted
  Lead 84:  0 actual + 84 predicted
  Lead 90: previous 84 predicted days (days 7-90)

VERIFICATION
  Actual DHW  : 84 observed days ending on the forecast valid date
  Persistence : observed DHW on the day before lead 1, held constant
  Sample      : chains defined at every lead step (same chains for
                model and persistence at all leads)

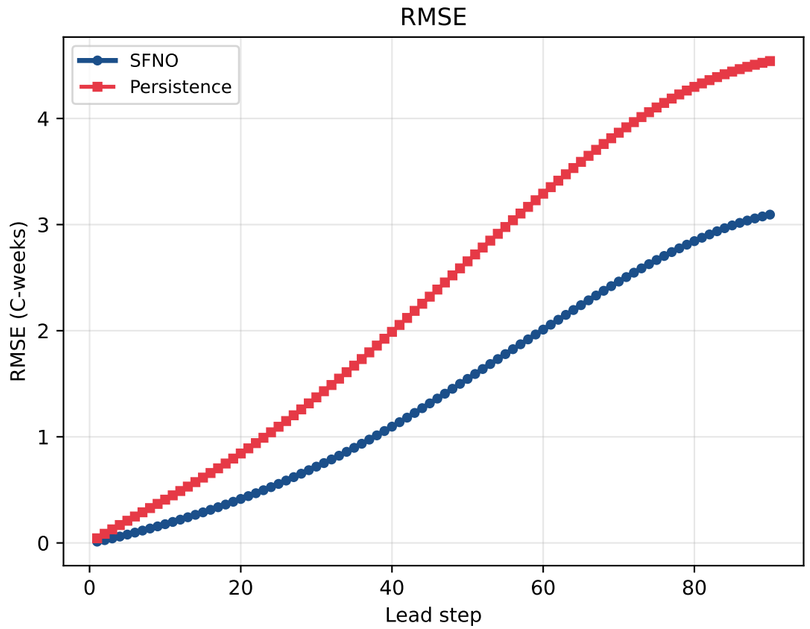
BLEACHING ALERT THRESHOLDS
  DHW  0-4  : No Stress
  DHW  4-8  : Warning
  DHW  8-12 : Alert Level 1
  DHW 12-16 : Alert Level 2
  DHW 16-20 : Alert Level 3
  DHW 20->= : Alert Level 4-5

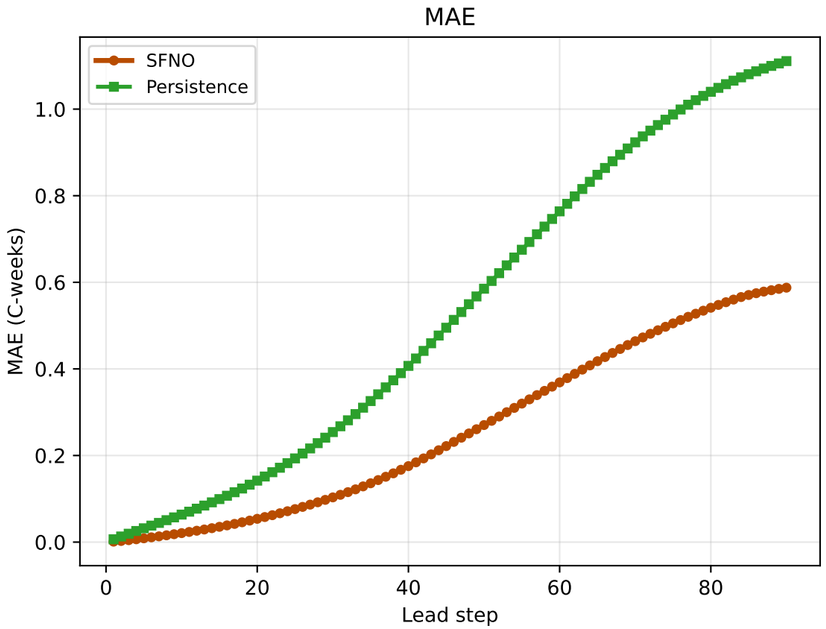
DATASETS
  Test          19 chains x 90 leads x 64 lat x 128 lon
  Val           16 chains x 90 leads x 64 lat x 128 lon
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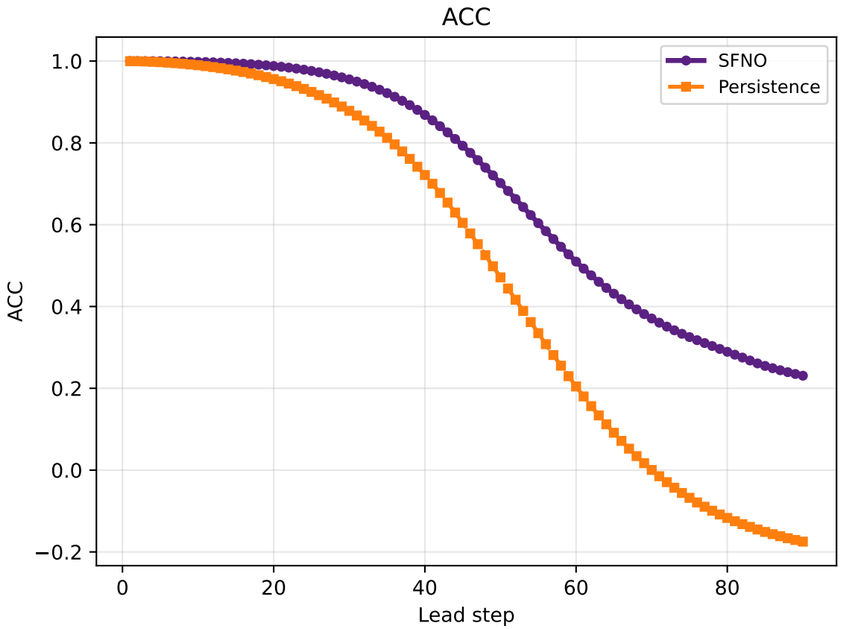
DHW Metrics by Lead Step - Test (18 chains scored at every lead)

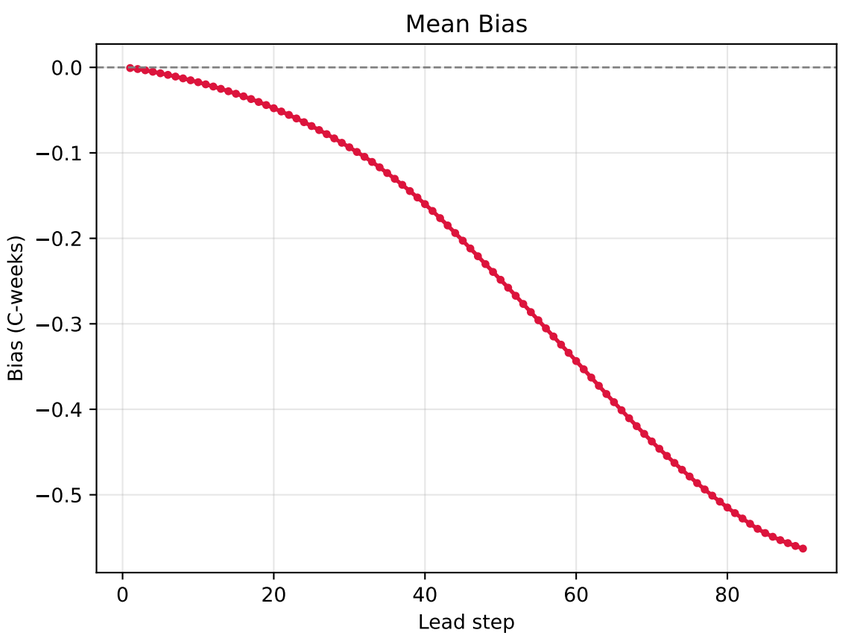
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0124	0.0437	71.6%	0.0011	0.0065	-0.0008	1.0000	0.9999	1.0000
2	0.0264	0.0862	69.4%	0.0026	0.0130	-0.0019	1.0000	0.9996	0.9999
3	0.0425	0.1278	66.7%	0.0045	0.0193	-0.0034	0.9999	0.9990	0.9998
4	0.0603	0.1687	64.3%	0.0066	0.0257	-0.0051	0.9998	0.9983	0.9996
5	0.0788	0.2091	62.3%	0.0088	0.0319	-0.0069	0.9996	0.9973	0.9994
6	0.0974	0.2491	60.9%	0.0110	0.0381	-0.0088	0.9994	0.9962	0.9990
7	0.1164	0.2888	59.7%	0.0133	0.0443	-0.0107	0.9992	0.9949	0.9986
8	0.1360	0.3285	58.6%	0.0157	0.0507	-0.0129	0.9989	0.9934	0.9981
9	0.1561	0.3680	57.6%	0.0182	0.0571	-0.0151	0.9985	0.9917	0.9975
10	0.1767	0.4079	56.7%	0.0207	0.0635	-0.0174	0.9981	0.9898	0.9967
11	0.1979	0.4484	55.9%	0.0235	0.0702	-0.0198	0.9976	0.9877	0.9959
12	0.2195	0.4893	55.1%	0.0262	0.0772	-0.0224	0.9970	0.9853	0.9949
13	0.2419	0.5310	54.4%	0.0291	0.0844	-0.0251	0.9963	0.9827	0.9937
14	0.2650	0.5733	53.8%	0.0321	0.0918	-0.0279	0.9955	0.9798	0.9924
15	0.2881	0.6158	53.2%	0.0353	0.0993	-0.0309	0.9947	0.9767	0.9909
16	0.3120	0.6590	52.7%	0.0386	0.1071	-0.0339	0.9936	0.9733	0.9893
17	0.3365	0.7031	52.1%	0.0421	0.1152	-0.0371	0.9925	0.9695	0.9874
18	0.3618	0.7484	51.7%	0.0458	0.1237	-0.0405	0.9912	0.9654	0.9852
19	0.3880	0.7951	51.2%	0.0497	0.1326	-0.0441	0.9896	0.9609	0.9827
20	0.4147	0.8429	50.8%	0.0538	0.1419	-0.0478	0.9880	0.9560	0.9800
21	0.4414	0.8913	50.5%	0.0579	0.1515	-0.0516	0.9861	0.9507	0.9770
22	0.4686	0.9406	50.2%	0.0621	0.1614	-0.0556	0.9840	0.9450	0.9737
23	0.4967	0.9912	49.9%	0.0666	0.1717	-0.0598	0.9817	0.9387	0.9700
24	0.5260	1.0427	49.6%	0.0713	0.1824	-0.0641	0.9790	0.9319	0.9659
25	0.5563	1.0951	49.2%	0.0762	0.1933	-0.0686	0.9760	0.9246	0.9612
26	0.5879	1.1486	48.8%	0.0813	0.2046	-0.0733	0.9727	0.9167	0.9560
27	0.6200	1.2029	48.5%	0.0865	0.2163	-0.0781	0.9689	0.9081	0.9503
28	0.6525	1.2585	48.2%	0.0919	0.2285	-0.0831	0.9648	0.8989	0.9441
29	0.6856	1.3152	47.9%	0.0975	0.2411	-0.0882	0.9603	0.8889	0.9372
30	0.7191	1.3721	47.6%	0.1033	0.2541	-0.0935	0.9553	0.8783	0.9297
31	0.7526	1.4298	47.4%	0.1092	0.2675	-0.0990	0.9499	0.8669	0.9217
32	0.7867	1.4883	47.1%	0.1154	0.2813	-0.1046	0.9439	0.8547	0.9130
33	0.8220	1.5482	46.9%	0.1218	0.2956	-0.1106	0.9373	0.8416	0.9035
34	0.8586	1.6090	46.6%	0.1287	0.3104	-0.1170	0.9299	0.8274	0.8930
35	0.8964	1.6707	46.3%	0.1359	0.3257	-0.1236	0.9217	0.8123	0.8816
36	0.9347	1.7328	46.1%	0.1432	0.3414	-0.1304	0.9128	0.7963	0.8692
37	0.9738	1.7960	45.8%	0.1509	0.3573	-0.1374	0.9030	0.7792	0.8559
38	1.0137	1.8599	45.5%	0.1588	0.3737	-0.1447	0.8923	0.7609	0.8415
39	1.0547	1.9244	45.2%	0.1671	0.3902	-0.1522	0.8807	0.7417	0.8262
40	1.0960	1.9890	44.9%	0.1755	0.4070	-0.1600	0.8683	0.7214	0.8100
41	1.1379	2.0544	44.6%	0.1842	0.4241	-0.1680	0.8551	0.7001	0.7931
42	1.1808	2.1205	44.3%	0.1933	0.4416	-0.1764	0.8408	0.6776	0.7751
43	1.2250	2.1870	44.0%	0.2026	0.4592	-0.1849	0.8256	0.6541	0.7564
44	1.2700	2.2540	43.7%	0.2121	0.4771	-0.1938	0.8096	0.6297	0.7369
45	1.3151	2.3209	43.3%	0.2217	0.4951	-0.2028	0.7929	0.6045	0.7171
46	1.3604	2.3878	43.0%	0.2314	0.5132	-0.2118	0.7757	0.5787	0.6969
47	1.4063	2.4546	42.7%	0.2411	0.5313	-0.2209	0.7579	0.5524	0.6763
48	1.4527	2.5213	42.4%	0.2509	0.5494	-0.2301	0.7395	0.5255	0.6555
49	1.4991	2.5876	42.1%	0.2607	0.5673	-0.2393	0.7208	0.4985	0.6345
50	1.5458	2.6535	41.7%	0.2703	0.5852	-0.2485	0.7017	0.4712	0.6136
51	1.5920	2.7188	41.4%	0.2801	0.6031	-0.2578	0.6825	0.4439	0.5929
52	1.6389	2.7840	41.1%	0.2900	0.6211	-0.2672	0.6629	0.4166	0.5721
53	1.6858	2.8487	40.8%	0.3000	0.6391	-0.2767	0.6432	0.3892	0.5516
54	1.7326	2.9131	40.5%	0.3100	0.6572	-0.2863	0.6234	0.3620	0.5313
55	1.7794	2.9772	40.2%	0.3199	0.6752	-0.2958	0.6037	0.3348	0.5113
56	1.8258	3.0409	40.0%	0.3297	0.6931	-0.3053	0.5842	0.3079	0.4918
57	1.8720	3.1042	39.7%	0.3395	0.7110	-0.3148	0.5649	0.2813	0.4728
58	1.9182	3.1668	39.4%	0.3493	0.7287	-0.3244	0.5460	0.2552	0.4544
59	1.9643	3.2292	39.2%	0.3591	0.7463	-0.3339	0.5276	0.2296	0.4365
60	2.0104	3.2912	38.9%	0.3689	0.7638	-0.3436	0.5098	0.2045	0.4194
61	2.0567	3.3528	38.7%	0.3788	0.7813	-0.3532	0.4926	0.1801	0.4031
62	2.1031	3.4139	38.4%	0.3886	0.7985	-0.3629	0.4761	0.1565	0.3874
63	2.1492	3.4741	38.1%	0.3984	0.8155	-0.3725	0.4605	0.1338	0.3726
64	2.1949	3.5329	37.9%	0.4081	0.8320	-0.3821	0.4455	0.1121	0.3585
65	2.2408	3.5910	37.6%	0.4178	0.8483	-0.3916	0.4313	0.0913	0.3451
66	2.2864	3.6481	37.3%	0.4274	0.8641	-0.4011	0.4179	0.0715	0.3325
67	2.3315	3.7045	37.1%	0.4368	0.8795	-0.4105	0.4051	0.0524	0.3205
68	2.3762	3.7596	36.8%	0.4461	0.8945	-0.4197	0.3930	0.0343	0.3091
69	2.4201	3.8136	36.5%	0.4551	0.9091	-0.4288	0.3815	0.0169	0.2983
70	2.4631	3.8659	36.3%	0.4640	0.9233	-0.4376	0.3707	0.0005	0.2881
71	2.5053	3.9166	36.0%	0.4727	0.9370	-0.4462	0.3605	-0.0149	0.2786
72	2.5467	3.9656	35.8%	0.4811	0.9503	-0.4545	0.3509	-0.0294	0.2695
73	2.5875	4.0132	35.5%	0.4893	0.9631	-0.4627	0.3418	-0.0431	0.2610
74	2.6271	4.0591	35.3%	0.4973	0.9755	-0.4707	0.3333	-0.0558	0.2530
75	2.6663	4.1036	35.0%	0.5051	0.9875	-0.4786	0.3253	-0.0678	0.2454
76	2.7043	4.1466	34.8%	0.5128	0.9991	-0.4863	0.3177	-0.0790	0.2383
77	2.7409	4.1876	34.5%	0.5203	1.0101	-0.4938	0.3106	-0.0895	0.2315
78	2.7762	4.2264	34.3%	0.5274	1.0206	-0.5010	0.3035	-0.0993	0.2248
79	2.8106	4.2630	34.1%	0.5344	1.0305	-0.5080	0.2963	-0.1084	0.2181
80	2.8440	4.2978	33.8%	0.5412	1.0400	-0.5149	0.2893	-0.1168	0.2115
81	2.8761	4.3305	33.6%	0.5478	1.0491	-0.5215	0.2822	-0.1248	0.2049
82	2.9071	4.3612	33.3%	0.5540	1.0576	-0.5278	0.2752	-0.1321	0.1985
83	2.9371	4.3898	33.1%	0.5601	1.0657	-0.5339	0.2680	-0.1388	0.1919
84	2.9659	4.4167	32.8%	0.5659	1.0733	-0.5398	0.2609	-0.1452	0.1854
85	2.9922	4.4418	32.6%	0.5707	1.0805	-0.5448	0.2547	-0.1511	0.1798
86	3.0161	4.4649	32.4%	0.5748	1.0873	-0.5491	0.2492	-0.1565	0.1748
87	3.0381	4.4865	32.3%	0.5785	1.0937	-0.5531	0.2442	-0.1616	0.1703
88	3.0580	4.5062	32.1%	0.5818	1.0998	-0.5566	0.2395	-0.1664	0.1662
89	3.0768	4.5244	32.0%	0.5848	1.1055	-0.5598	0.2350	-0.1708	0.1624
90	3.0939	4.5408	31.9%	0.5876	1.1109	-0.5629	0.2308	-0.1748	0.1588

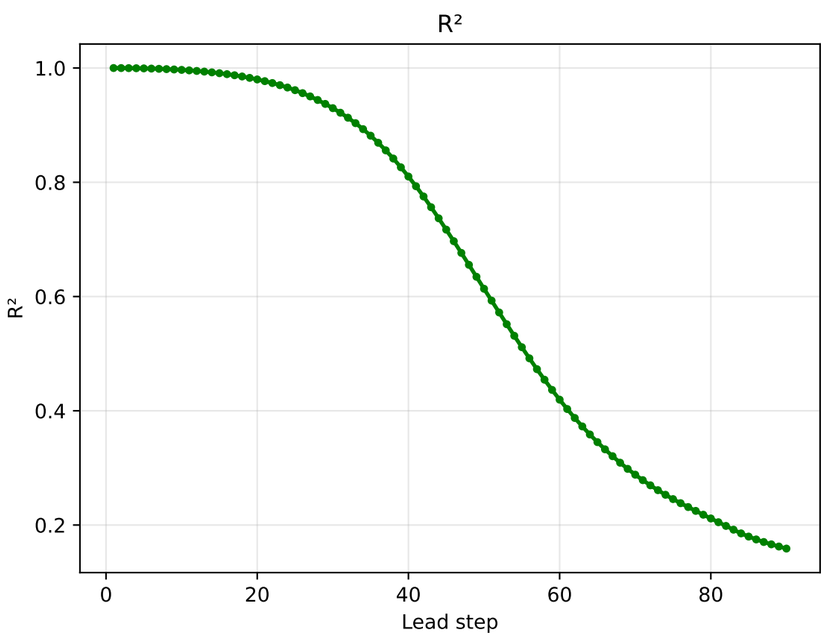
DHW Forecast Skill vs Lead Step - Test

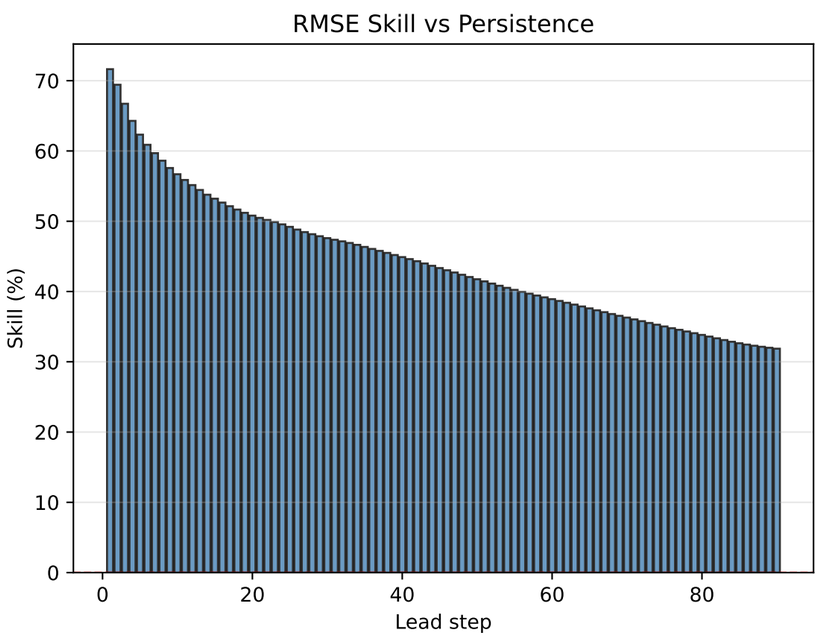


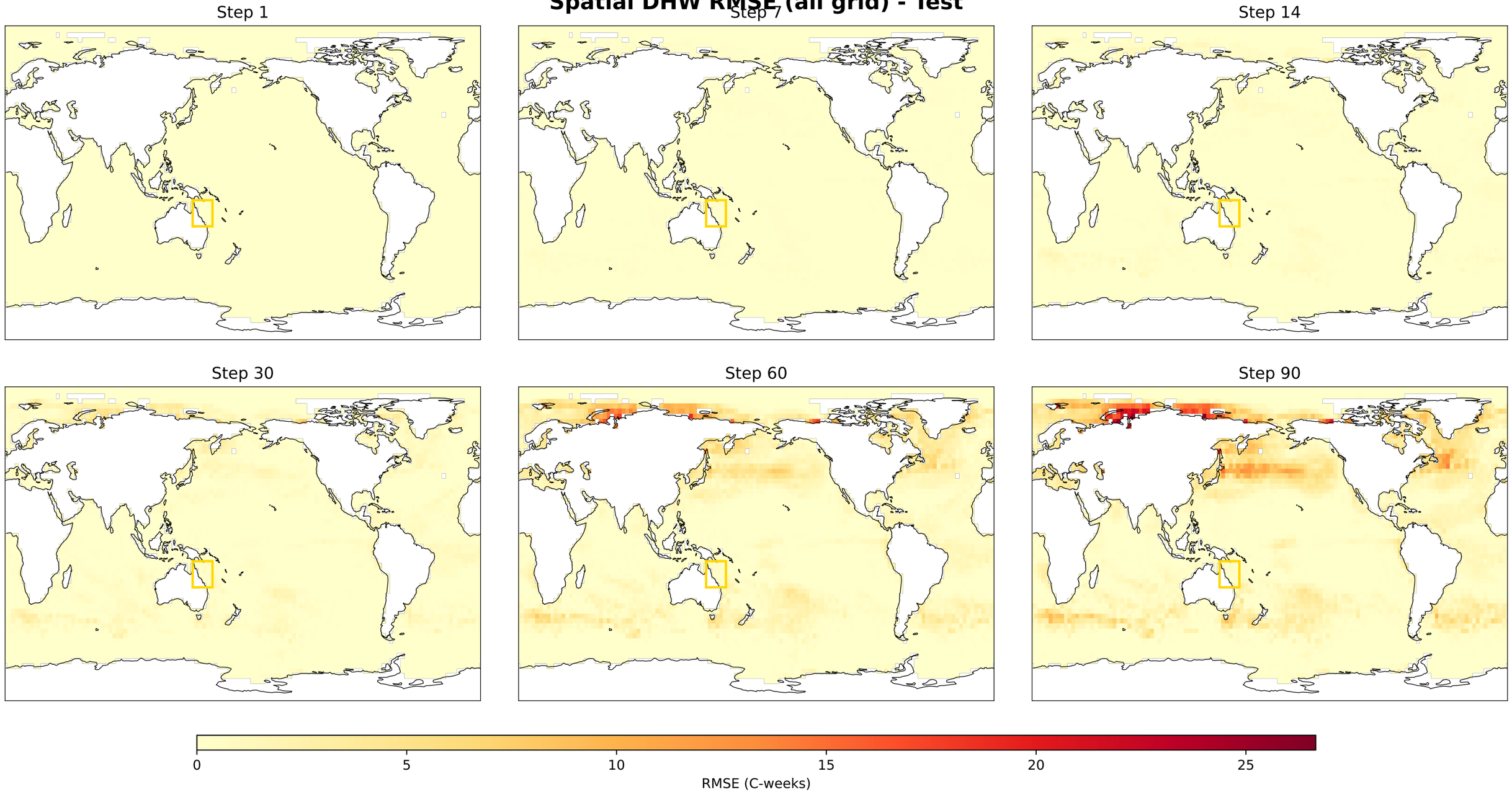


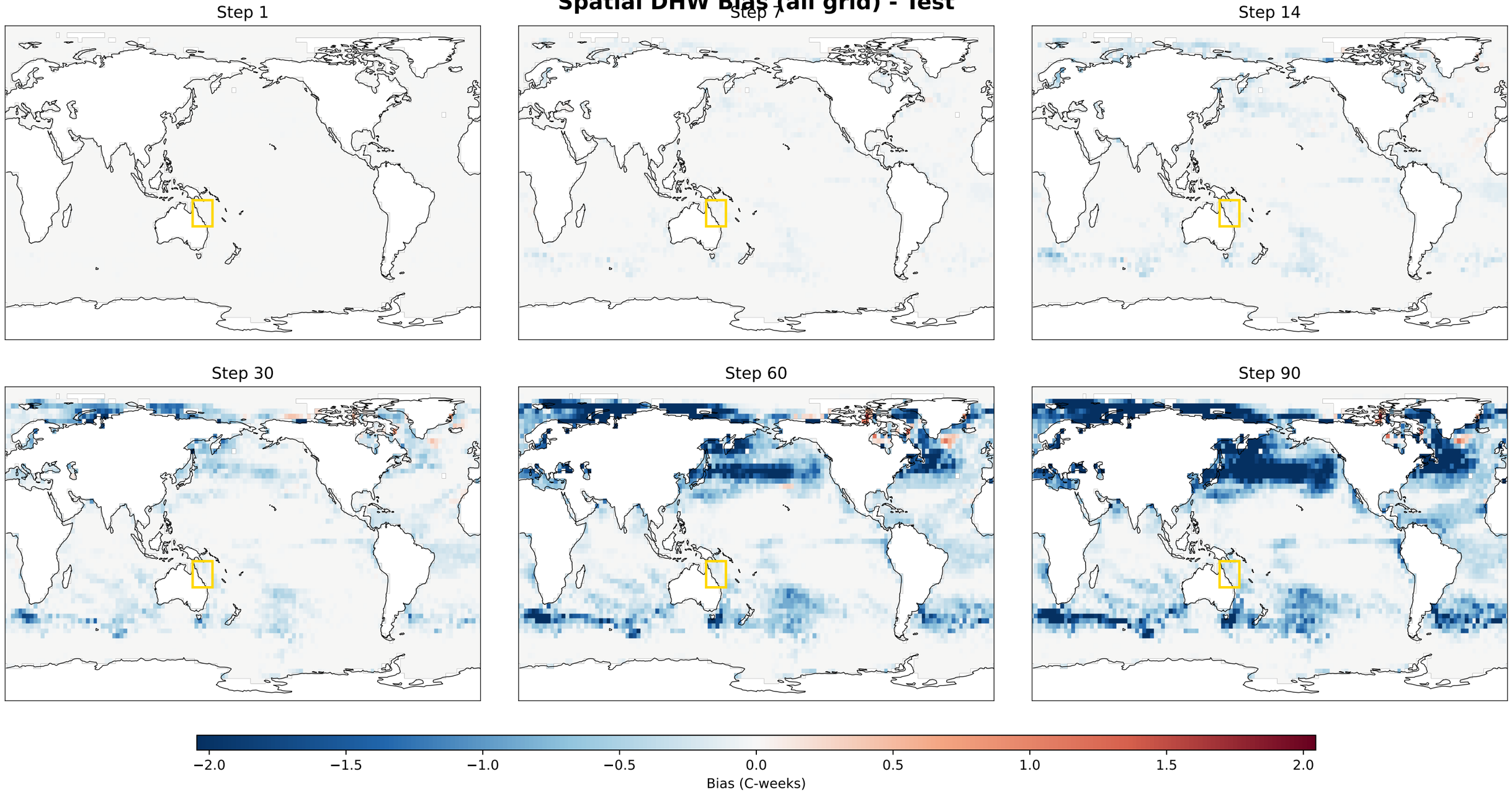


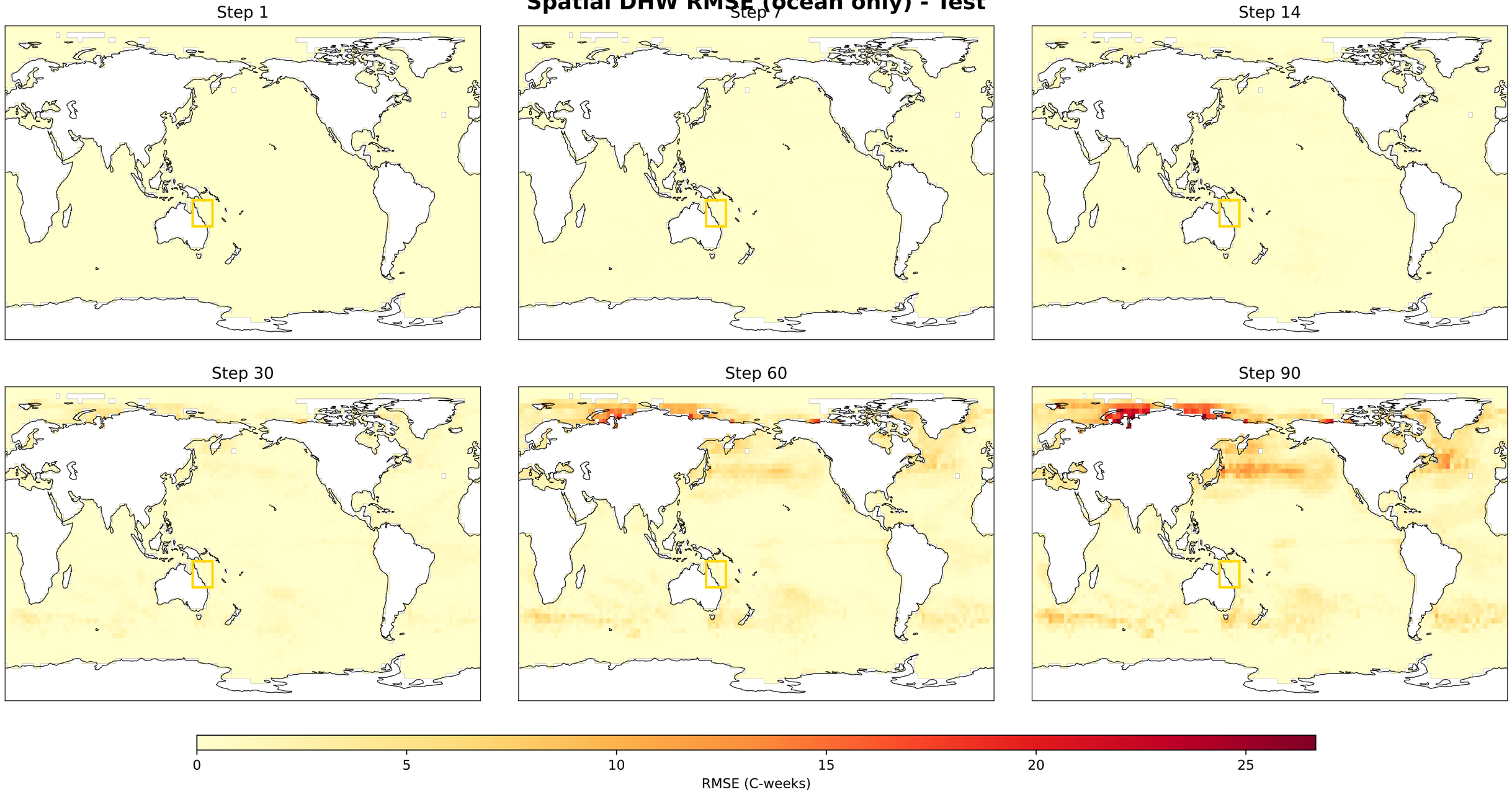


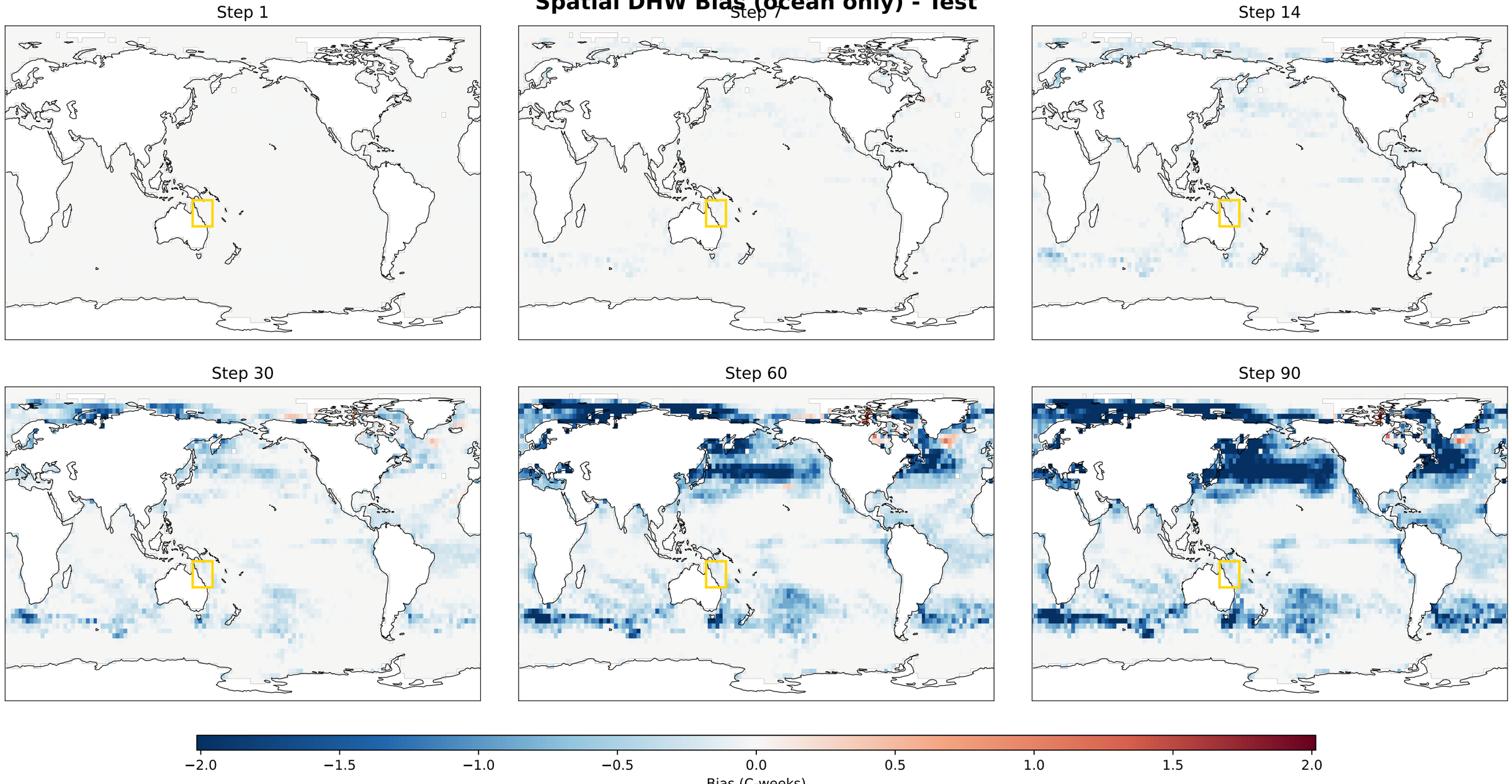





Spatial DHW RMSE (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

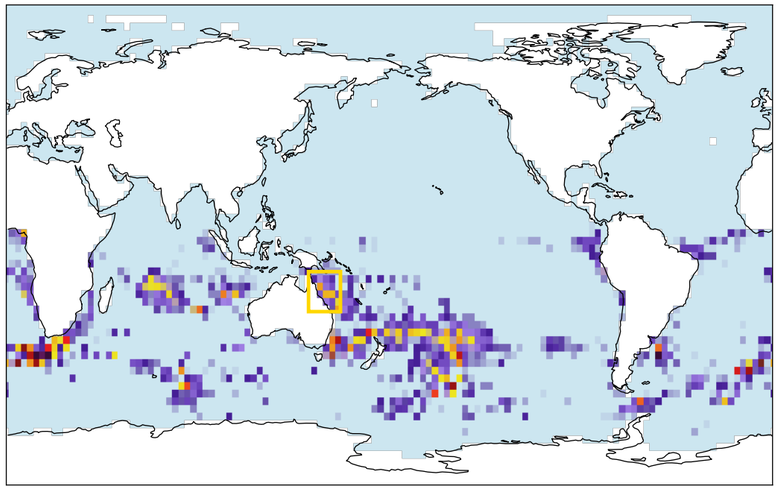

Spatial DHW Bias (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

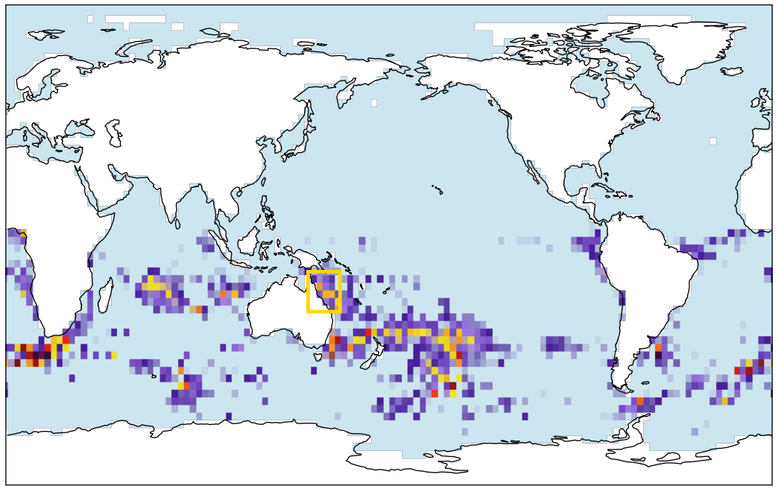

Spatial DHW Bias (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test (C-weeks)

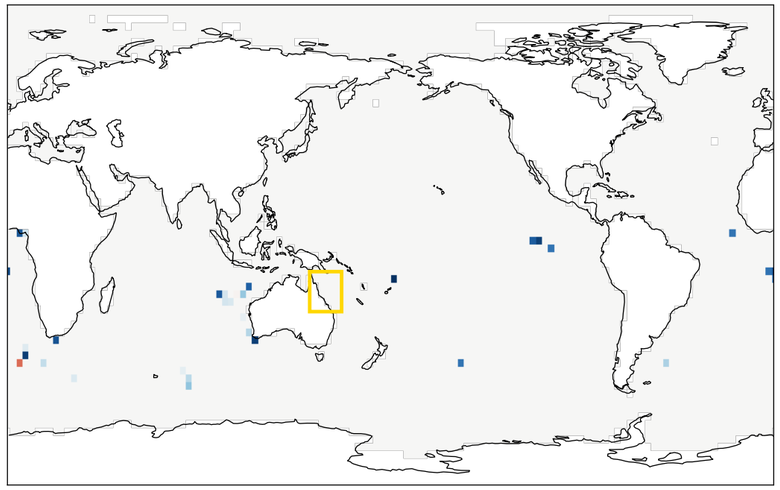
Actual DHW - Step 1



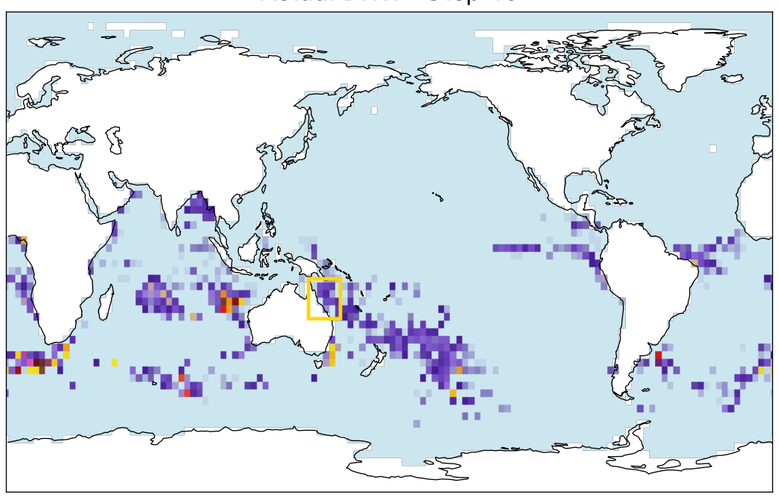
Predicted DHW - Step 1



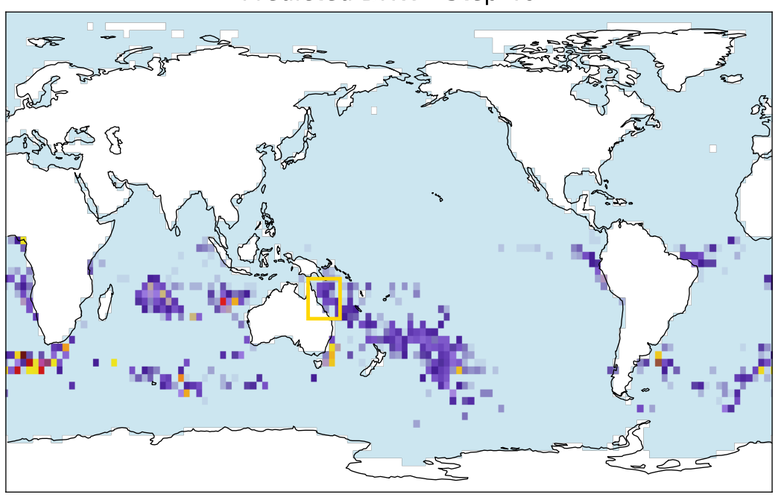
Error (Pred - Actual) - Step 1 [scale ±0.24]



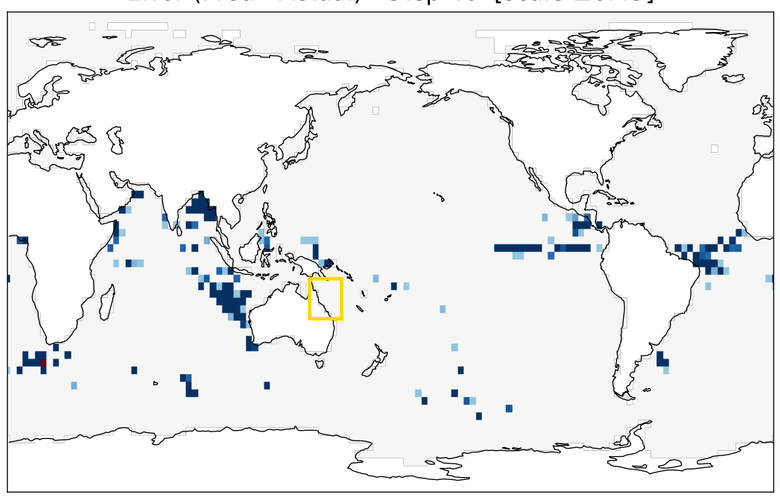
Actual DHW - Step 46



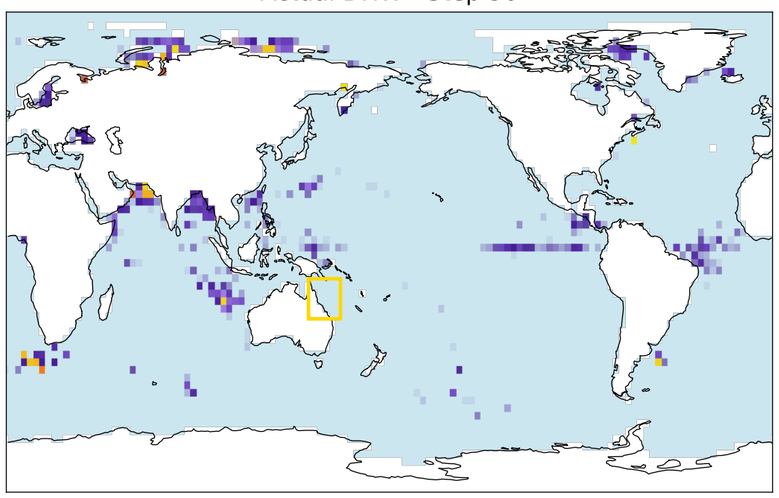
Predicted DHW - Step 46



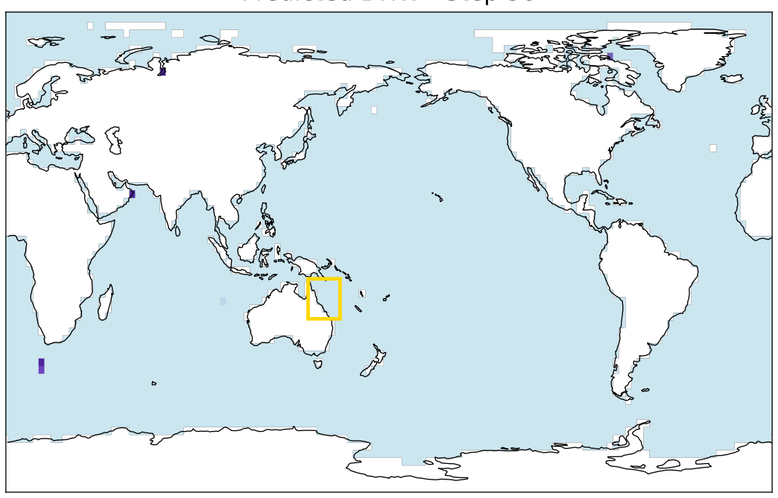
Error (Pred - Actual) - Step 46 [scale ±0.45]



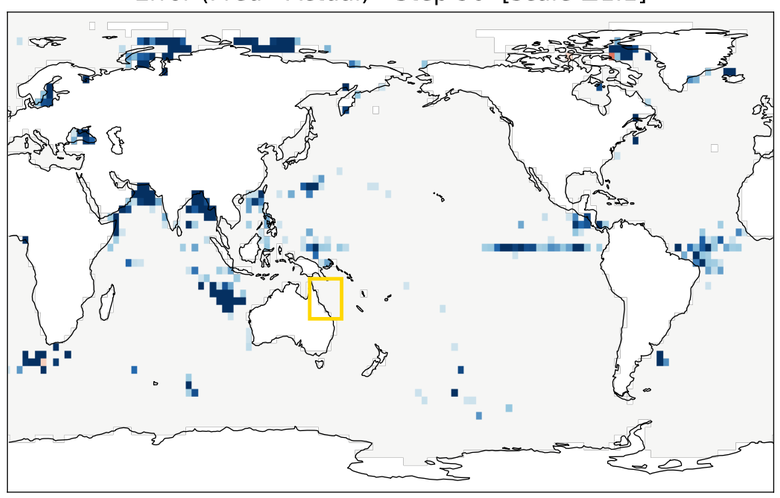
Actual DHW - Step 90

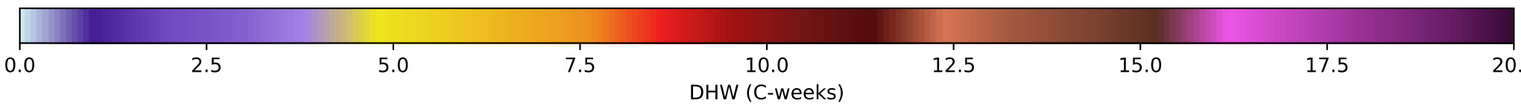


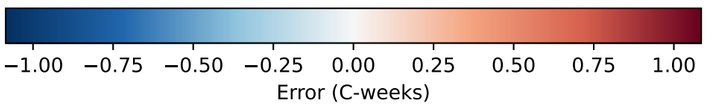
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±1.1]

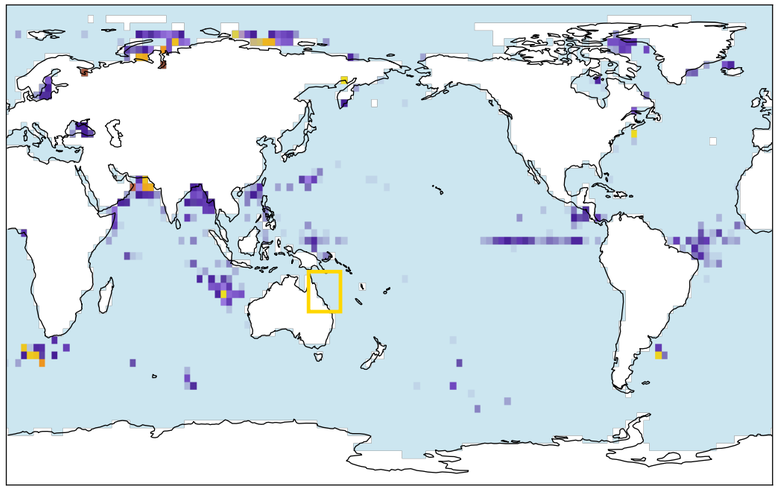




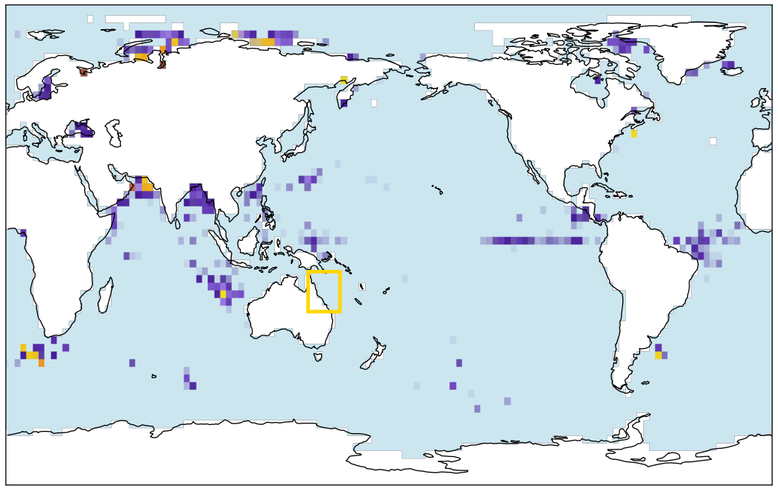


DHW Chain init: 2020-07-09 - Test (C-weeks)

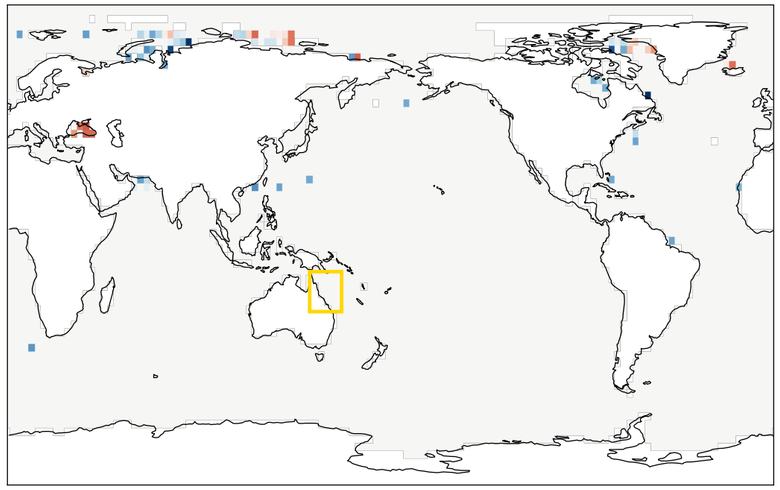
Actual DHW - Step 1



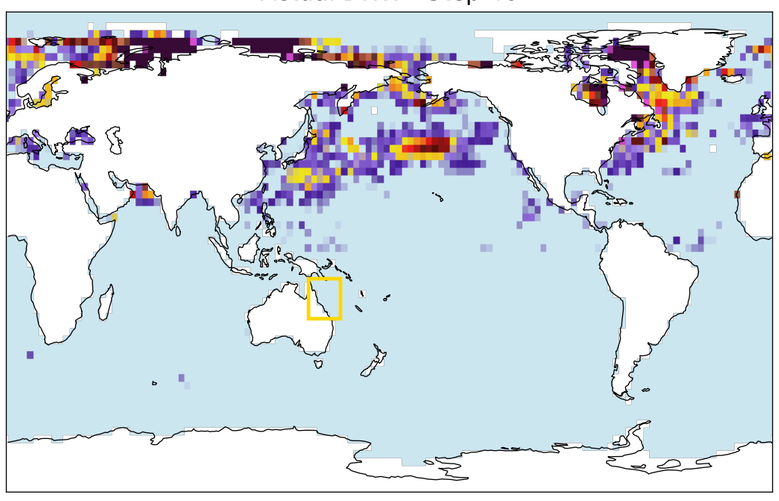
Predicted DHW - Step 1



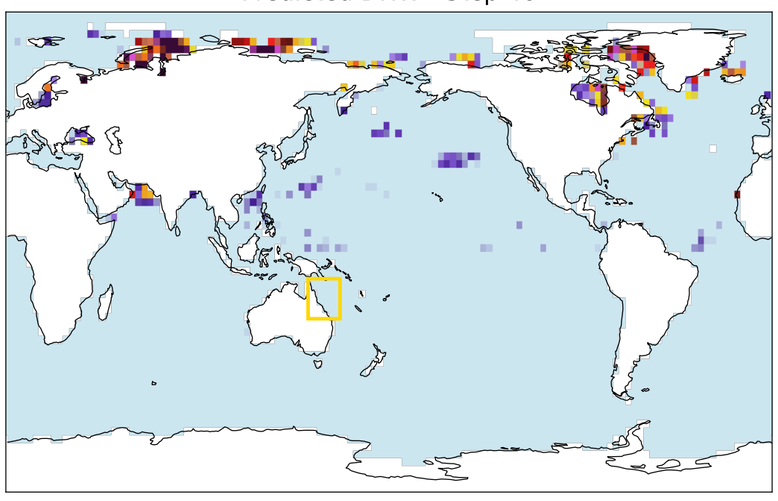
Error (Pred - Actual) - Step 1 [scale ±0.36]



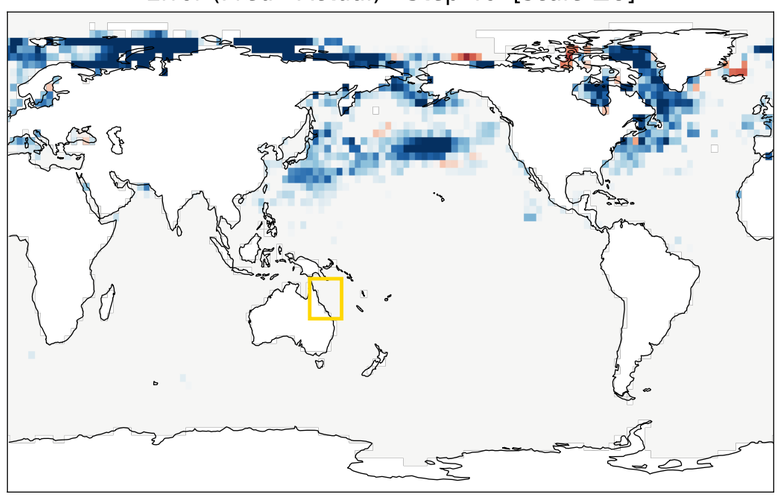
Actual DHW - Step 46



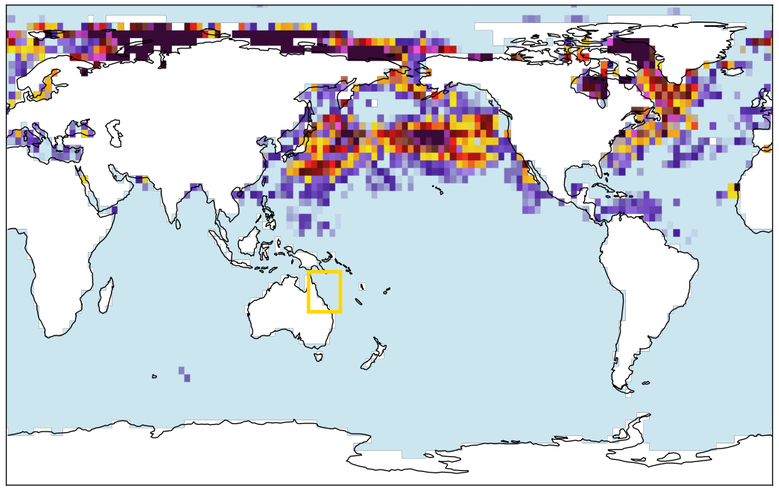
Predicted DHW - Step 46



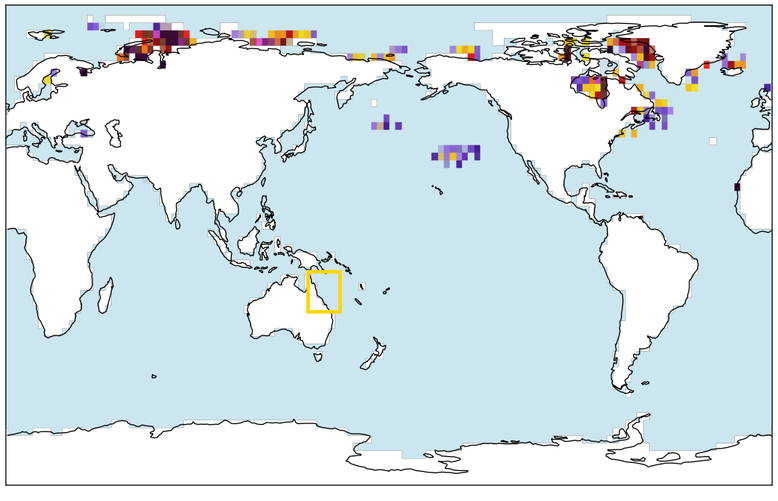
Error (Pred - Actual) - Step 46 [scale ±8]



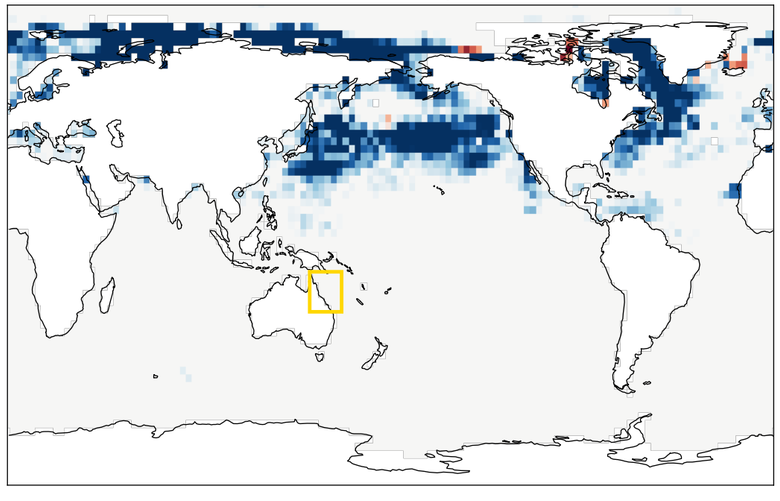
Actual DHW - Step 90

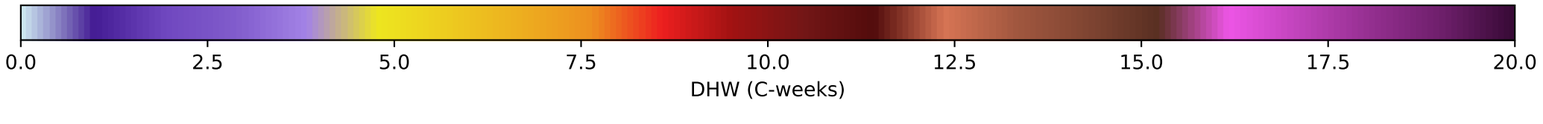


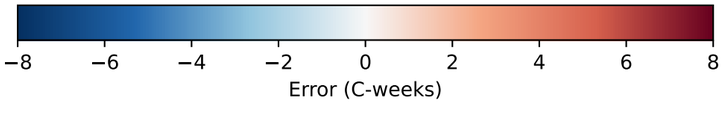
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

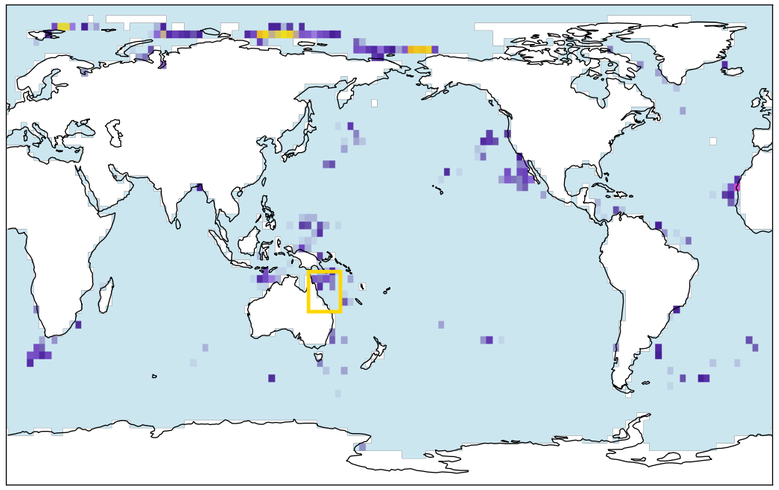




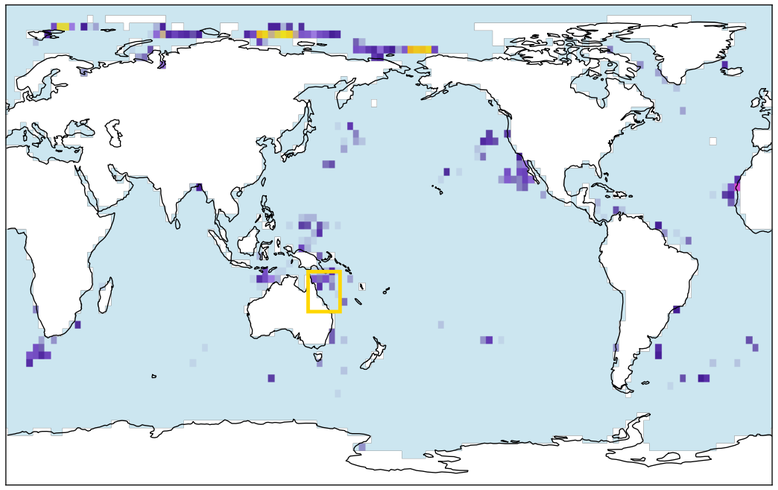


DHW Chain init: 2021-01-06 - Test (C-weeks)

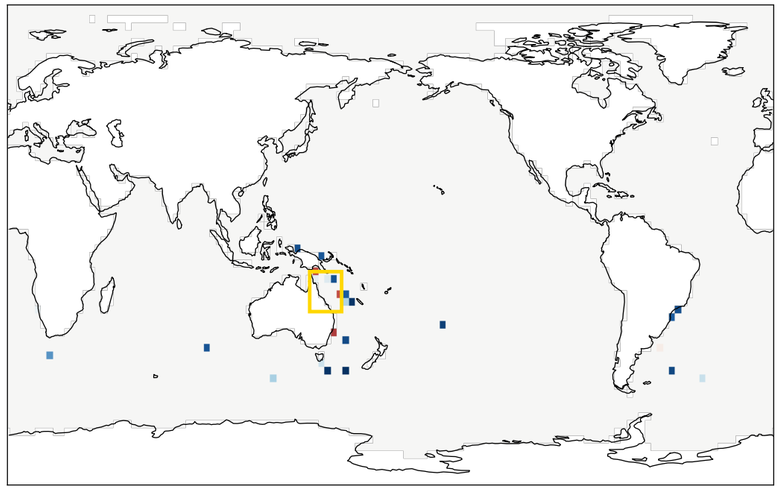
Actual DHW - Step 1



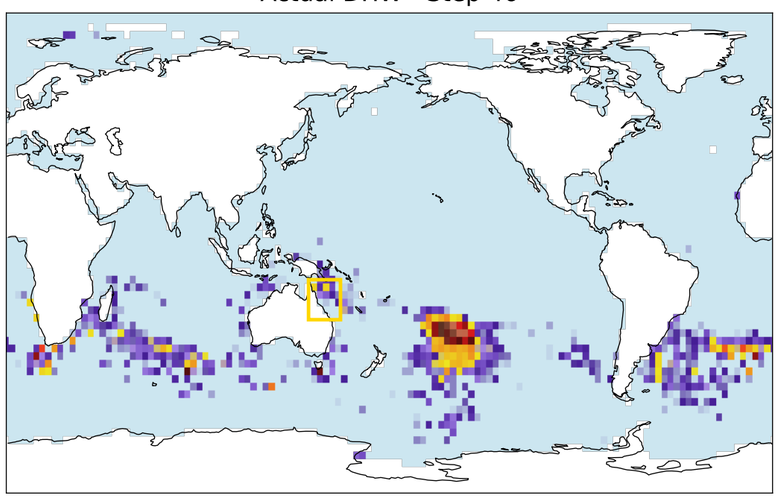
Predicted DHW - Step 1



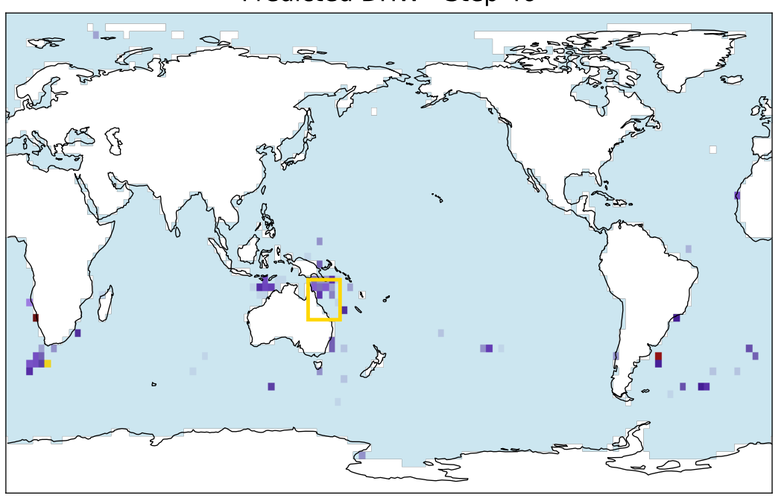
Error (Pred - Actual) - Step 1 [scale ±0.2]



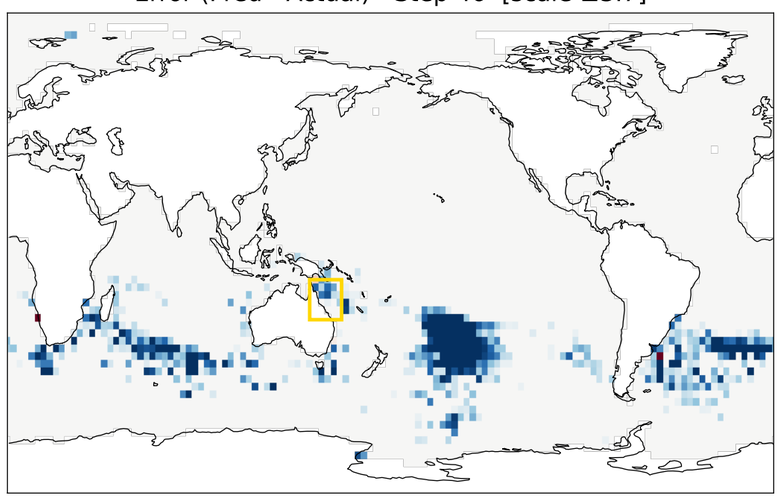
Actual DHW - Step 46



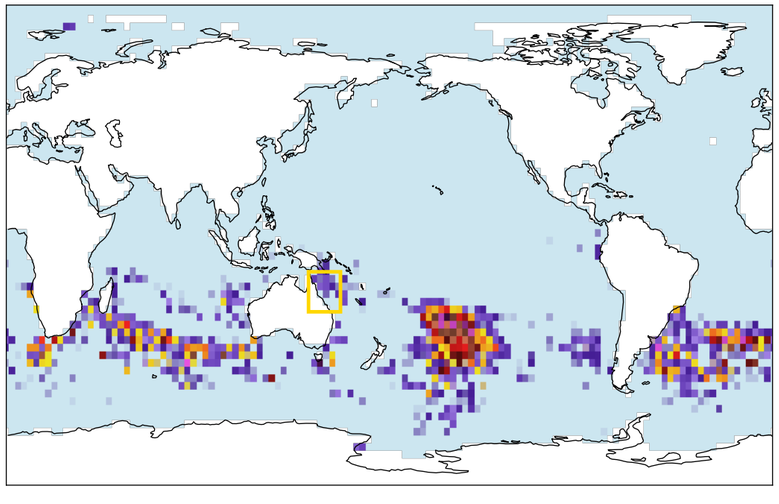
Predicted DHW - Step 46



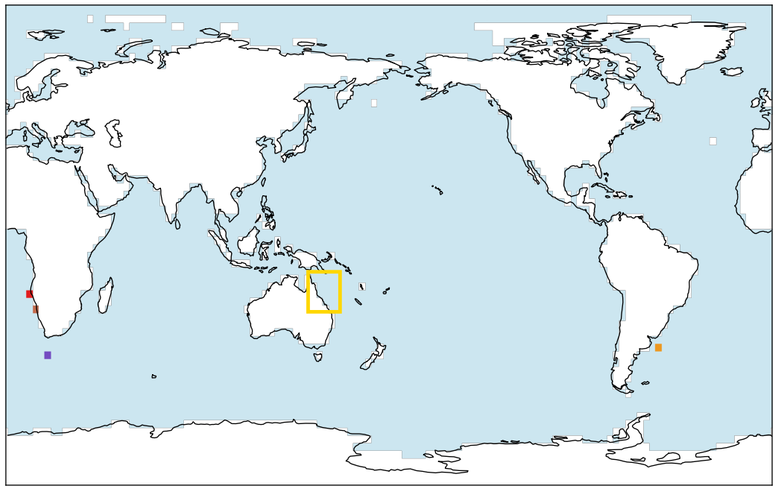
Error (Pred - Actual) - Step 46 [scale ±3.7]



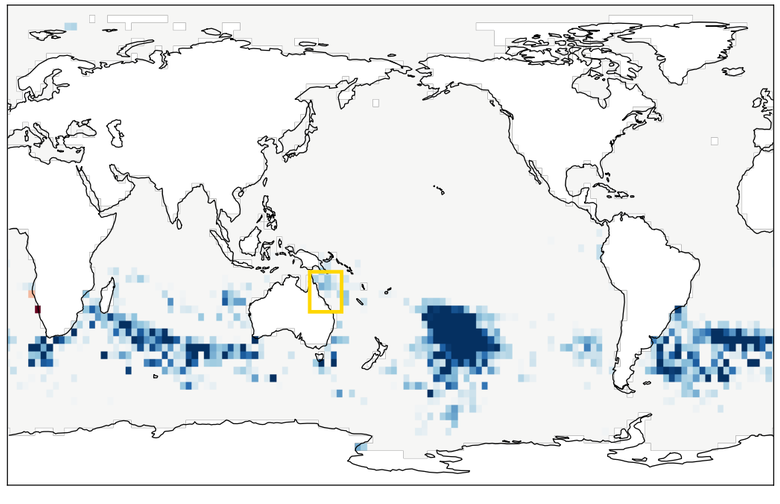
Actual DHW - Step 90

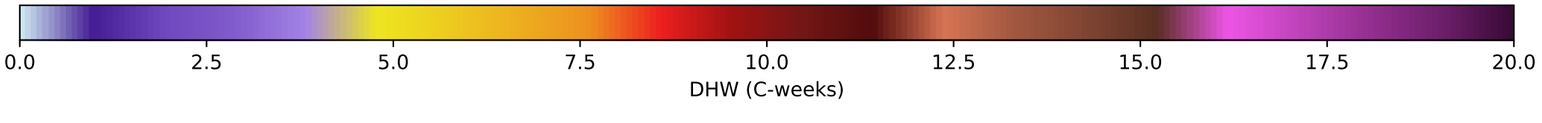


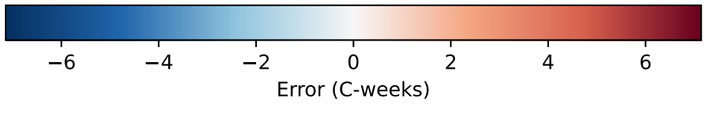
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±7.1]

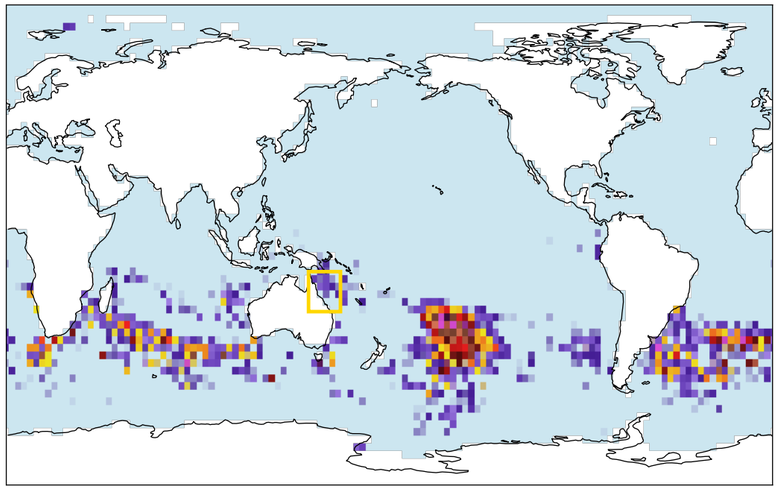




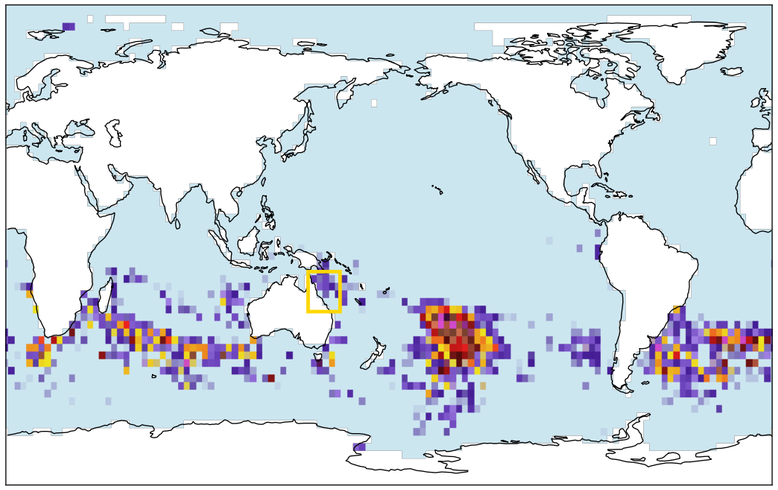


DHW Chain init: 2021-04-06 - Test (C-weeks)

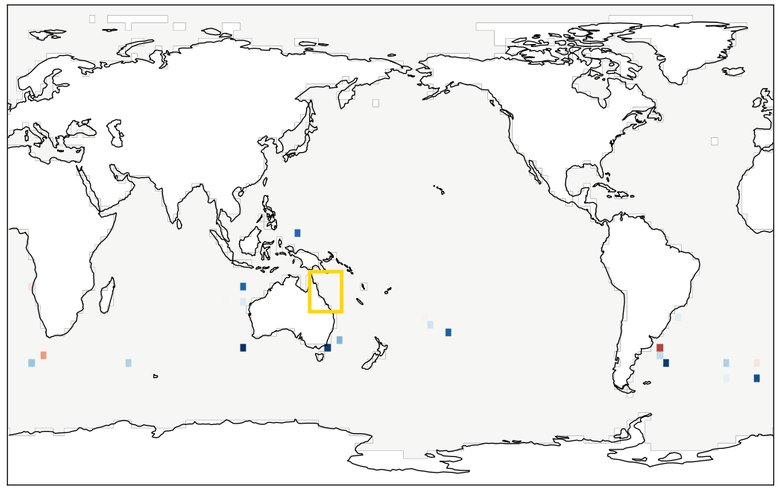
Actual DHW - Step 1



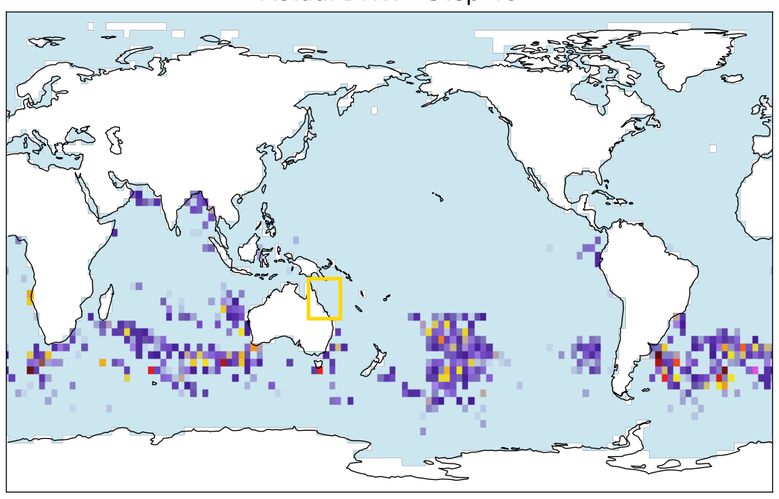
Predicted DHW - Step 1



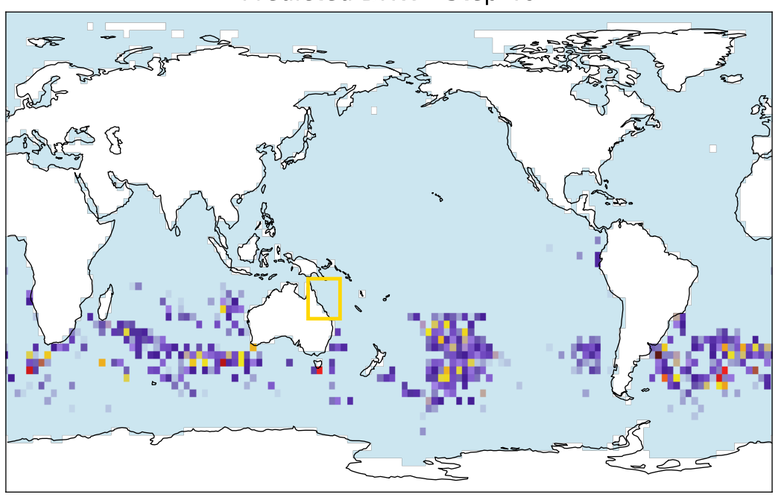
Error (Pred - Actual) - Step 1 [scale ±0.23]



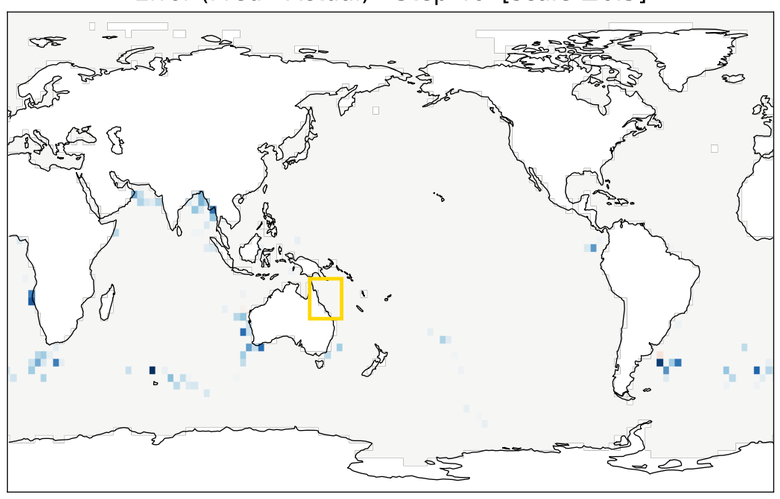
Actual DHW - Step 46



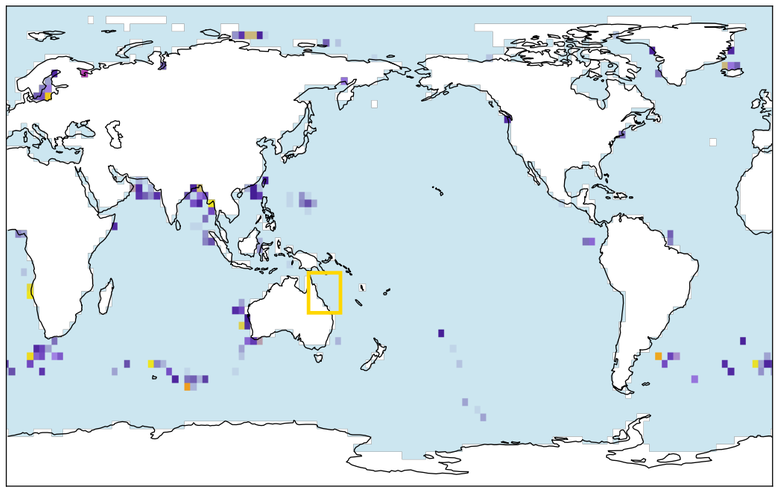
Predicted DHW - Step 46



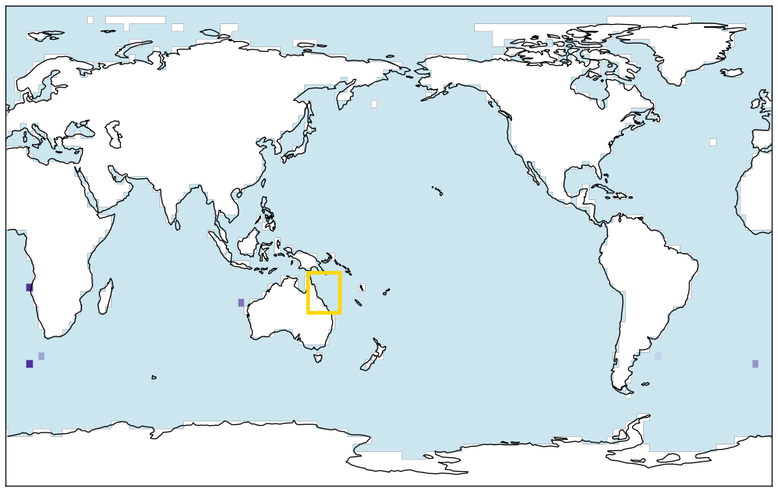
Error (Pred - Actual) - Step 46 [scale ±6.5]



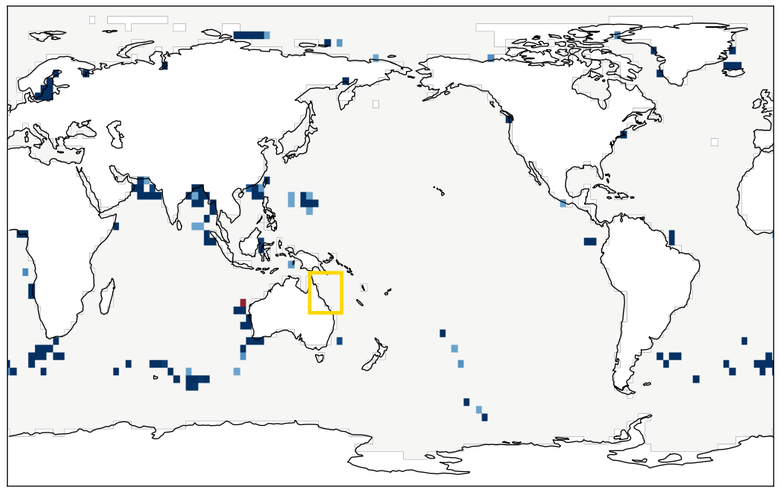
Actual DHW - Step 90

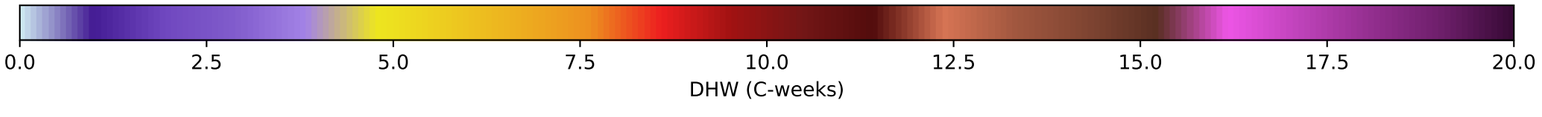


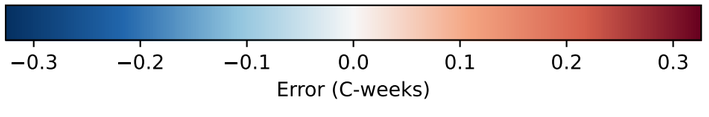
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

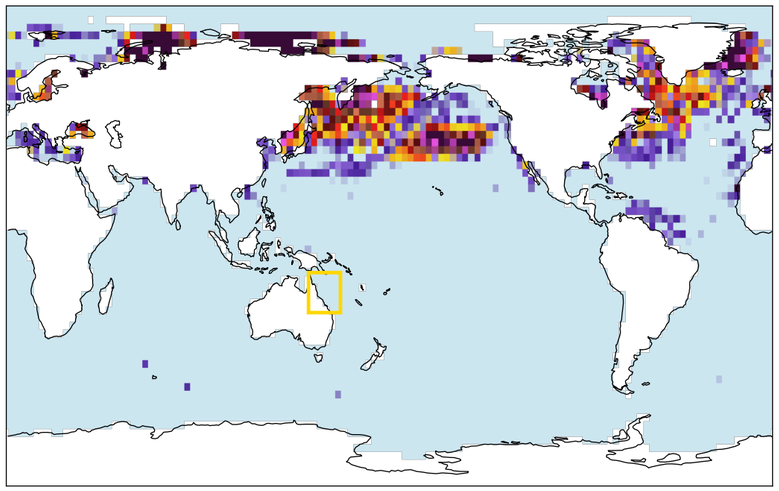




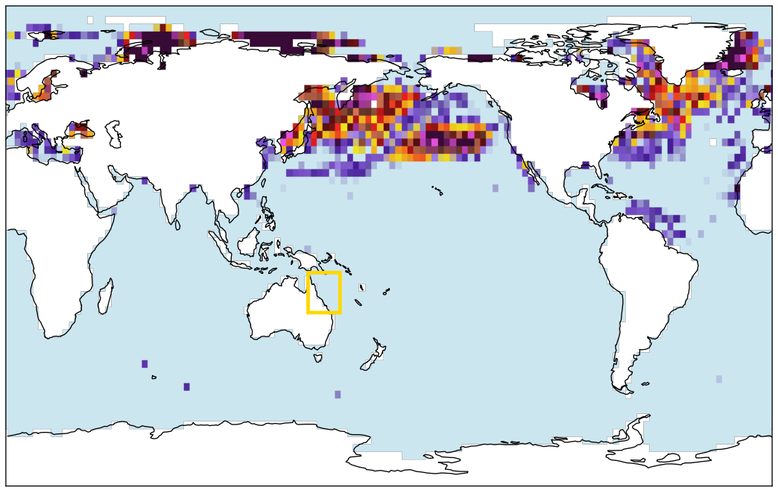


DHW Chain init: 2021-10-03 - Test (C-weeks)

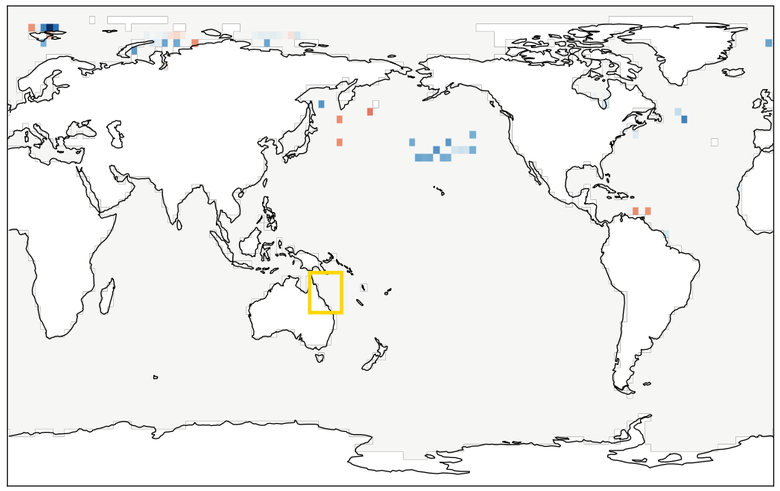
Actual DHW - Step 1



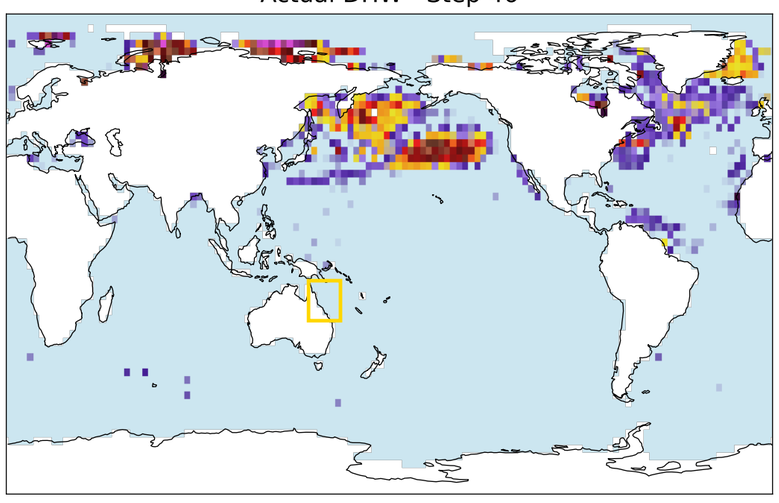
Predicted DHW - Step 1



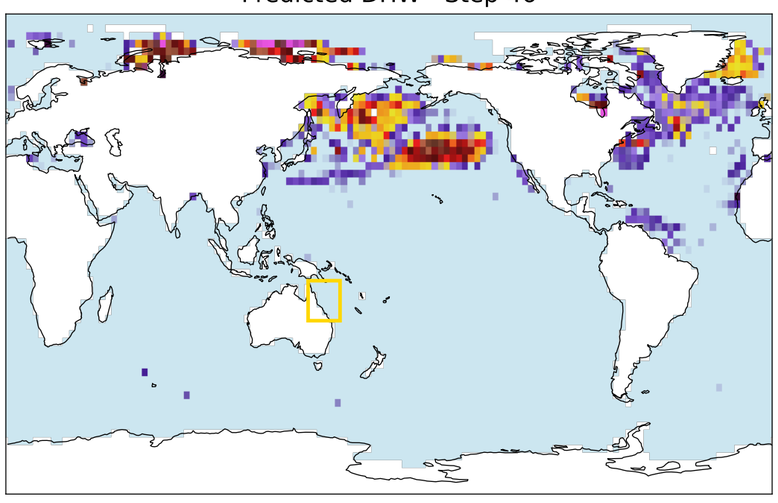
Error (Pred - Actual) - Step 1 [scale ±0.36]



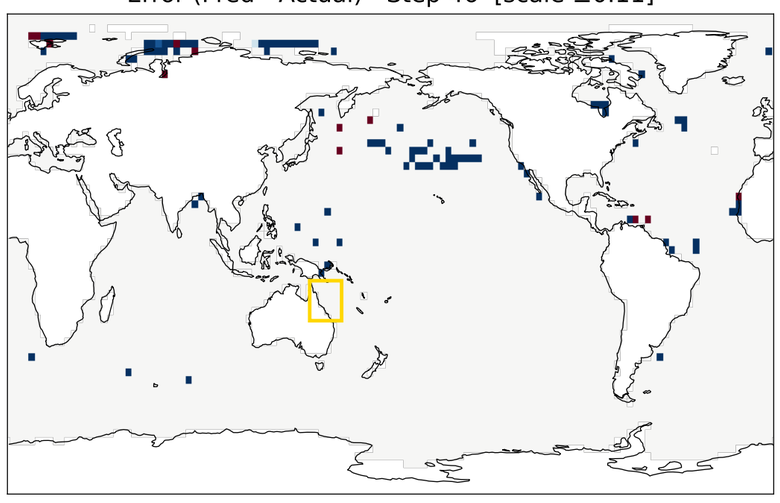
Actual DHW - Step 46



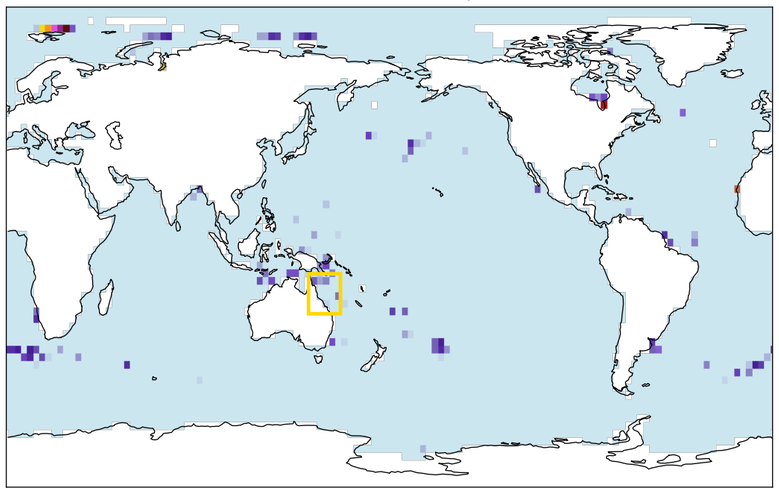
Predicted DHW - Step 46



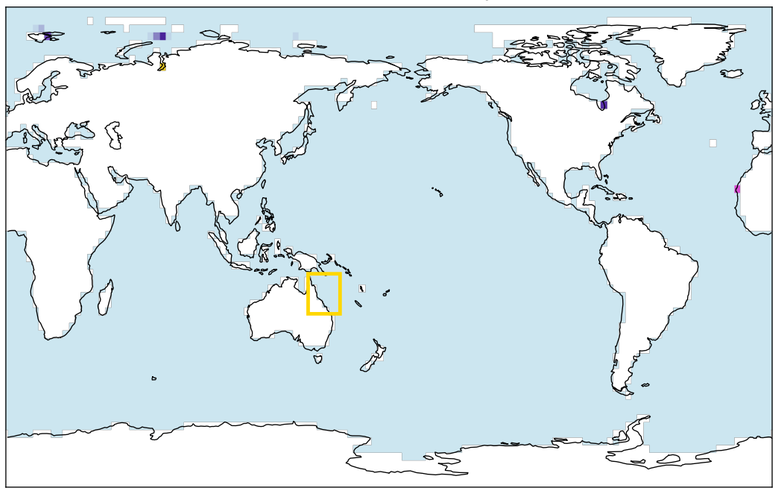
Error (Pred - Actual) - Step 46 [scale ±0.11]



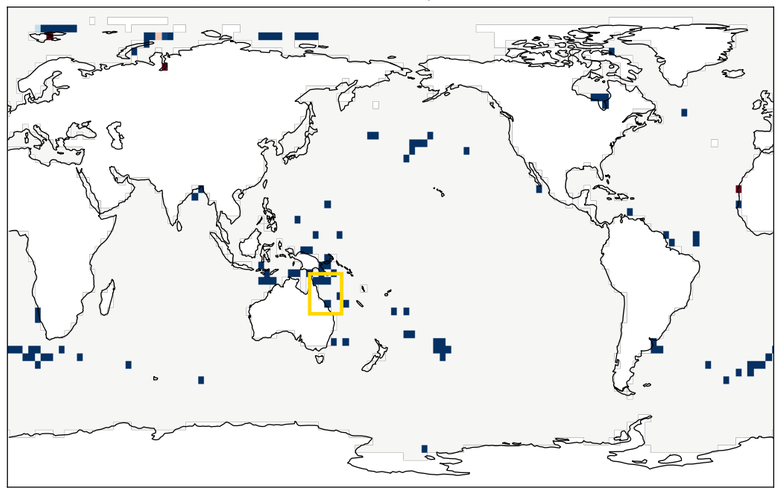
Actual DHW - Step 90

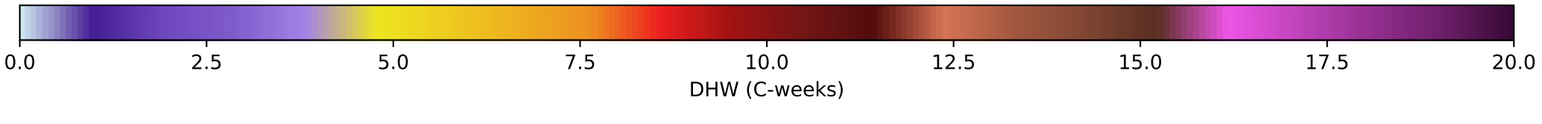


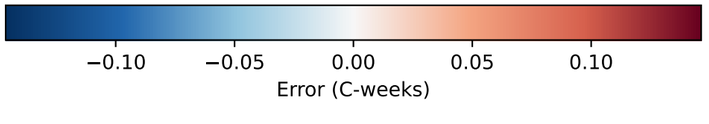
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.15]

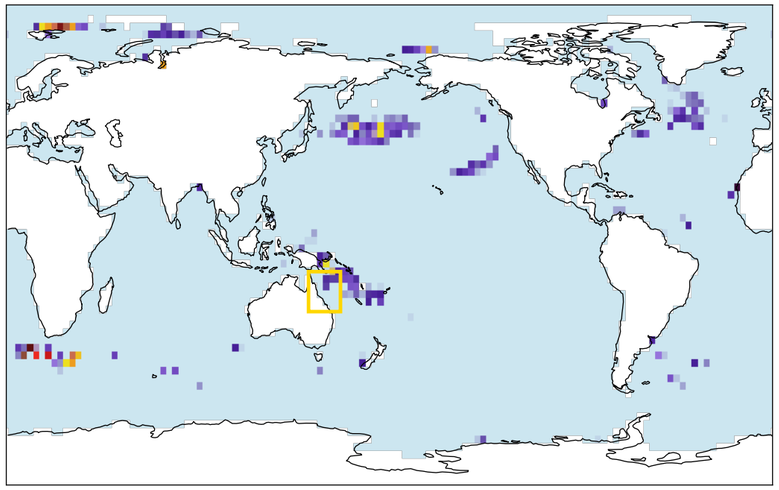




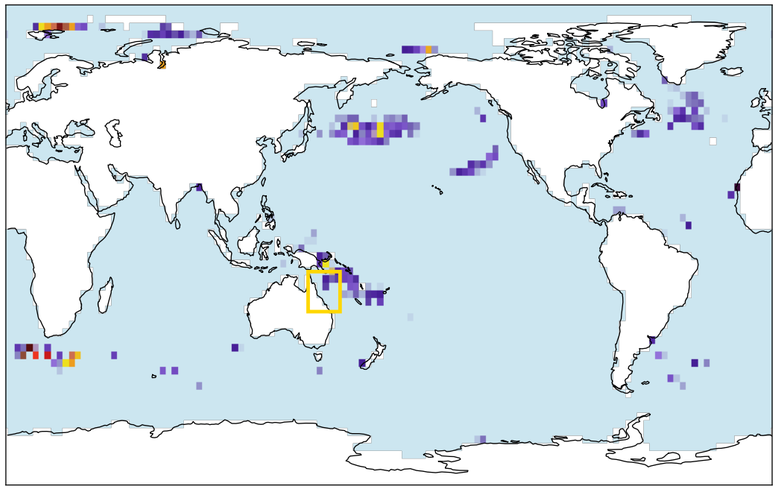


DHW Chain init: 2022-12-27 - Test (C-weeks)

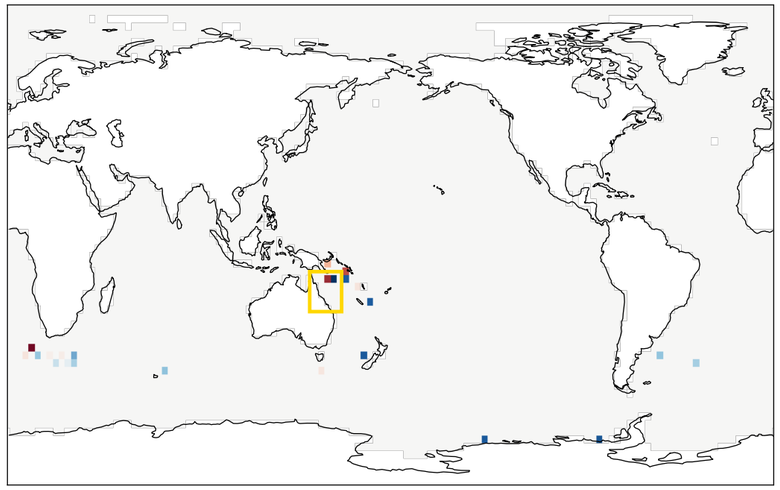
Actual DHW - Step 1



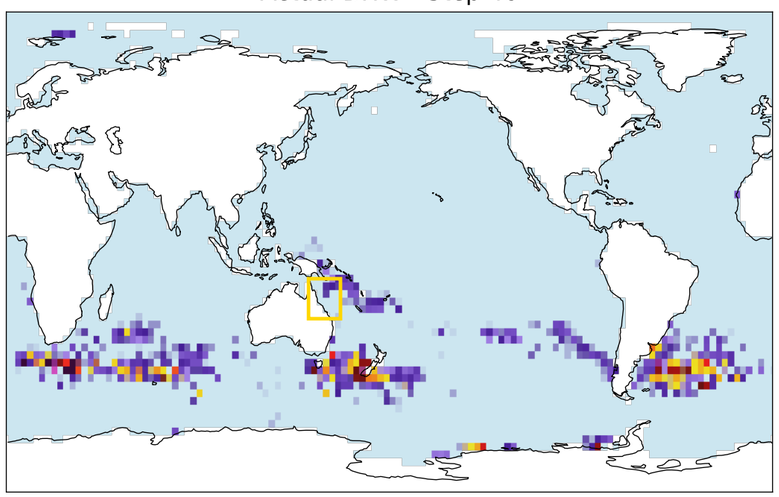
Predicted DHW - Step 1



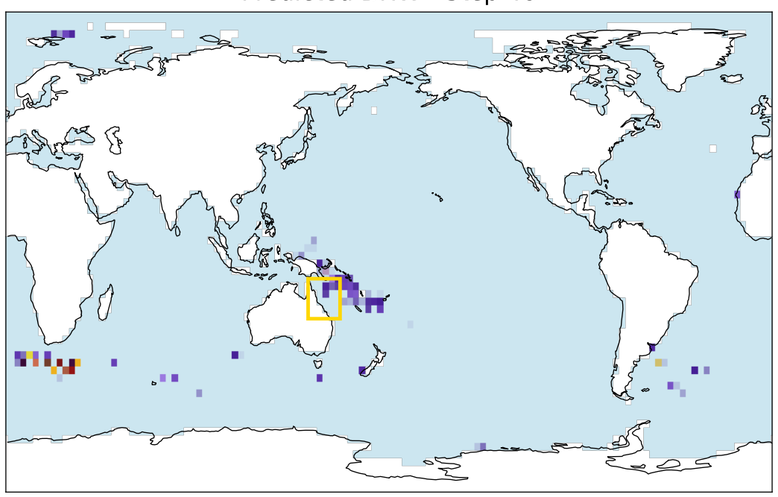
Error (Pred - Actual) - Step 1 [scale ±0.2]



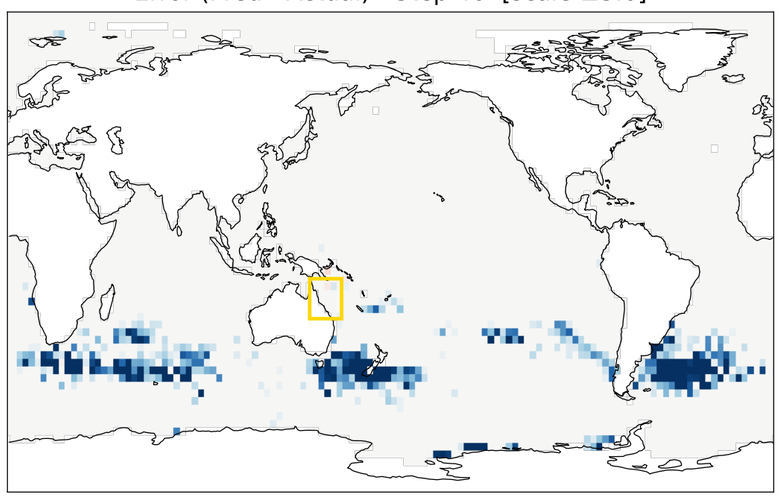
Actual DHW - Step 46



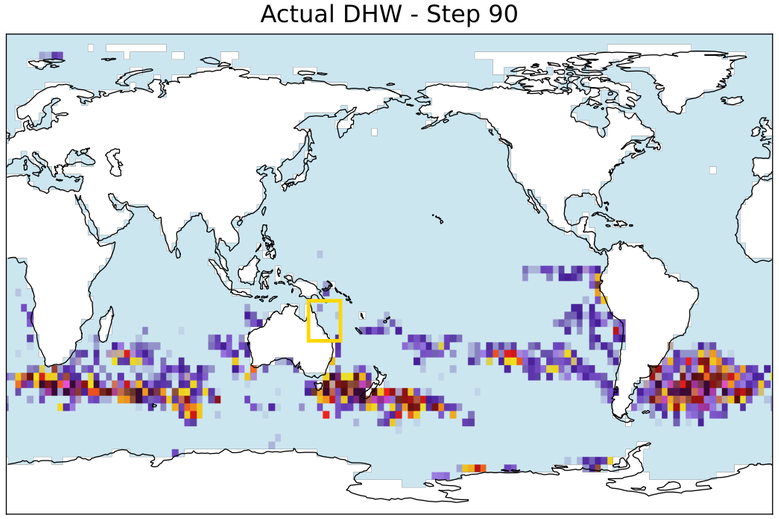
Predicted DHW - Step 46



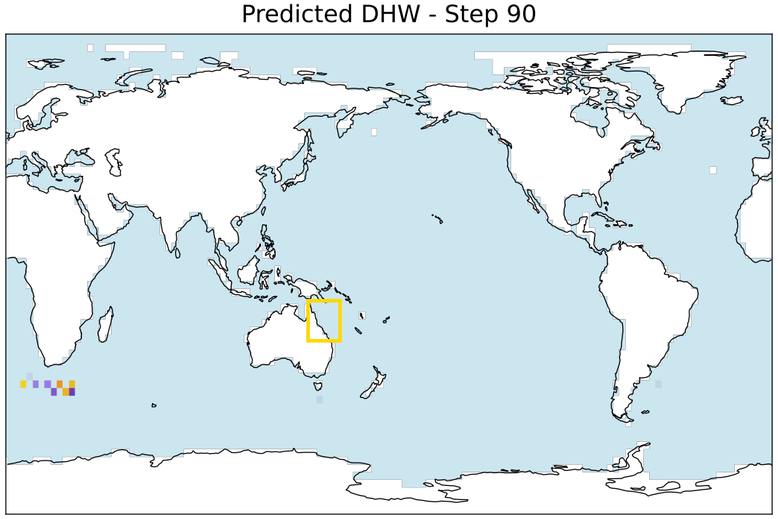
Error (Pred - Actual) - Step 46 [scale ±3.6]



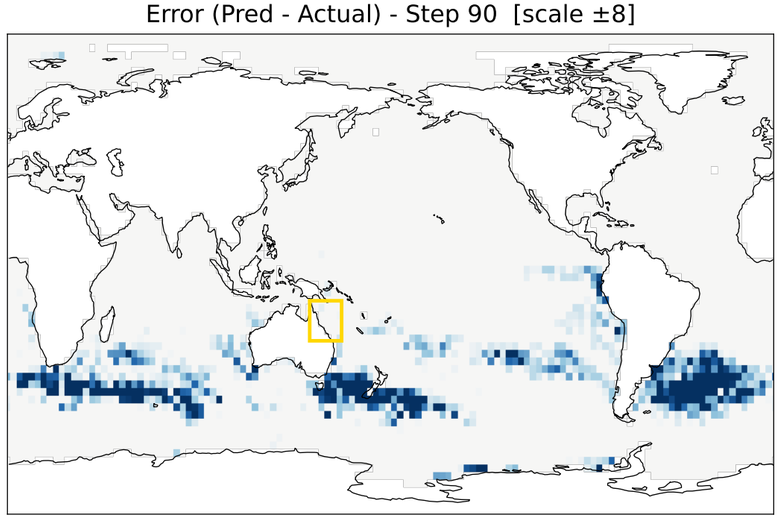
Actual DHW - Step 90

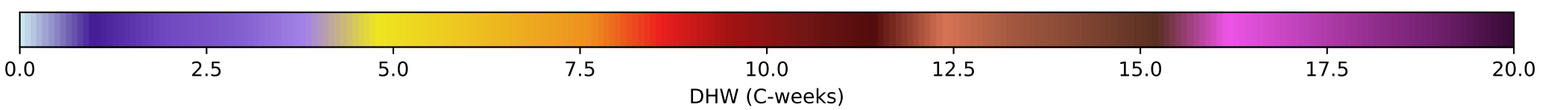


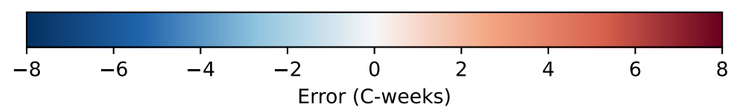
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

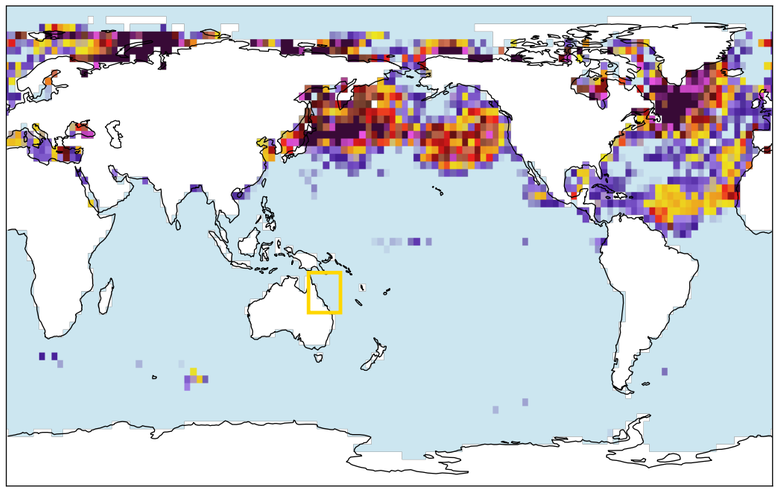




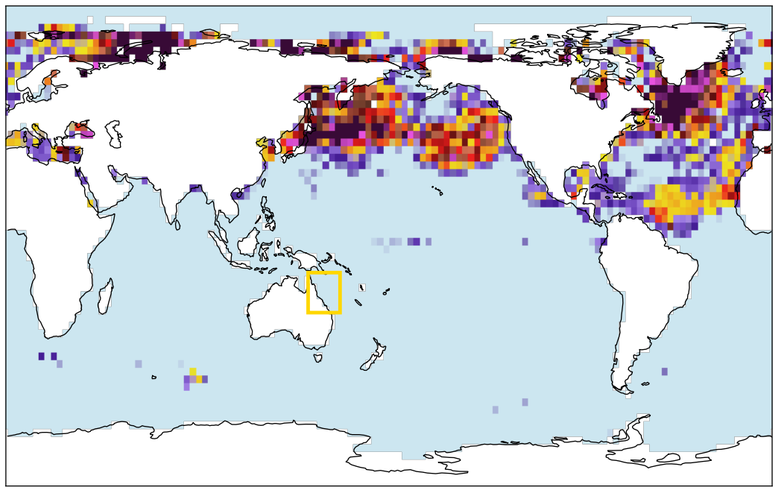


DHW Chain init: 2023-09-23 - Test (C-weeks)

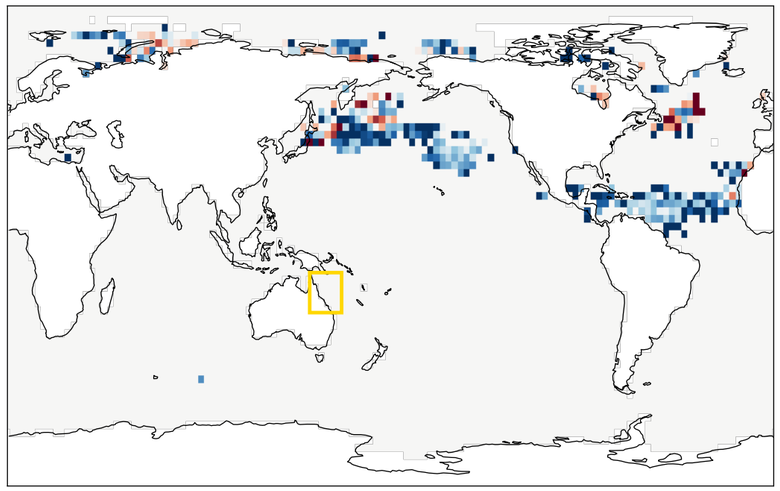
Actual DHW - Step 1



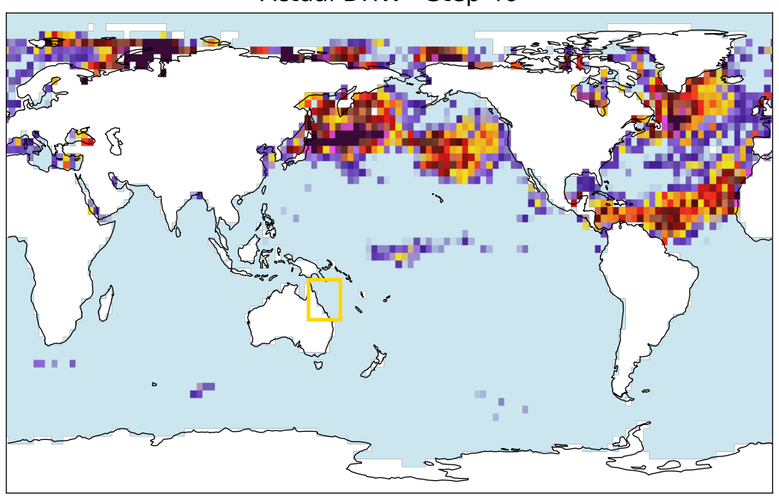
Predicted DHW - Step 1



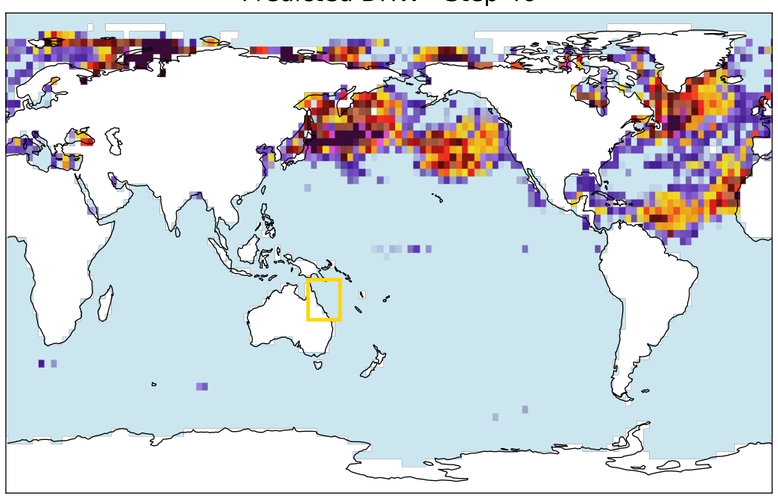
Error (Pred - Actual) - Step 1 [scale ±0.046]



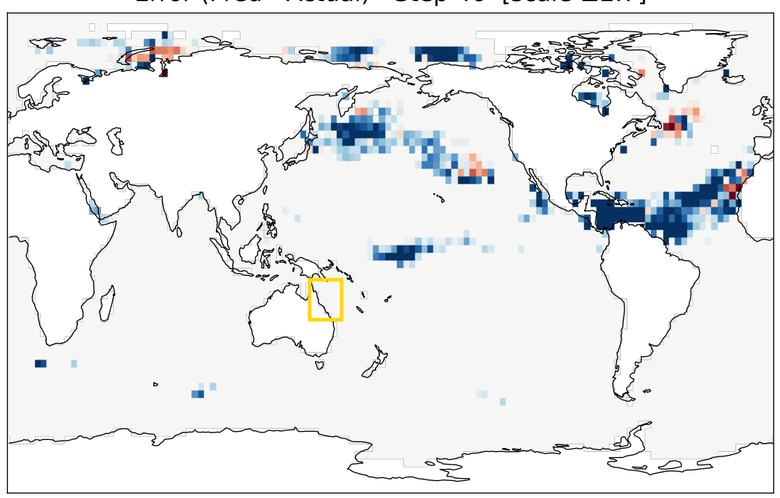
Actual DHW - Step 46



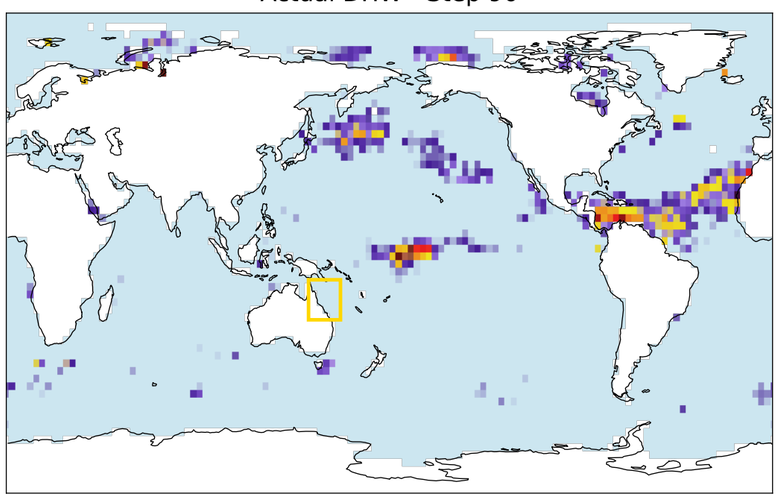
Predicted DHW - Step 46



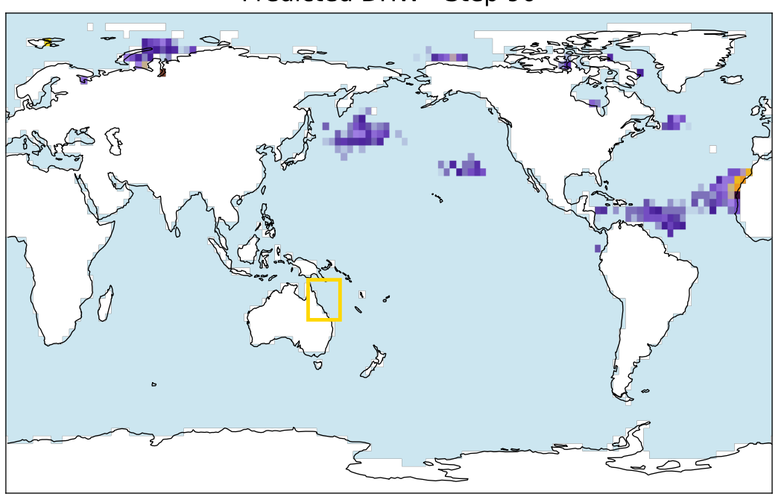
Error (Pred - Actual) - Step 46 [scale ±2.7]



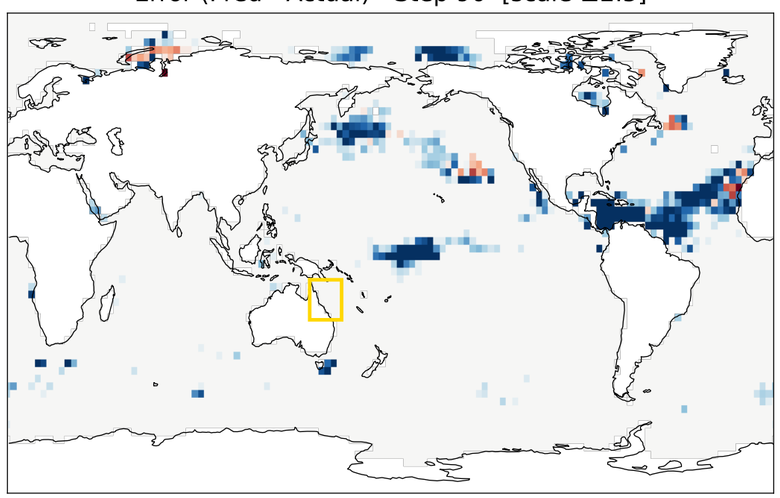
Actual DHW - Step 90

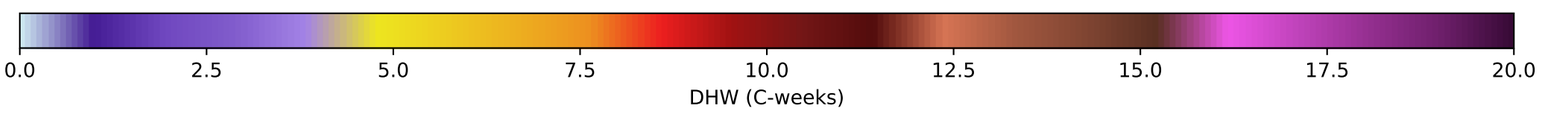


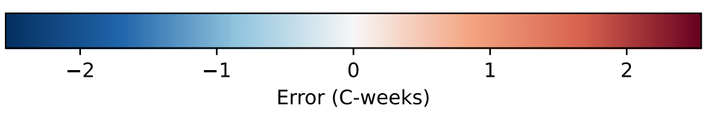
Predicted DHW - Step 90



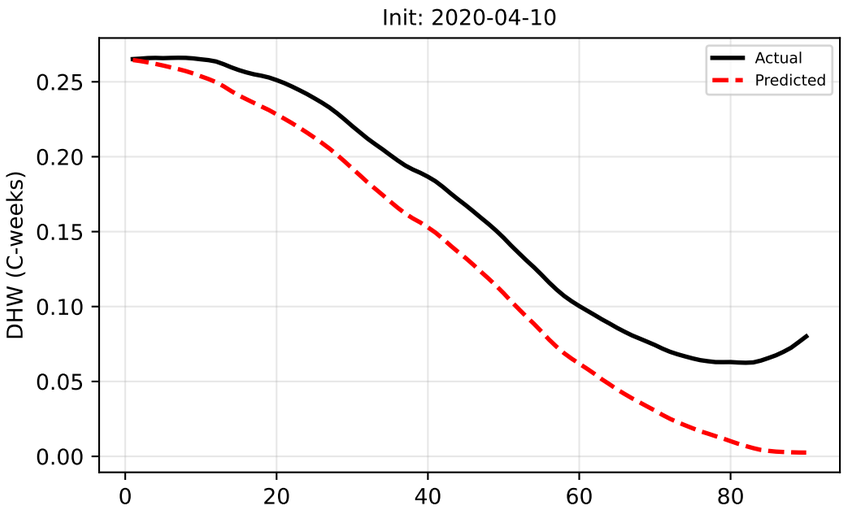
Error (Pred - Actual) - Step 90 [scale ±2.5]

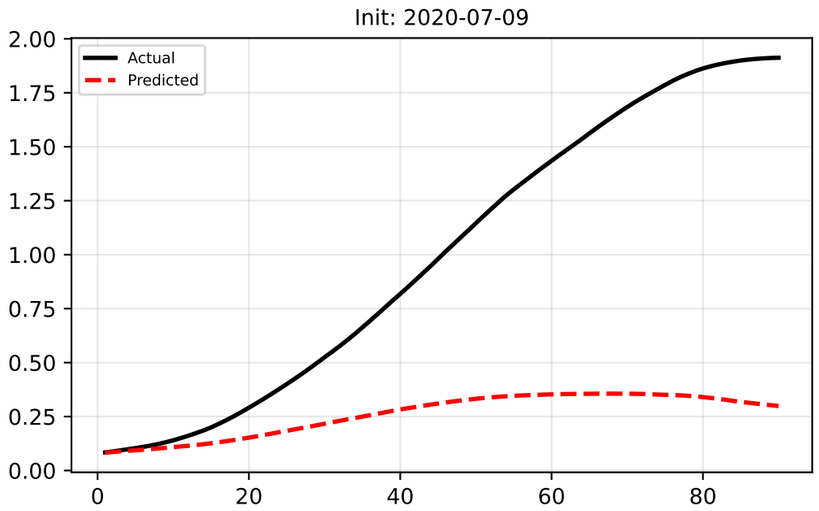


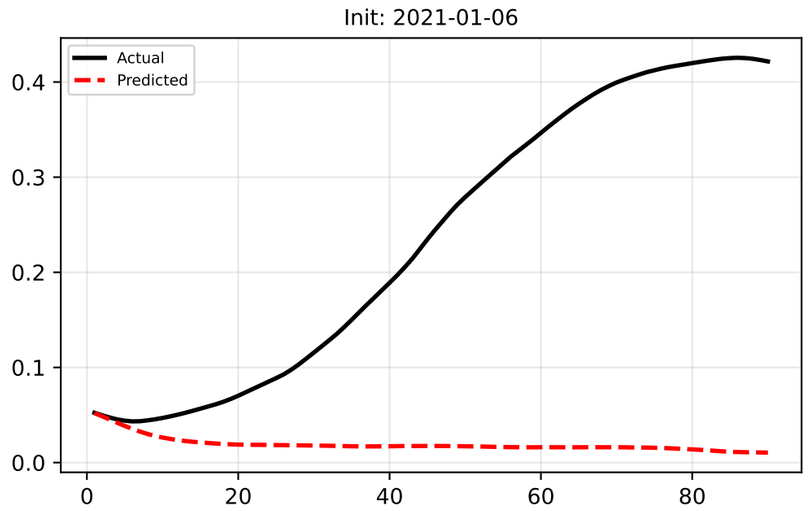


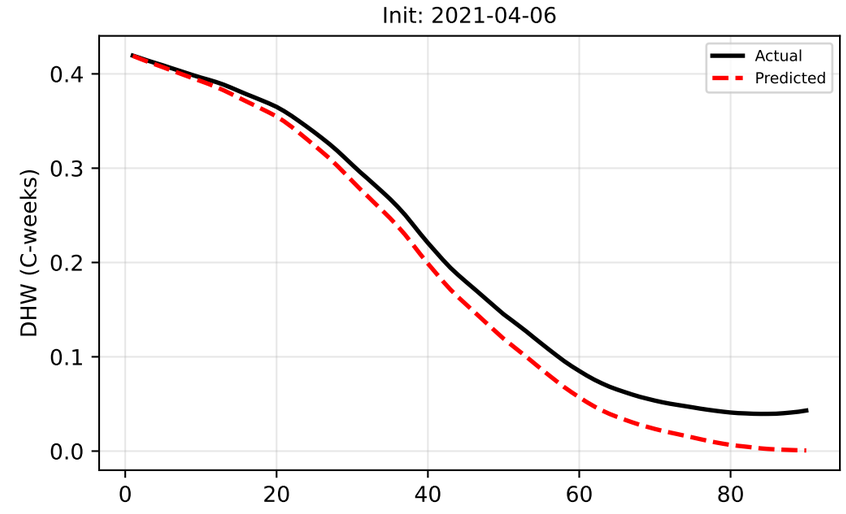


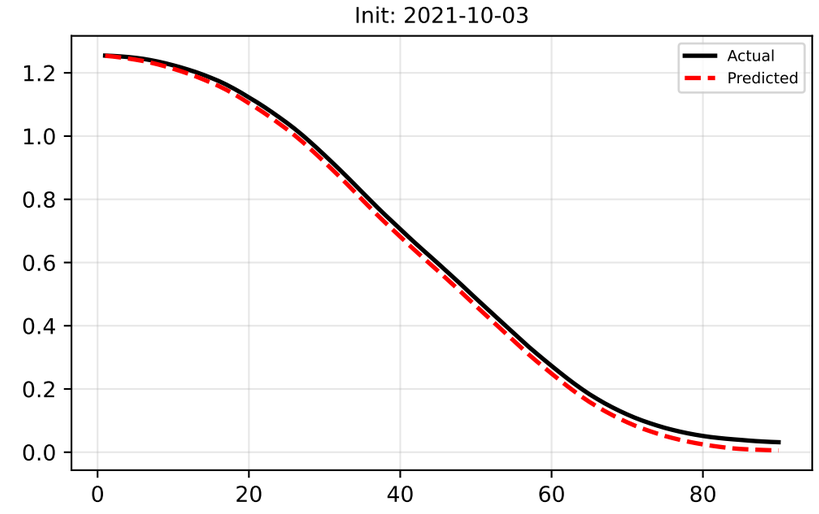
Global-Mean DHW - Test (C-weeks)

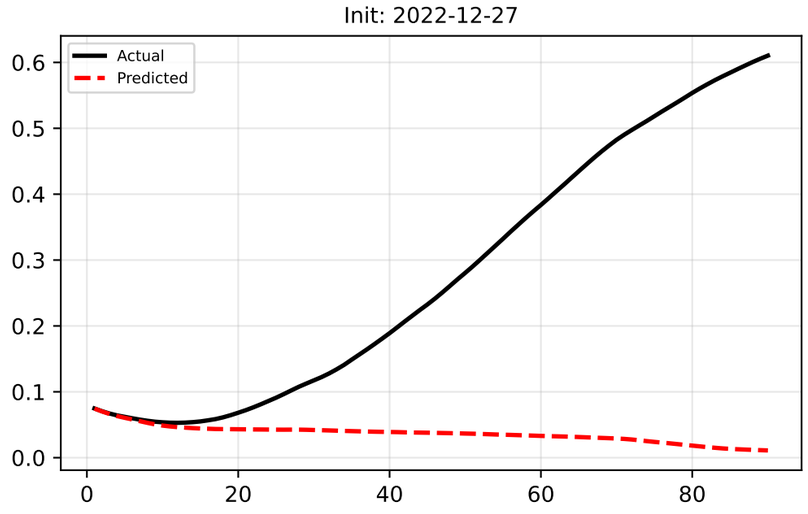


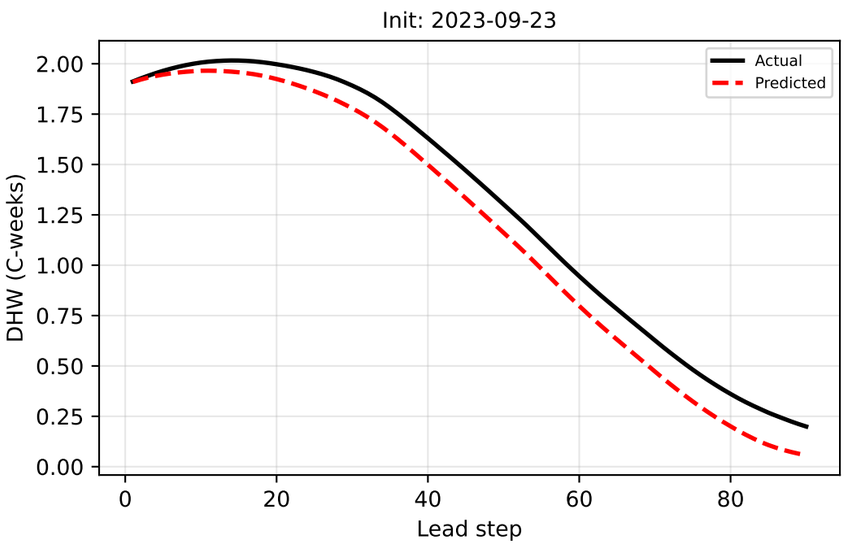




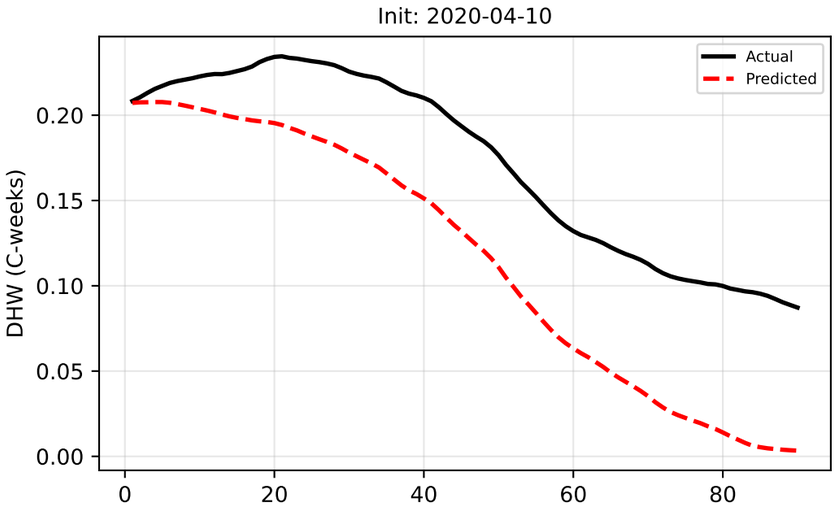


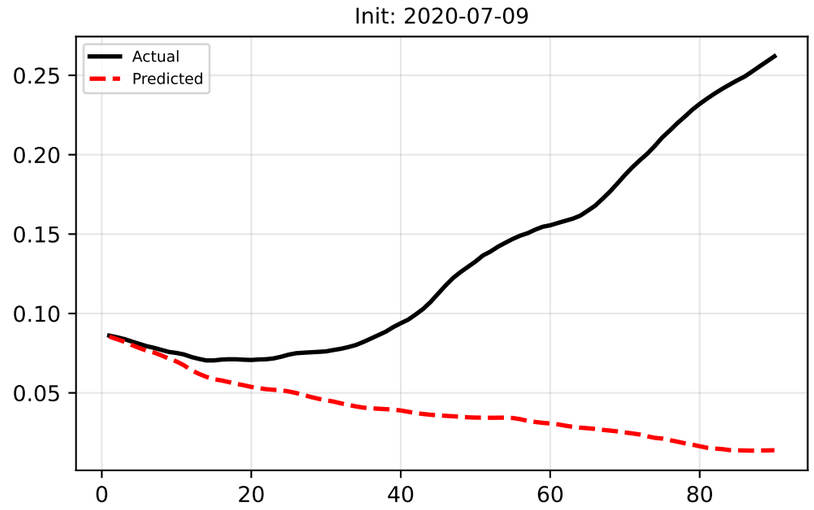


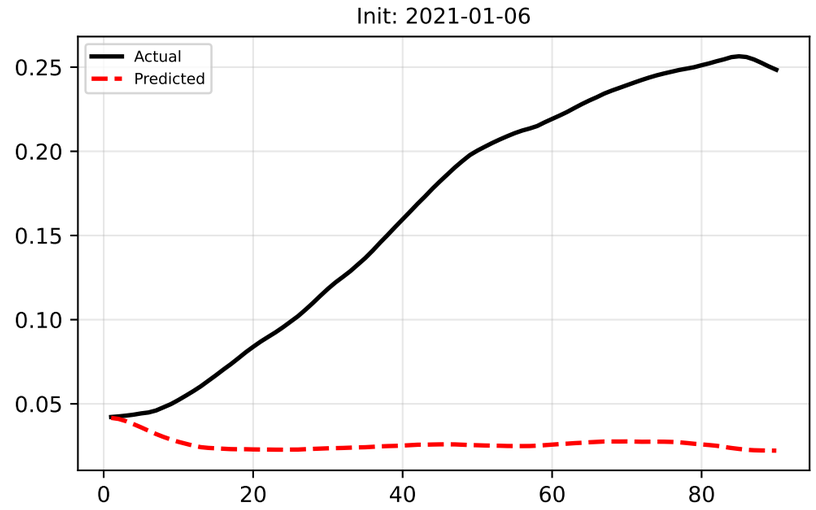


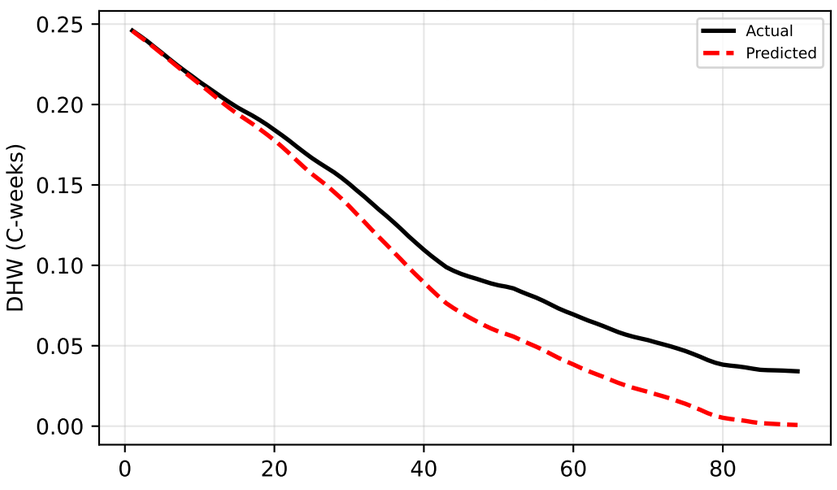


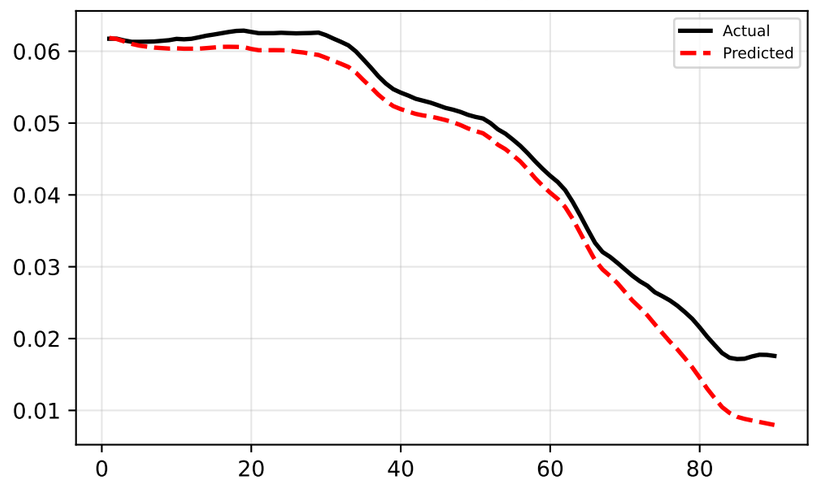
Tropical-Mean DHW (|lat|<=30) - Test (C-weeks)

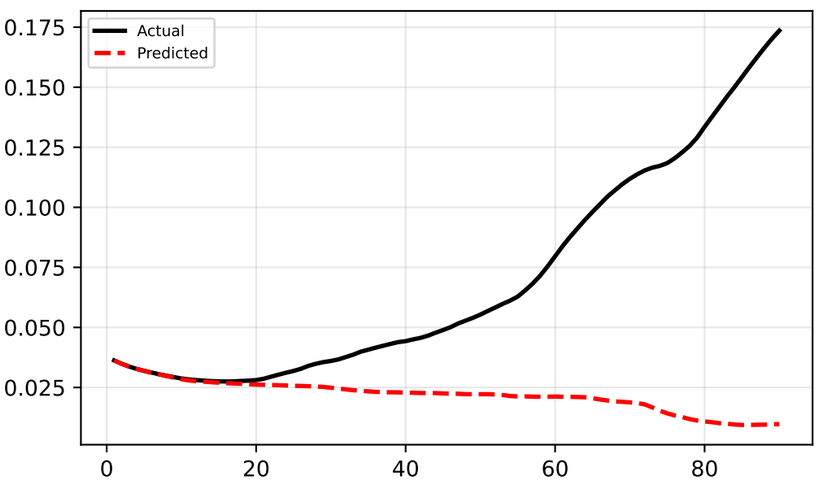


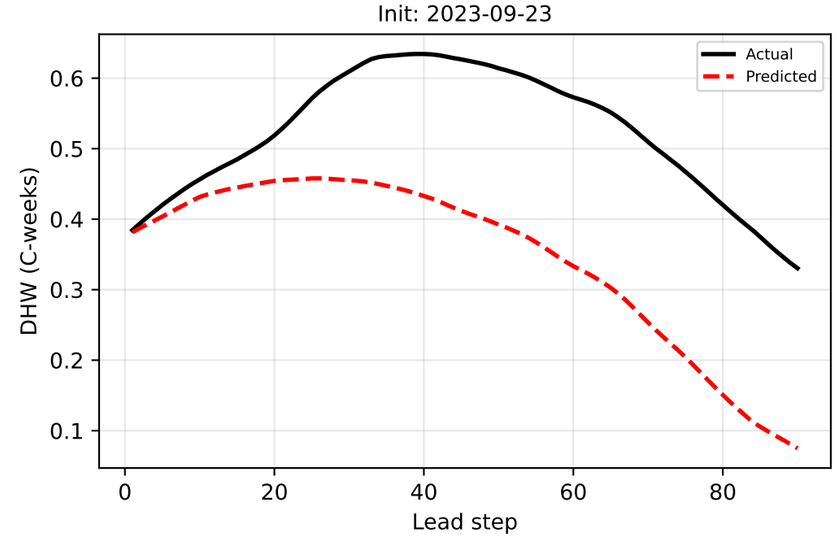




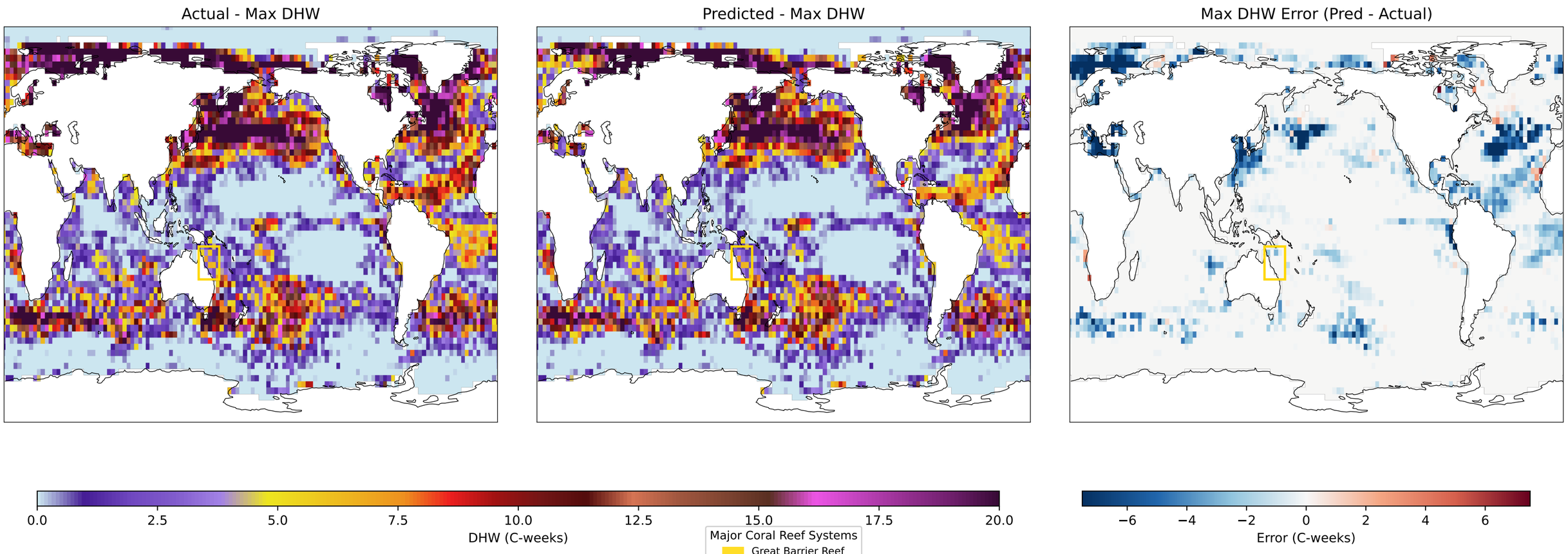


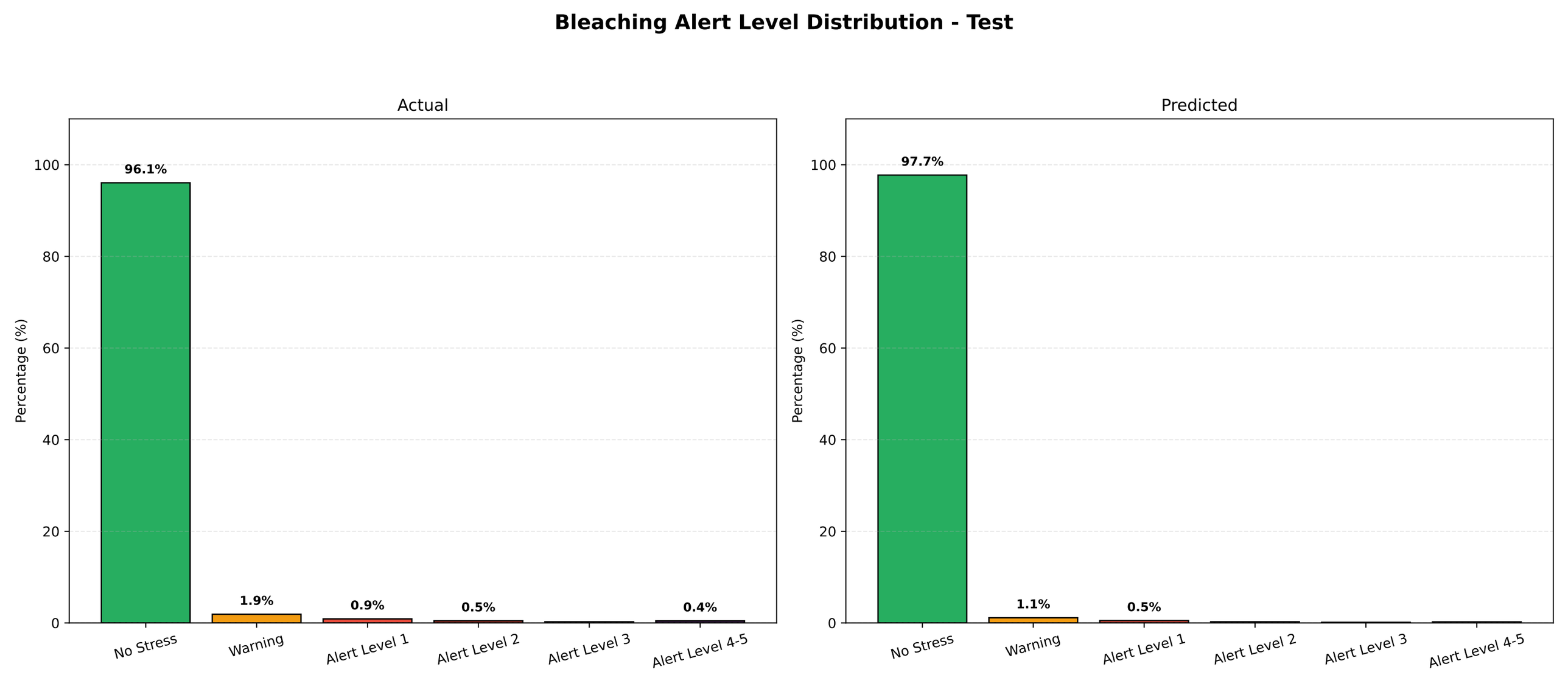






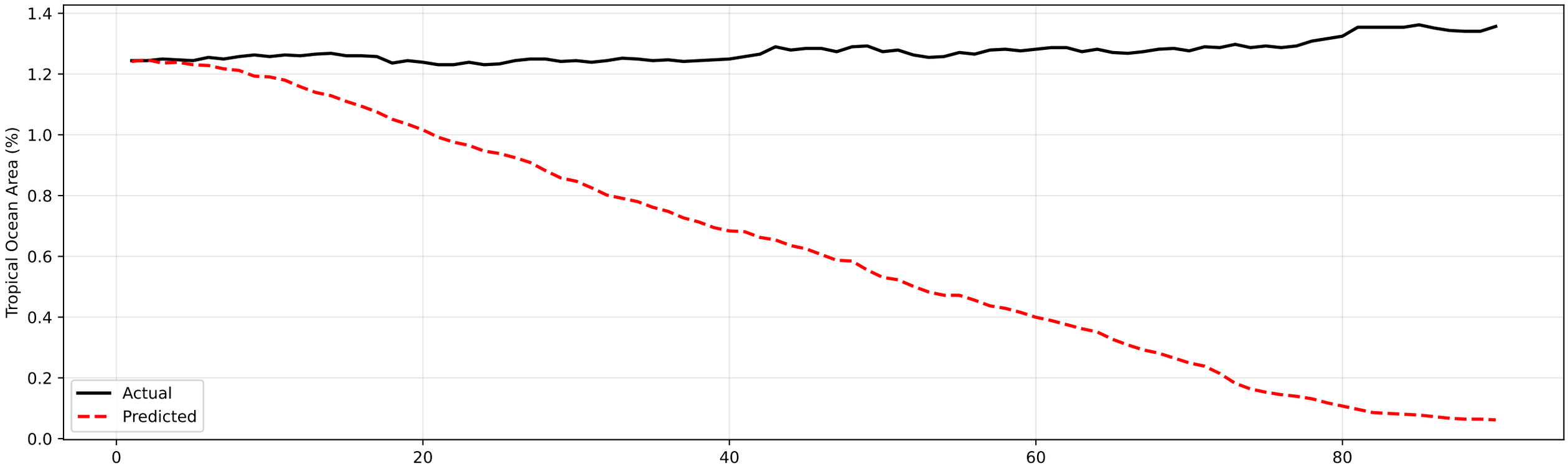
Peak DHW (max over all chains & leads) - Test (C-weeks)



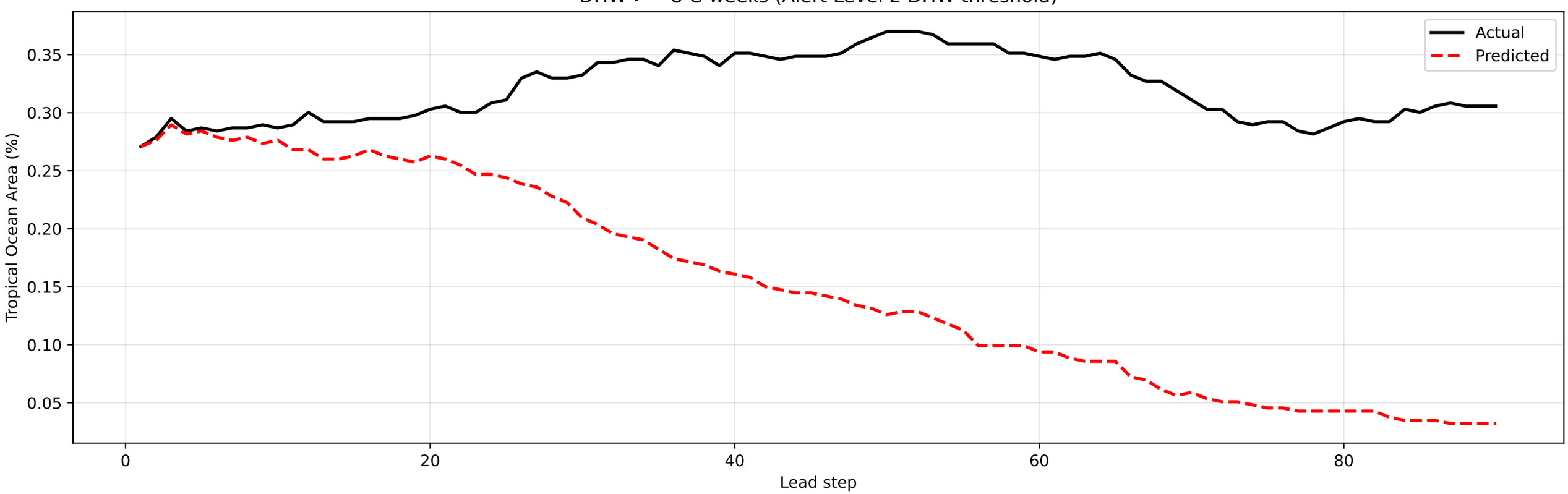

Bleaching Alert Level Distribution - Test
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.1%
1.9%
0.9%
0.5%
0.4%
97.7%
1.1%
0.5%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



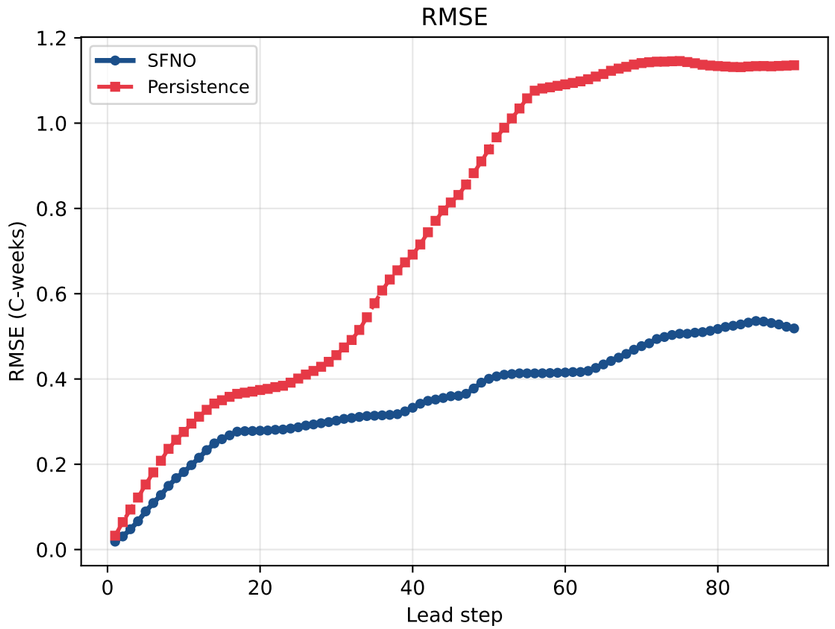
Great Barrier Reef Analysis - Test

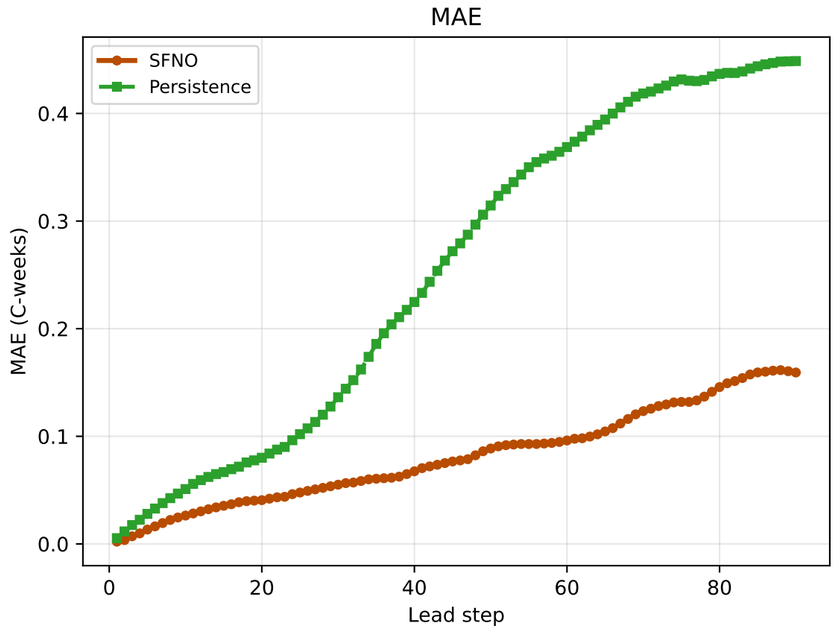
```
                    GREAT BARRIER REEF - TEST
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                      Grid: 5 lat x 5 lon points
```

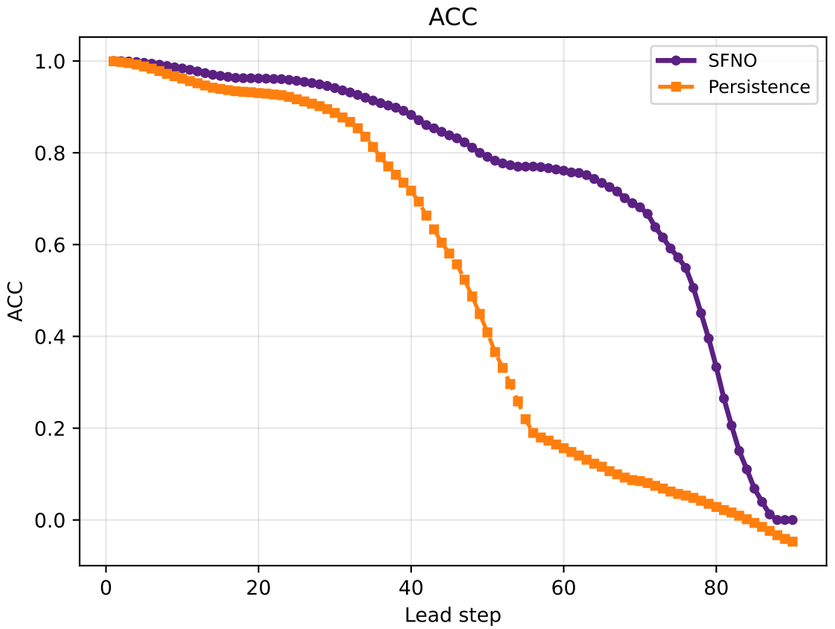
DHW Metrics by Lead Step - Test - GBR (18 chains scored at every lead)

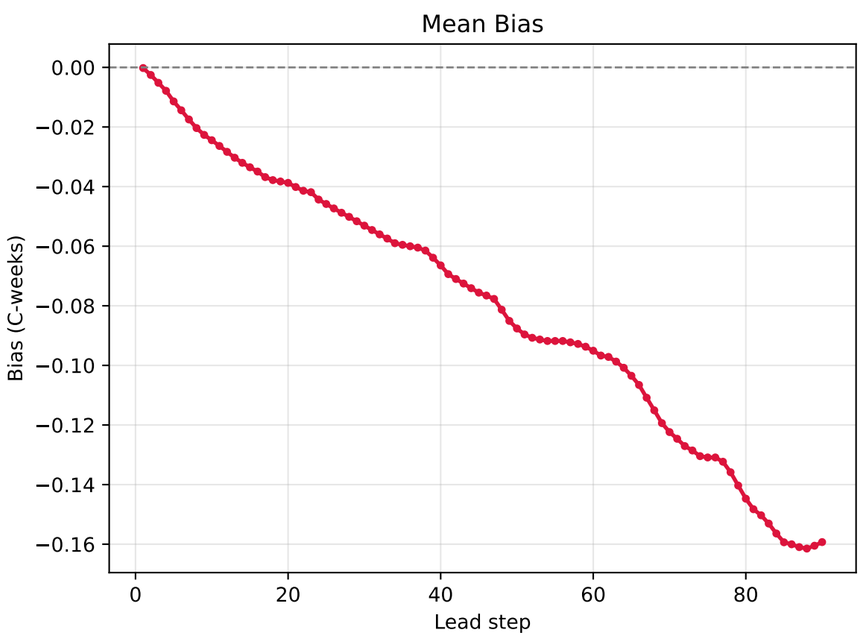
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0186	0.0325	42.8%	0.0021	0.0054	-0.0002	0.9998	0.9995	0.9997
2	0.0308	0.0643	52.2%	0.0036	0.0116	-0.0025	0.9995	0.9979	0.9991
3	0.0477	0.0939	49.1%	0.0071	0.0175	-0.0052	0.9988	0.9955	0.9977
4	0.0663	0.1219	45.6%	0.0098	0.0224	-0.0079	0.9978	0.9924	0.9957
5	0.0895	0.1525	41.4%	0.0133	0.0279	-0.0114	0.9960	0.9881	0.9922
6	0.1091	0.1810	39.7%	0.0163	0.0329	-0.0144	0.9940	0.9834	0.9884
7	0.1278	0.2083	38.6%	0.0194	0.0379	-0.0175	0.9919	0.9781	0.9843
8	0.1493	0.2362	36.8%	0.0223	0.0425	-0.0204	0.9890	0.9721	0.9787
9	0.1675	0.2574	34.9%	0.0246	0.0468	-0.0227	0.9863	0.9669	0.9734
10	0.1818	0.2759	34.1%	0.0264	0.0510	-0.0244	0.9838	0.9619	0.9685
11	0.1981	0.2955	33.0%	0.0283	0.0558	-0.0264	0.9808	0.9562	0.9627
12	0.2153	0.3116	30.9%	0.0303	0.0593	-0.0283	0.9774	0.9515	0.9561
13	0.2332	0.3277	28.8%	0.0322	0.0622	-0.0303	0.9736	0.9466	0.9489
14	0.2490	0.3424	27.3%	0.0339	0.0649	-0.0320	0.9700	0.9419	0.9421
15	0.2588	0.3502	26.1%	0.0354	0.0668	-0.0335	0.9676	0.9393	0.9375
16	0.2679	0.3582	25.2%	0.0369	0.0694	-0.0350	0.9654	0.9366	0.9332
17	0.2765	0.3652	24.3%	0.0387	0.0718	-0.0368	0.9632	0.9343	0.9291
18	0.2777	0.3680	24.5%	0.0397	0.0757	-0.0378	0.9626	0.9329	0.9279
19	0.2781	0.3705	24.9%	0.0402	0.0777	-0.0383	0.9625	0.9319	0.9277
20	0.2787	0.3742	25.5%	0.0407	0.0801	-0.0387	0.9620	0.9302	0.9267
21	0.2793	0.3768	25.9%	0.0421	0.0840	-0.0401	0.9616	0.9289	0.9261
22	0.2809	0.3810	26.3%	0.0433	0.0876	-0.0414	0.9609	0.9271	0.9249
23	0.2815	0.3841	26.7%	0.0438	0.0901	-0.0419	0.9605	0.9256	0.9241
24	0.2839	0.3914	27.5%	0.0463	0.0963	-0.0444	0.9588	0.9217	0.9211
25	0.2868	0.4011	28.5%	0.0478	0.1020	-0.0458	0.9568	0.9167	0.9173
26	0.2909	0.4104	29.1%	0.0493	0.1074	-0.0473	0.9544	0.9118	0.9129
27	0.2934	0.4189	30.0%	0.0507	0.1133	-0.0488	0.9520	0.9070	0.9085
28	0.2963	0.4287	30.9%	0.0521	0.1201	-0.0502	0.9492	0.9017	0.9034
29	0.2990	0.4402	32.1%	0.0536	0.1277	-0.0516	0.9456	0.8952	0.8969
30	0.3024	0.4558	33.7%	0.0550	0.1363	-0.0531	0.9410	0.8868	0.8888
31	0.3065	0.4739	35.3%	0.0565	0.1442	-0.0546	0.9360	0.8770	0.8798
32	0.3085	0.4915	37.2%	0.0571	0.1522	-0.0560	0.9315	0.8672	0.8718
33	0.3110	0.5150	39.6%	0.0585	0.1621	-0.0574	0.9262	0.8535	0.8623
34	0.3131	0.5448	42.5%	0.0600	0.1738	-0.0590	0.9200	0.8349	0.8514
35	0.3139	0.5777	45.7%	0.0606	0.1858	-0.0595	0.9138	0.8127	0.8405
36	0.3148	0.6076	48.2%	0.0611	0.1958	-0.0600	0.9082	0.7908	0.8307
37	0.3158	0.6333	50.1%	0.0615	0.2040	-0.0605	0.9031	0.7703	0.8219
38	0.3175	0.6548	51.5%	0.0625	0.2107	-0.0615	0.8983	0.7521	0.8137
39	0.3241	0.6735	51.9%	0.0649	0.2175	-0.0639	0.8916	0.7349	0.8024
40	0.3326	0.6918	51.9%	0.0675	0.2248	-0.0664	0.8826	0.7175	0.7871
41	0.3420	0.7155	52.2%	0.0704	0.2332	-0.0694	0.8711	0.6935	0.7677
42	0.3487	0.7441	53.1%	0.0720	0.2436	-0.0710	0.8605	0.6632	0.7494
43	0.3518	0.7709	54.4%	0.0736	0.2537	-0.0725	0.8532	0.6331	0.7363
44	0.3555	0.7952	55.3%	0.0751	0.2632	-0.0741	0.8457	0.6042	0.7225
45	0.3596	0.8139	55.8%	0.0766	0.2720	-0.0756	0.8381	0.5806	0.7079
46	0.3605	0.8315	56.6%	0.0776	0.2793	-0.0766	0.8316	0.5570	0.6957
47	0.3653	0.8560	57.3%	0.0787	0.2874	-0.0777	0.8228	0.5236	0.6786
48	0.3777	0.8827	57.2%	0.0823	0.2967	-0.0813	0.8112	0.4870	0.6566
49	0.3914	0.9103	57.0%	0.0861	0.3059	-0.0851	0.7998	0.4490	0.6350
50	0.4005	0.9384	57.3%	0.0886	0.3145	-0.0876	0.7913	0.4087	0.6179
51	0.4061	0.9668	58.0%	0.0907	0.3234	-0.0896	0.7829	0.3658	0.6026
52	0.4100	0.9892	58.5%	0.0918	0.3297	-0.0907	0.7769	0.3313	0.5922
53	0.4116	1.0111	59.3%	0.0923	0.3362	-0.0913	0.7731	0.2963	0.5843
54	0.4132	1.0344	60.1%	0.0928	0.3432	-0.0918	0.7698	0.2584	0.5769
55	0.4132	1.0582	61.0%	0.0928	0.3501	-0.0918	0.7698	0.2195	0.5744
56	0.4132	1.0761	61.6%	0.0928	0.3548	-0.0918	0.7700	0.1896	0.5726
57	0.4134	1.0812	61.8%	0.0933	0.3581	-0.0923	0.7688	0.1797	0.5681
58	0.4139	1.0841	61.8%	0.0938	0.3607	-0.0928	0.7664	0.1728	0.5624
59	0.4146	1.0875	61.9%	0.0948	0.3643	-0.0937	0.7637	0.1645	0.5555
60	0.4152	1.0909	61.9%	0.0961	0.3688	-0.0951	0.7611	0.1564	0.5489
61	0.4163	1.0943	62.0%	0.0977	0.3738	-0.0967	0.7573	0.1481	0.5394
62	0.4164	1.0980	62.1%	0.0982	0.3785	-0.0972	0.7561	0.1404	0.5333
63	0.4189	1.1030	62.0%	0.0998	0.3843	-0.0987	0.7514	0.1313	0.5254
64	0.4259	1.1092	61.6%	0.1018	0.3894	-0.1008	0.7430	0.1227	0.5144
65	0.4341	1.1152	61.1%	0.1045	0.3943	-0.1035	0.7344	0.1160	0.5044
66	0.4422	1.1228	60.6%	0.1076	0.3999	-0.1066	0.7253	0.1063	0.4939
67	0.4501	1.1280	60.1%	0.1119	0.4056	-0.1109	0.7157	0.0994	0.4834
68	0.4588	1.1321	59.5%	0.1161	0.4108	-0.1151	0.7011	0.0924	0.4664
69	0.4686	1.1374	58.8%	0.1204	0.4156	-0.1194	0.6902	0.0870	0.4552
70	0.4771	1.1409	58.2%	0.1234	0.4186	-0.1224	0.6813	0.0847	0.4461
71	0.4836	1.1428	57.7%	0.1257	0.4203	-0.1246	0.6670	0.0805	0.4303
72	0.4938	1.1442	56.8%	0.1281	0.4232	-0.1271	0.6380	0.0744	0.3989
73	0.4985	1.1443	56.4%	0.1296	0.4259	-0.1285	0.6154	0.0684	0.3727
74	0.5032	1.1451	56.1%	0.1315	0.4297	-0.1304	0.5915	0.0619	0.3481
75	0.5061	1.1456	55.8%	0.1319	0.4316	-0.1309	0.5720	0.0567	0.3280
76	0.5061	1.1433	55.7%	0.1319	0.4305	-0.1309	0.5491	0.0532	0.3074
77	0.5086	1.1405	55.4%	0.1334	0.4300	-0.1323	0.5056	0.0482	0.2716
78	0.5099	1.1370	55.1%	0.1369	0.4312	-0.1359	0.4506	0.0419	0.2302
79	0.5129	1.1351	54.8%	0.1413	0.4344	-0.1403	0.3954	0.0353	0.1914
80	0.5173	1.1339	54.4%	0.1458	0.4368	-0.1447	0.3330	0.0284	0.1483
81	0.5221	1.1328	53.9%	0.1493	0.4377	-0.1483	0.2646	0.0216	0.1013
82	0.5249	1.1317	53.6%	0.1513	0.4376	-0.1503	0.2057	0.0162	0.0603
83	0.5280	1.1313	53.3%	0.1541	0.4390	-0.1531	0.1505	0.0092	0.0209
84	0.5322	1.1328	53.0%	0.1574	0.4417	-0.1564	0.1102	0.0015	-0.0089
85	0.5360	1.1337	52.7%	0.1594	0.4438	-0.1594	0.0682	-0.0065	-0.0419
86	0.5348	1.1340	52.8%	0.1600	0.4457	-0.1600	0.0394	-0.0150	-0.0656
87	0.5313	1.1333	53.1%	0.1610	0.4469	-0.1610	0.0122	-0.0240	-0.0902
88	0.5279	1.1341	53.5%	0.1615	0.4482	-0.1615	0.0000	-0.0332	-0.1032
89	0.5225	1.1349	54.0%	0.1605	0.4484	-0.1605	-0.0000	-0.0412	-0.1042
90	0.5186	1.1357	54.3%	0.1593	0.4487	-0.1593	0.0000	-0.0472	-0.1042

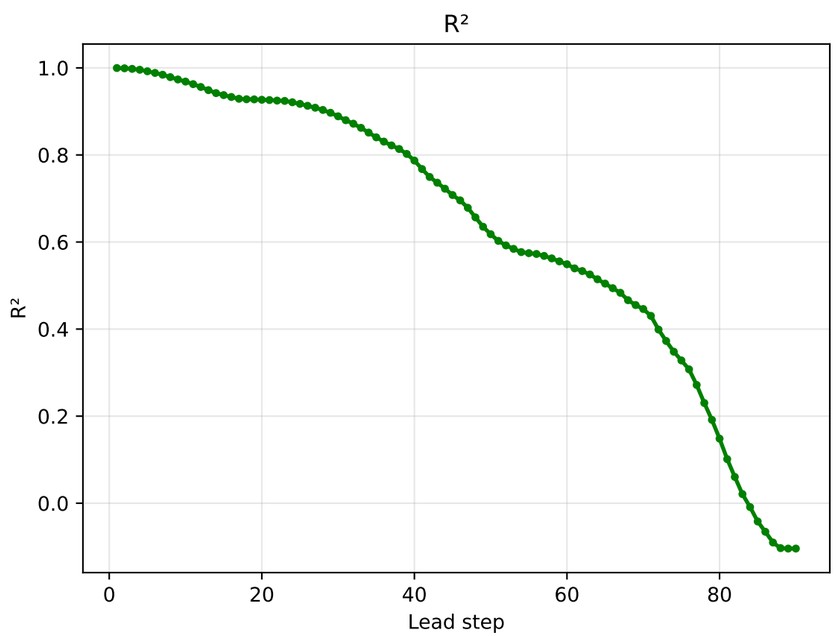
DHW Forecast Skill vs Lead Step - Test - GBR

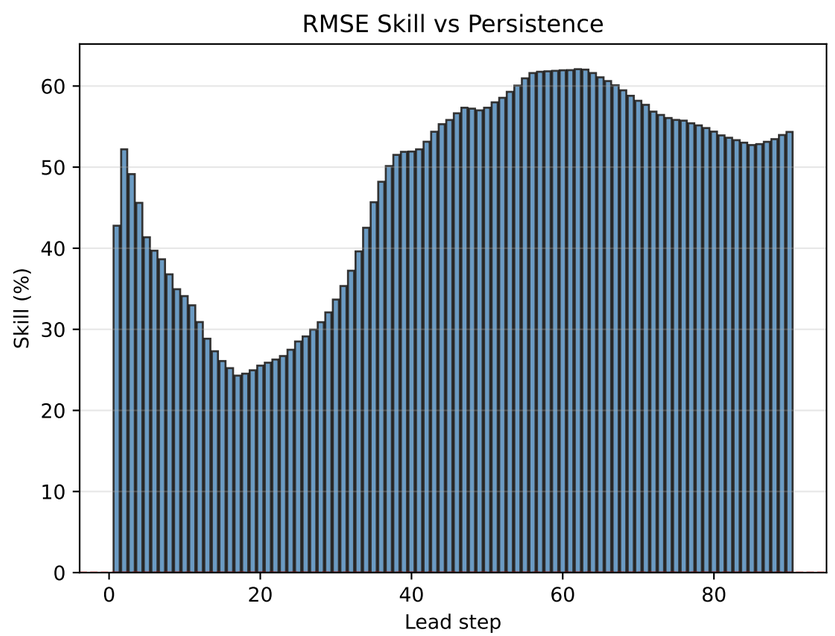


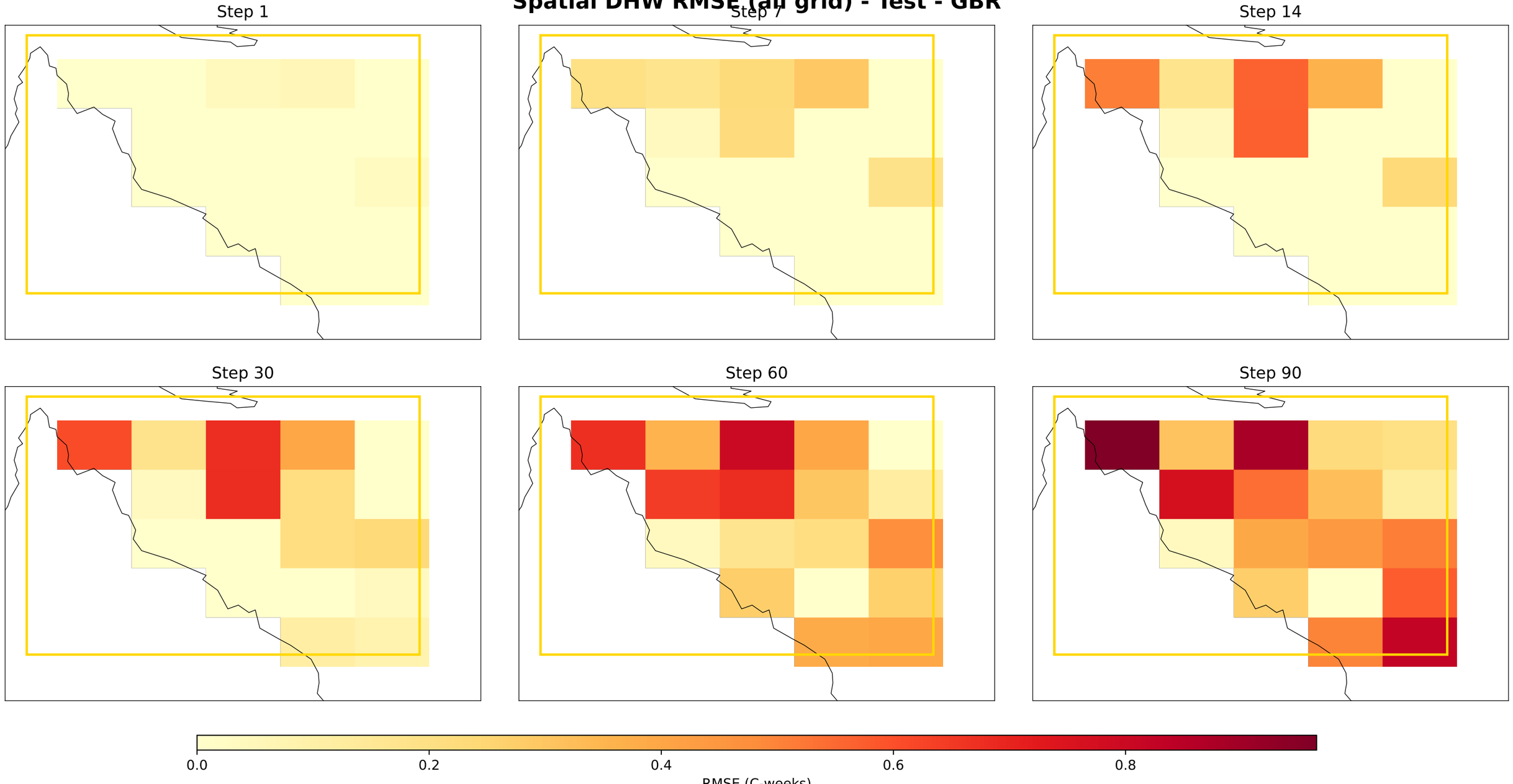


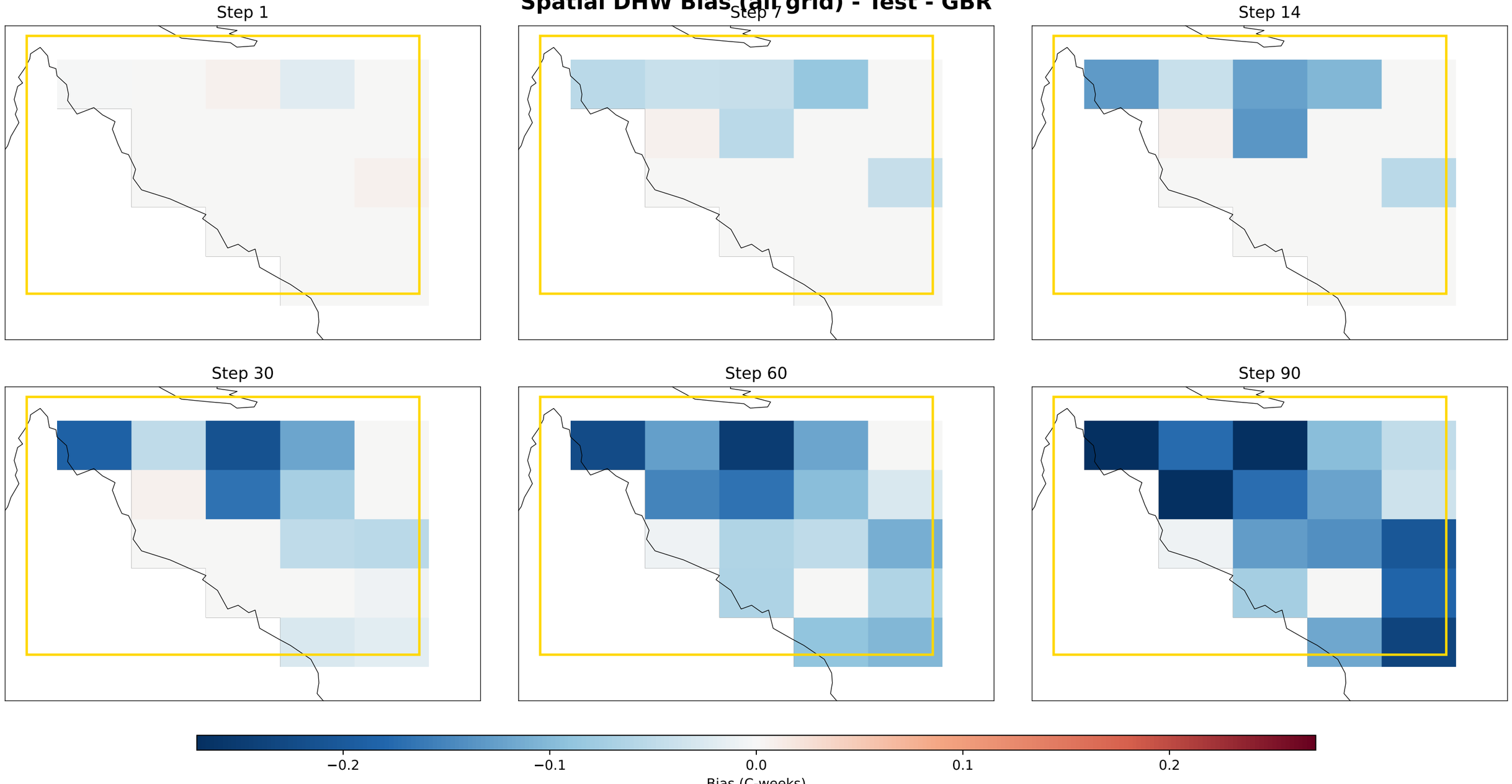


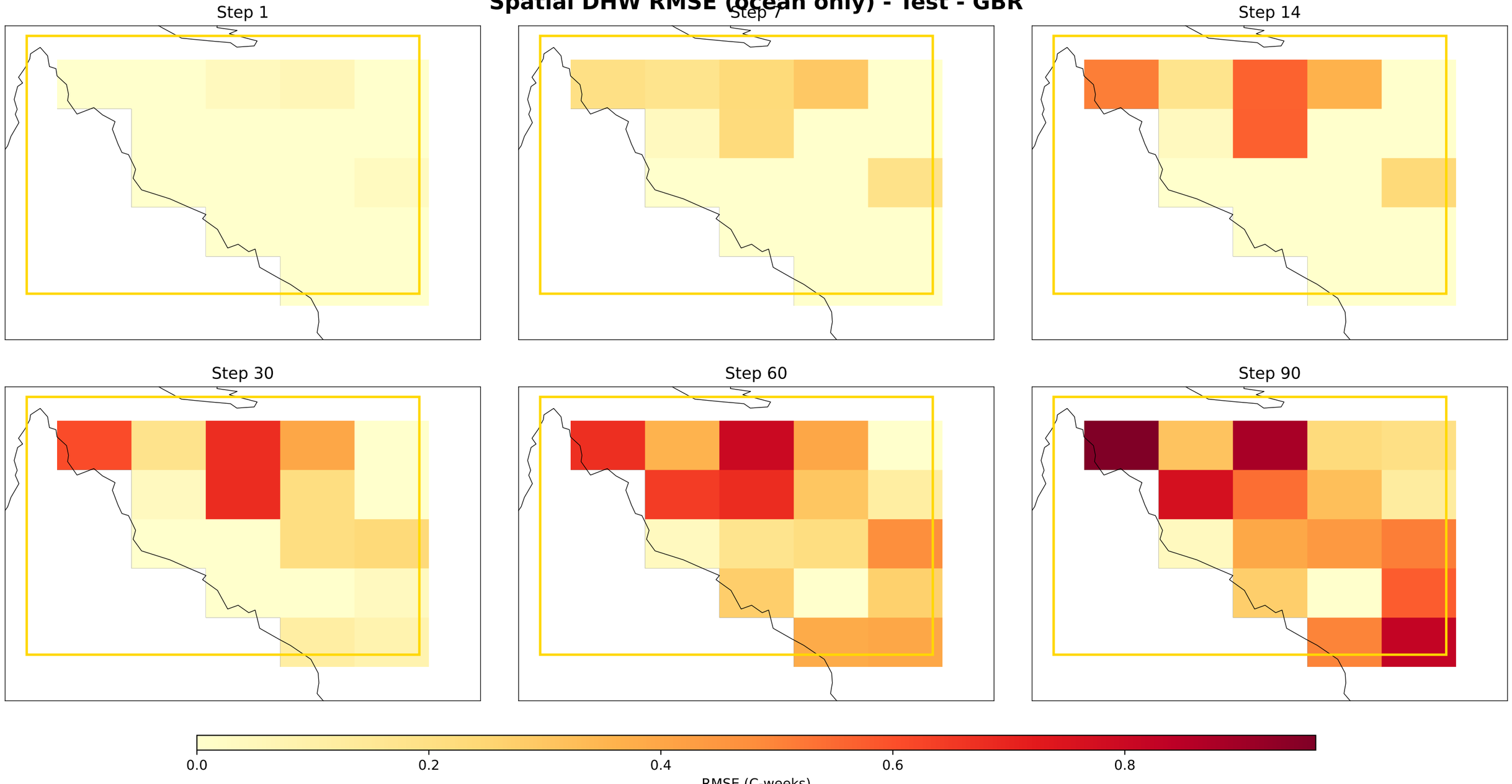


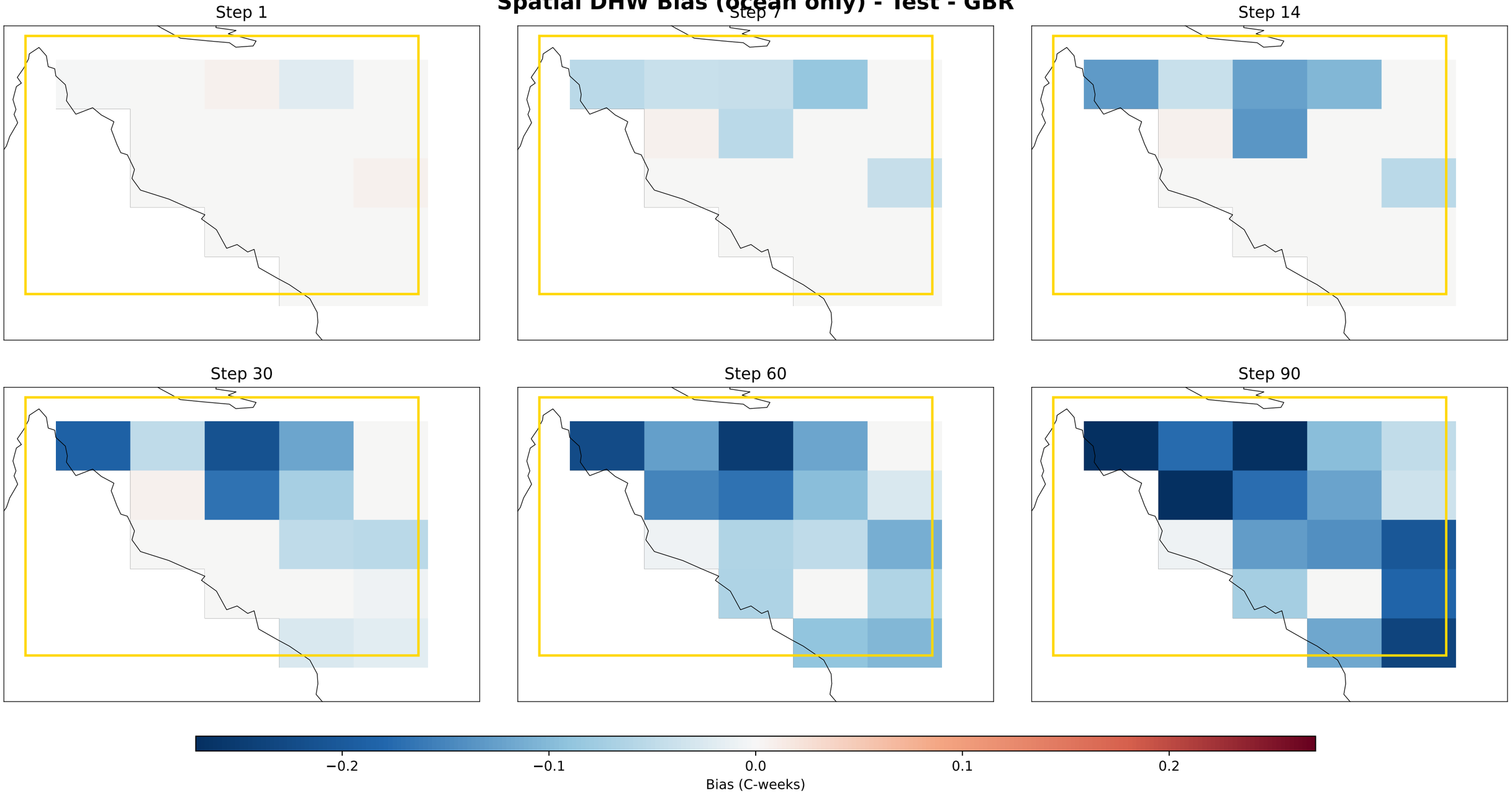





Spatial DHW RMSE (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

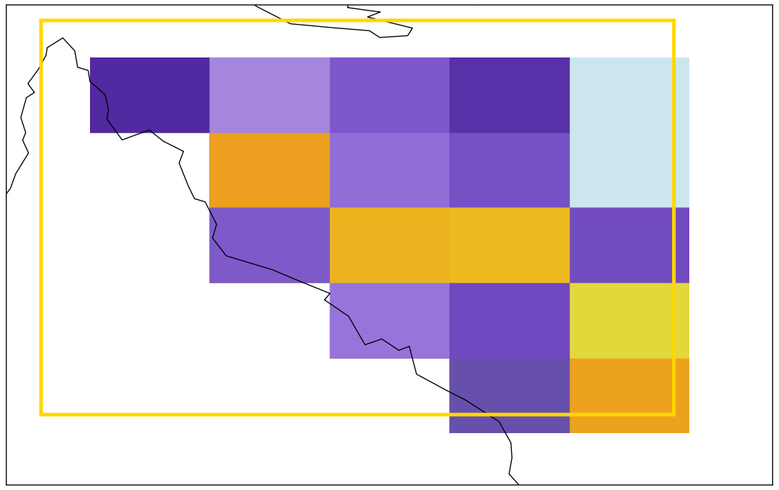

Spatial DHW Bias (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

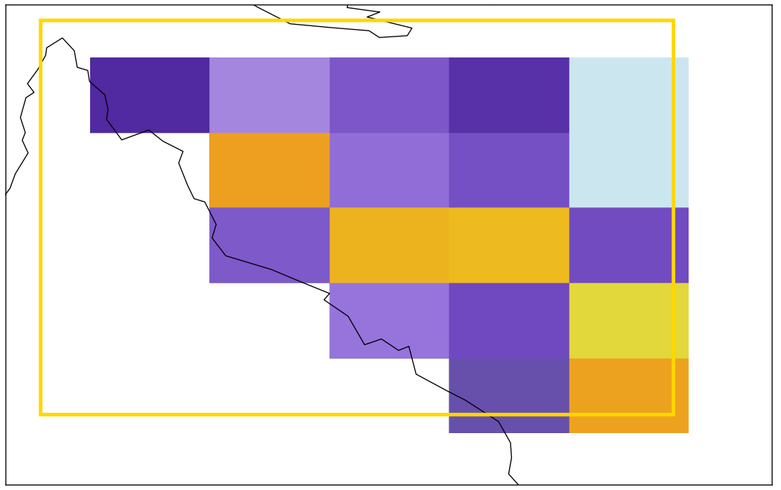

Spatial DHW Bias (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test - GBR (C-weeks)

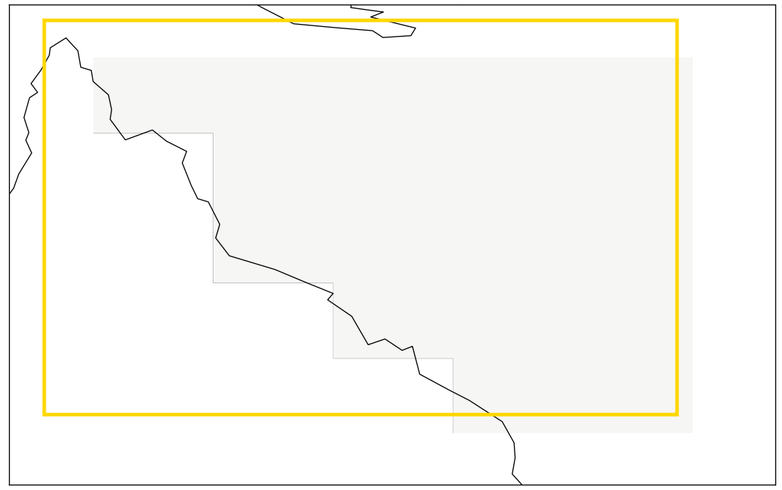
Actual DHW - Step 1



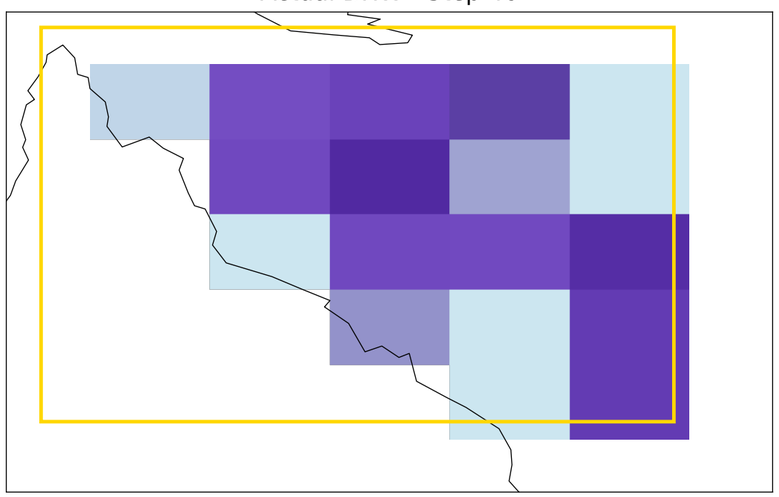
Predicted DHW - Step 1



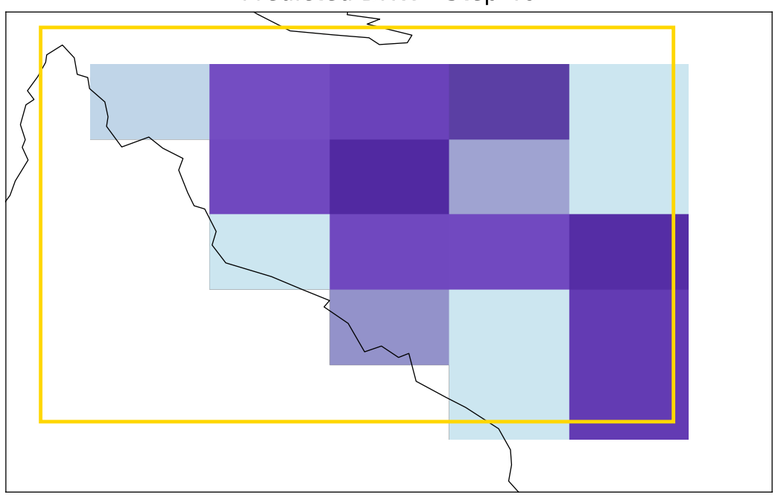
Error (Pred - Actual) - Step 1 [scale ±8]



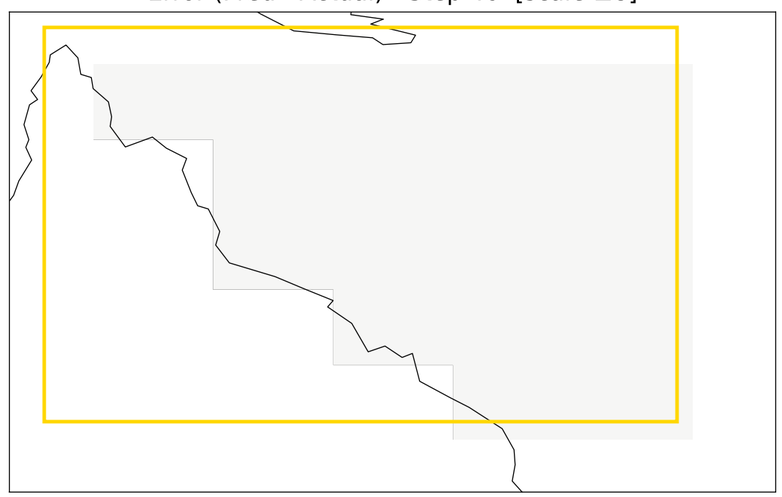
Actual DHW - Step 46



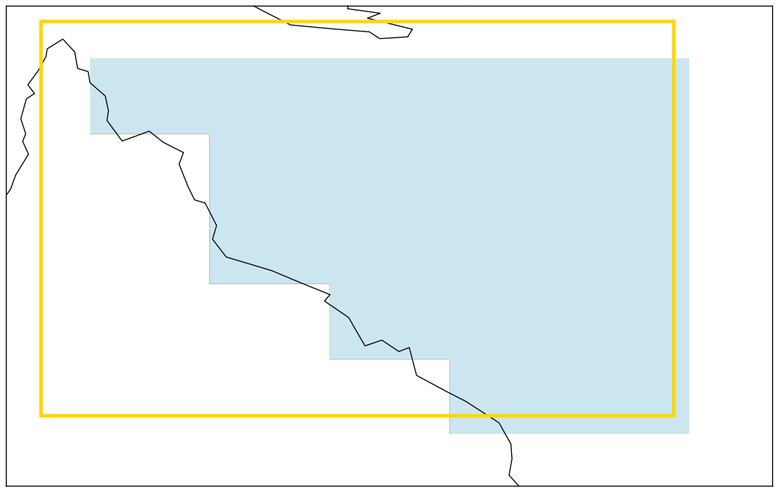
Predicted DHW - Step 46



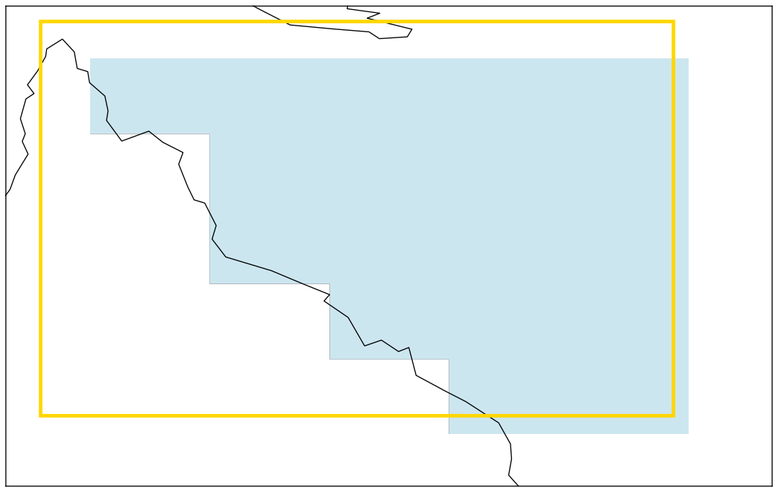
Error (Pred - Actual) - Step 46 [scale ±8]



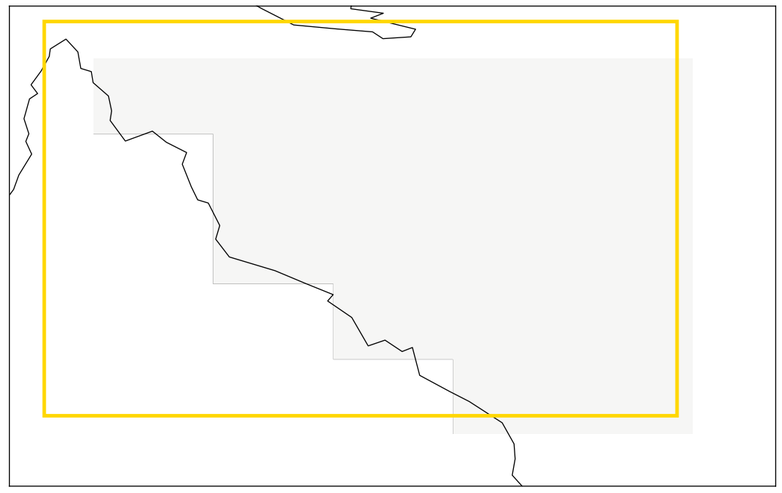
Actual DHW - Step 90

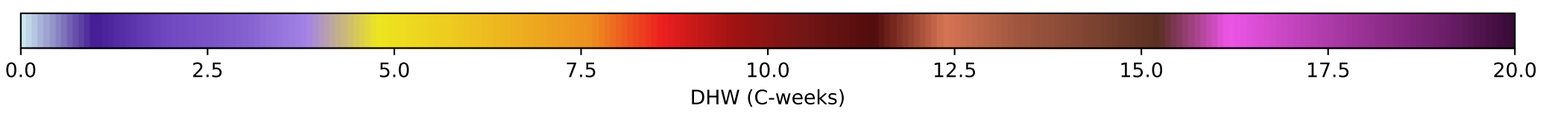


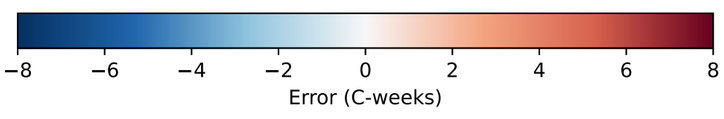
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

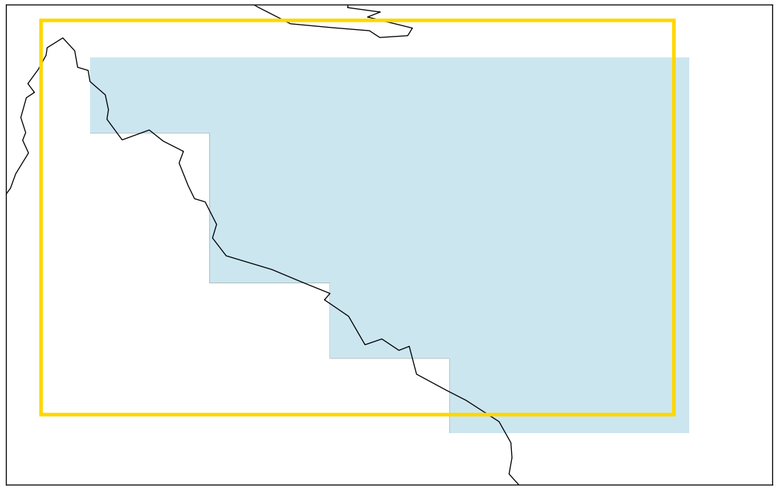




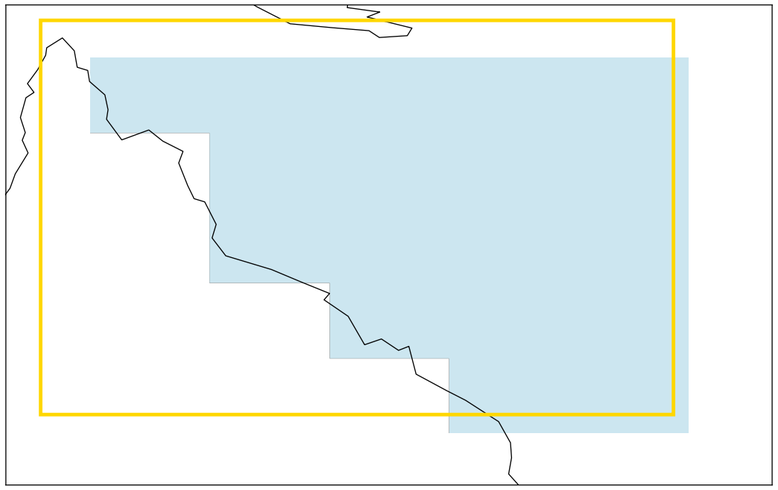


DHW Chain init: 2020-07-09 - Test - GBR (C-weeks)

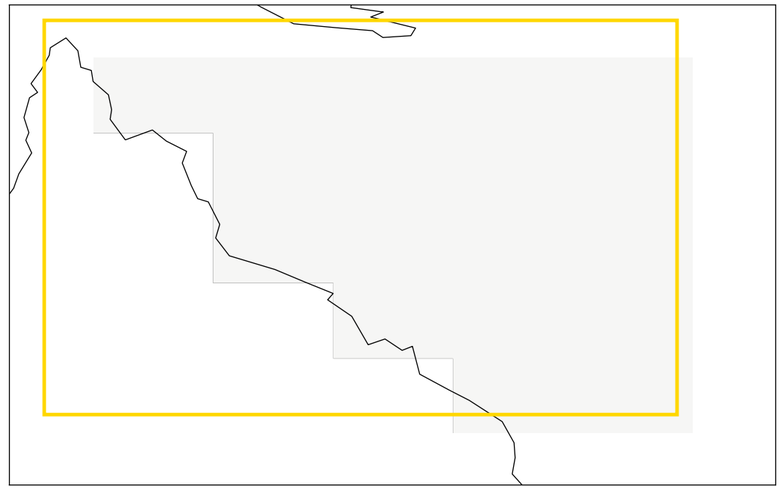
Actual DHW - Step 1



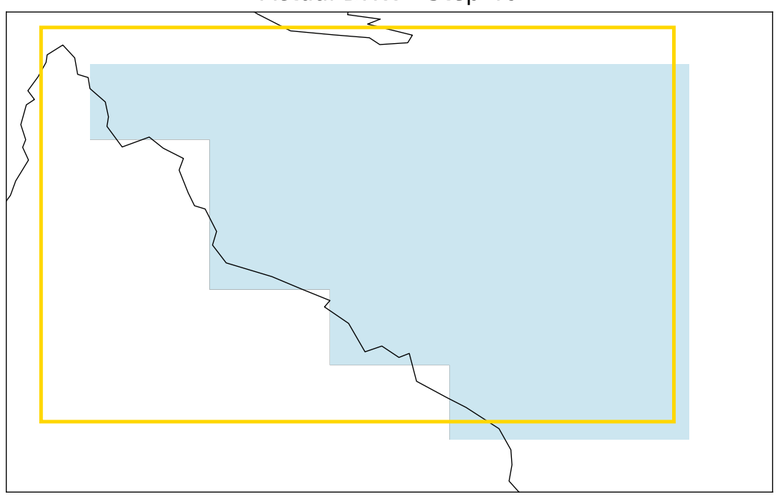
Predicted DHW - Step 1



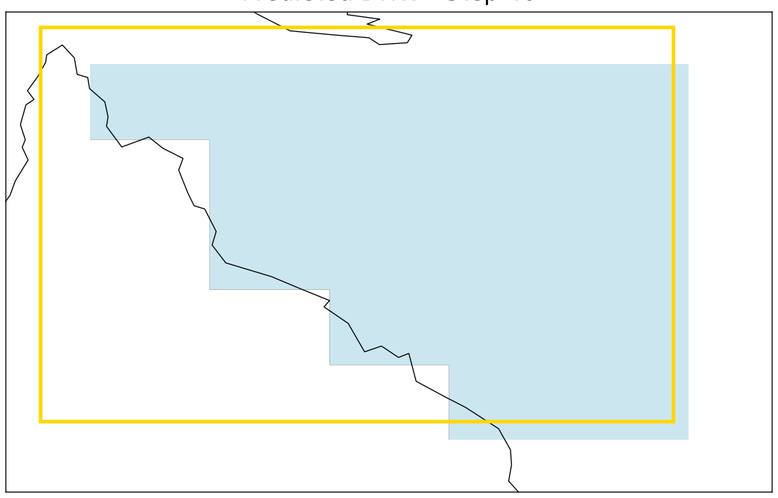
Error (Pred - Actual) - Step 1 [scale ±8]



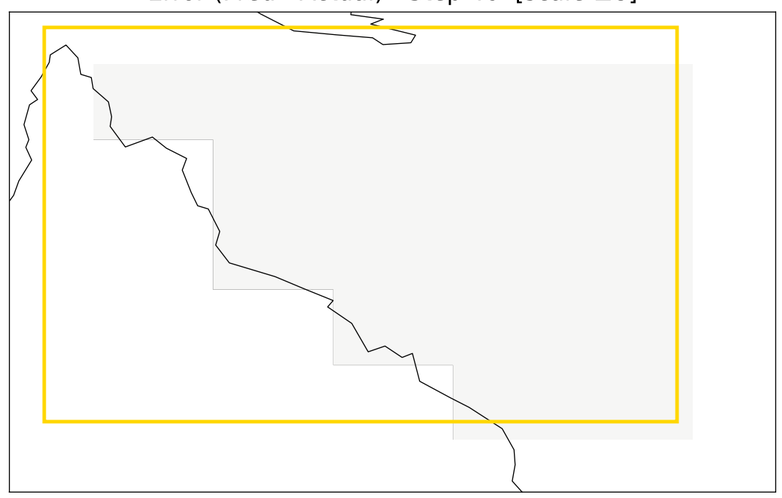
Actual DHW - Step 46



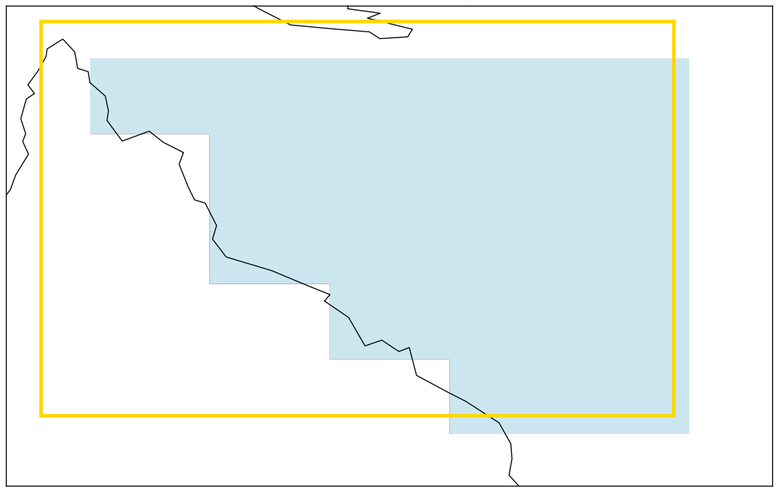
Predicted DHW - Step 46



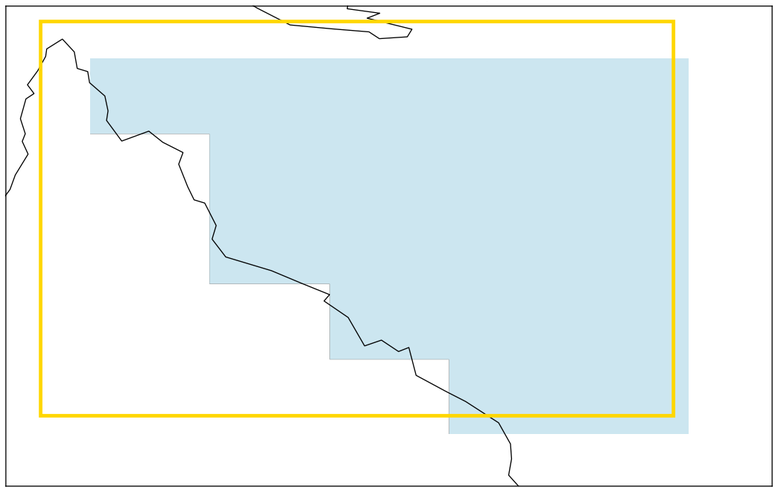
Error (Pred - Actual) - Step 46 [scale ±8]



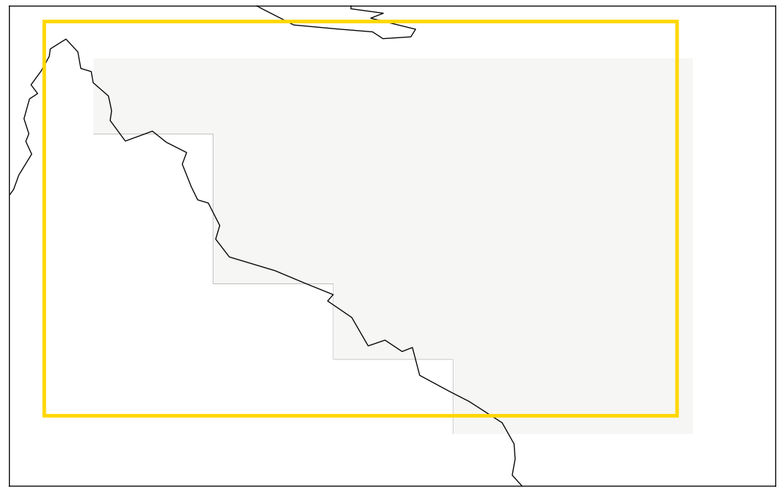
Actual DHW - Step 90

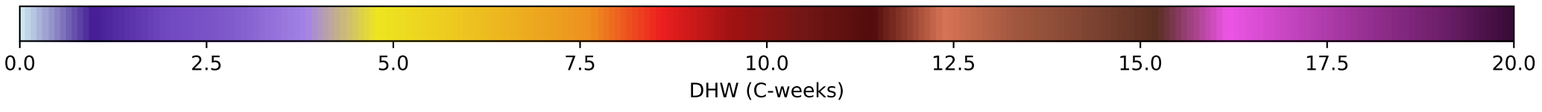


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

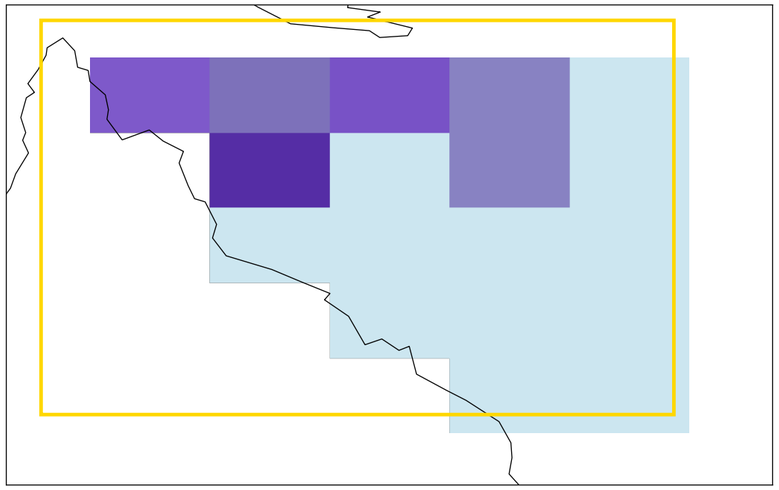




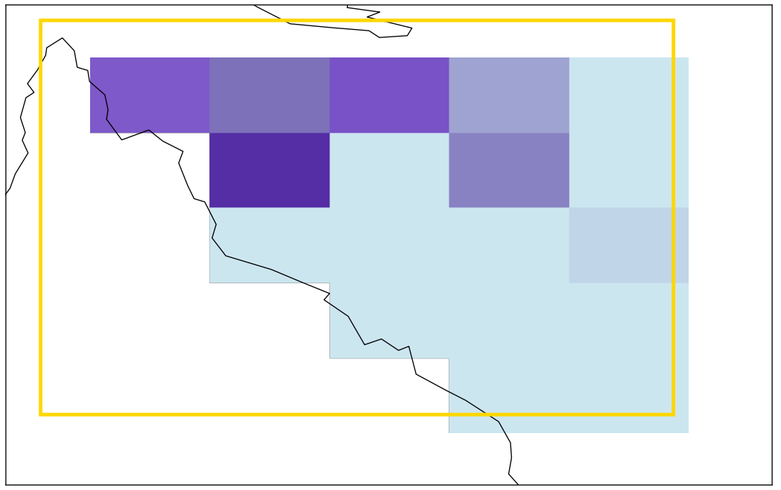


DHW Chain init: 2021-01-06 - Test - GBR (C-weeks)

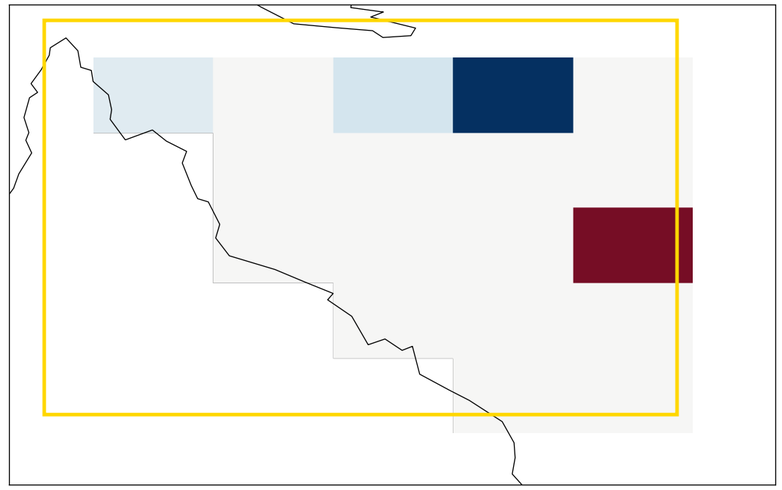
Actual DHW - Step 1



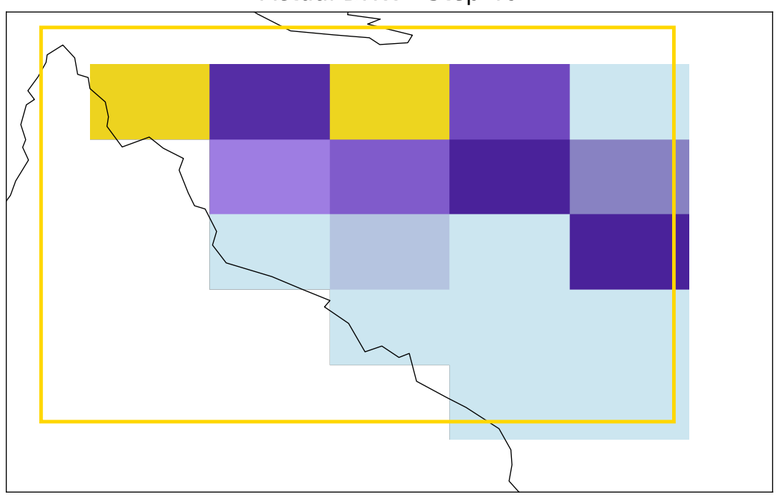
Predicted DHW - Step 1



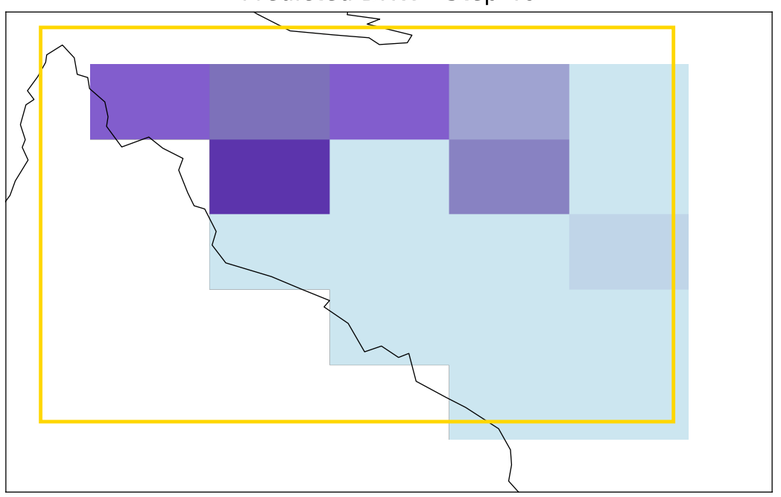
Error (Pred - Actual) - Step 1 [scale ±0.15]



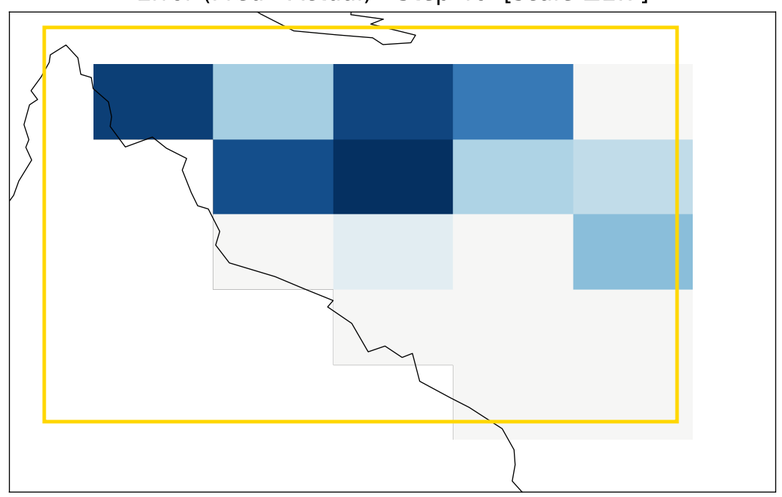
Actual DHW - Step 46



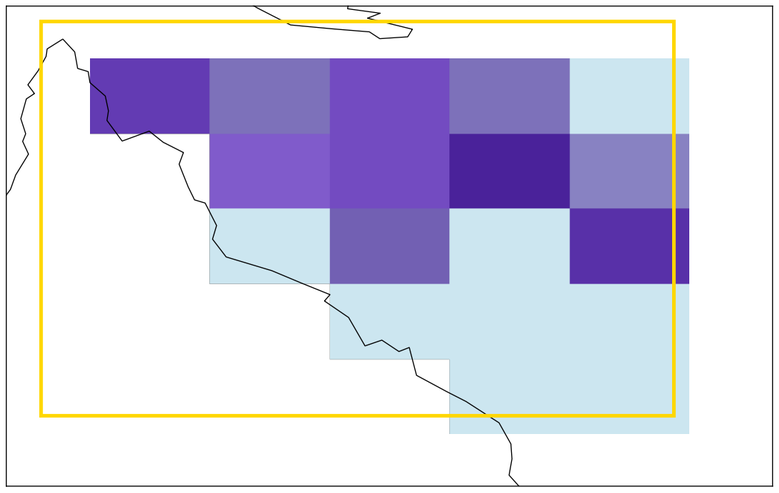
Predicted DHW - Step 46



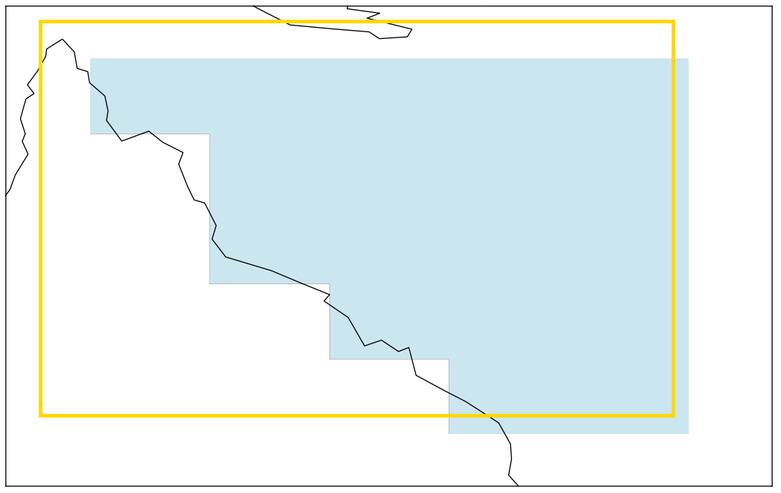
Error (Pred - Actual) - Step 46 [scale ±2.7]



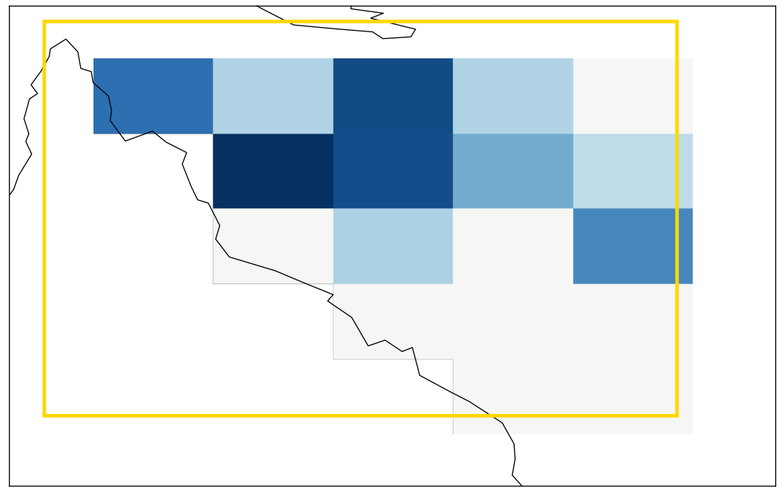
Actual DHW - Step 90

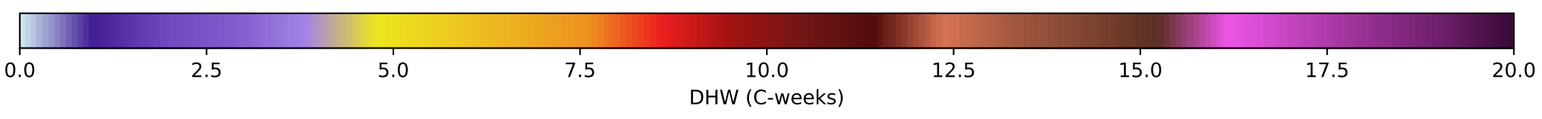


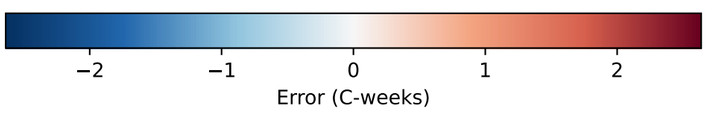
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.6]

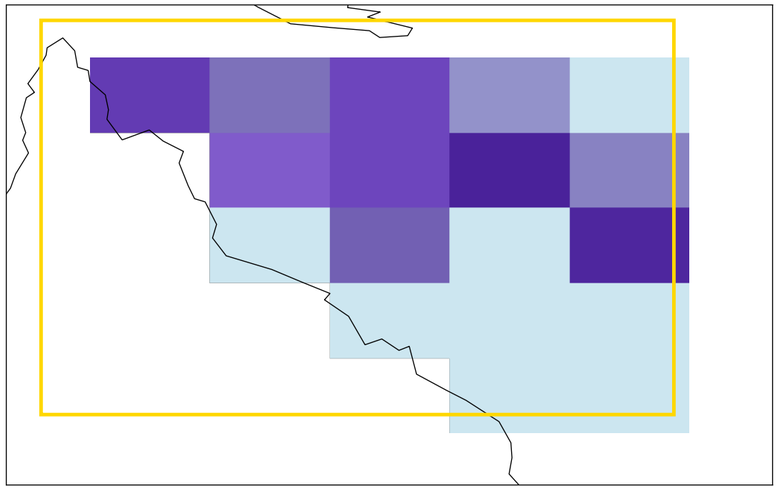




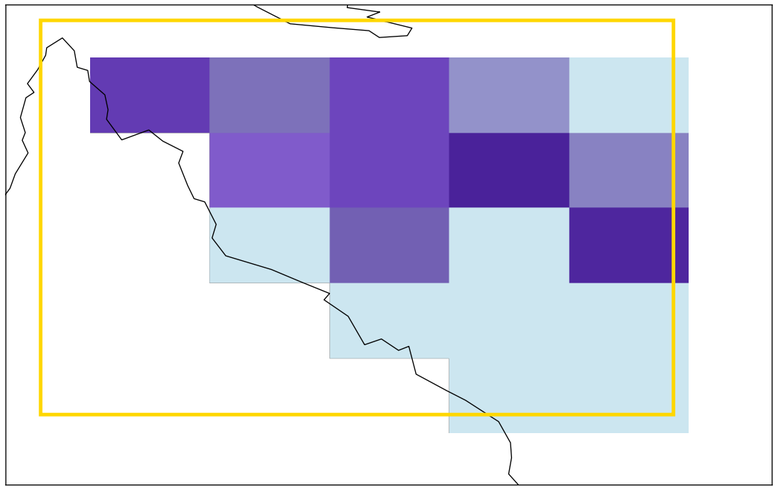


DHW Chain init: 2021-04-06 - Test - GBR (C-weeks)

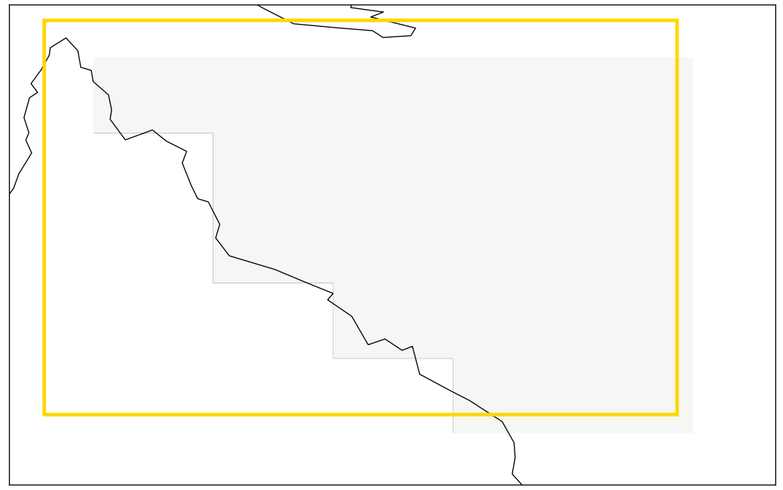
Actual DHW - Step 1



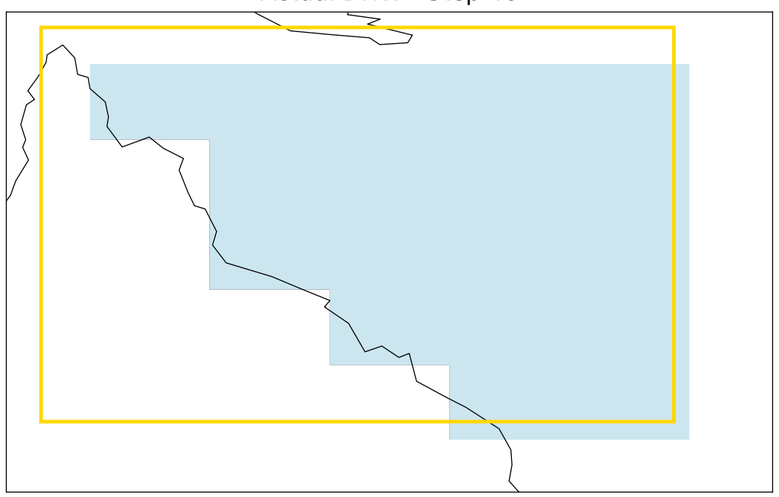
Predicted DHW - Step 1



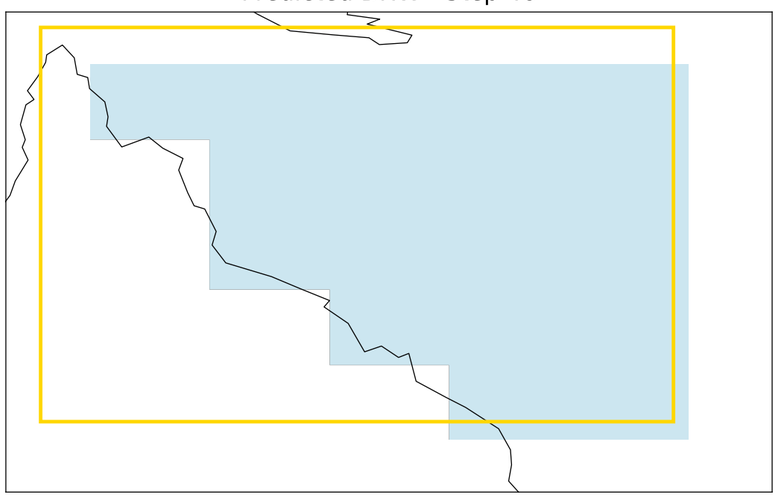
Error (Pred - Actual) - Step 1 [scale ±8]



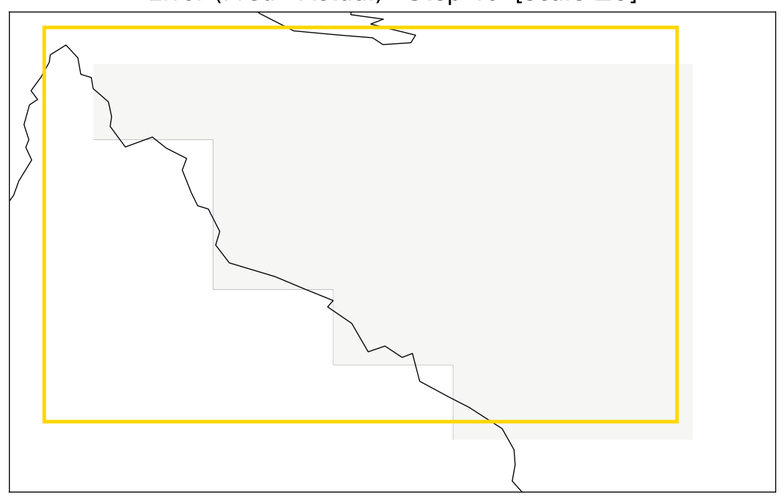
Actual DHW - Step 46



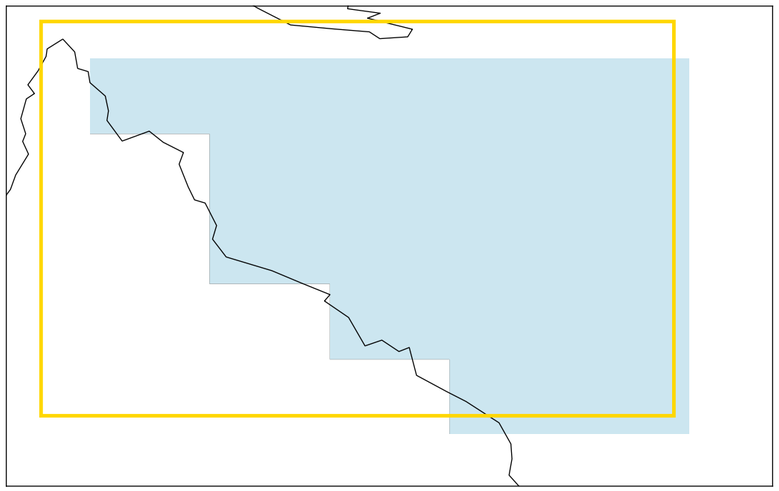
Predicted DHW - Step 46



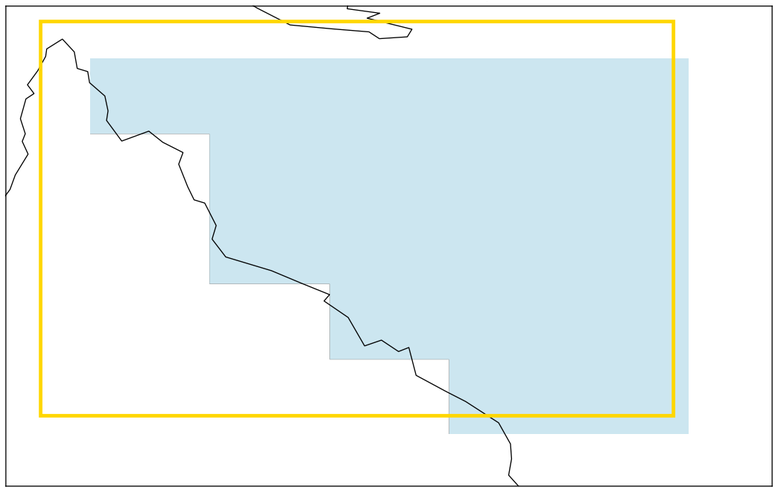
Error (Pred - Actual) - Step 46 [scale ±8]



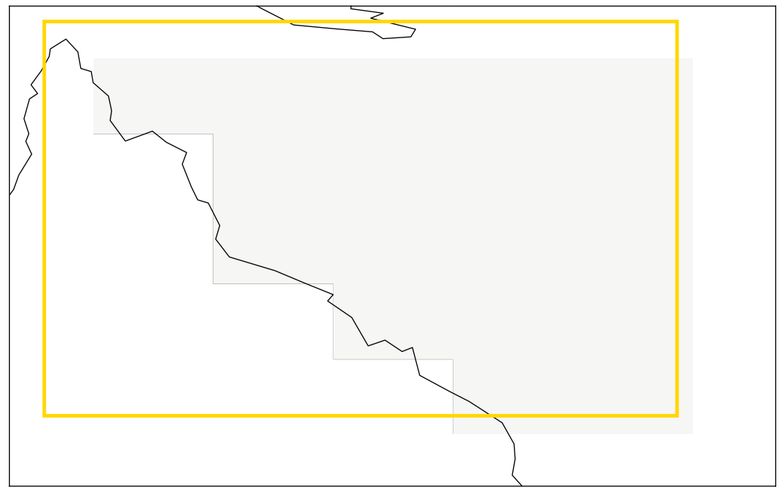
Actual DHW - Step 90

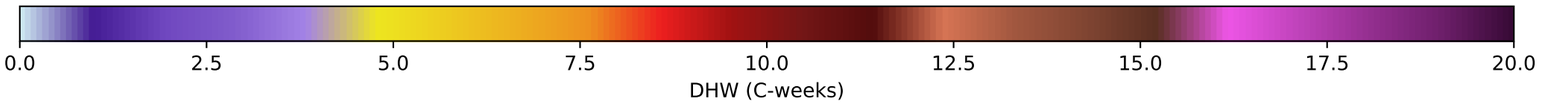


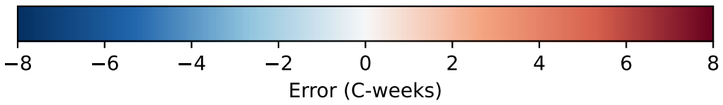
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

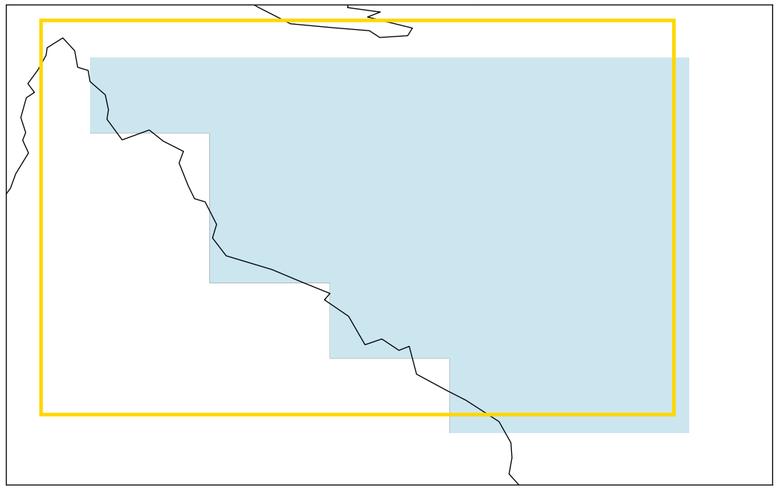




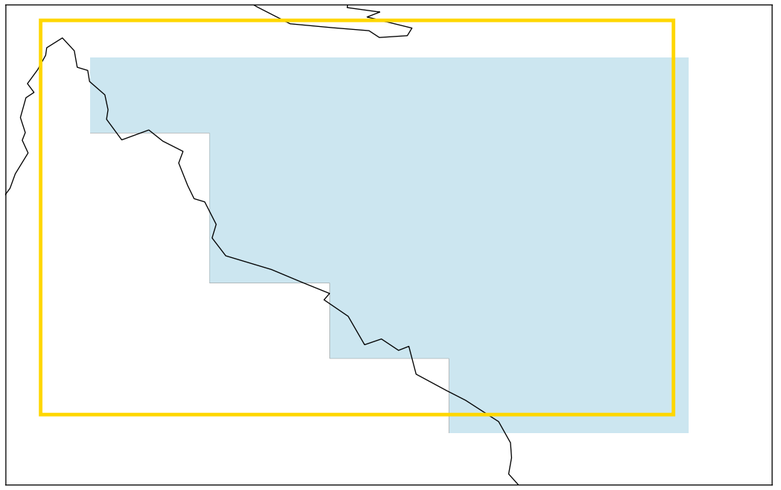


DHW Chain init: 2021-10-03 - Test - GBR (C-weeks)

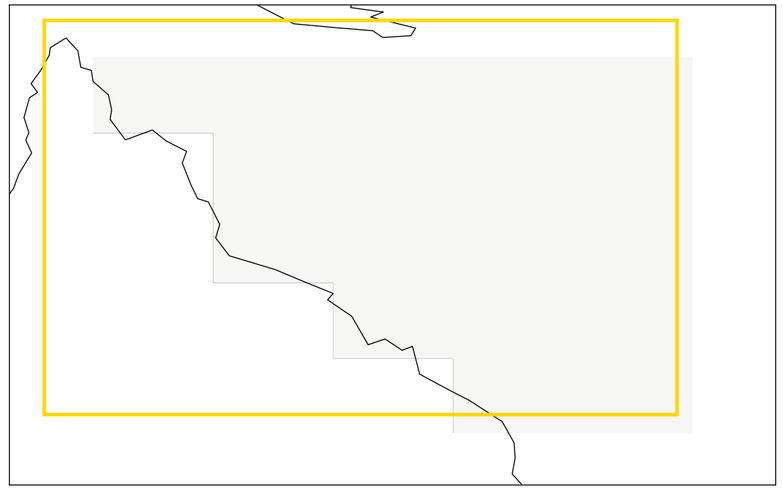
Actual DHW - Step 1



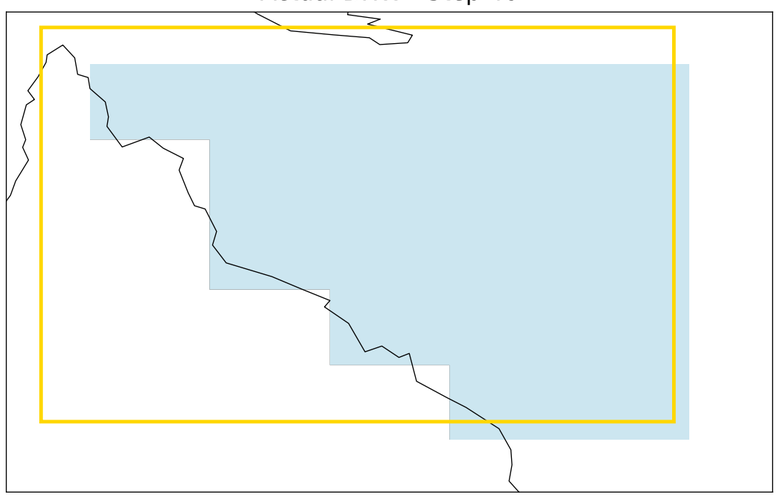
Predicted DHW - Step 1



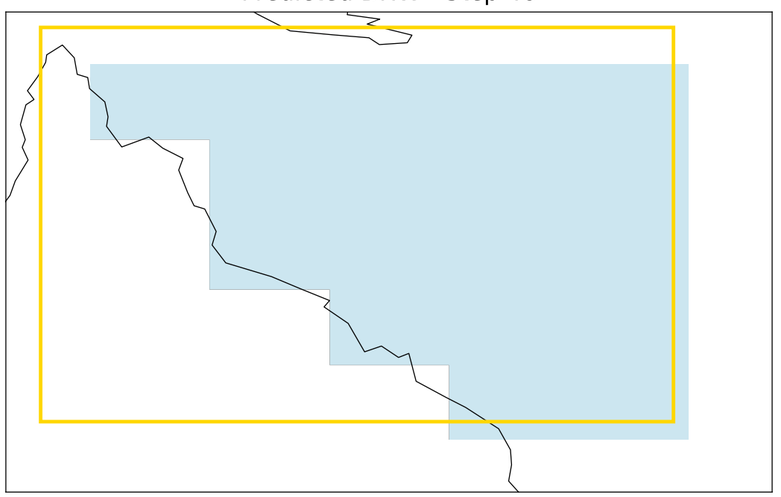
Error (Pred - Actual) - Step 1 [scale ±8]



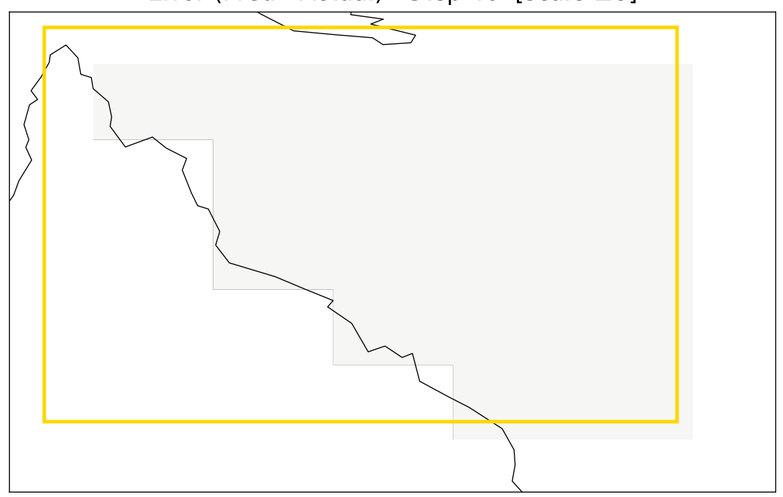
Actual DHW - Step 46



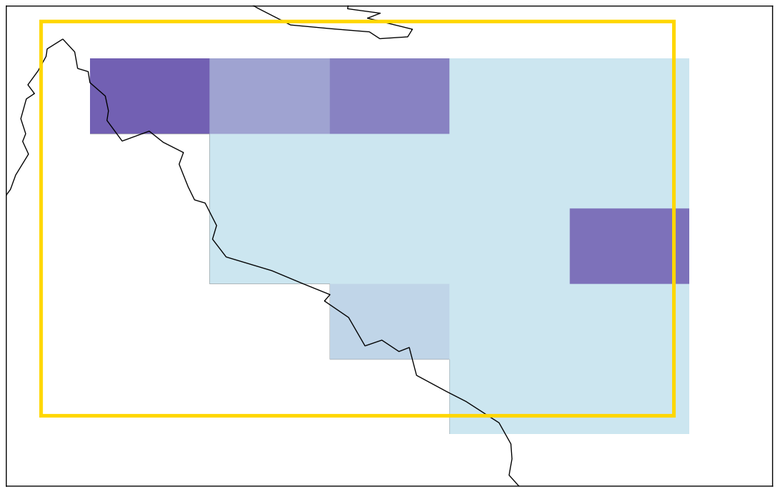
Predicted DHW - Step 46



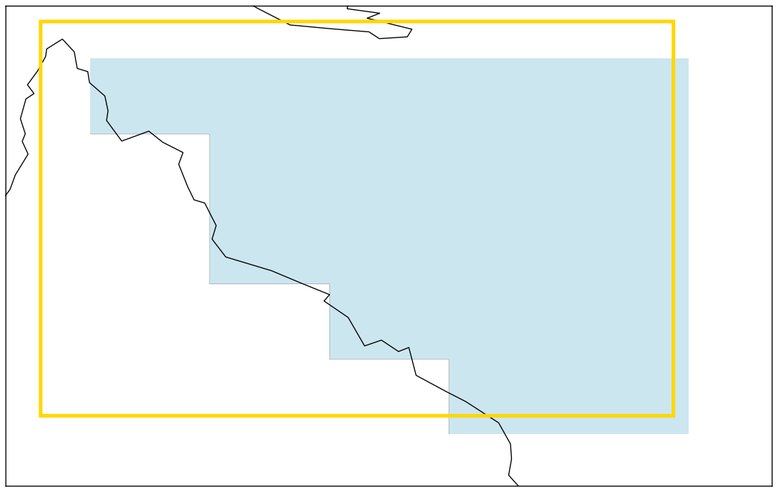
Error (Pred - Actual) - Step 46 [scale ±8]



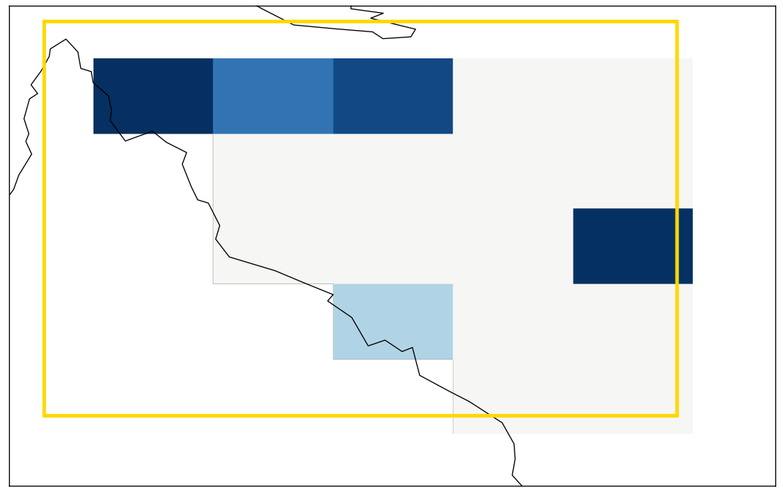
Actual DHW - Step 90

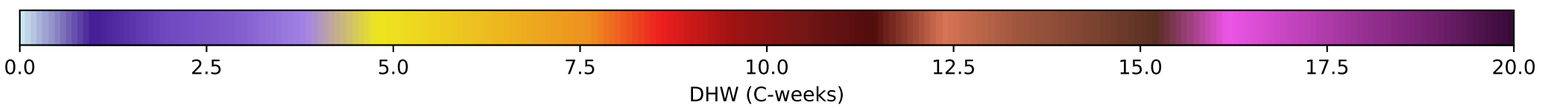


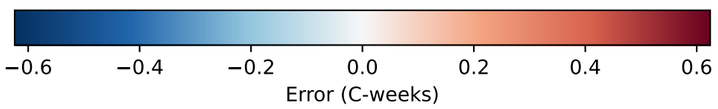
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.62]

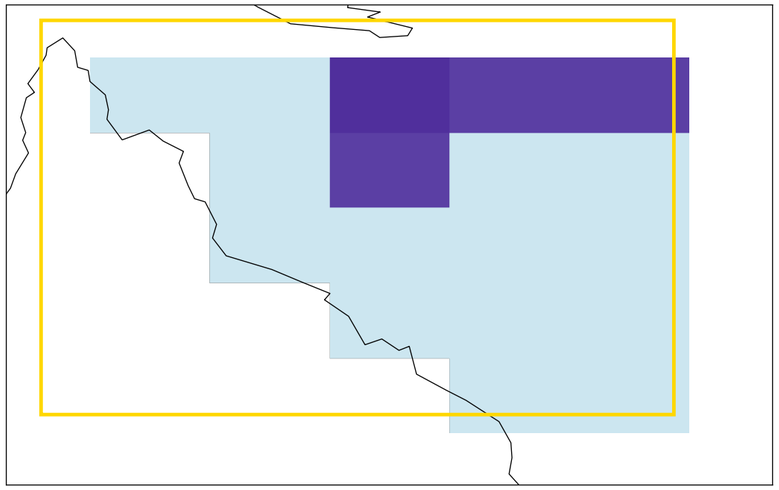




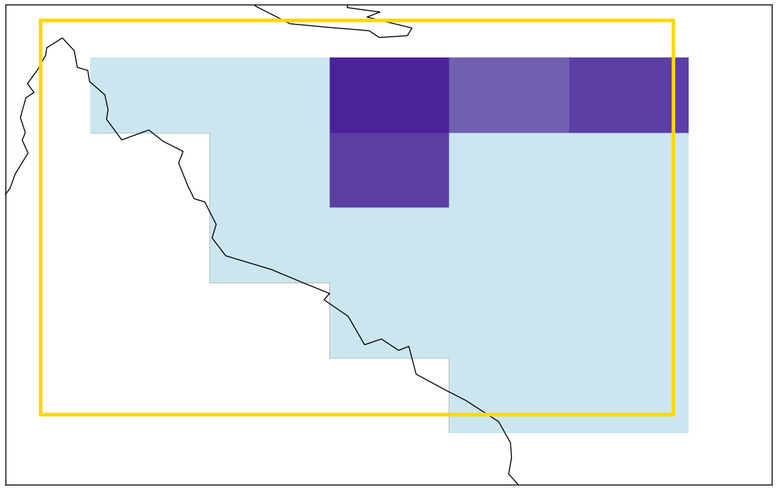


DHW Chain init: 2022-12-27 - Test - GBR (C-weeks)

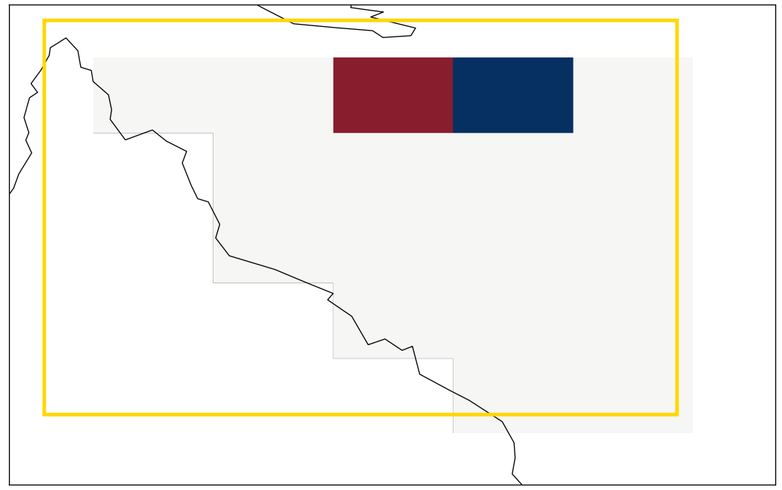
Actual DHW - Step 1



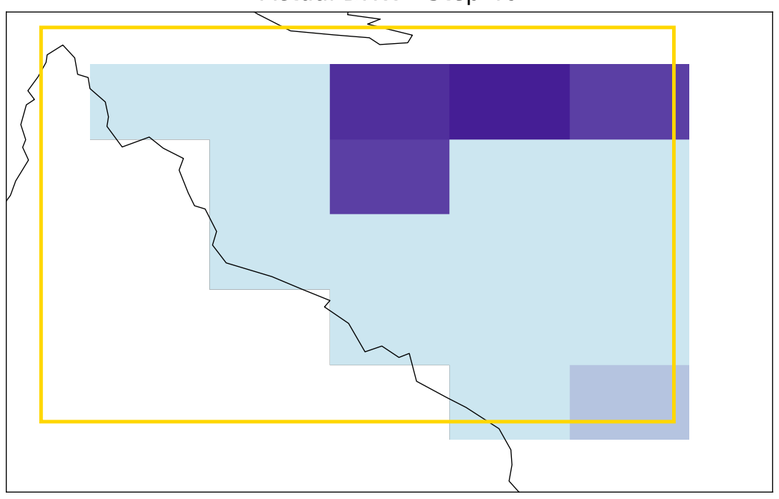
Predicted DHW - Step 1



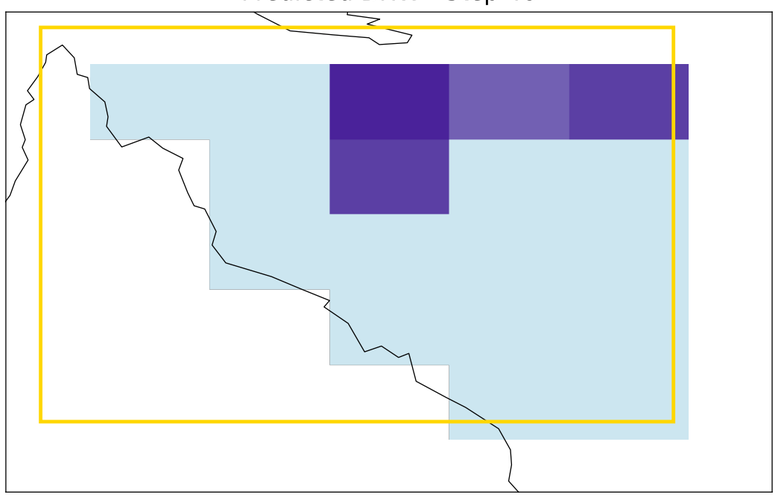
Error (Pred - Actual) - Step 1 [scale ±0.19]



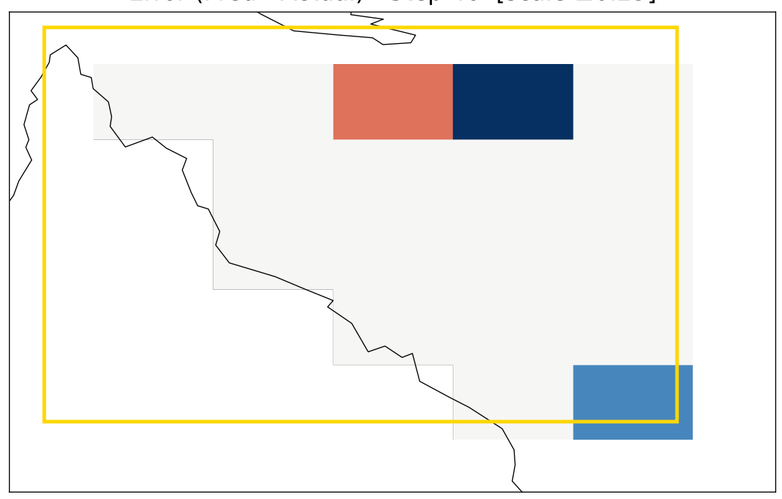
Actual DHW - Step 46



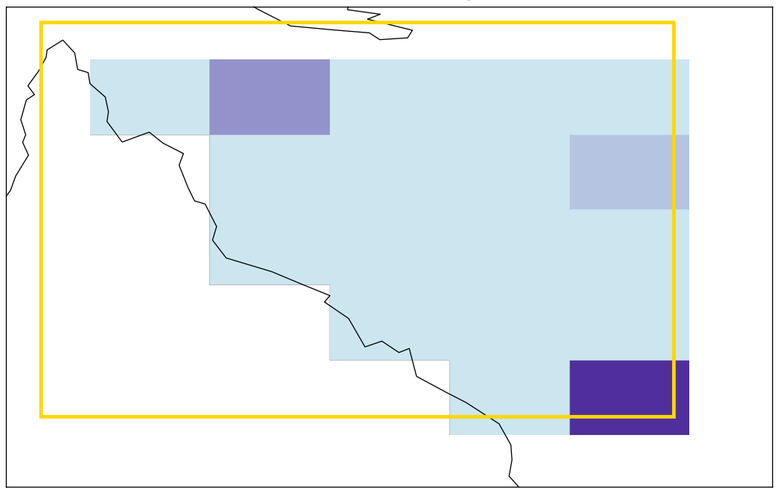
Predicted DHW - Step 46



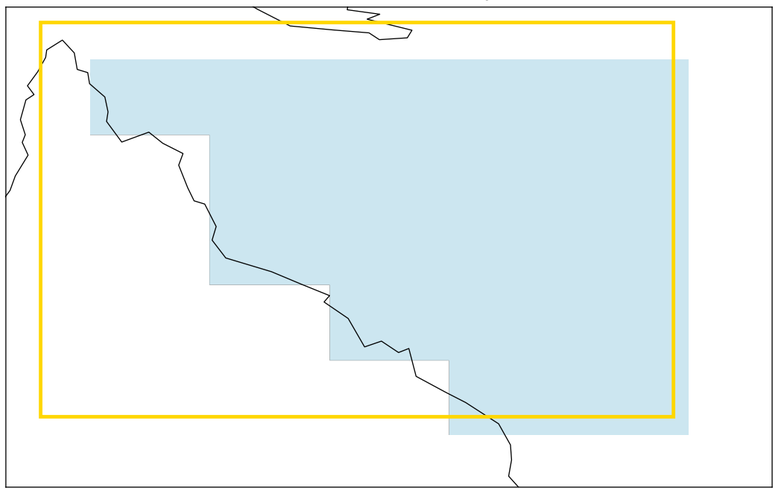
Error (Pred - Actual) - Step 46 [scale ±0.29]



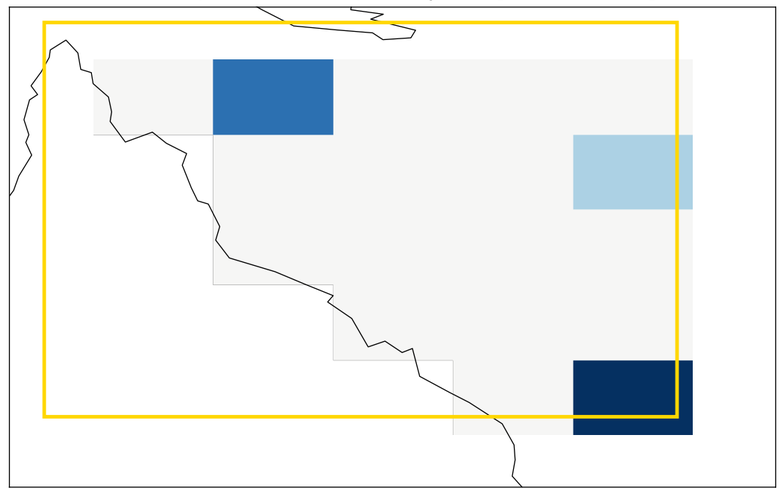
Actual DHW - Step 90

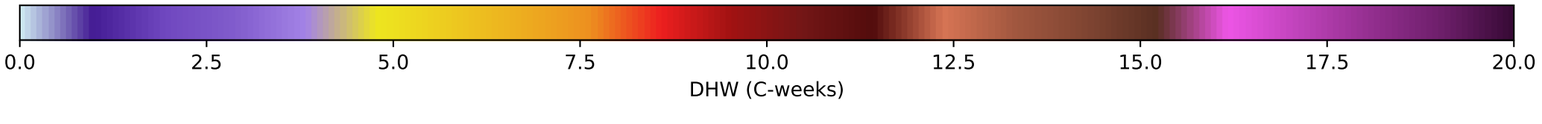


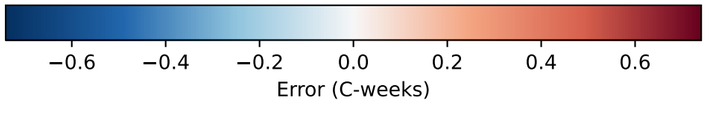
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.74]

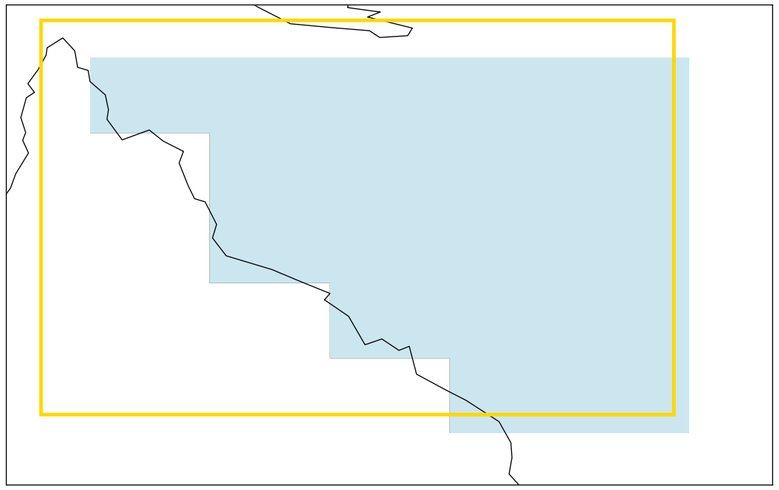




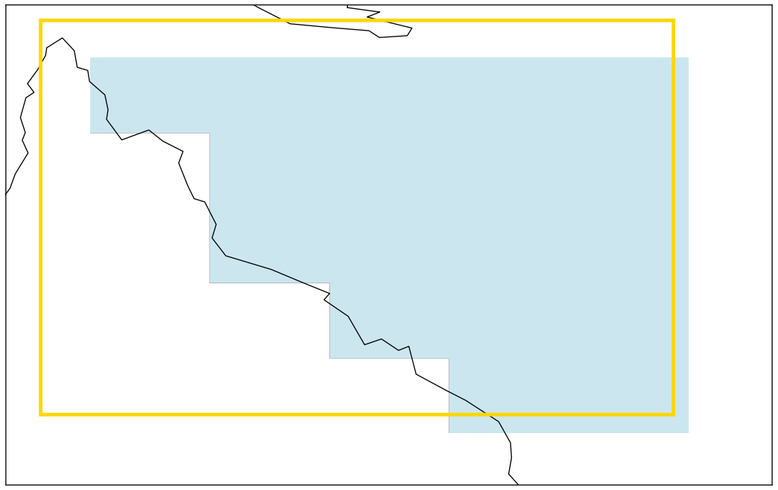


DHW Chain init: 2023-09-23 - Test - GBR (C-weeks)

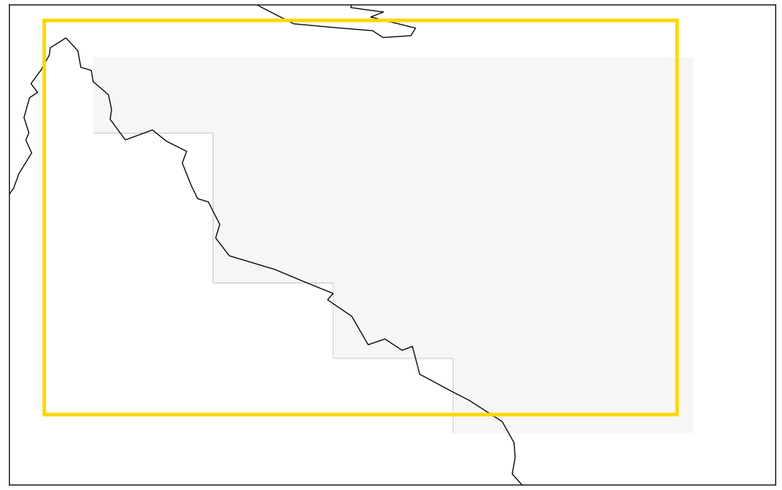
Actual DHW - Step 1



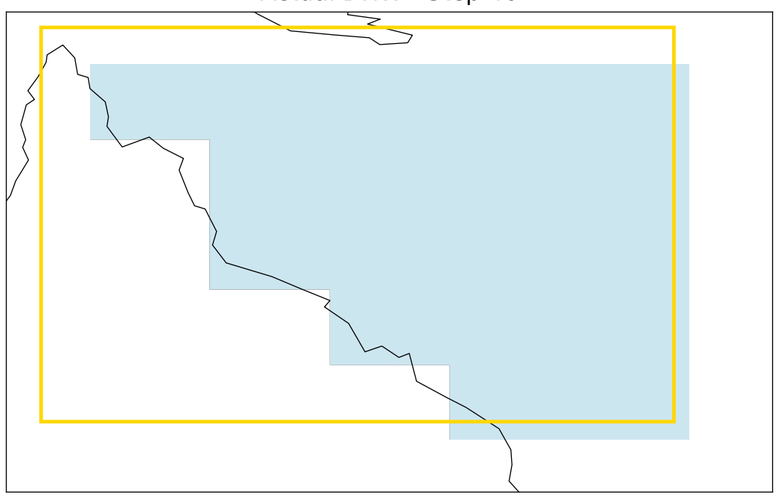
Predicted DHW - Step 1



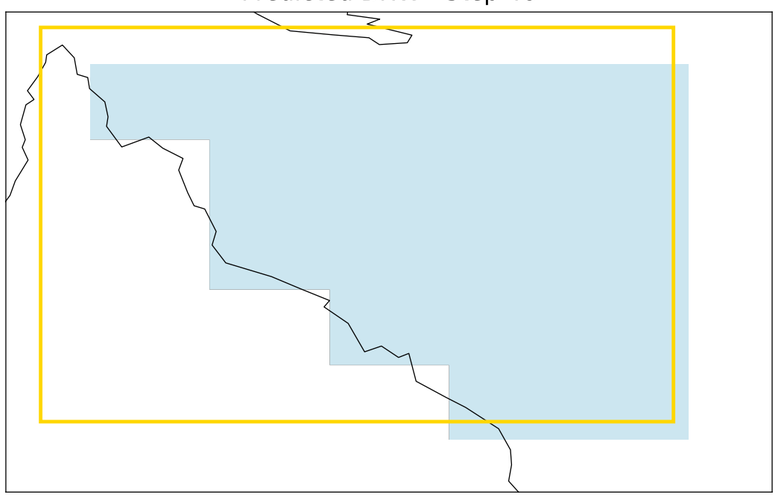
Error (Pred - Actual) - Step 1 [scale ±8]



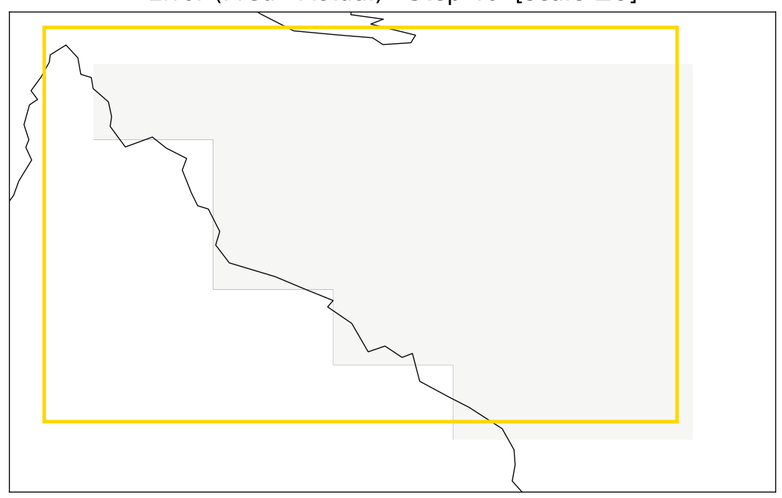
Actual DHW - Step 46



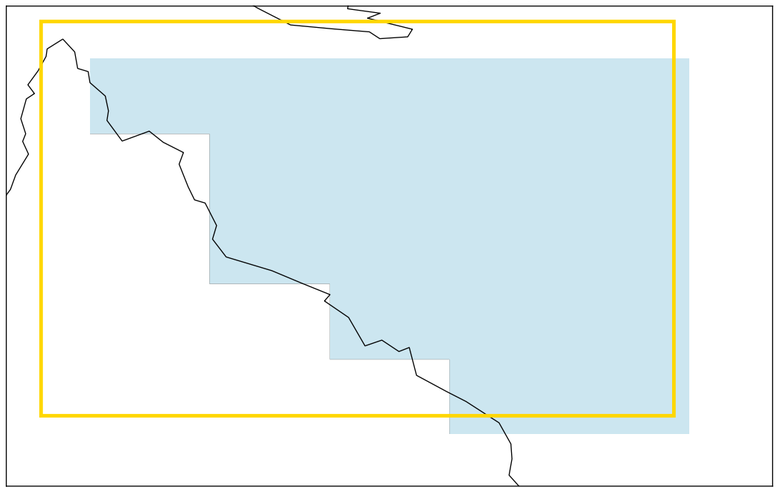
Predicted DHW - Step 46



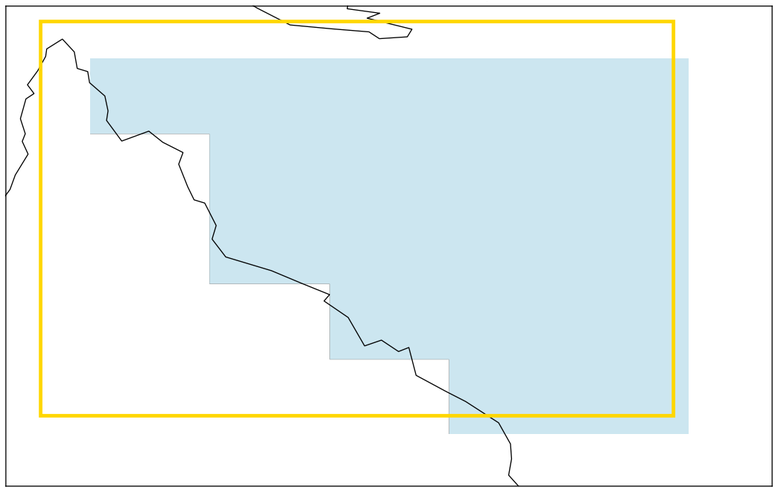
Error (Pred - Actual) - Step 46 [scale ±8]



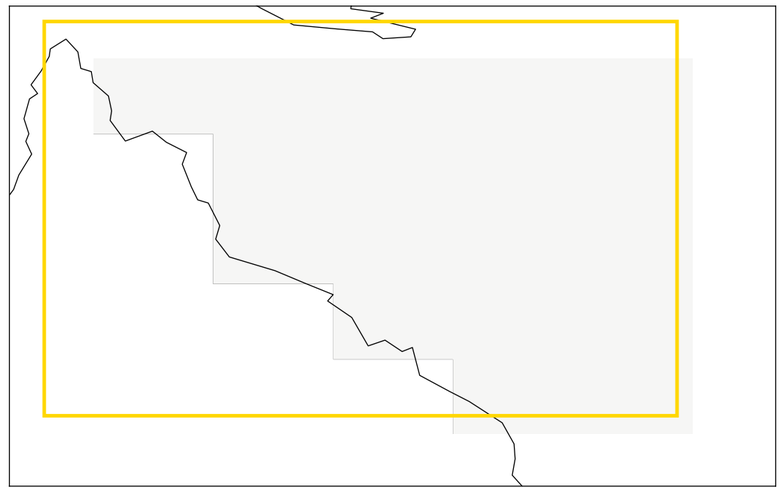
Actual DHW - Step 90

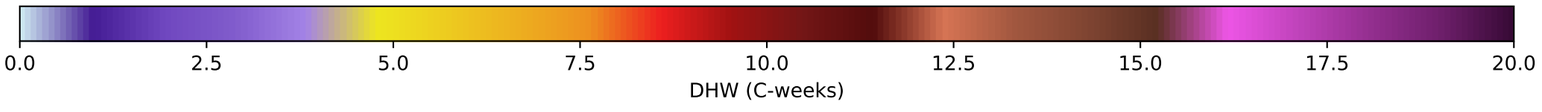


Predicted DHW - Step 90



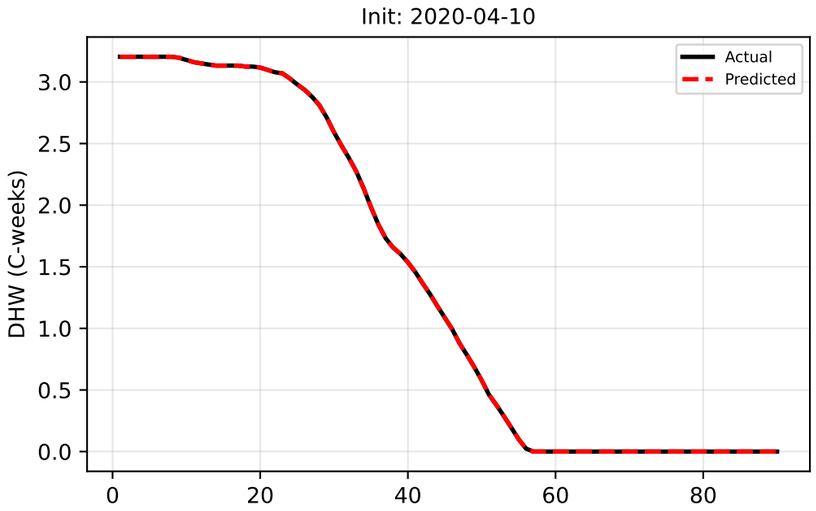
Error (Pred - Actual) - Step 90 [scale ±8]

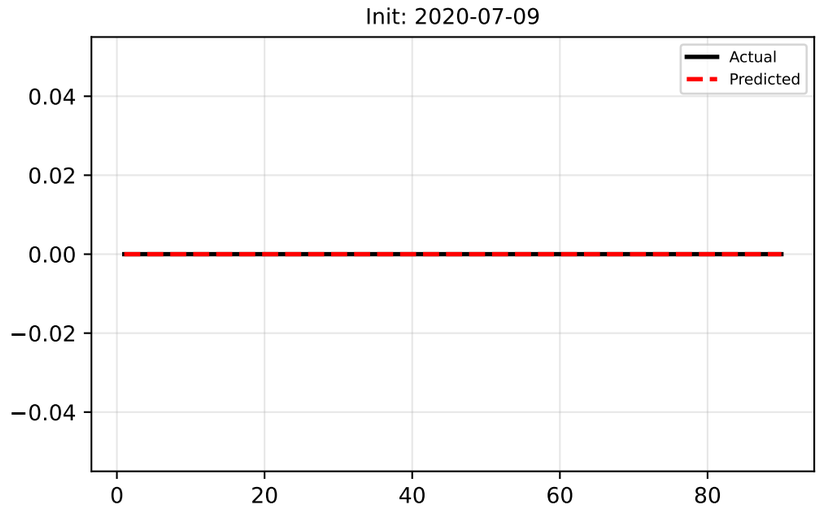


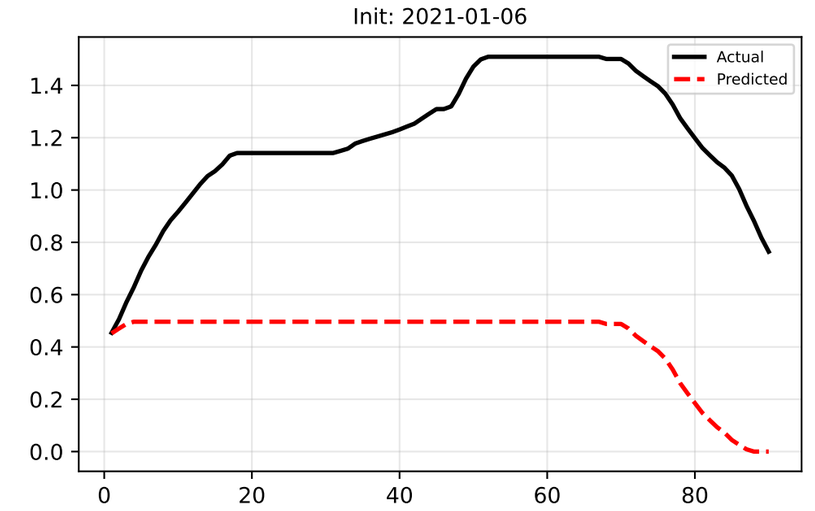


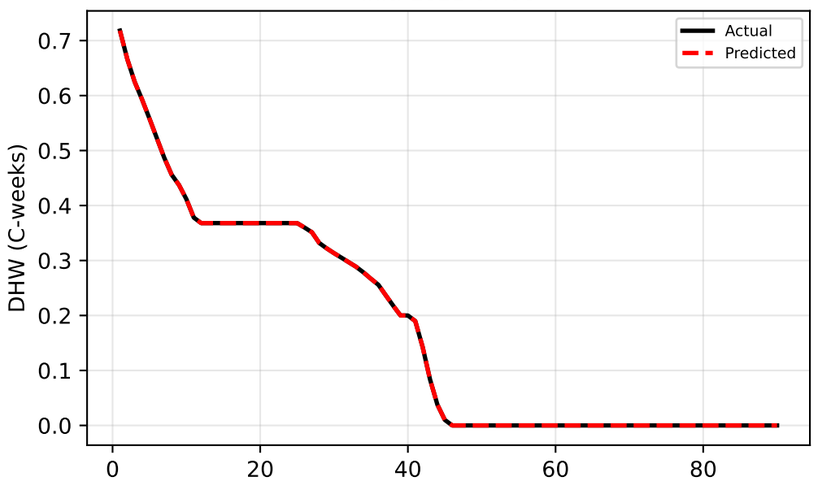


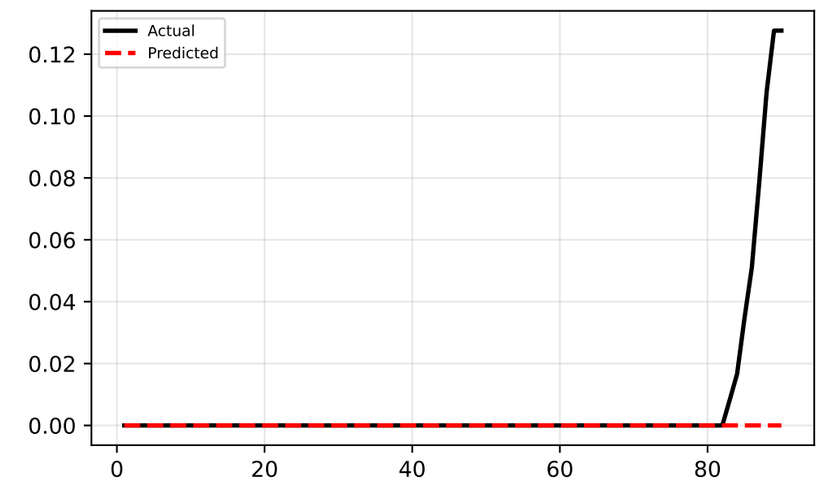
Global-Mean DHW - Test - GBR (C-weeks)

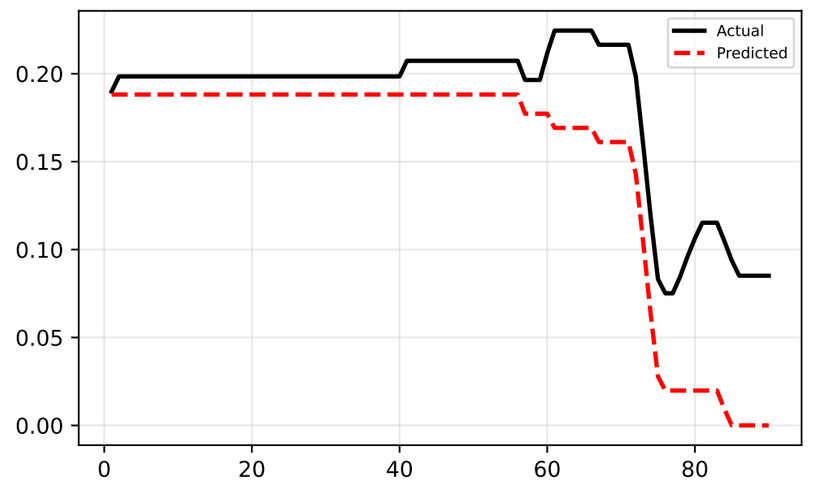


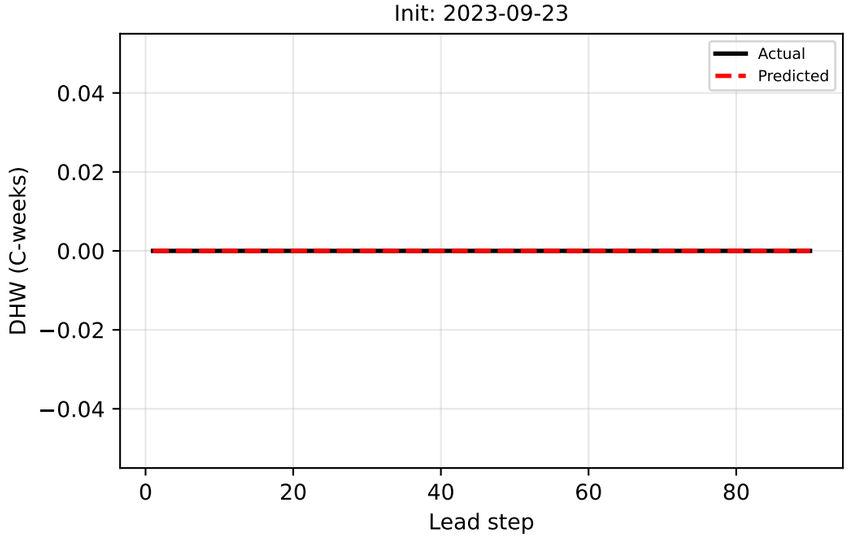


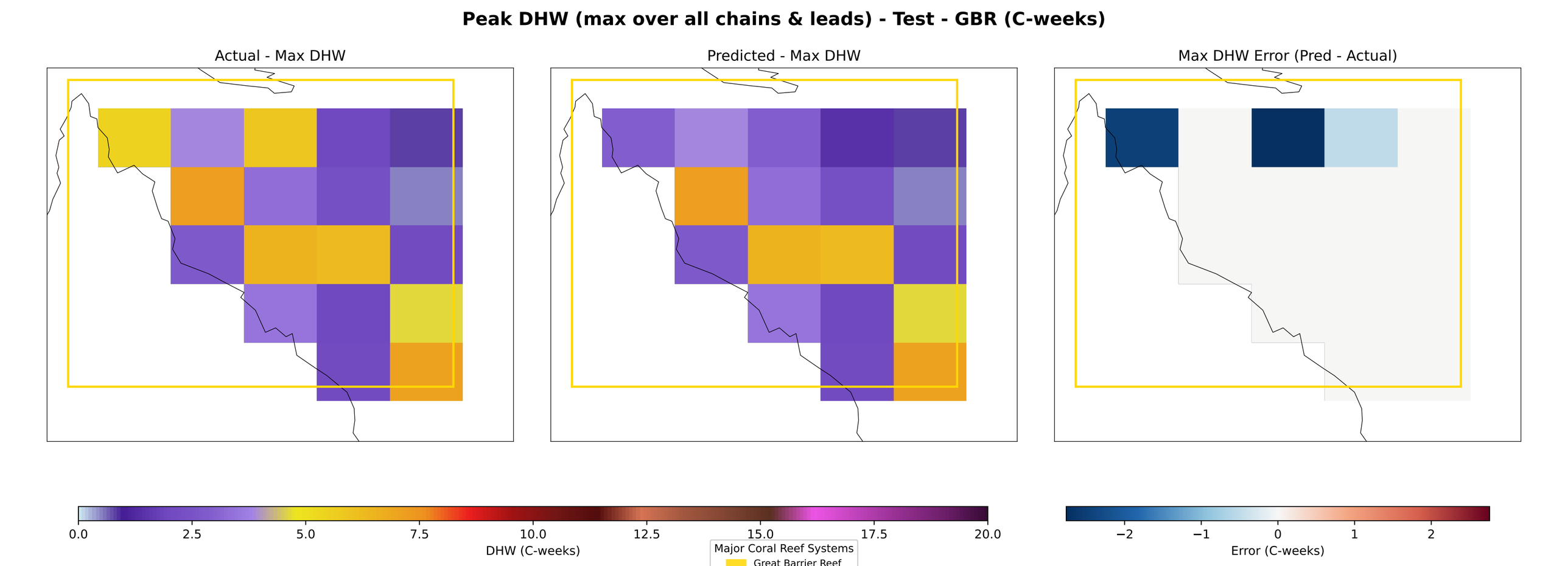


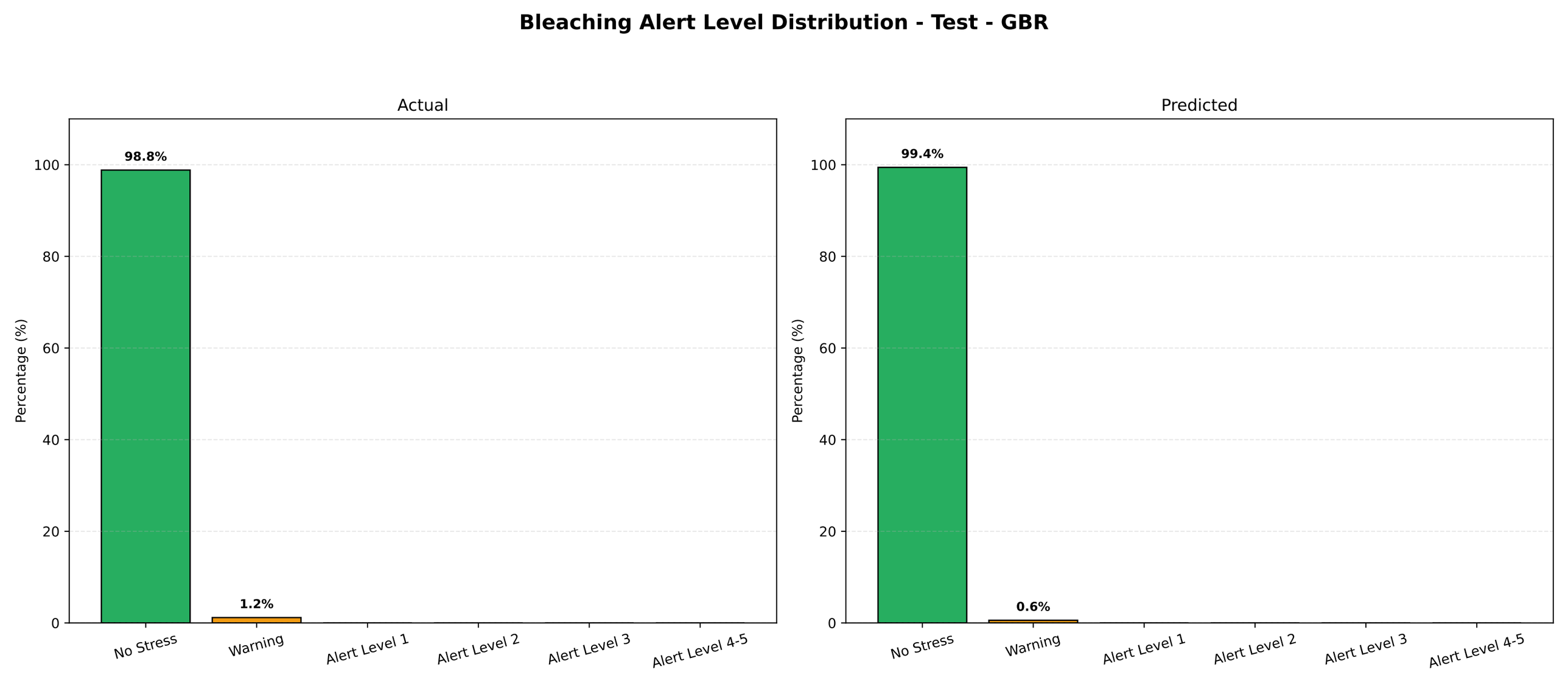




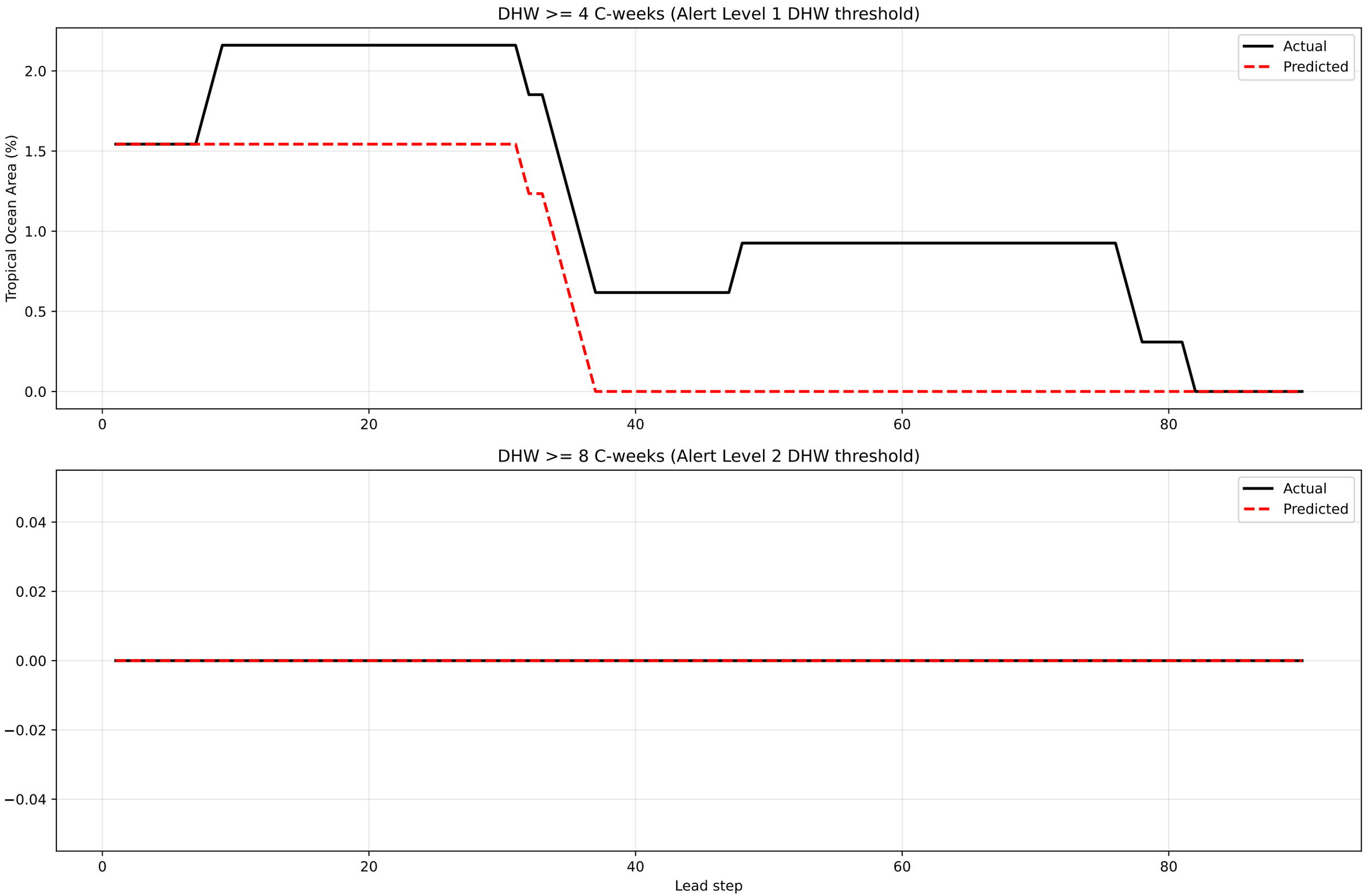





Peak DHW (max over all chains & leads) - Test - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−2
−1
0
1
2
Error (C-weeks)


Bleaching Alert Level Distribution - Test - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
98.8%
1.2%
99.4%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

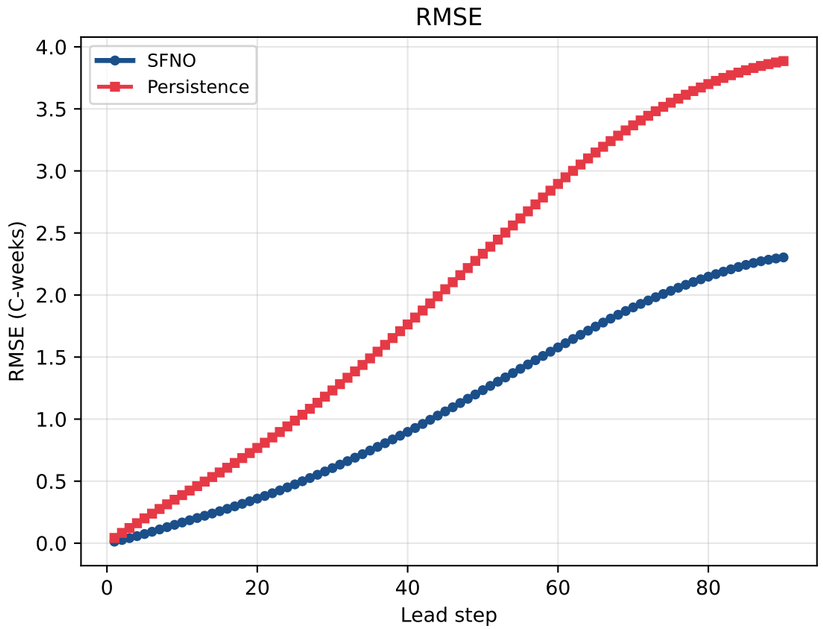
Tropical Ocean Area Under Bleaching Stress - Test - GBR

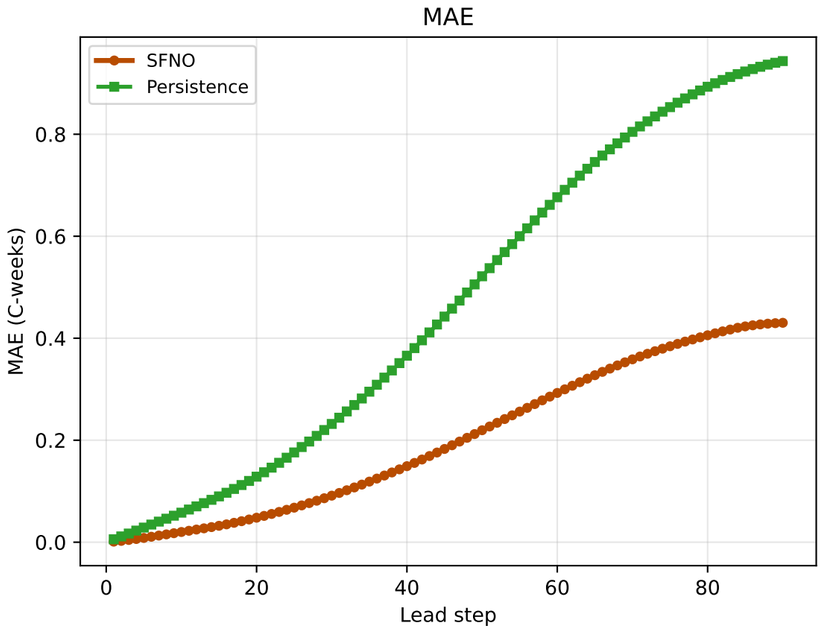


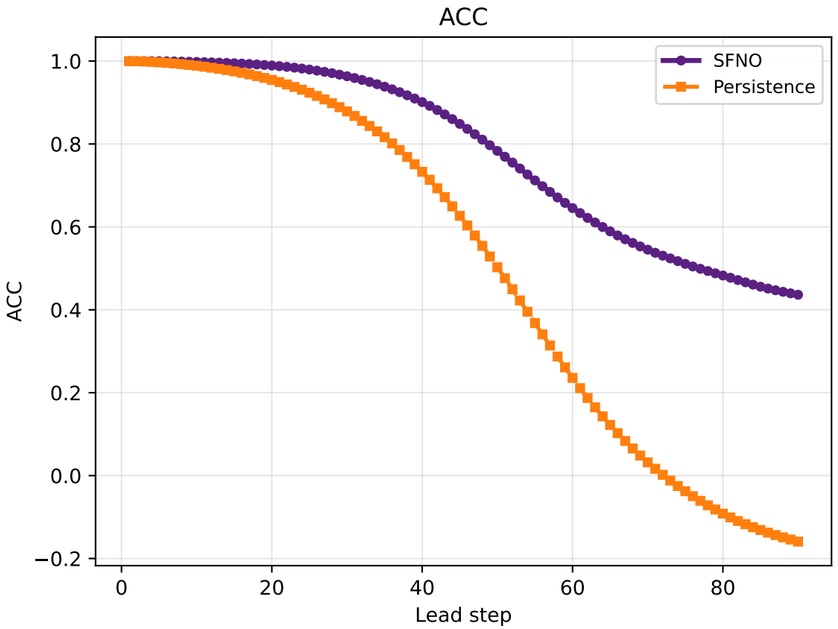
DHW Metrics by Lead Step - Val (15 chains scored at every lead)

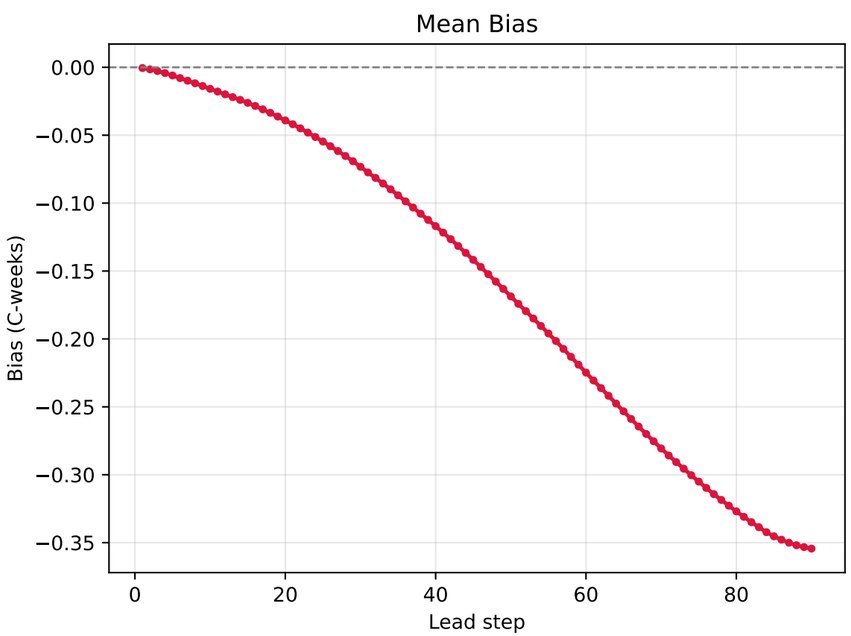
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0127	0.0425	70.1%	0.0011	0.0057	-0.0005	1.0000	0.9999	1.0000
2	0.0261	0.0829	68.6%	0.0026	0.0114	-0.0015	0.9999	0.9995	0.9999
3	0.0414	0.1222	66.1%	0.0044	0.0170	-0.0028	0.9999	0.9989	0.9998
4	0.0571	0.1613	64.6%	0.0063	0.0227	-0.0043	0.9998	0.9980	0.9996
5	0.0741	0.2000	62.9%	0.0084	0.0286	-0.0060	0.9996	0.9970	0.9993
6	0.0923	0.2381	61.3%	0.0107	0.0346	-0.0079	0.9994	0.9957	0.9989
7	0.1110	0.2758	59.8%	0.0130	0.0404	-0.0099	0.9991	0.9943	0.9985
8	0.1296	0.3133	58.6%	0.0153	0.0461	-0.0118	0.9987	0.9926	0.9979
9	0.1481	0.3504	57.7%	0.0176	0.0520	-0.0138	0.9984	0.9907	0.9973
10	0.1668	0.3874	56.9%	0.0200	0.0579	-0.0159	0.9979	0.9886	0.9965
11	0.1850	0.4241	56.4%	0.0224	0.0641	-0.0179	0.9974	0.9863	0.9957
12	0.2030	0.4601	55.9%	0.0248	0.0702	-0.0199	0.9969	0.9839	0.9948
13	0.2211	0.4965	55.5%	0.0272	0.0766	-0.0220	0.9963	0.9812	0.9938
14	0.2392	0.5332	55.1%	0.0296	0.0831	-0.0240	0.9956	0.9783	0.9927
15	0.2577	0.5701	54.8%	0.0322	0.0898	-0.0262	0.9948	0.9752	0.9914
16	0.2769	0.6080	54.4%	0.0350	0.0970	-0.0285	0.9939	0.9717	0.9900
17	0.2970	0.6467	54.1%	0.0380	0.1044	-0.0310	0.9929	0.9680	0.9884
18	0.3172	0.6861	53.8%	0.0412	0.1121	-0.0335	0.9919	0.9639	0.9867
19	0.3381	0.7264	53.5%	0.0445	0.1201	-0.0363	0.9906	0.9594	0.9847
20	0.3593	0.7676	53.2%	0.0480	0.1285	-0.0391	0.9892	0.9546	0.9826
21	0.3807	0.8096	53.0%	0.0516	0.1371	-0.0419	0.9878	0.9494	0.9802
22	0.4027	0.8524	52.8%	0.0554	0.1462	-0.0449	0.9861	0.9437	0.9776
23	0.4255	0.8961	52.5%	0.0593	0.1557	-0.0481	0.9842	0.9376	0.9747
24	0.4497	0.9412	52.2%	0.0635	0.1657	-0.0513	0.9820	0.9309	0.9713
25	0.4740	0.9875	52.0%	0.0677	0.1760	-0.0546	0.9796	0.9237	0.9677
26	0.4992	1.0350	51.8%	0.0721	0.1866	-0.0581	0.9770	0.9158	0.9636
27	0.5252	1.0830	51.5%	0.0767	0.1974	-0.0617	0.9740	0.9074	0.9592
28	0.5514	1.1316	51.3%	0.0813	0.2084	-0.0653	0.9708	0.8984	0.9544
29	0.5781	1.1811	51.1%	0.0862	0.2200	-0.0692	0.9673	0.8888	0.9492
30	0.6054	1.2311	50.8%	0.0913	0.2319	-0.0732	0.9634	0.8785	0.9435
31	0.6334	1.2818	50.6%	0.0966	0.2441	-0.0774	0.9591	0.8675	0.9372
32	0.6614	1.3332	50.4%	0.1019	0.2564	-0.0814	0.9545	0.8558	0.9306
33	0.6892	1.3846	50.2%	0.1073	0.2690	-0.0856	0.9496	0.8434	0.9235
34	0.7175	1.4364	50.1%	0.1129	0.2818	-0.0898	0.9442	0.8304	0.9159
35	0.7467	1.4894	49.9%	0.1188	0.2951	-0.0942	0.9383	0.8164	0.9076
36	0.7762	1.5434	49.7%	0.1247	0.3088	-0.0987	0.9319	0.8014	0.8987
37	0.8060	1.5981	49.6%	0.1307	0.3227	-0.1032	0.9250	0.7854	0.8892
38	0.8361	1.6531	49.4%	0.1368	0.3368	-0.1078	0.9175	0.7687	0.8791
39	0.8663	1.7081	49.3%	0.1429	0.3512	-0.1124	0.9096	0.7511	0.8685
40	0.8969	1.7629	49.1%	0.1491	0.3657	-0.1170	0.9011	0.7329	0.8573
41	0.9284	1.8185	48.9%	0.1555	0.3806	-0.1217	0.8919	0.7136	0.8453
42	0.9610	1.8750	48.7%	0.1621	0.3959	-0.1265	0.8819	0.6932	0.8325
43	0.9946	1.9323	48.5%	0.1690	0.4113	-0.1316	0.8713	0.6717	0.8189
44	1.0282	1.9898	48.3%	0.1758	0.4269	-0.1366	0.8601	0.6495	0.8048
45	1.0619	2.0466	48.1%	0.1829	0.4424	-0.1418	0.8485	0.6267	0.7904
46	1.0955	2.1032	47.9%	0.1900	0.4581	-0.1470	0.8363	0.6033	0.7754
47	1.1296	2.1599	47.7%	0.1973	0.4738	-0.1524	0.8236	0.5792	0.7599
48	1.1639	2.2173	47.5%	0.2047	0.4897	-0.1578	0.8105	0.5542	0.7441
49	1.1984	2.2749	47.3%	0.2120	0.5055	-0.1632	0.7970	0.5286	0.7281
50	1.2332	2.3325	47.1%	0.2195	0.5215	-0.1687	0.7833	0.5026	0.7117
51	1.2676	2.3896	47.0%	0.2269	0.5374	-0.1742	0.7693	0.4762	0.6952
52	1.3017	2.4464	46.8%	0.2342	0.5532	-0.1795	0.7552	0.4495	0.6788
53	1.3360	2.5032	46.6%	0.2415	0.5689	-0.1850	0.7409	0.4226	0.6622
54	1.3707	2.5604	46.5%	0.2488	0.5845	-0.1904	0.7264	0.3952	0.6456
55	1.4055	2.6175	46.3%	0.2561	0.6001	-0.1959	0.7122	0.3679	0.6293
56	1.4401	2.6741	46.1%	0.2635	0.6155	-0.2015	0.6981	0.3407	0.6134
57	1.4747	2.7302	46.0%	0.2708	0.6310	-0.2073	0.6844	0.3138	0.5979
58	1.5090	2.7859	45.8%	0.2782	0.6464	-0.2132	0.6710	0.2872	0.5830
59	1.5434	2.8409	45.7%	0.2855	0.6616	-0.2190	0.6579	0.2611	0.5684
60	1.5778	2.8953	45.5%	0.2927	0.6764	-0.2248	0.6453	0.2357	0.5543
61	1.6119	2.9487	45.3%	0.2998	0.6909	-0.2306	0.6333	0.2110	0.5409
62	1.6456	3.0007	45.2%	0.3068	0.7050	-0.2362	0.6218	0.1873	0.5280
63	1.6792	3.0513	45.0%	0.3138	0.7188	-0.2419	0.6106	0.1647	0.5155
64	1.7126	3.1006	44.8%	0.3207	0.7323	-0.2476	0.5997	0.1429	0.5033
65	1.7457	3.1484	44.6%	0.3274	0.7456	-0.2533	0.5892	0.1222	0.4916
66	1.7781	3.1948	44.3%	0.3340	0.7582	-0.2589	0.5793	0.1024	0.4805
67	1.8098	3.2402	44.1%	0.3404	0.7704	-0.2645	0.5701	0.0835	0.4700
68	1.8408	3.2842	43.9%	0.3466	0.7822	-0.2700	0.5613	0.0655	0.4600
69	1.8710	3.3266	43.8%	0.3527	0.7936	-0.2754	0.5529	0.0483	0.4504
70	1.9003	3.3677	43.6%	0.3586	0.8047	-0.2807	0.5450	0.0319	0.4413
71	1.9284	3.4069	43.4%	0.3642	0.8153	-0.2858	0.5375	0.0165	0.4327
72	1.9556	3.4445	43.2%	0.3695	0.8253	-0.2907	0.5306	0.0020	0.4247
73	1.9823	3.4811	43.1%	0.3745	0.8349	-0.2956	0.5240	-0.0119	0.4171
74	2.0082	3.5164	42.9%	0.3795	0.8443	-0.3003	0.5174	-0.0252	0.4097
75	2.0336	3.5499	42.7%	0.3842	0.8532	-0.3050	0.5108	-0.0377	0.4024
76	2.0585	3.5823	42.5%	0.3889	0.8619	-0.3097	0.5045	-0.0496	0.3954
77	2.0822	3.6139	42.4%	0.3934	0.8703	-0.3143	0.4989	-0.0609	0.3892
78	2.1047	3.6437	42.2%	0.3976	0.8782	-0.3185	0.4935	-0.0716	0.3832
79	2.1267	3.6724	42.1%	0.4017	0.8858	-0.3228	0.4881	-0.0819	0.3772
80	2.1480	3.6998	41.9%	0.4058	0.8931	-0.3270	0.4827	-0.0917	0.3714
81	2.1684	3.7255	41.8%	0.4097	0.9000	-0.3310	0.4773	-0.1009	0.3655
82	2.1883	3.7494	41.6%	0.4135	0.9064	-0.3349	0.4717	-0.1095	0.3597
83	2.2072	3.7716	41.5%	0.4170	0.9123	-0.3386	0.4663	-0.1173	0.3540
84	2.2257	3.7922	41.3%	0.4205	0.9179	-0.3423	0.4608	-0.1245	0.3483
85	2.2427	3.8114	41.2%	0.4232	0.9231	-0.3453	0.4557	-0.1313	0.3432
86	2.2582	3.8295	41.0%	0.4253	0.9278	-0.3478	0.4512	-0.1376	0.3387
87	2.2723	3.8461	40.9%	0.4271	0.9323	-0.3500	0.4471	-0.1434	0.3347
88	2.2843	3.8609	40.8%	0.4285	0.9365	-0.3519	0.4434	-0.1489	0.3310
89	2.2945	3.8738	40.8%	0.4295	0.9402	-0.3533	0.4398	-0.1542	0.3274
90	2.3031	3.8852	40.7%	0.4302	0.9435	-0.3544	0.4363	-0.1593	0.3240

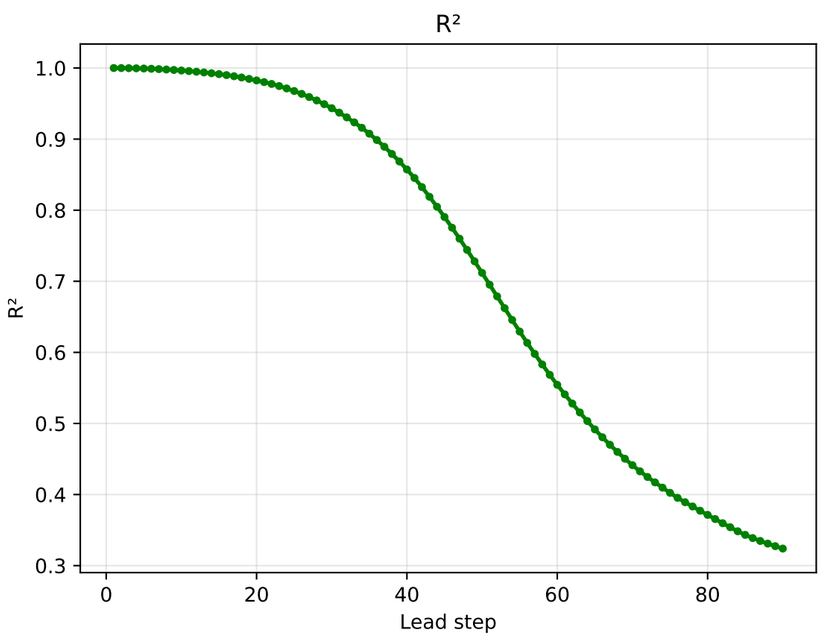
DHW Forecast Skill vs Lead Step - Val

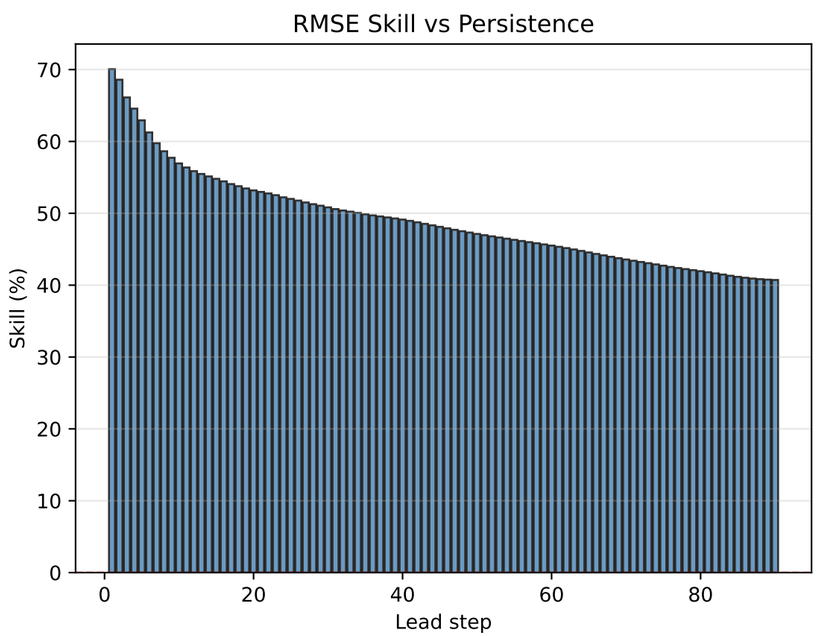


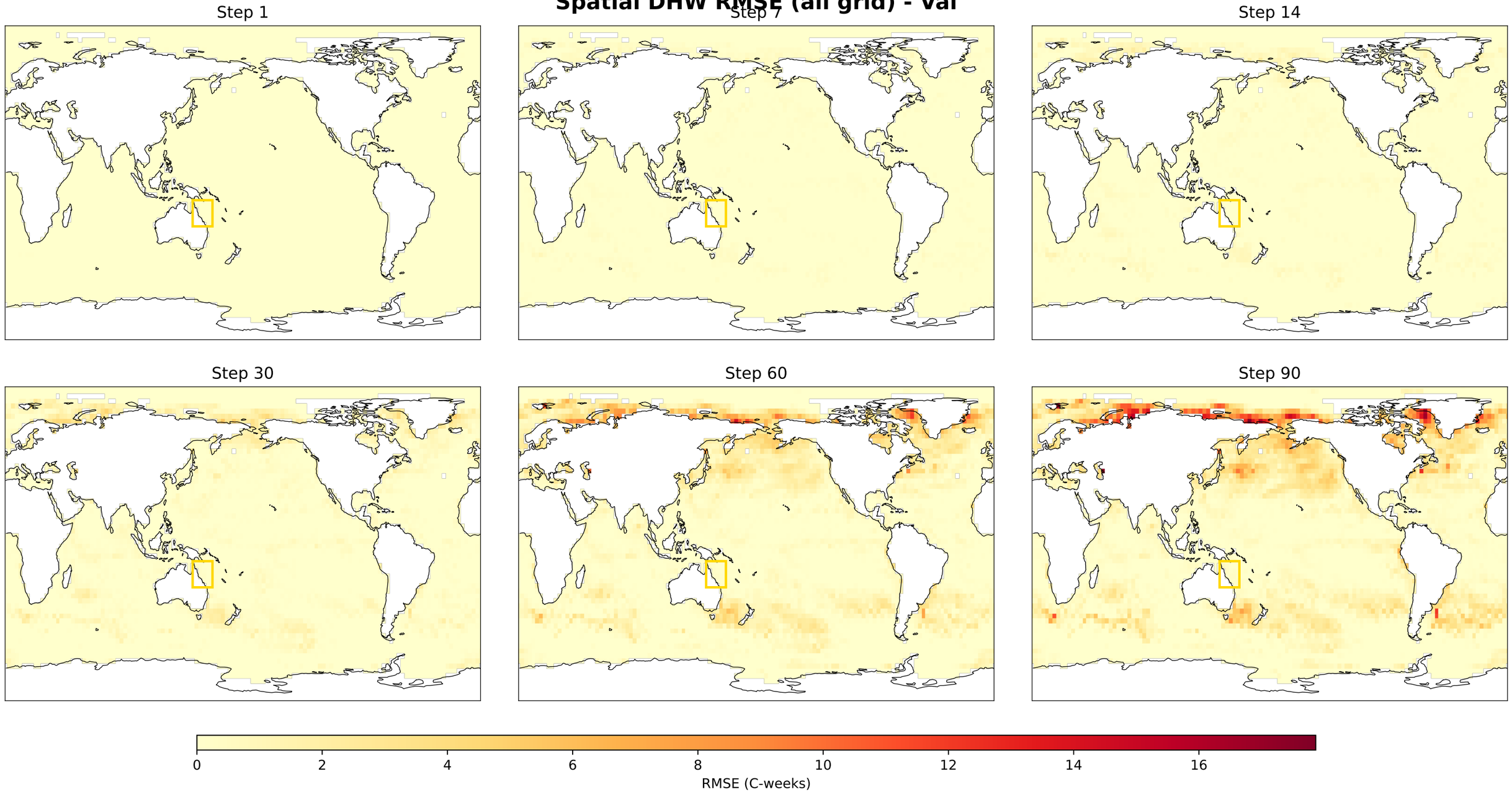


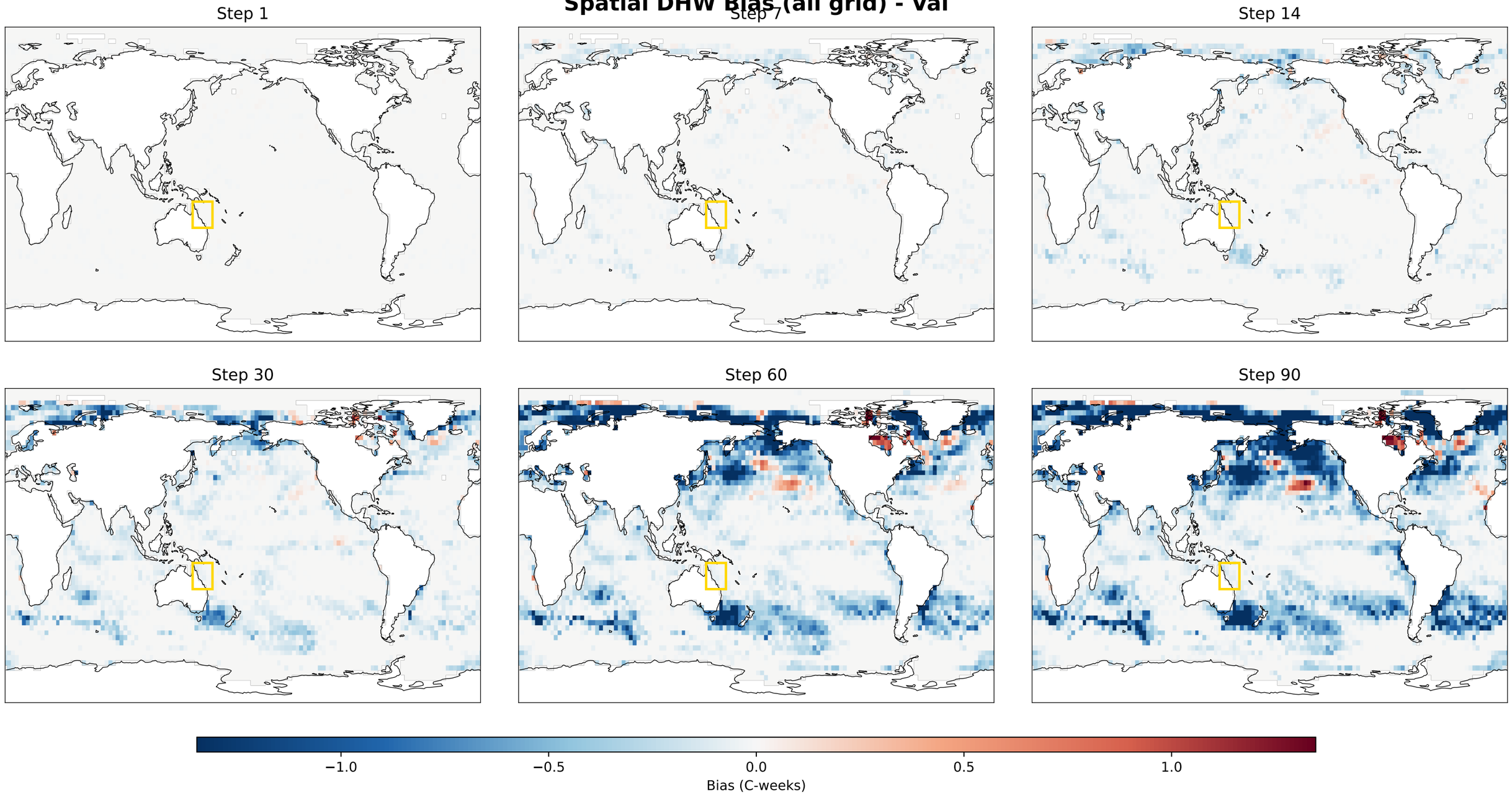


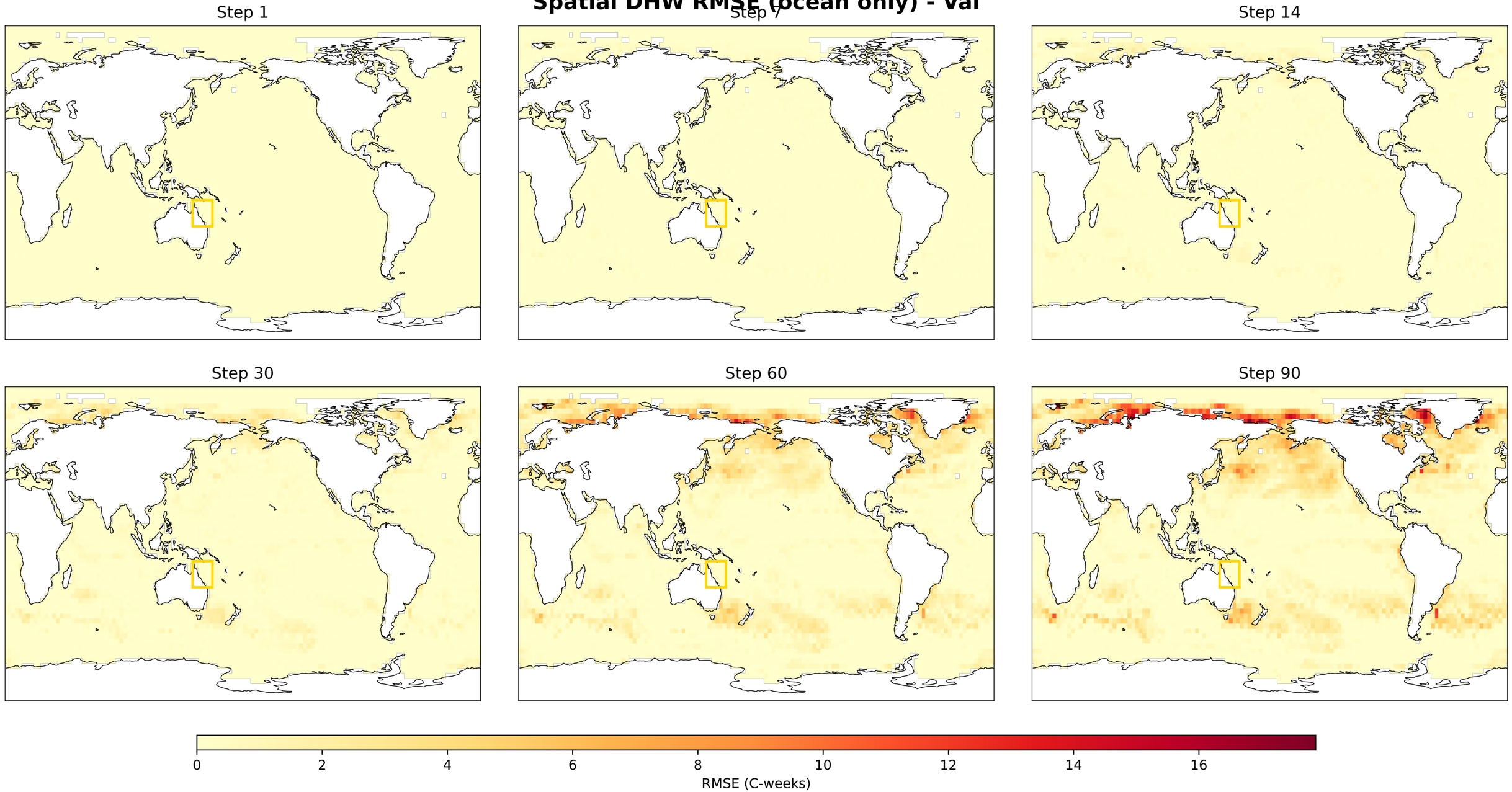


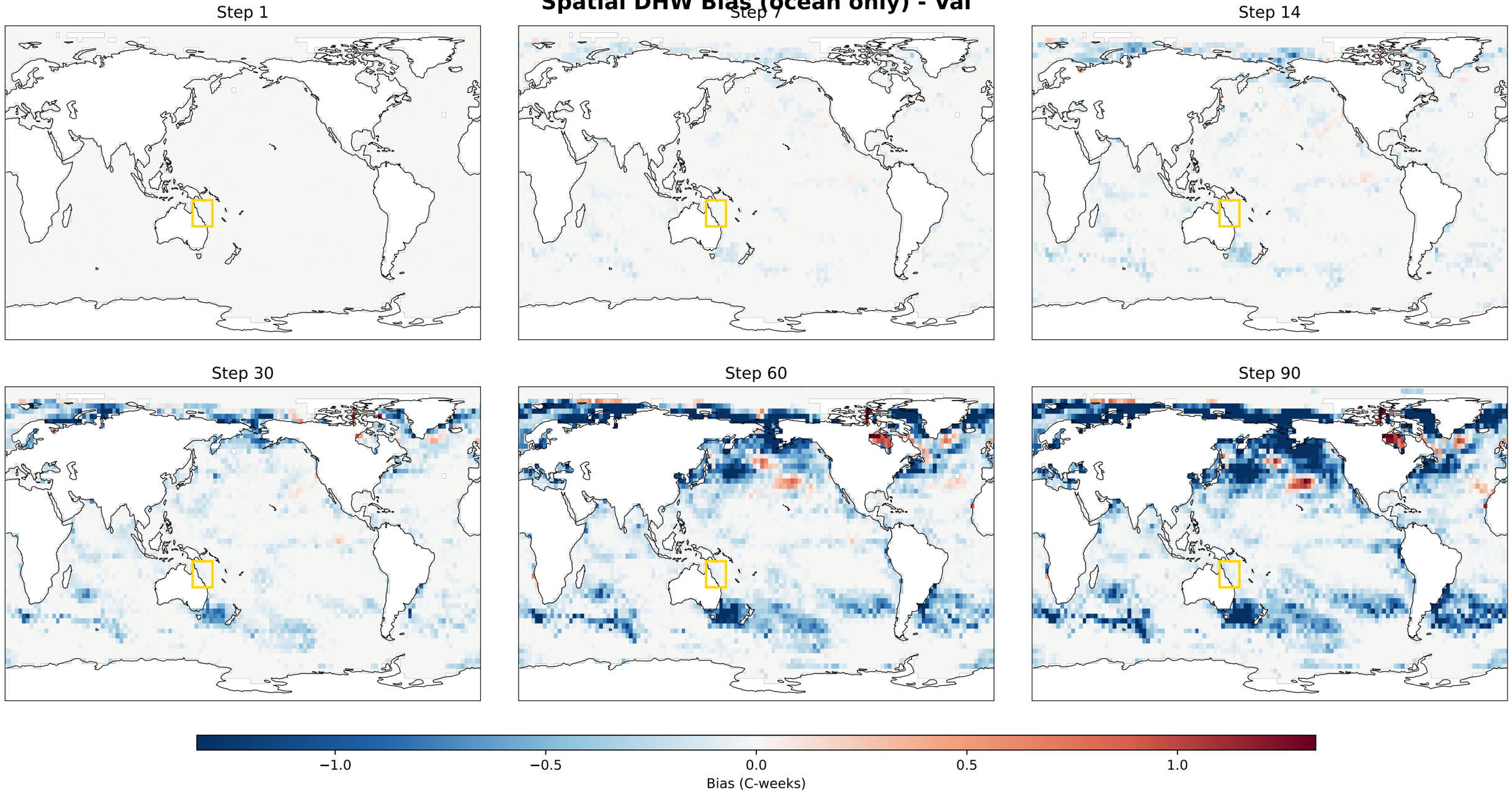





Spatial DHW RMSE (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
2
4
6
8
10
12
14
16
RMSE (C-weeks)

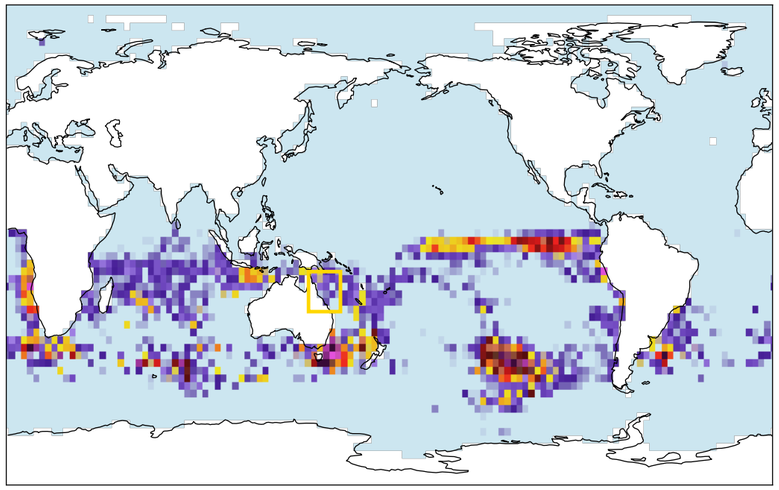

Spatial DHW Bias (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
2
4
6
8
10
12
14
16
RMSE (C-weeks)

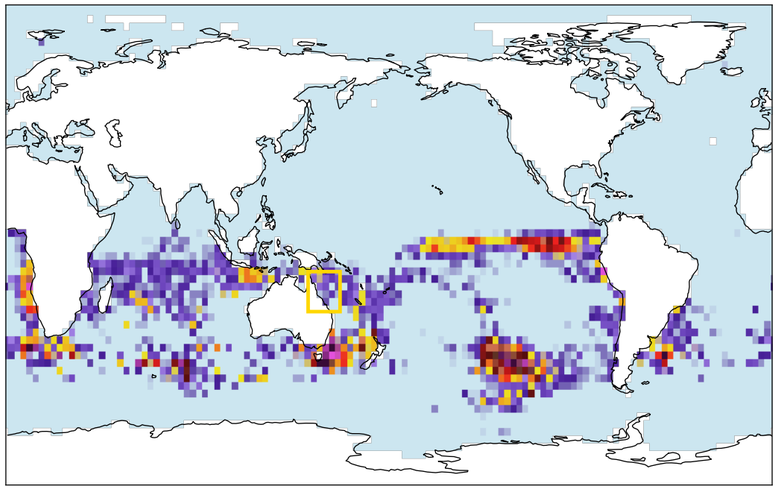

Spatial DHW Bias (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val (C-weeks)

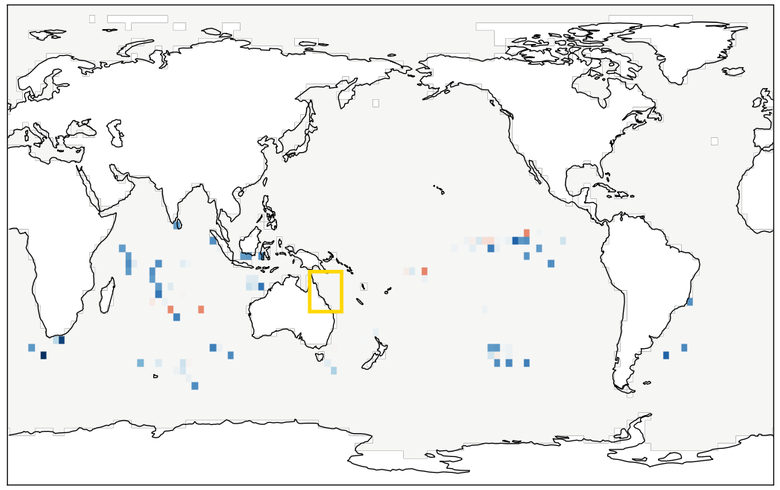
Actual DHW - Step 1



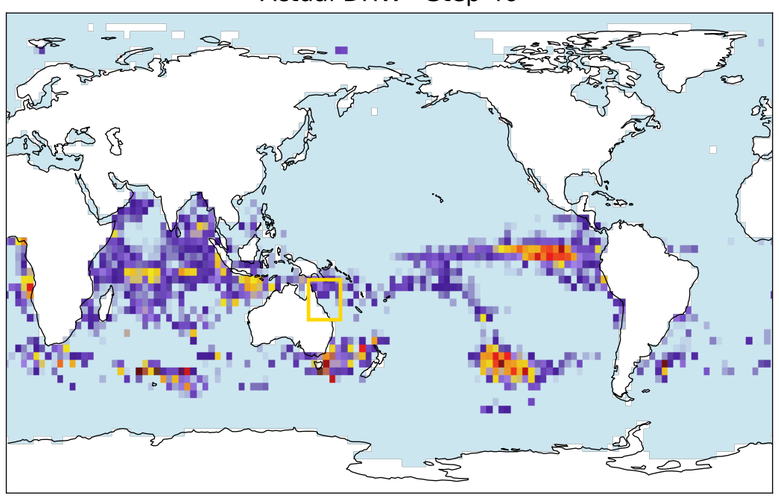
Predicted DHW - Step 1



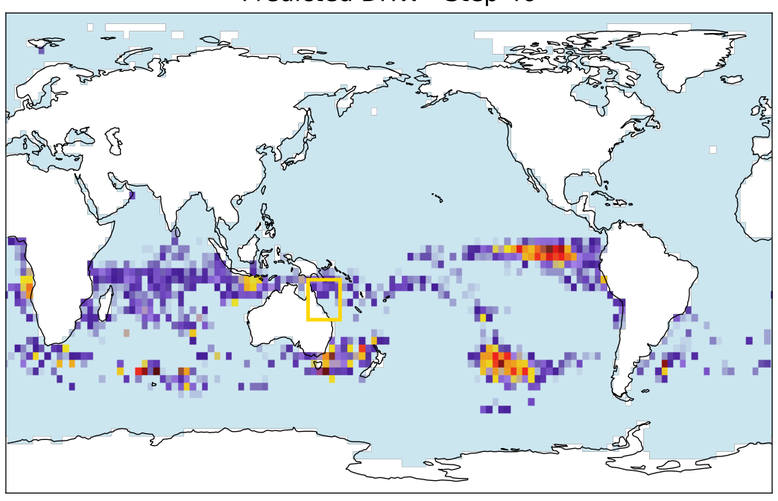
Error (Pred - Actual) - Step 1 [scale ±0.31]



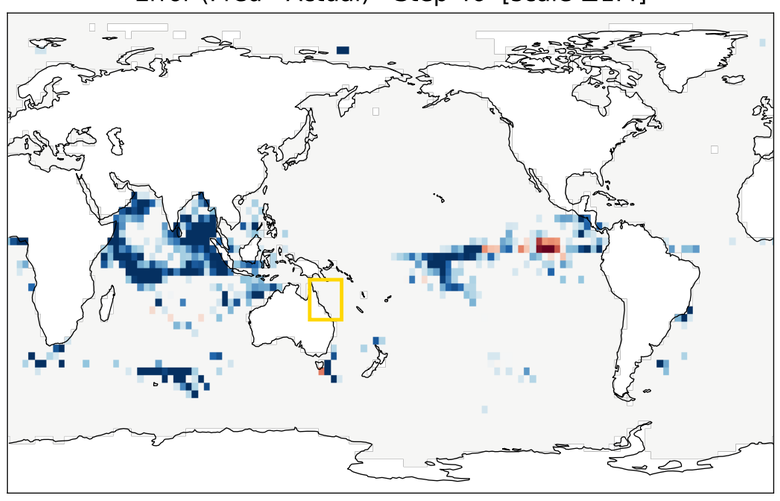
Actual DHW - Step 46



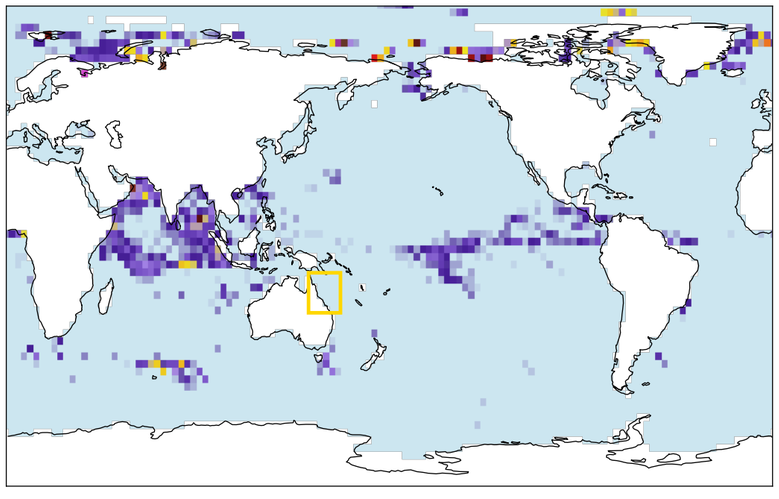
Predicted DHW - Step 46



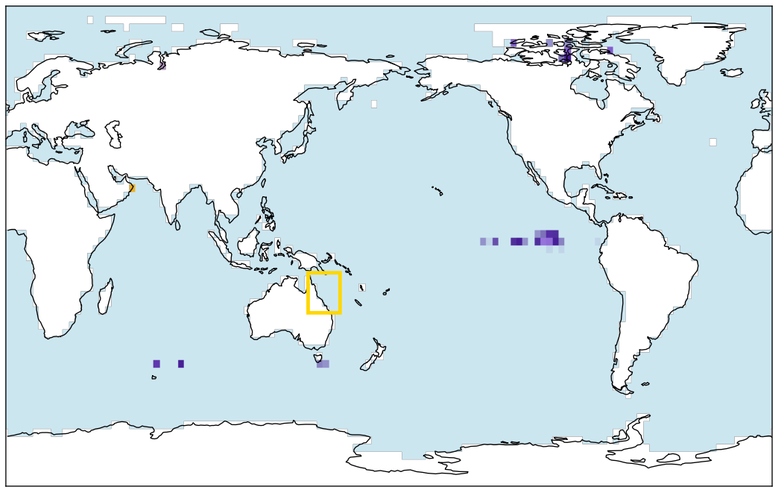
Error (Pred - Actual) - Step 46 [scale ±1.4]



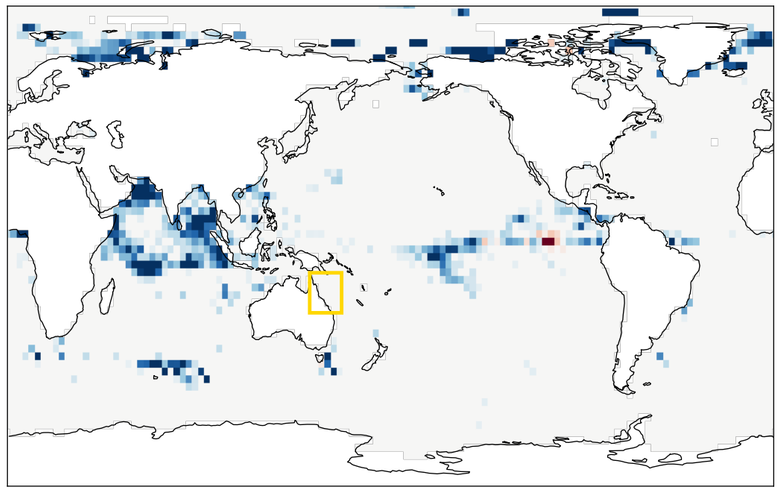
Actual DHW - Step 90

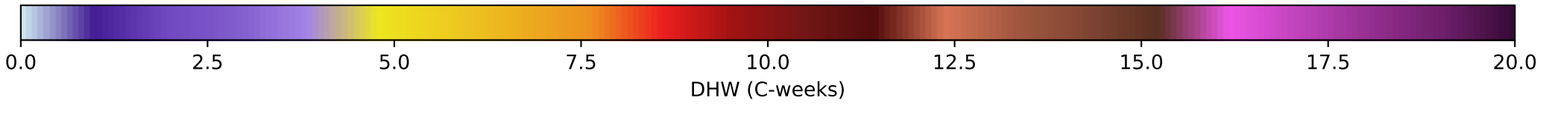


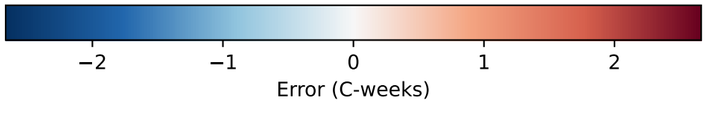
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.7]

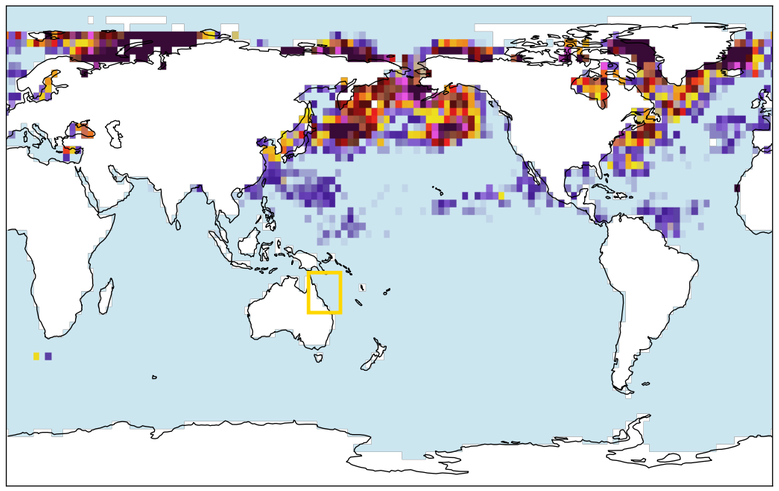




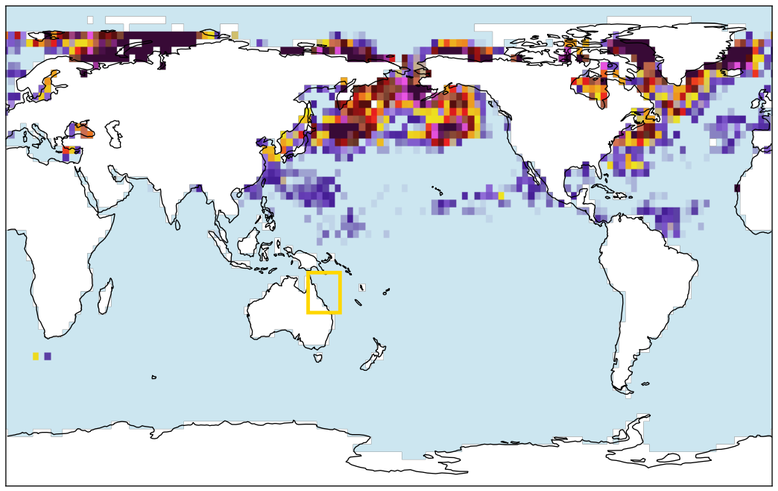


DHW Chain init: 2016-10-07 - Val (C-weeks)

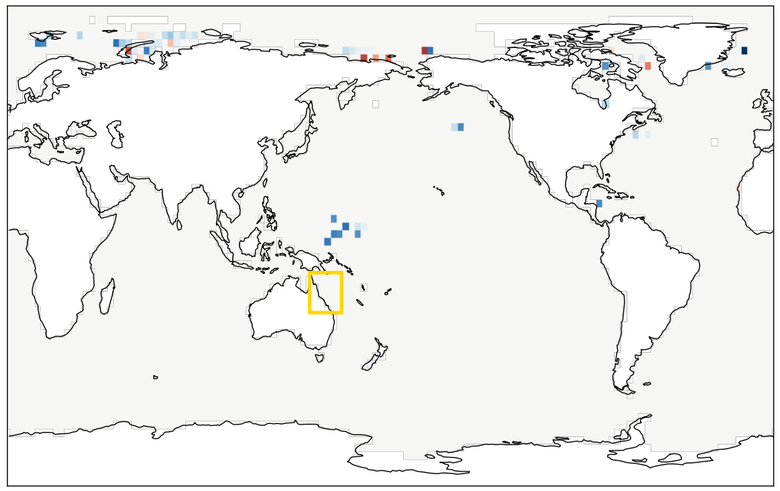
Actual DHW - Step 1



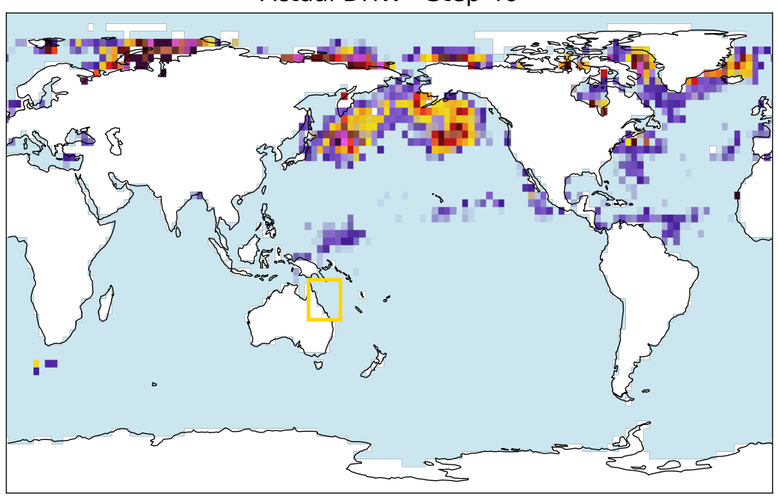
Predicted DHW - Step 1



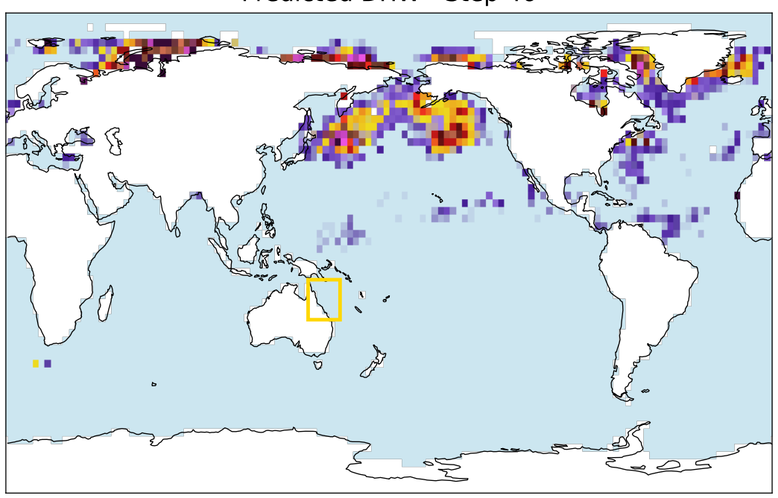
Error (Pred - Actual) - Step 1 [scale ±0.28]



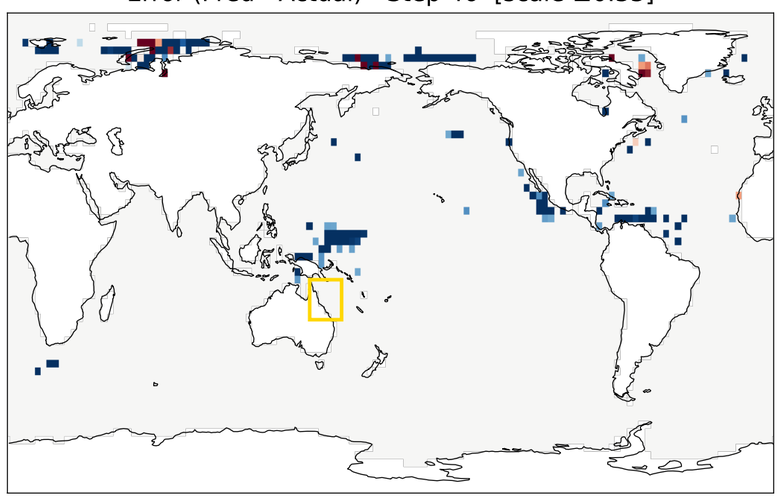
Actual DHW - Step 46



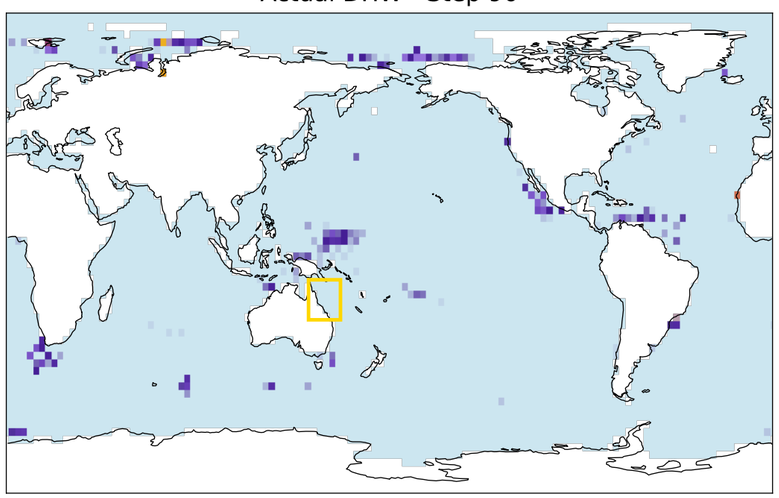
Predicted DHW - Step 46



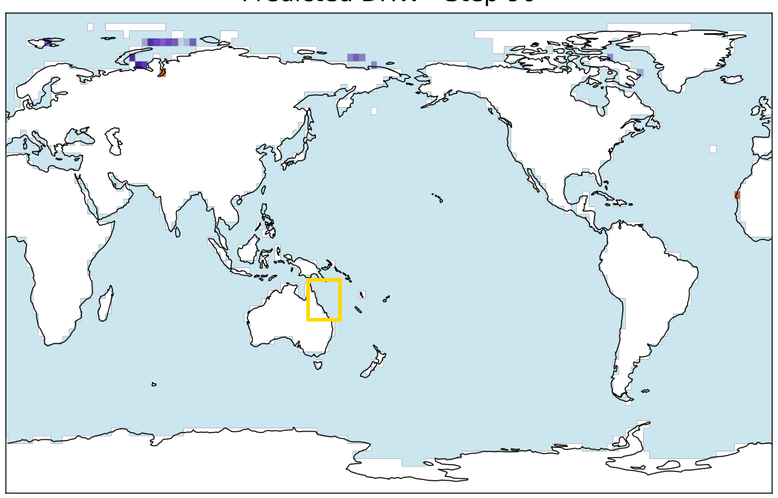
Error (Pred - Actual) - Step 46 [scale ±0.33]



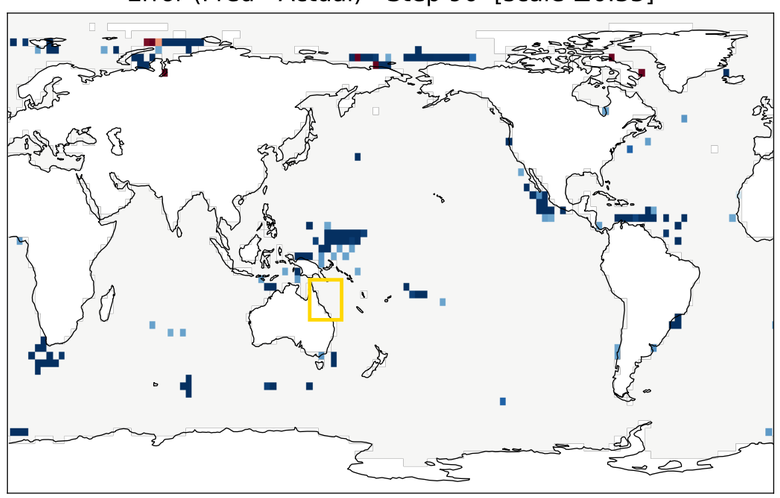
Actual DHW - Step 90

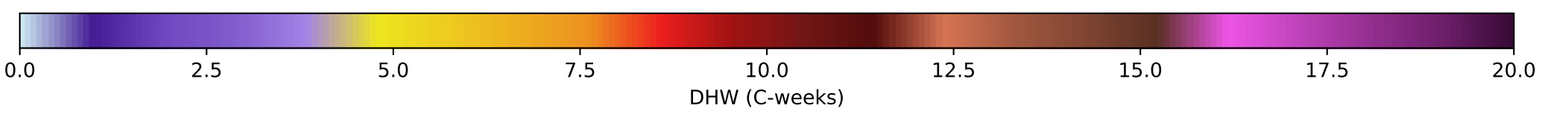


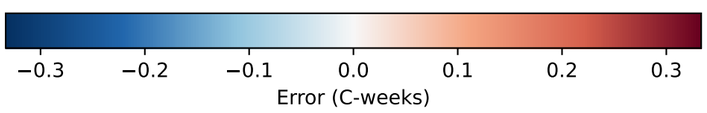
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

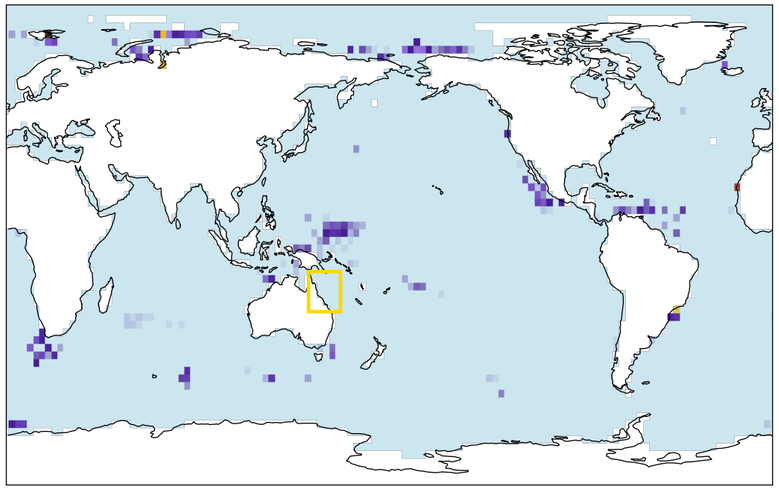




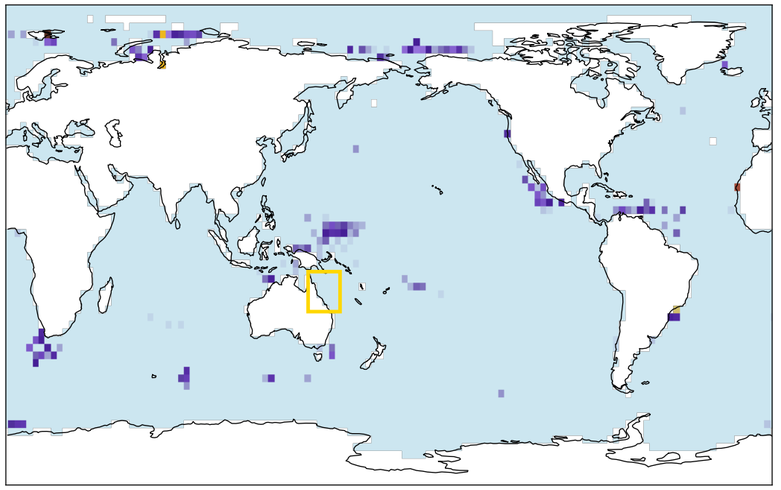


DHW Chain init: 2017-01-05 - Val (C-weeks)

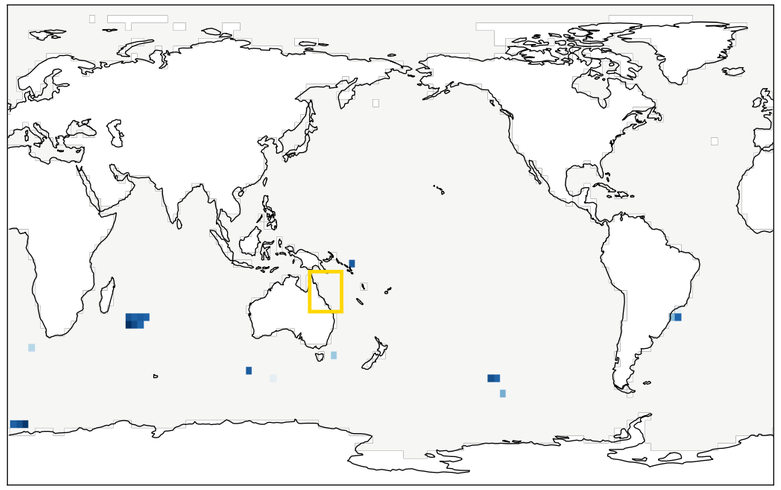
Actual DHW - Step 1



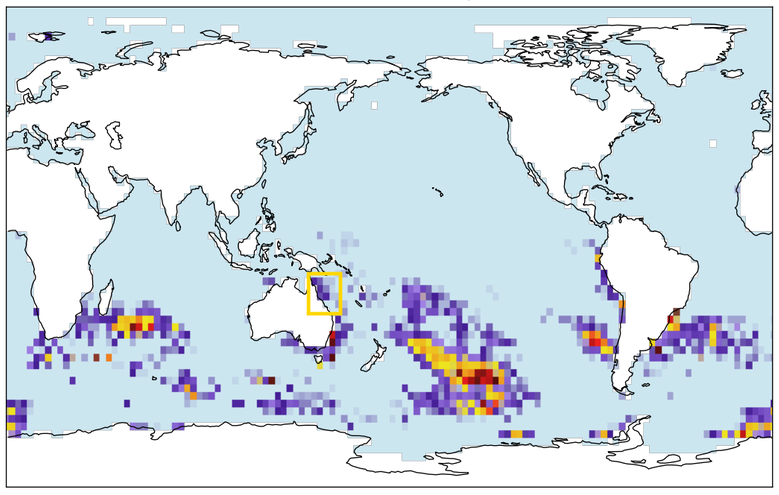
Predicted DHW - Step 1



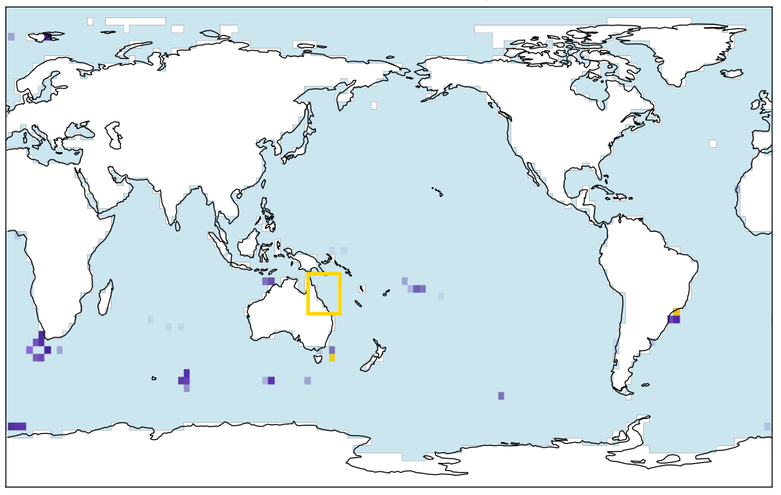
Error (Pred - Actual) - Step 1 [scale ±0.22]



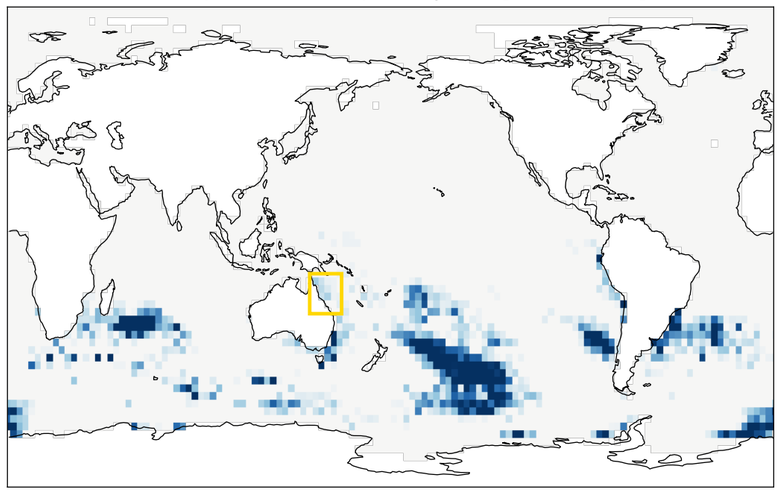
Actual DHW - Step 46



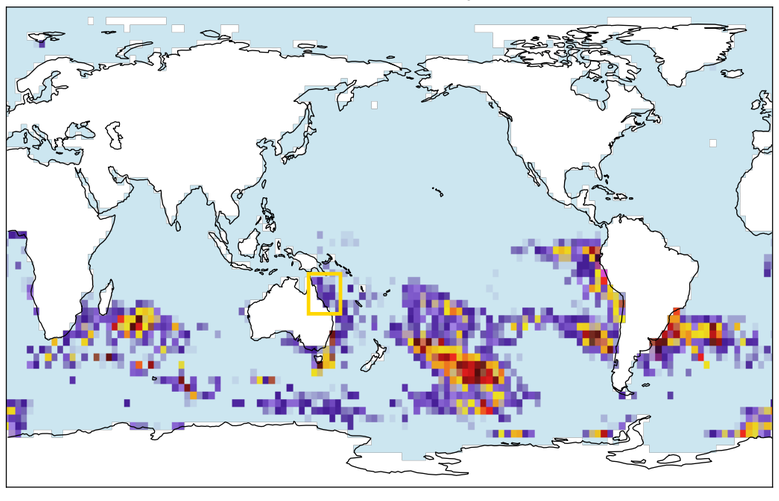
Predicted DHW - Step 46



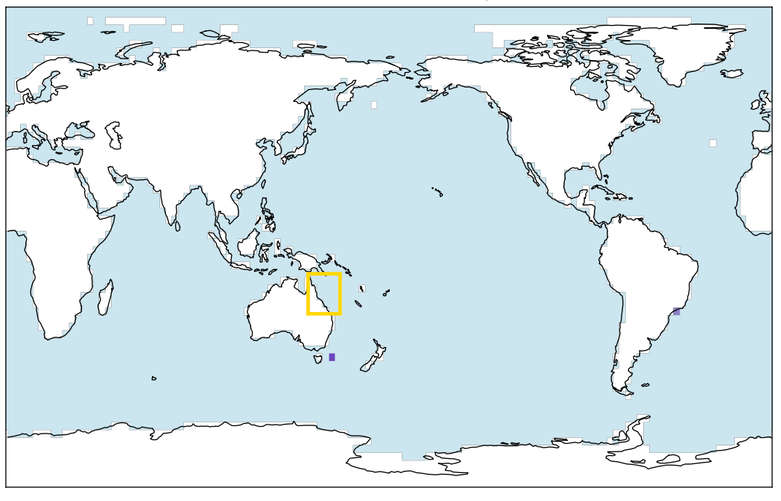
Error (Pred - Actual) - Step 46 [scale ±4.8]



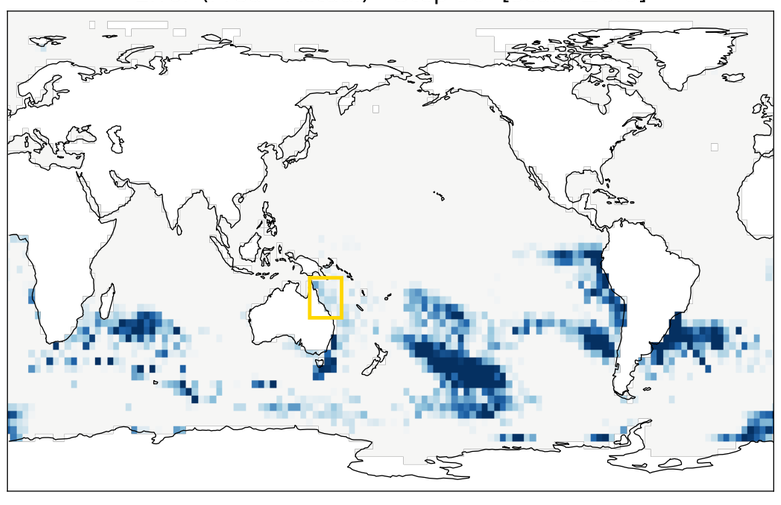
Actual DHW - Step 90

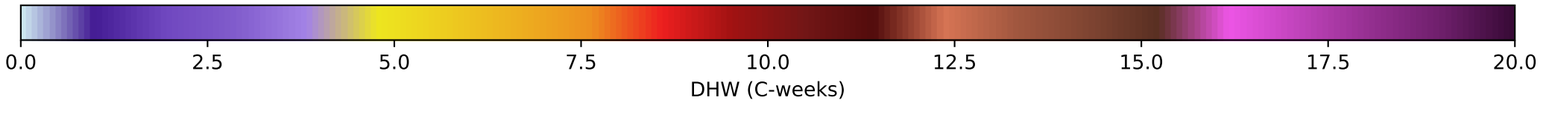


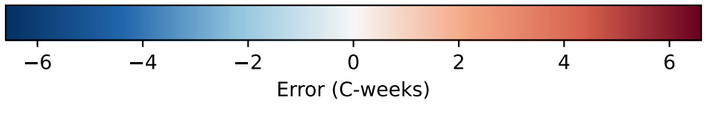
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.6]

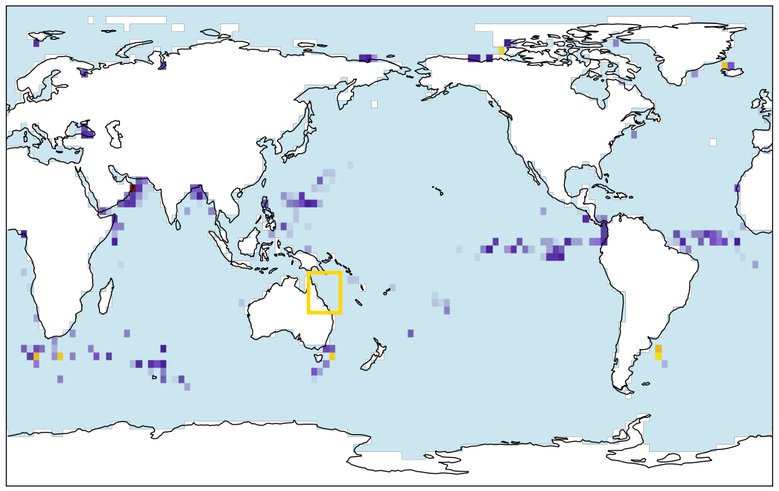




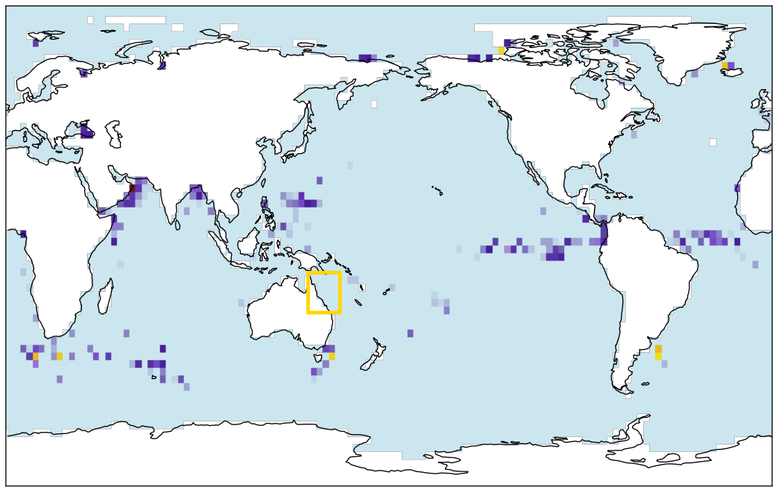


DHW Chain init: 2017-07-04 - Val (C-weeks)

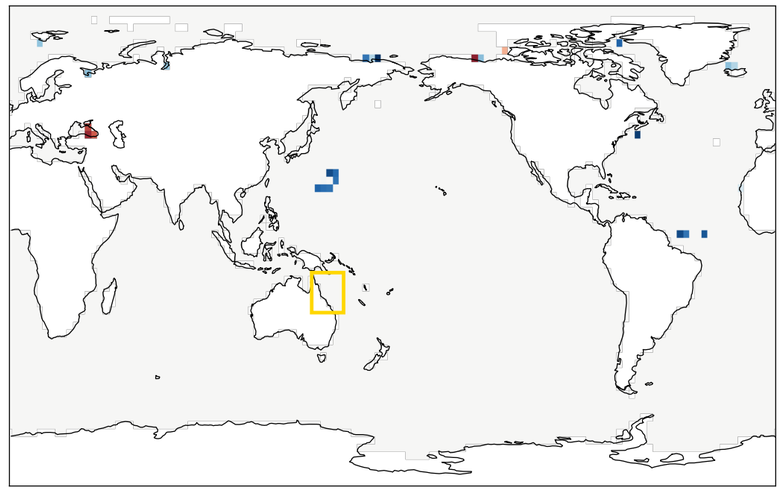
Actual DHW - Step 1



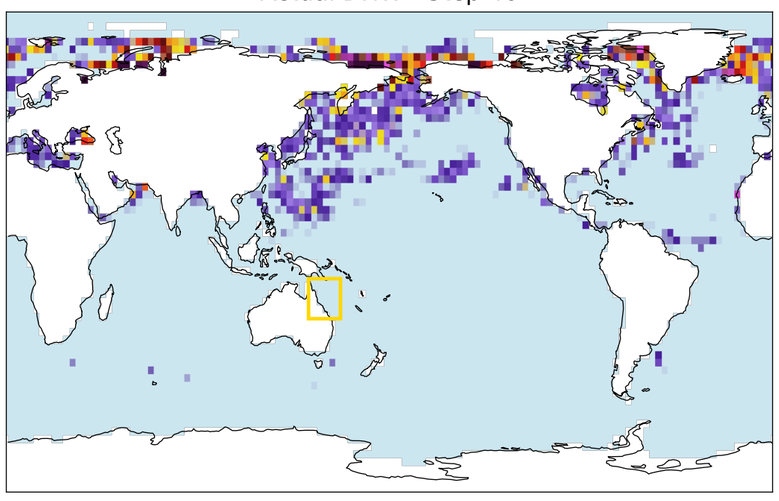
Predicted DHW - Step 1



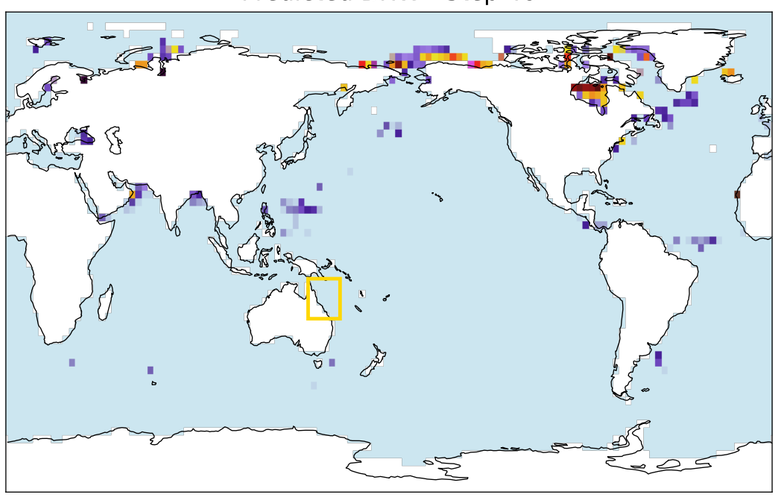
Error (Pred - Actual) - Step 1 [scale ±0.24]



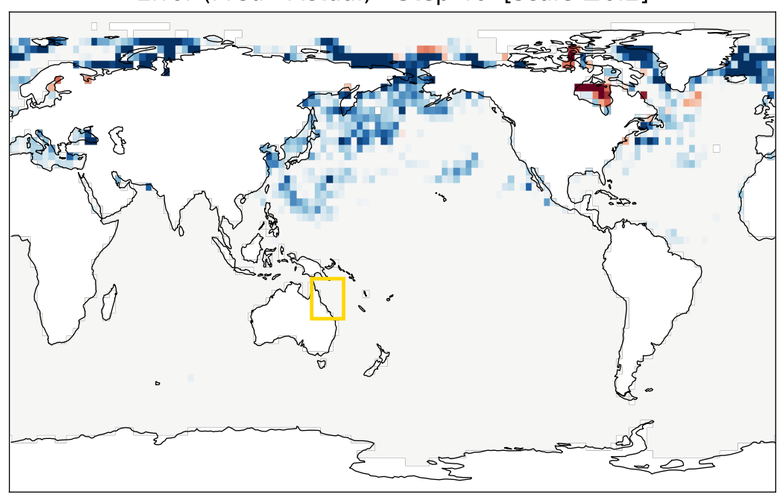
Actual DHW - Step 46



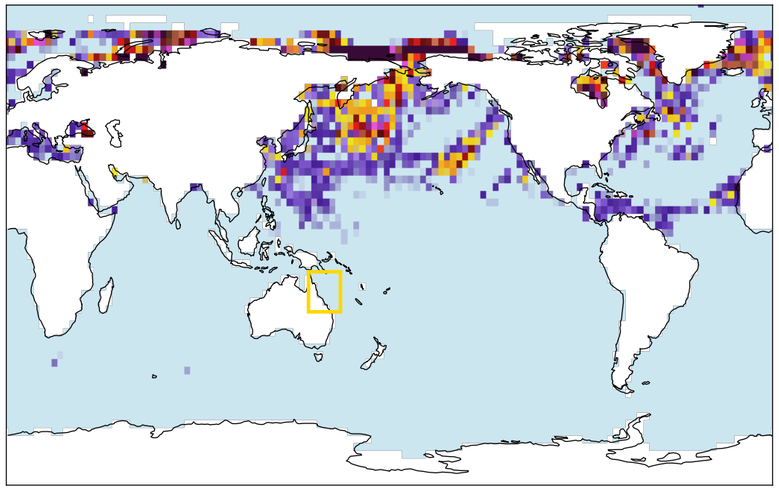
Predicted DHW - Step 46



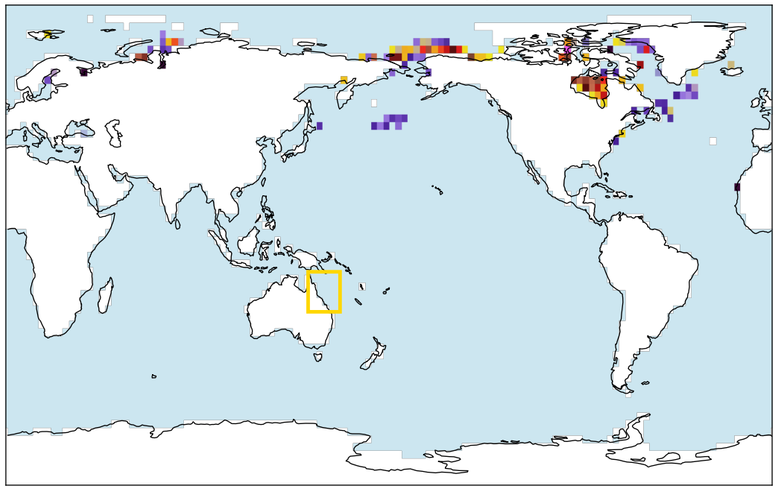
Error (Pred - Actual) - Step 46 [scale ±6.2]



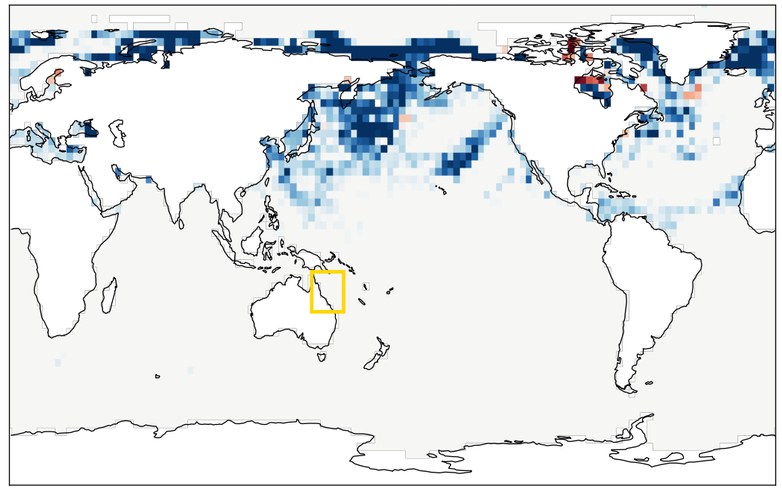
Actual DHW - Step 90

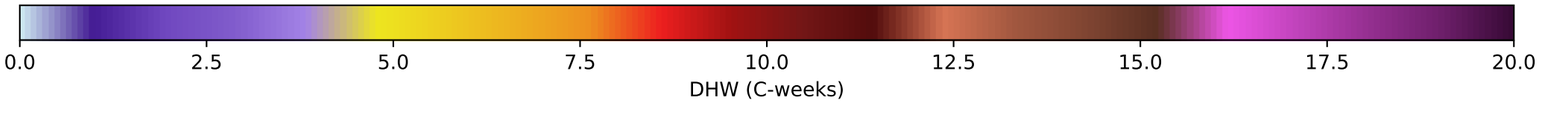


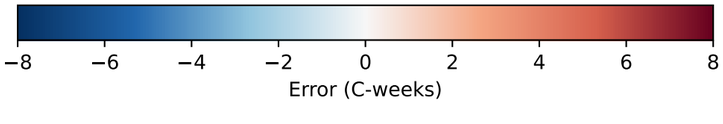
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

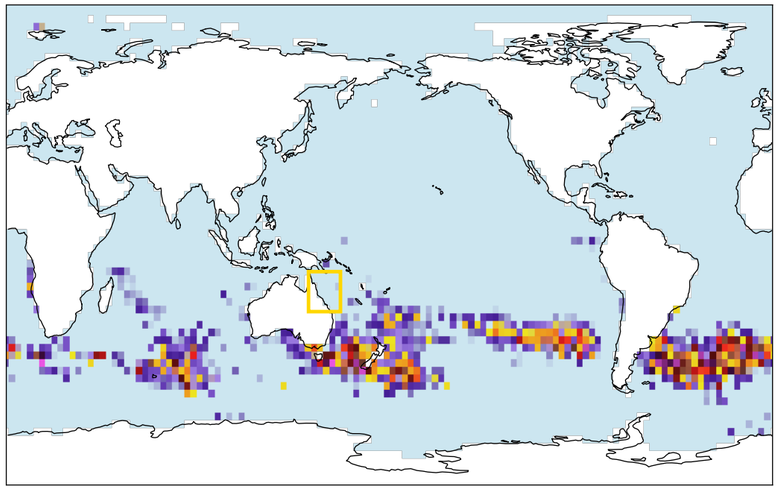




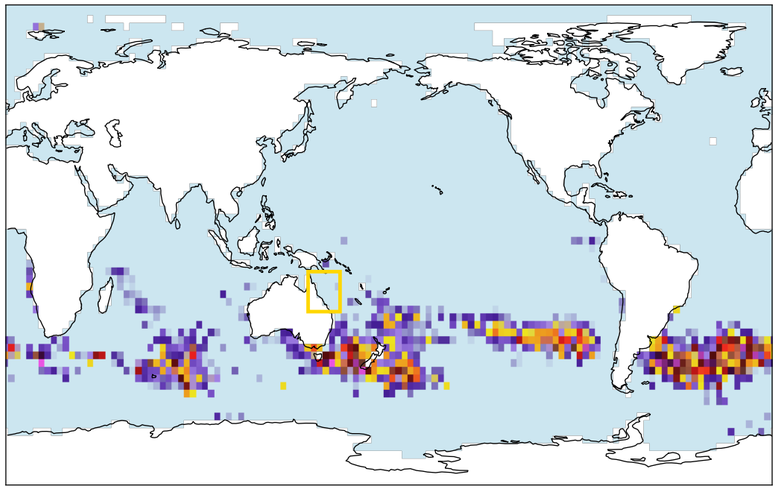


DHW Chain init: 2018-03-31 - Val (C-weeks)

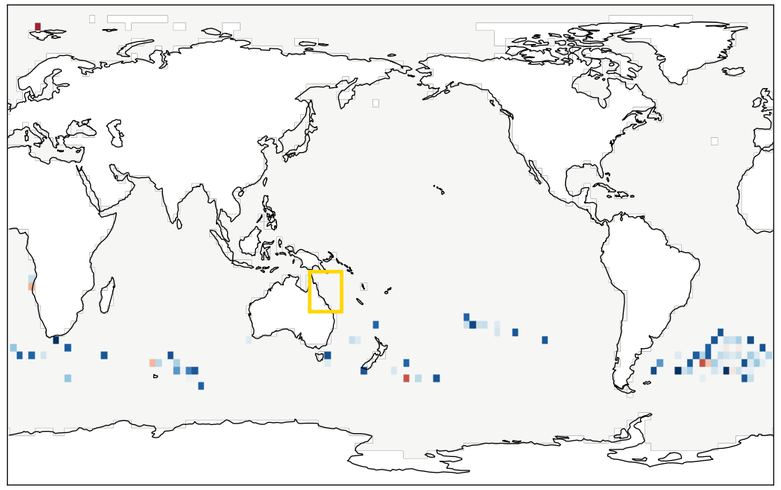
Actual DHW - Step 1



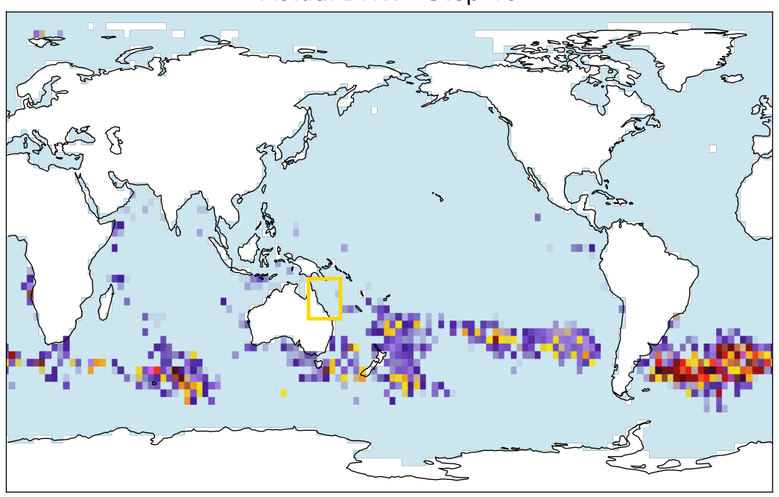
Predicted DHW - Step 1



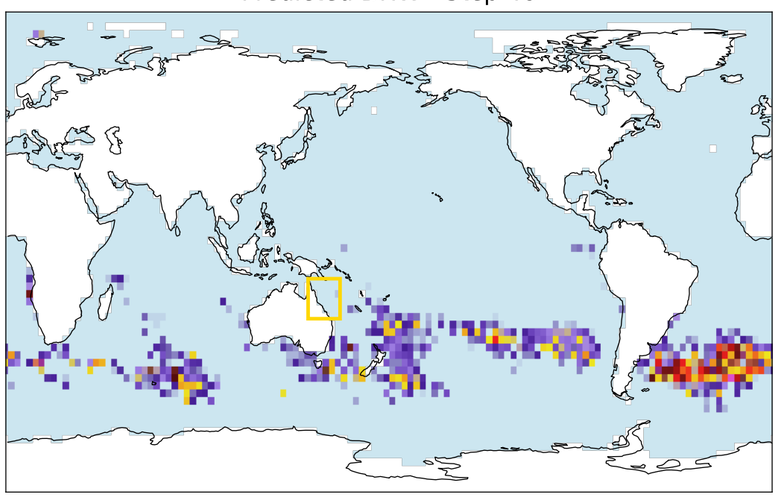
Error (Pred - Actual) - Step 1 [scale ±0.21]



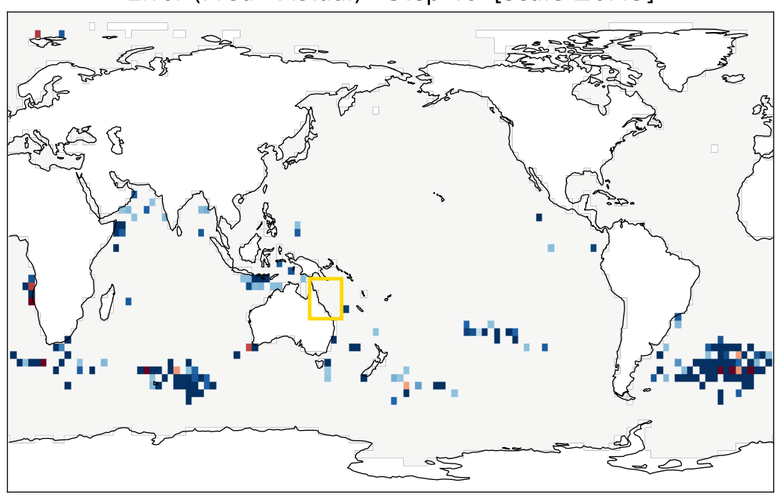
Actual DHW - Step 46



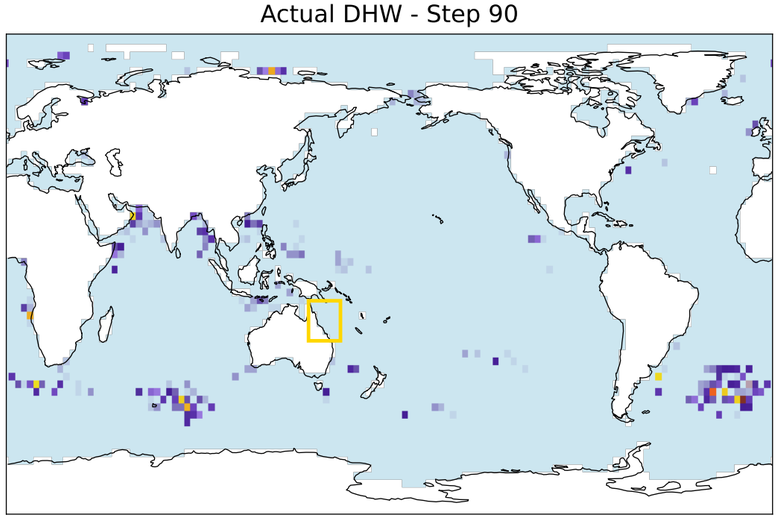
Predicted DHW - Step 46



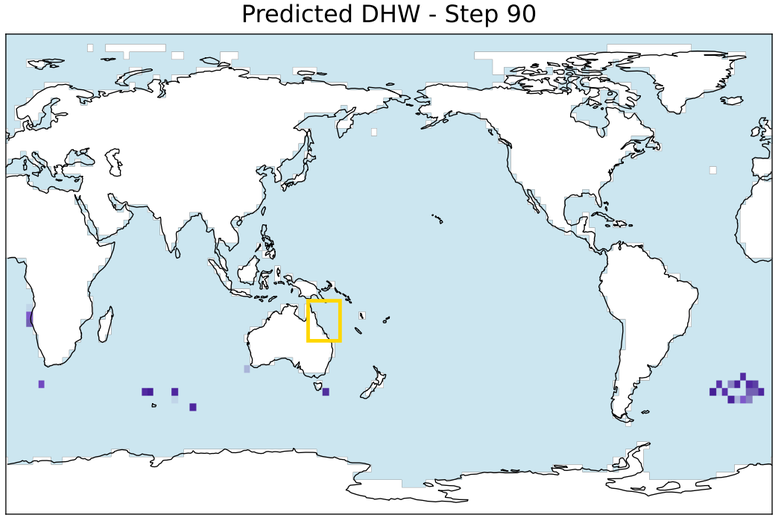
Error (Pred - Actual) - Step 46 [scale ±0.43]



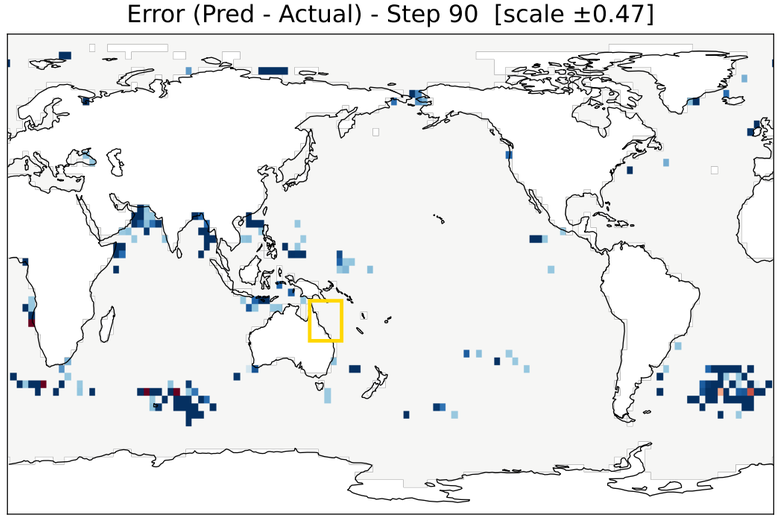
Actual DHW - Step 90

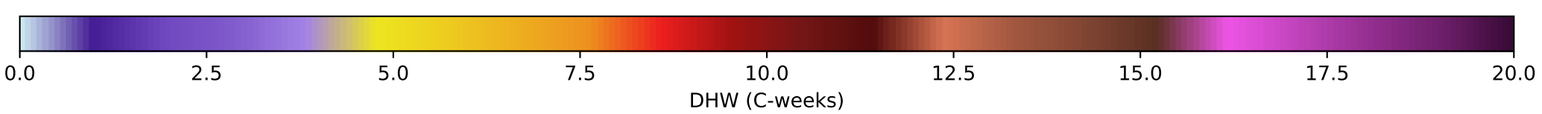


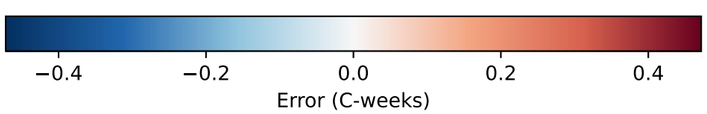
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.47]

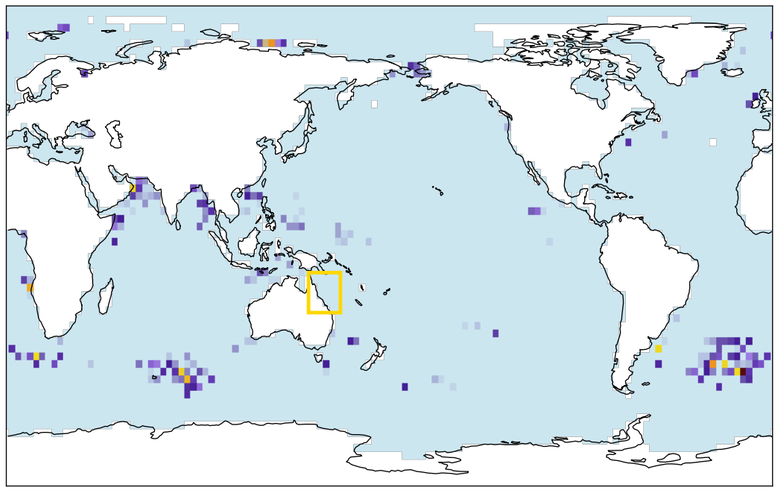




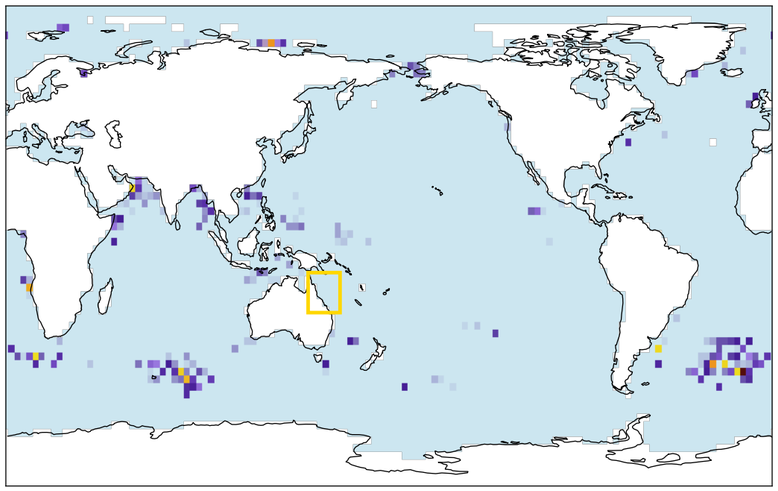


DHW Chain init: 2018-06-29 - Val (C-weeks)

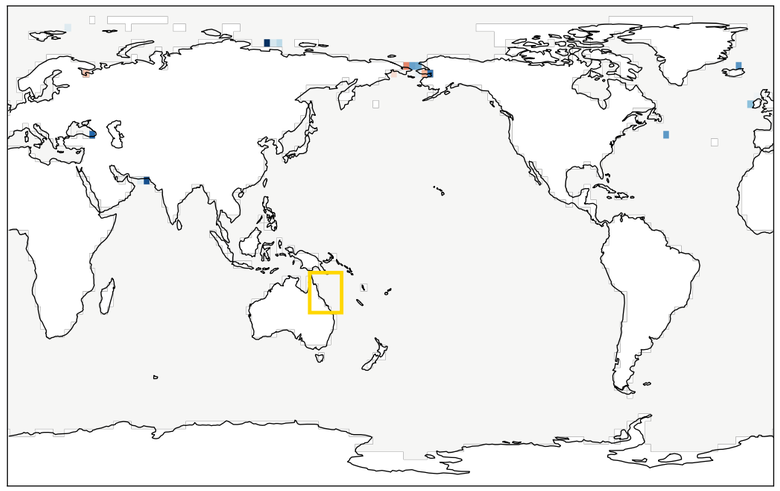
Actual DHW - Step 1



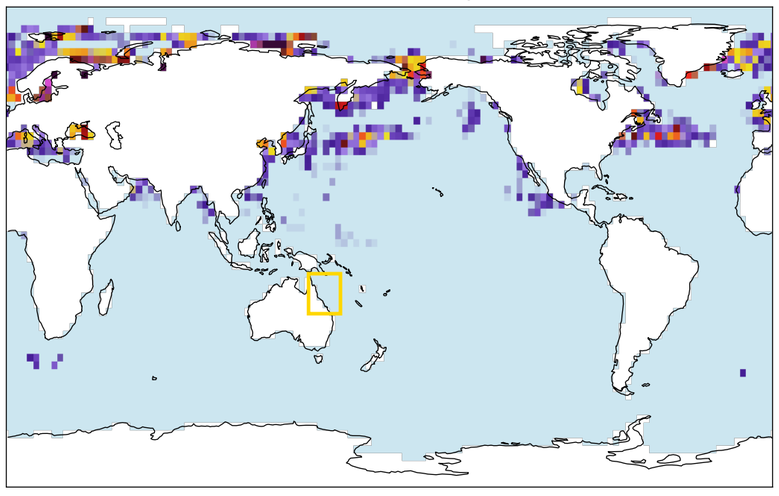
Predicted DHW - Step 1



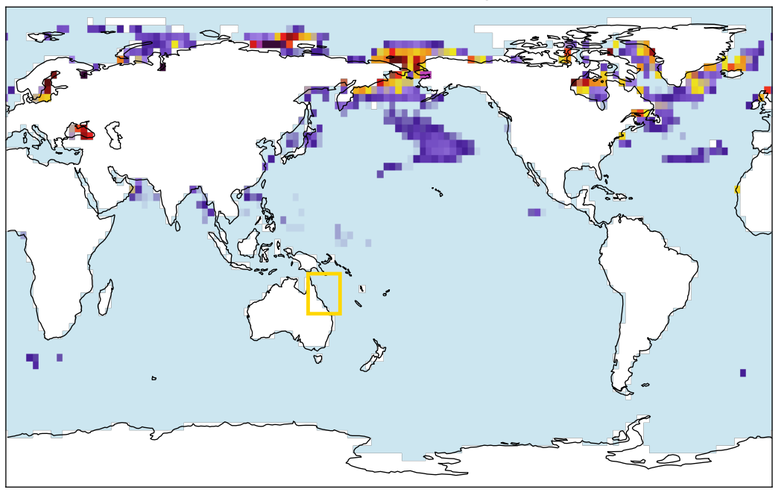
Error (Pred - Actual) - Step 1 [scale ±0.3]



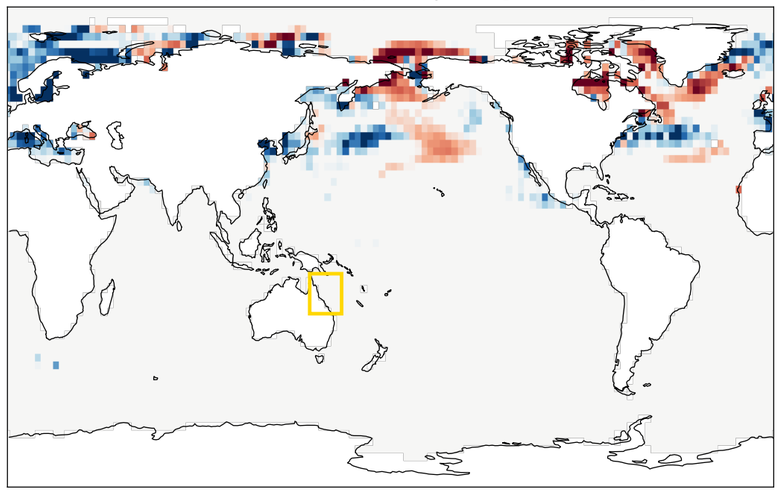
Actual DHW - Step 46



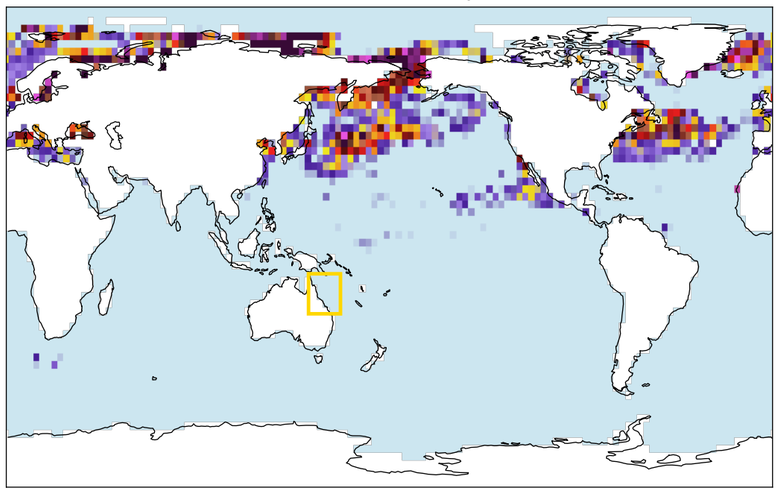
Predicted DHW - Step 46



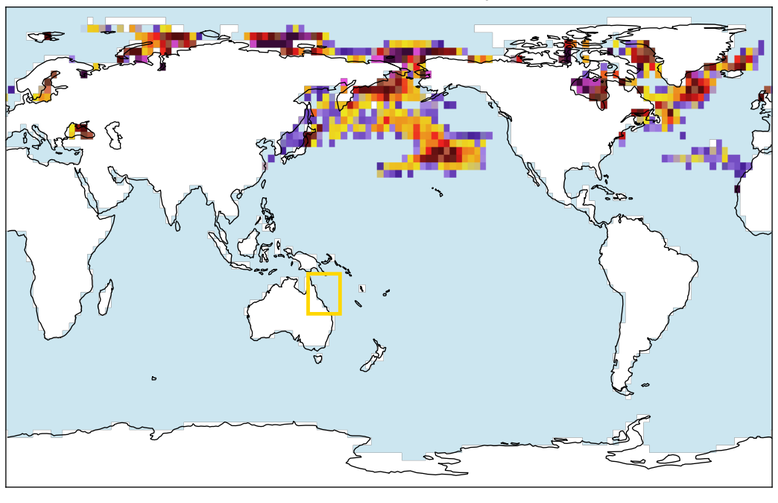
Error (Pred - Actual) - Step 46 [scale ±5.5]



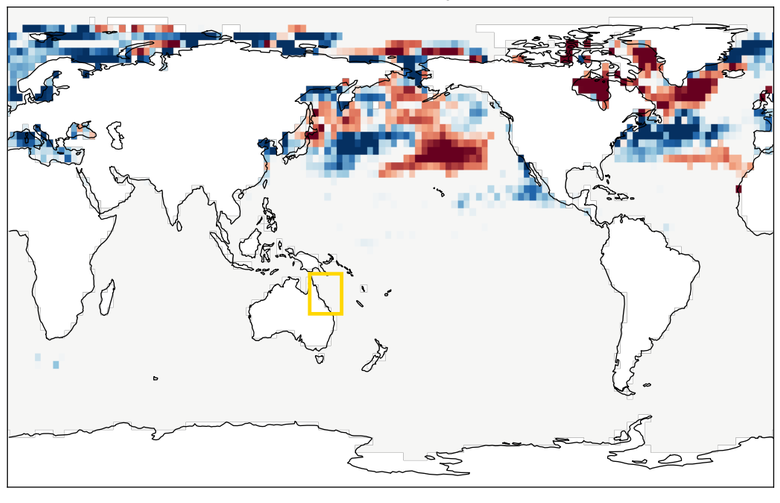
Actual DHW - Step 90

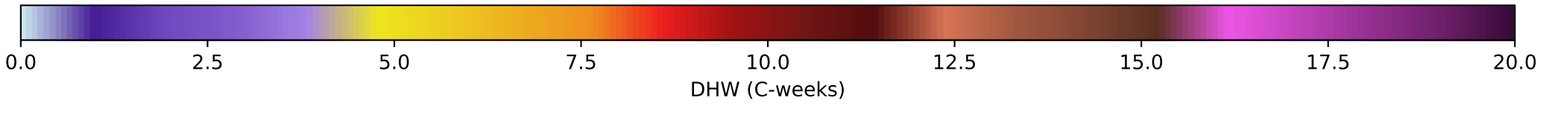


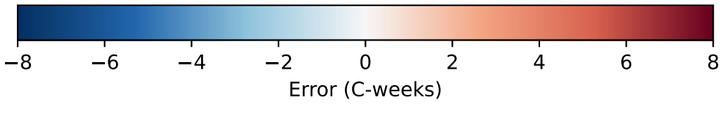
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

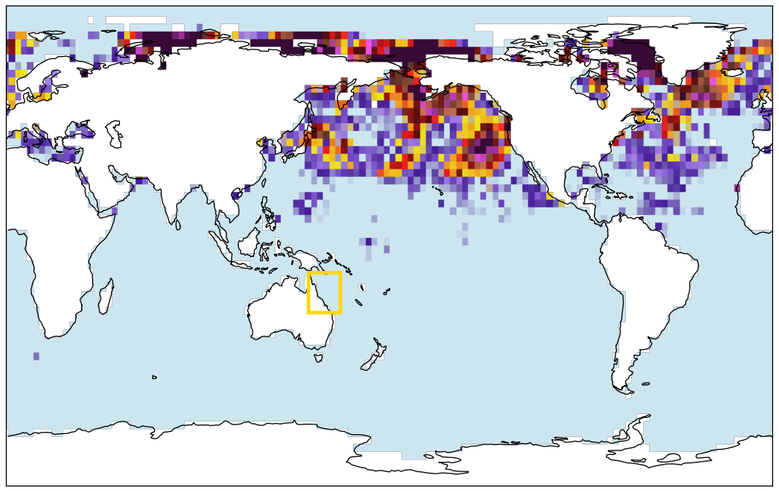




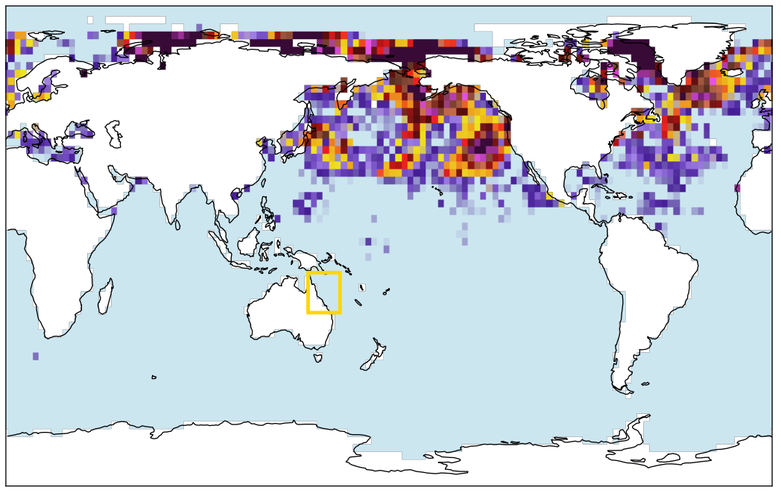


DHW Chain init: 2019-09-22 - Val (C-weeks)

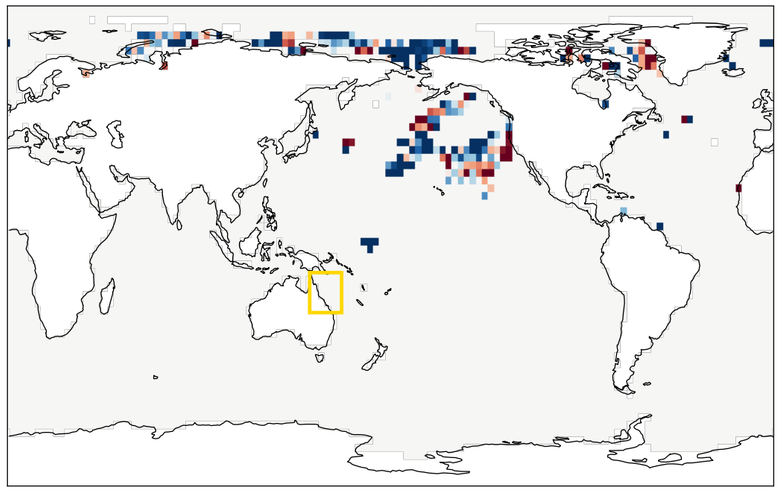
Actual DHW - Step 1



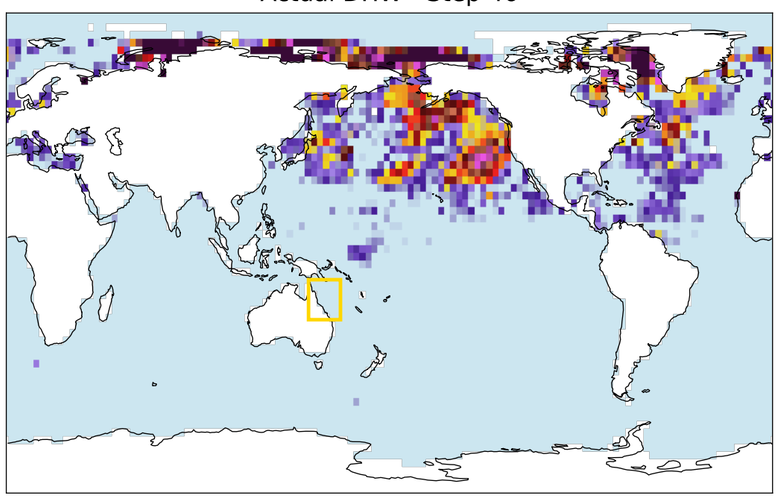
Predicted DHW - Step 1



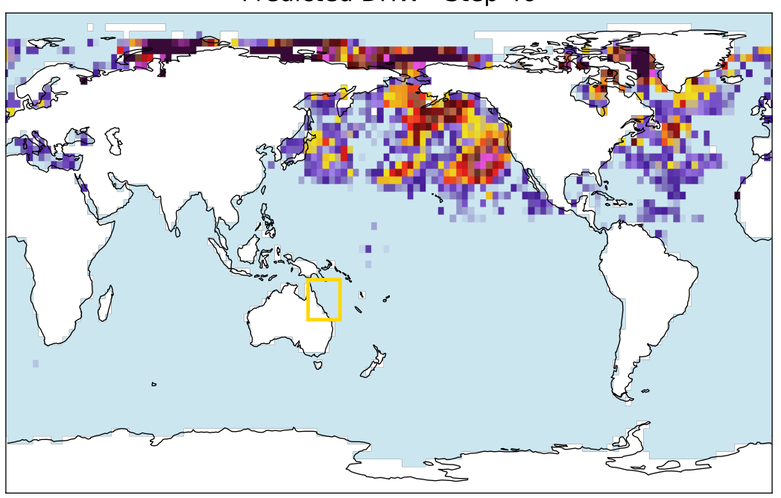
Error (Pred - Actual) - Step 1 [scale ±0.031]



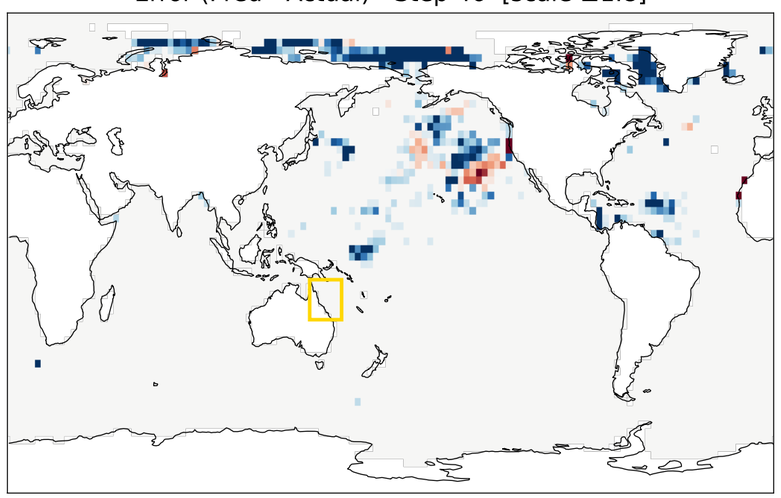
Actual DHW - Step 46



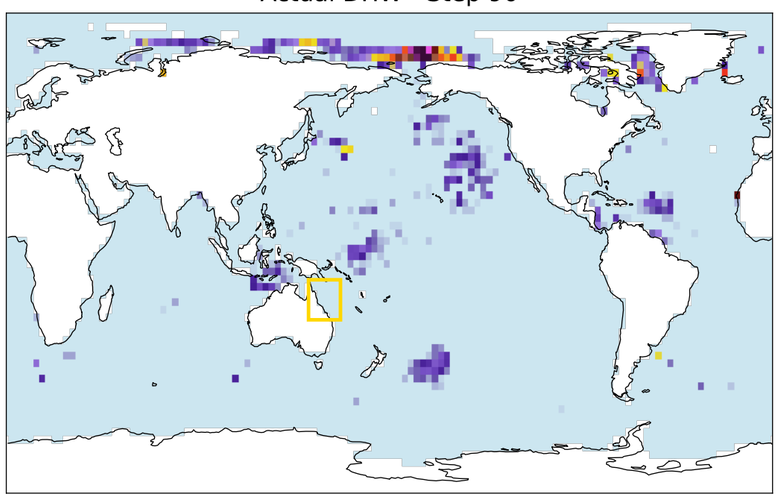
Predicted DHW - Step 46



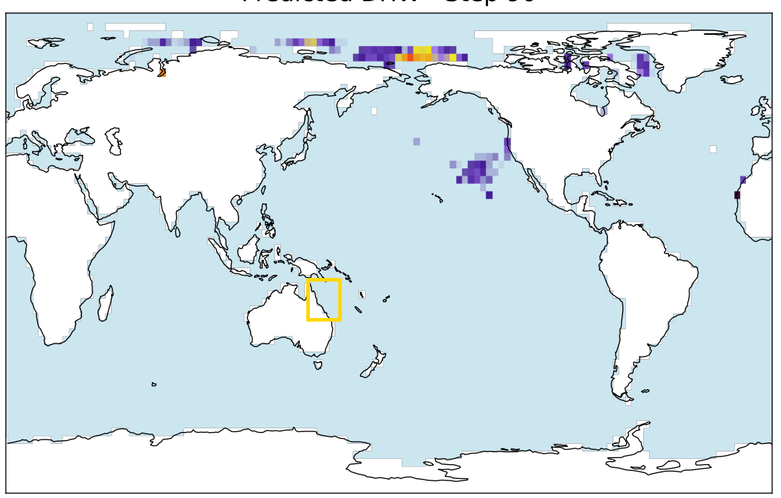
Error (Pred - Actual) - Step 46 [scale ±1.8]



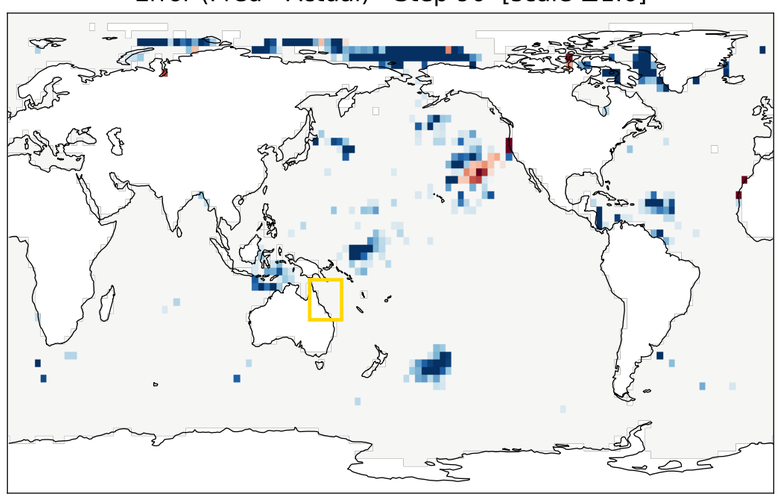
Actual DHW - Step 90

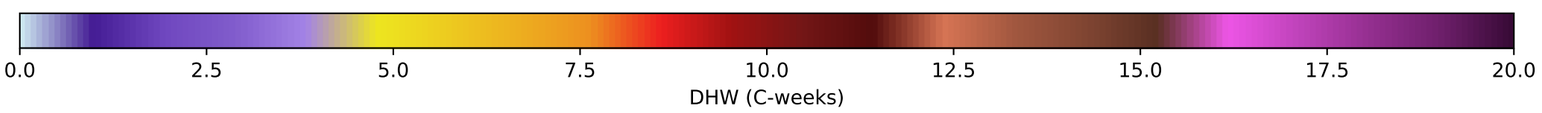


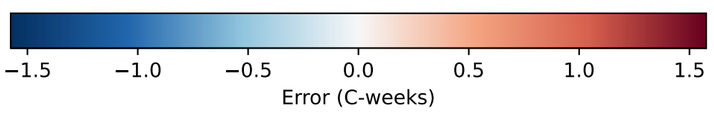
Predicted DHW - Step 90



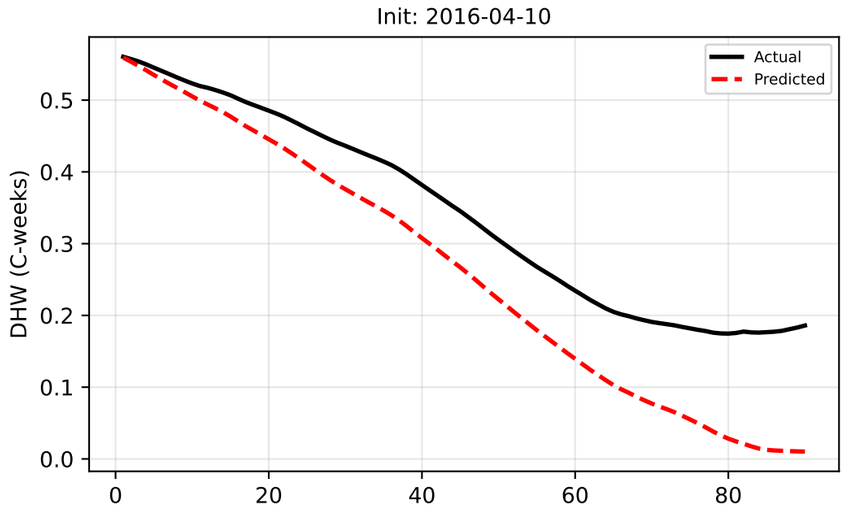
Error (Pred - Actual) - Step 90 [scale ±1.6]

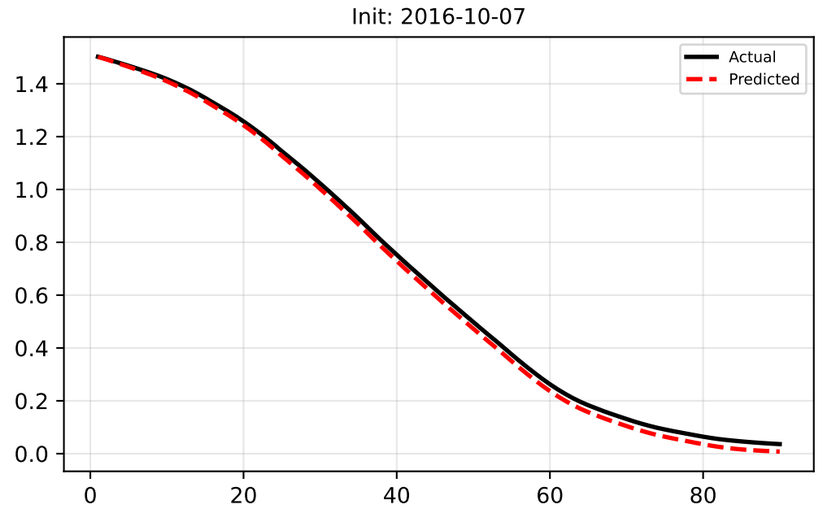


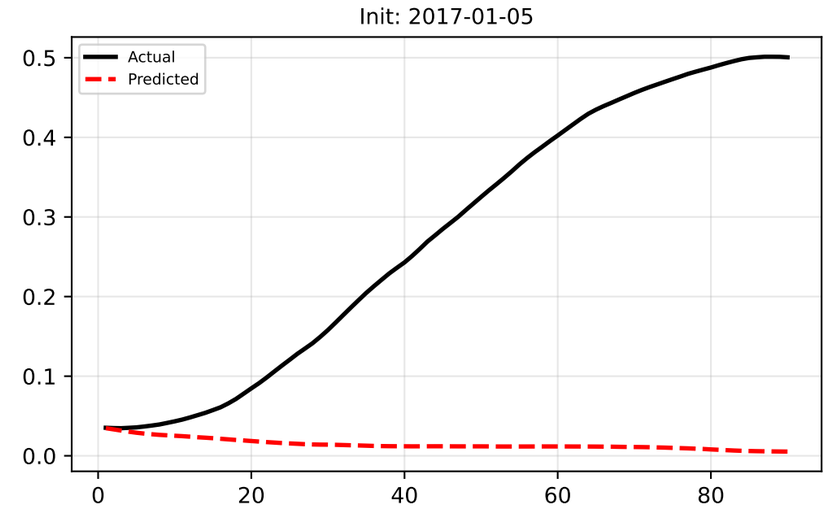


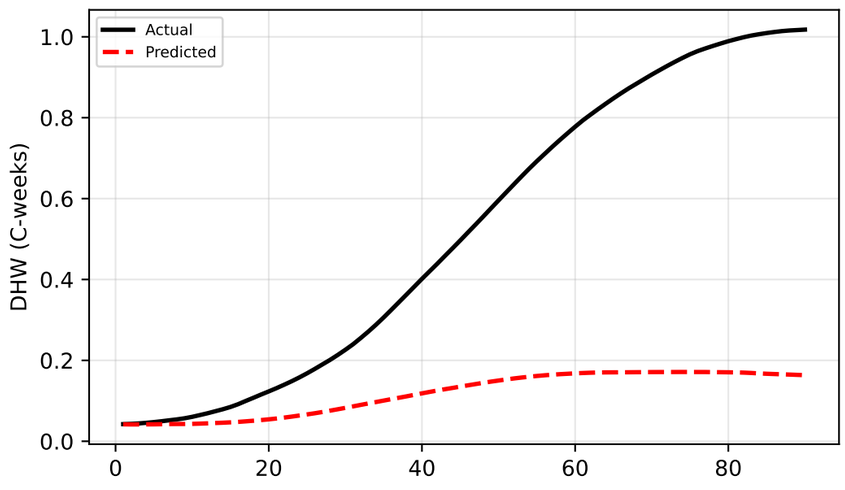


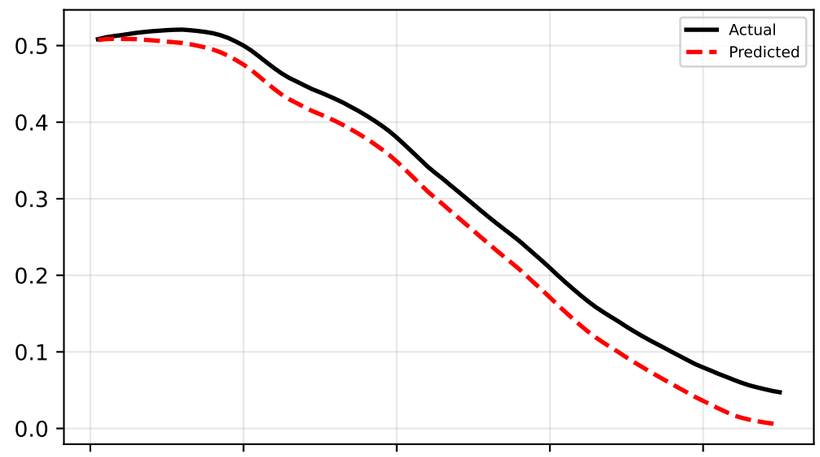
Global-Mean DHW - Val (C-weeks)

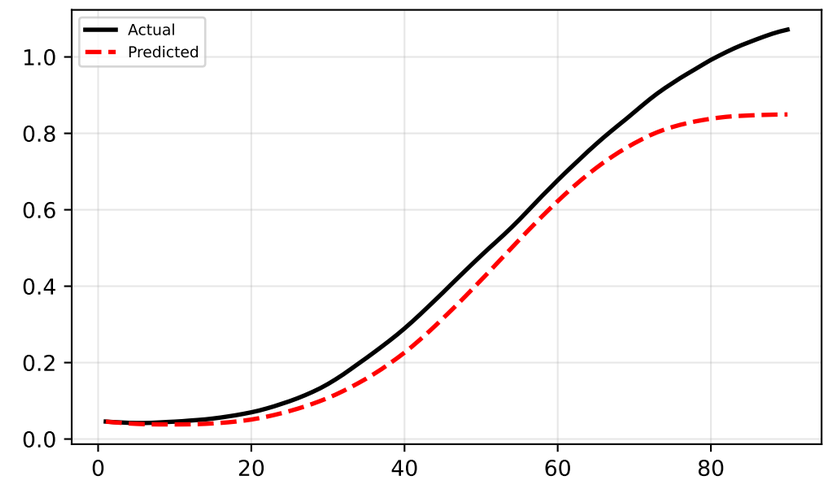


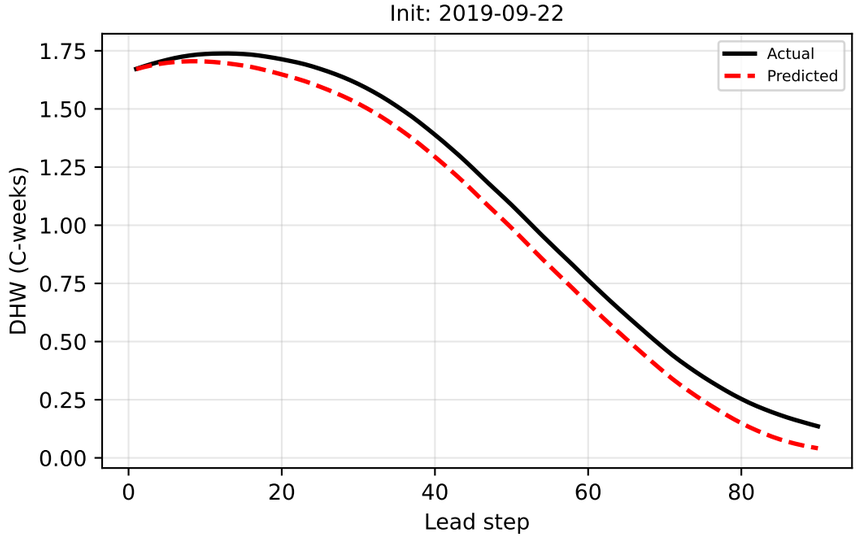




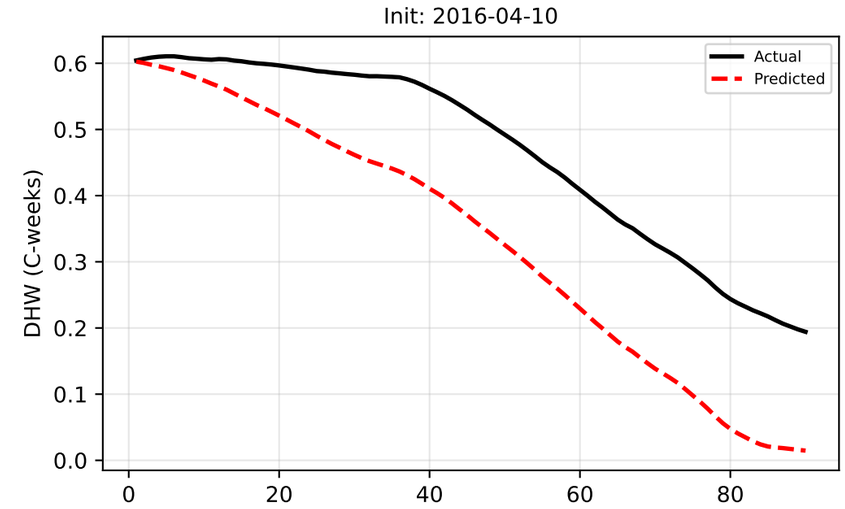


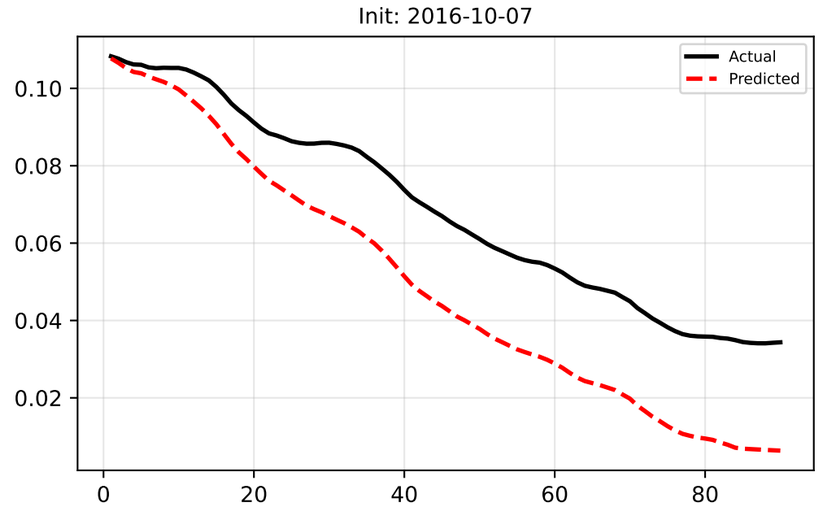


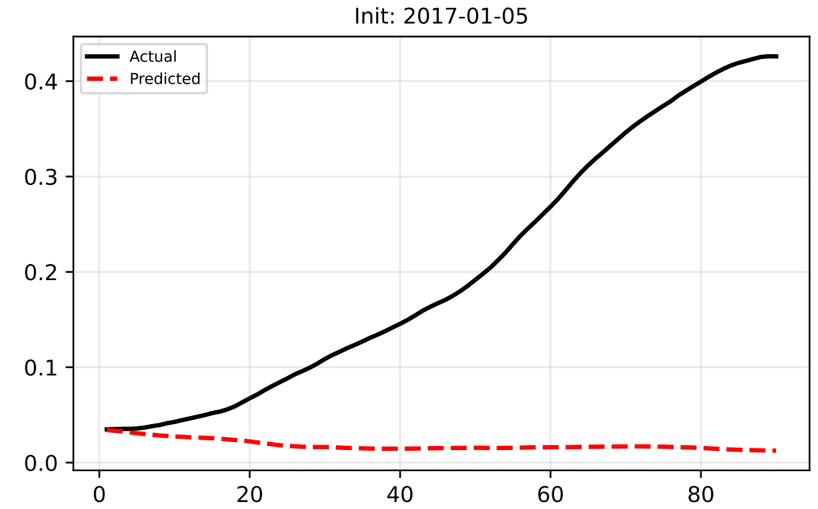


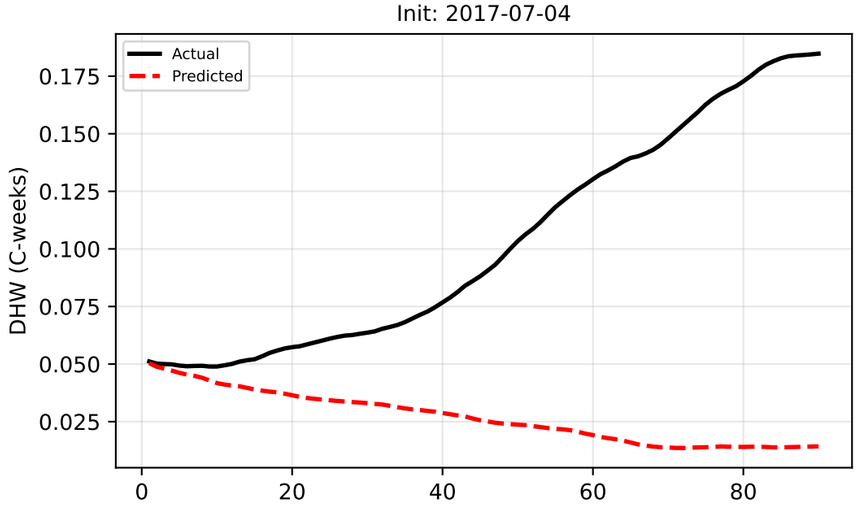


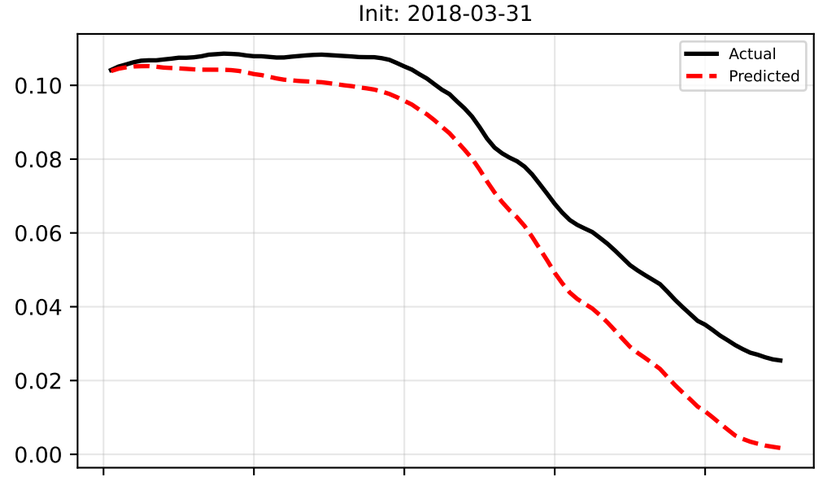
Tropical-Mean DHW (|lat|<=30) - Val (C-weeks)

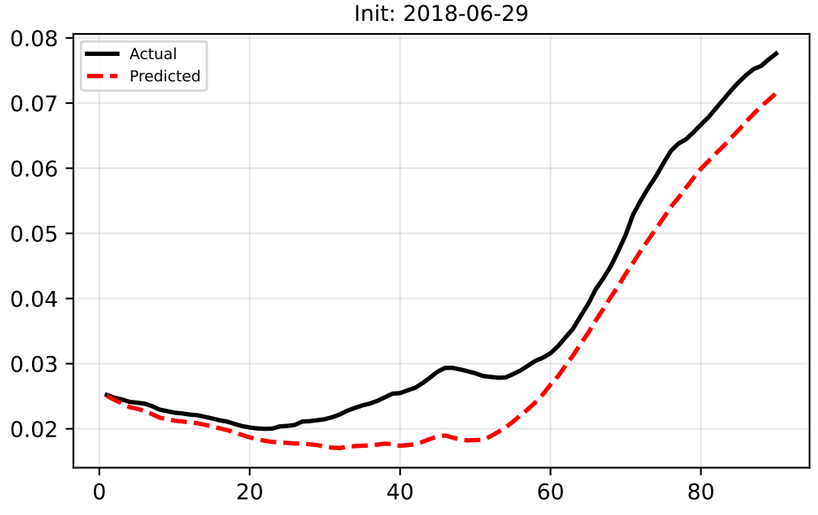


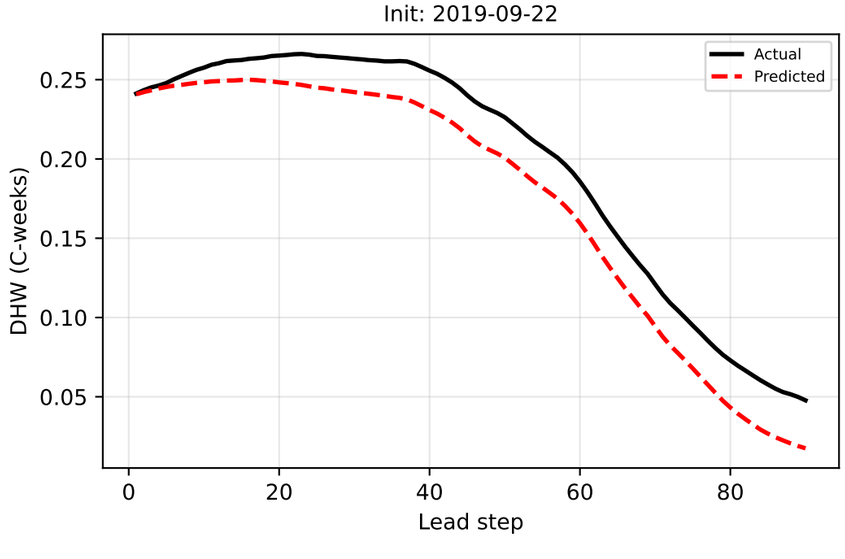


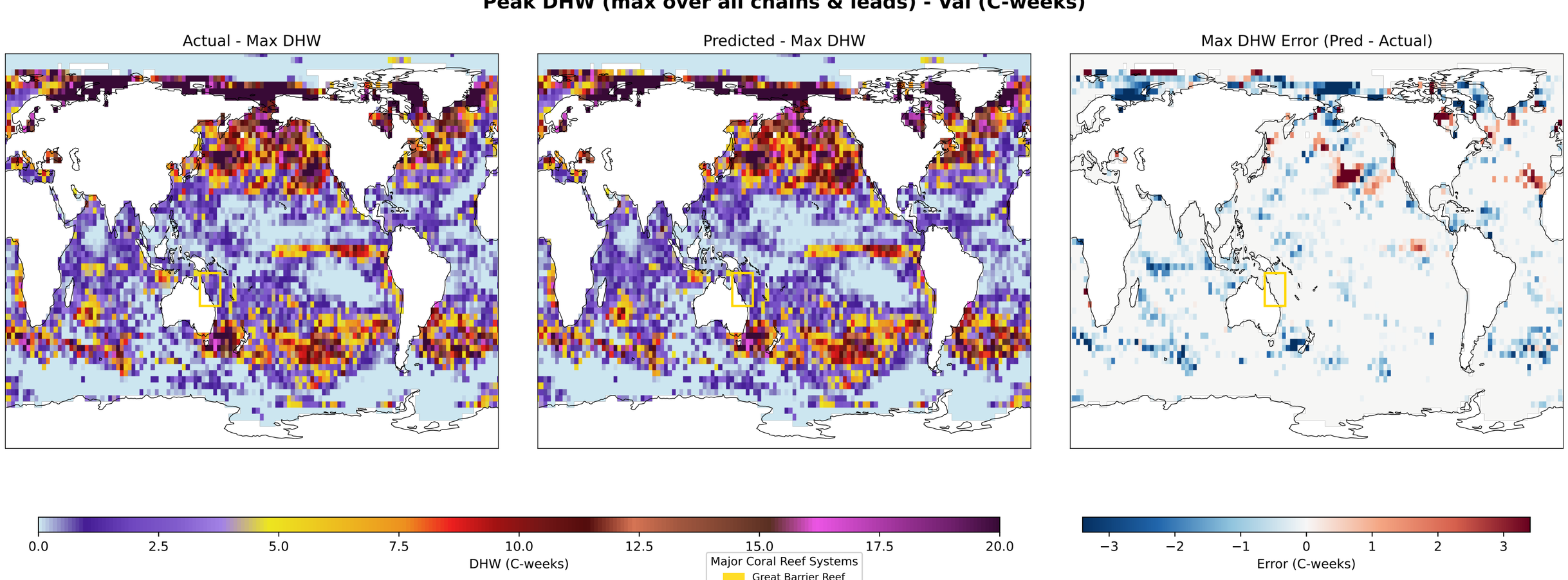


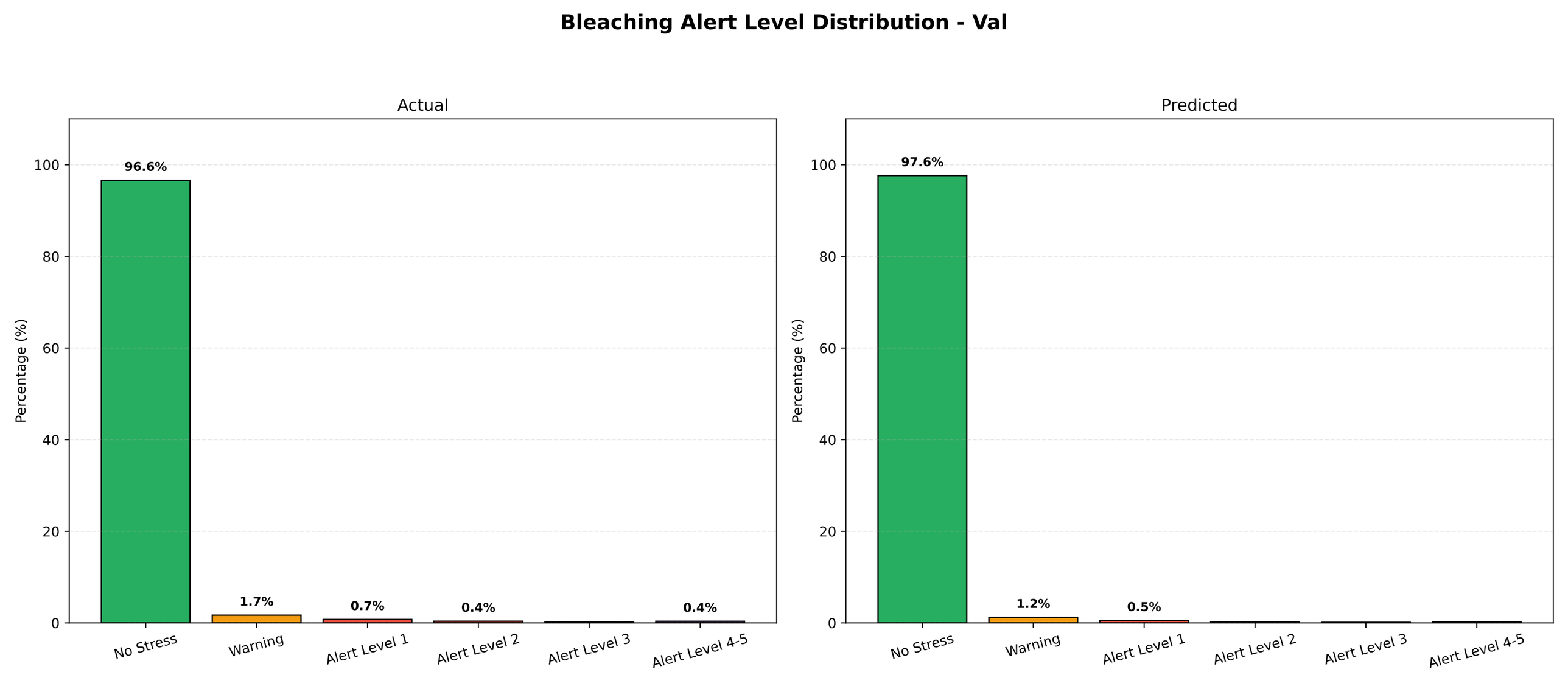






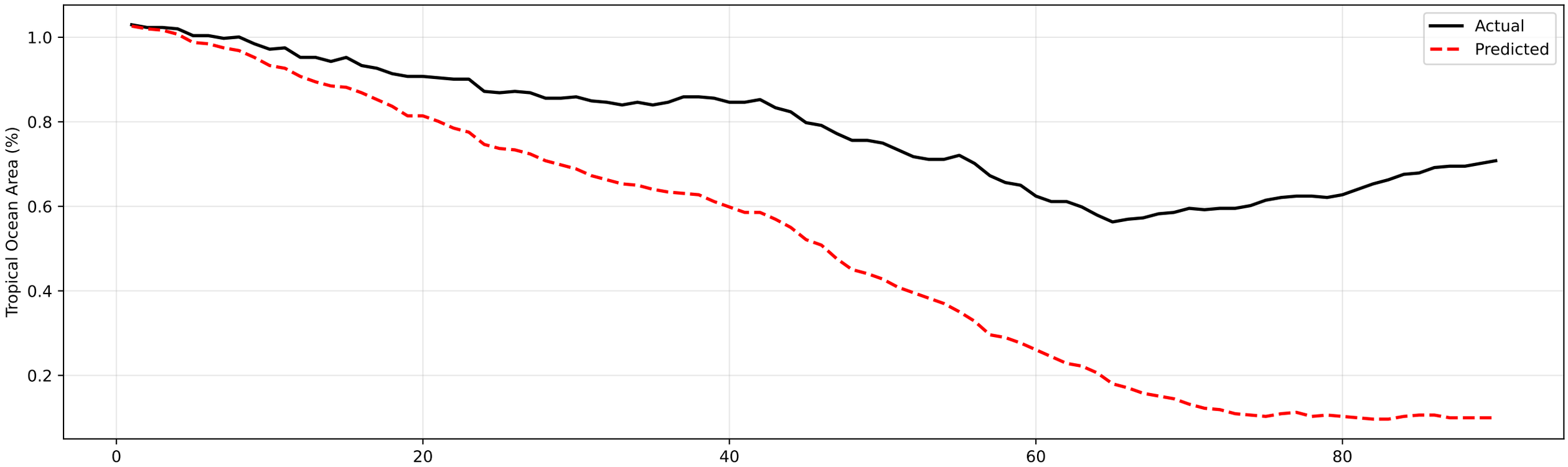



Peak DHW (max over all chains & leads) - Val (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−3
−2
−1
0
1
2
3
Error (C-weeks)

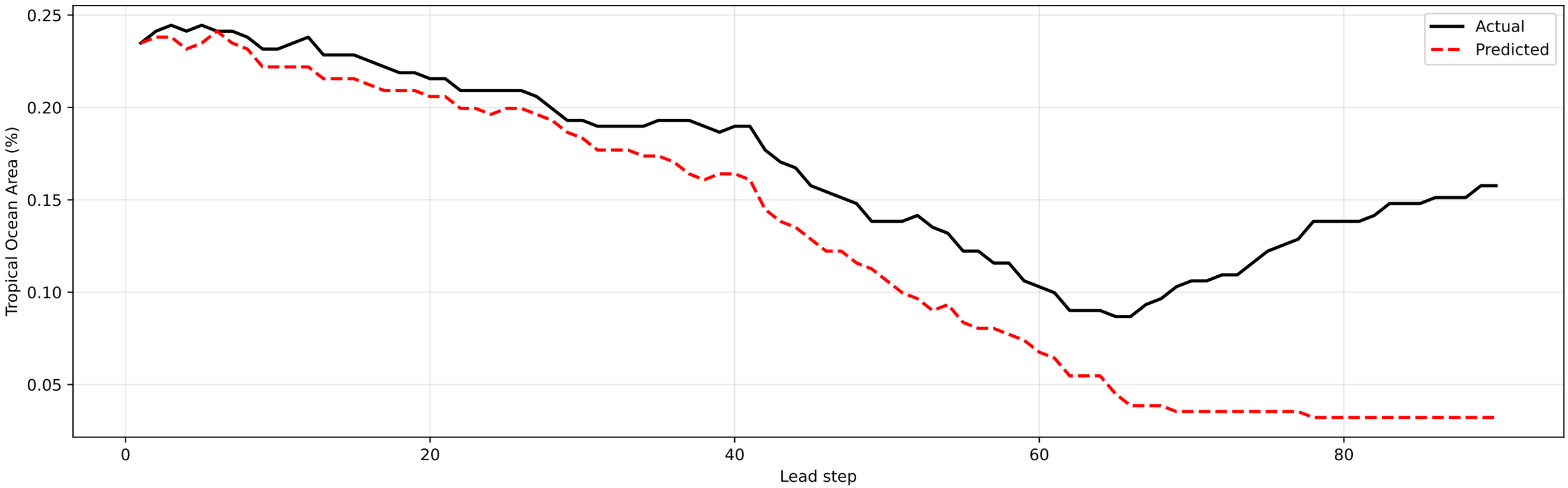

Bleaching Alert Level Distribution - Val
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.6%
1.7%
0.7%
0.4%
0.4%
97.6%
1.2%
0.5%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



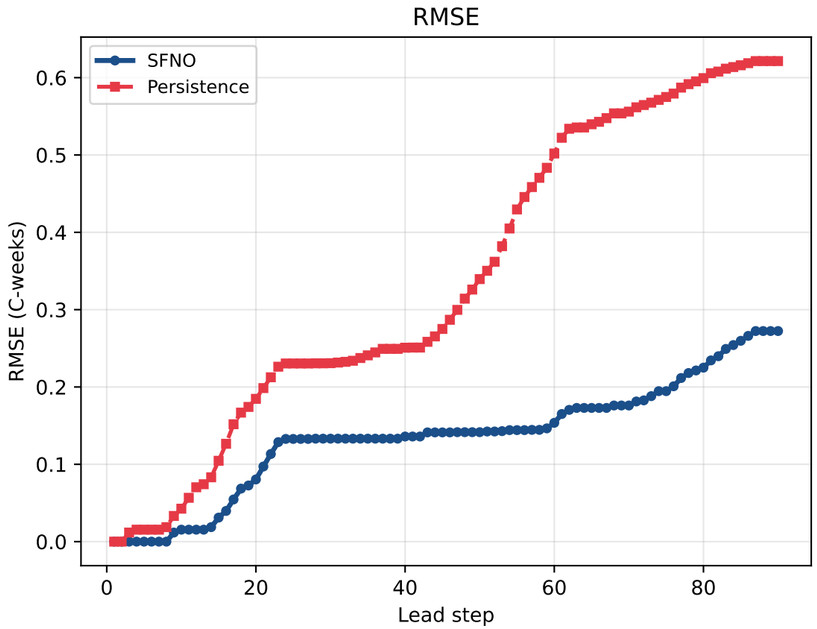
Great Barrier Reef Analysis - Val

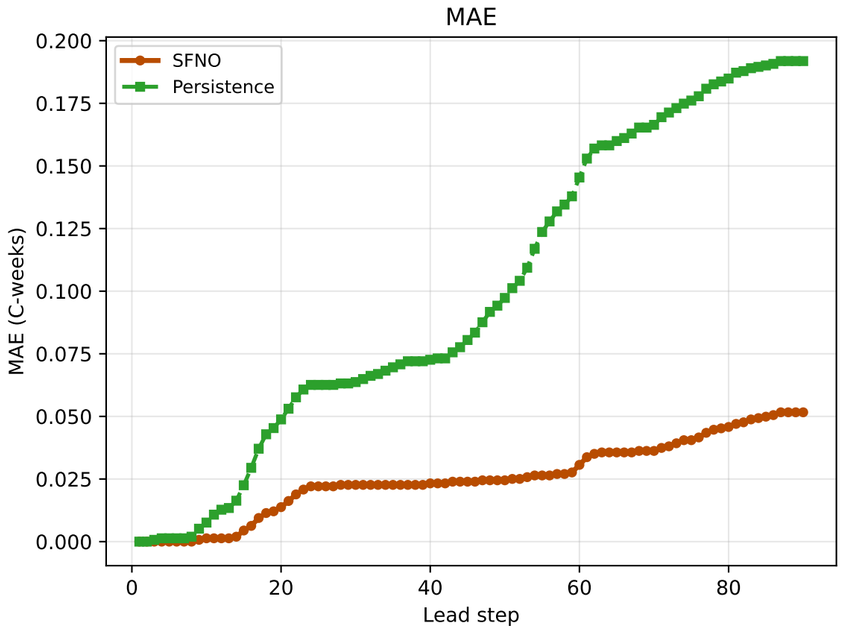
```
                    GREAT BARRIER REEF - VAL
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 5 lat x 5 lon points
```

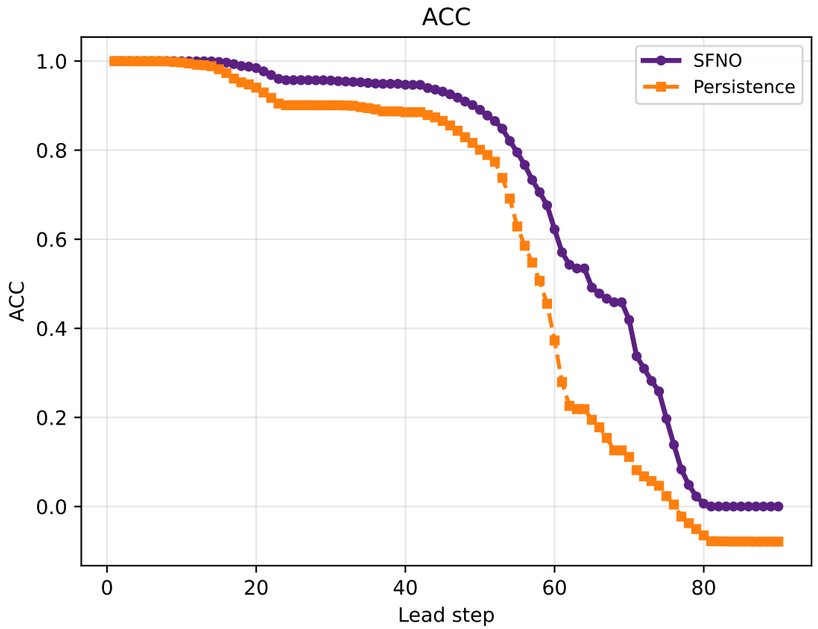
DHW Metrics by Lead Step - Val - GBR (15 chains scored at every lead)

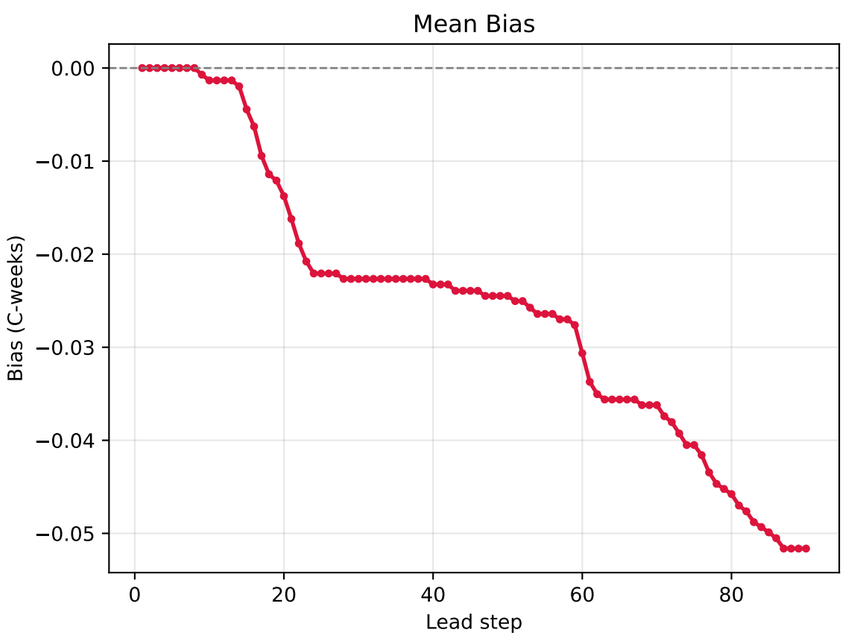
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
2	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
3	0.0000	0.0117	100.0%	0.0000	0.0007	0.0000	1.0000	0.9998	1.0000
4	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
5	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
6	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
7	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
8	0.0000	0.0187	100.0%	0.0000	0.0020	0.0000	1.0000	0.9994	1.0000
9	0.0117	0.0332	64.6%	0.0007	0.0052	-0.0007	0.9997	0.9981	0.9995
10	0.0155	0.0426	63.7%	0.0013	0.0076	-0.0013	0.9996	0.9969	0.9991
11	0.0155	0.0567	72.7%	0.0013	0.0108	-0.0013	0.9995	0.9947	0.9991
12	0.0155	0.0703	78.0%	0.0013	0.0127	-0.0013	0.9995	0.9917	0.9991
13	0.0155	0.0742	79.2%	0.0013	0.0134	-0.0013	0.9995	0.9907	0.9991
14	0.0187	0.0832	77.5%	0.0020	0.0163	-0.0020	0.9993	0.9884	0.9987
15	0.0311	0.1046	70.3%	0.0044	0.0225	-0.0044	0.9980	0.9817	0.9962
16	0.0397	0.1265	68.6%	0.0063	0.0295	-0.0063	0.9966	0.9732	0.9935
17	0.0545	0.1518	64.1%	0.0094	0.0371	-0.0094	0.9932	0.9608	0.9871
18	0.0686	0.1668	58.9%	0.0114	0.0429	-0.0114	0.9889	0.9523	0.9789
19	0.0726	0.1742	58.3%	0.0121	0.0453	-0.0121	0.9874	0.9478	0.9761
20	0.0804	0.1846	56.4%	0.0138	0.0488	-0.0138	0.9843	0.9404	0.9703
21	0.0971	0.1986	51.1%	0.0162	0.0531	-0.0162	0.9770	0.9296	0.9566
22	0.1133	0.2125	46.7%	0.0189	0.0576	-0.0189	0.9688	0.9174	0.9416
23	0.1287	0.2262	43.1%	0.0208	0.0607	-0.0208	0.9600	0.9047	0.9259
24	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
25	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
26	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
27	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
28	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
29	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
30	0.1332	0.2308	42.3%	0.0227	0.0637	-0.0227	0.9564	0.9008	0.9192
31	0.1332	0.2316	42.5%	0.0227	0.0649	-0.0227	0.9550	0.9008	0.9167
32	0.1332	0.2325	42.7%	0.0227	0.0662	-0.0227	0.9542	0.9004	0.9152
33	0.1332	0.2339	43.1%	0.0227	0.0669	-0.0227	0.9533	0.8996	0.9136
34	0.1332	0.2374	43.9%	0.0227	0.0683	-0.0227	0.9525	0.8967	0.9121
35	0.1332	0.2409	44.7%	0.0227	0.0696	-0.0227	0.9509	0.8944	0.9093
36	0.1332	0.2447	45.6%	0.0227	0.0708	-0.0227	0.9496	0.8915	0.9068
37	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
38	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
39	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
40	0.1359	0.2508	45.8%	0.0233	0.0726	-0.0233	0.9469	0.8852	0.9019
41	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
42	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
43	0.1413	0.2584	45.3%	0.0239	0.0756	-0.0239	0.9396	0.8788	0.8887
44	0.1413	0.2654	46.8%	0.0239	0.0776	-0.0239	0.9361	0.8735	0.8822
45	0.1413	0.2750	48.6%	0.0239	0.0805	-0.0239	0.9315	0.8656	0.8740
46	0.1413	0.2869	50.8%	0.0239	0.0834	-0.0239	0.9255	0.8554	0.8631
47	0.1415	0.2997	52.8%	0.0245	0.0876	-0.0245	0.9180	0.8435	0.8499
48	0.1415	0.3142	55.0%	0.0245	0.0917	-0.0245	0.9092	0.8291	0.8344
49	0.1415	0.3260	56.6%	0.0245	0.0942	-0.0245	0.9013	0.8163	0.8207
50	0.1415	0.3395	58.3%	0.0245	0.0973	-0.0245	0.8907	0.8009	0.8025
51	0.1424	0.3503	59.3%	0.0250	0.1012	-0.0250	0.8778	0.7892	0.7809
52	0.1424	0.3620	60.7%	0.0250	0.1041	-0.0250	0.8653	0.7741	0.7605
53	0.1429	0.3821	62.6%	0.0257	0.1093	-0.0257	0.8482	0.7377	0.7322
54	0.1442	0.4049	64.4%	0.0264	0.1169	-0.0264	0.8206	0.6912	0.6890
55	0.1442	0.4296	66.4%	0.0264	0.1236	-0.0264	0.7950	0.6287	0.6501
56	0.1442	0.4457	67.6%	0.0264	0.1278	-0.0264	0.7668	0.5854	0.6088
57	0.1445	0.4585	68.5%	0.0270	0.1318	-0.0270	0.7330	0.5477	0.5614
58	0.1445	0.4705	69.3%	0.0270	0.1346	-0.0270	0.7057	0.5066	0.5248
59	0.1465	0.4834	69.7%	0.0276	0.1379	-0.0276	0.6760	0.4551	0.4861
60	0.1537	0.5019	69.4%	0.0306	0.1454	-0.0306	0.6222	0.3728	0.4186
61	0.1651	0.5223	68.4%	0.0337	0.1529	-0.0337	0.5707	0.2794	0.3594
62	0.1704	0.5339	68.1%	0.0350	0.1569	-0.0350	0.5428	0.2259	0.3295
63	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
64	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
65	0.1729	0.5397	68.0%	0.0356	0.1599	-0.0356	0.4915	0.1947	0.2746
66	0.1729	0.5429	68.2%	0.0356	0.1611	-0.0356	0.4781	0.1779	0.2597
67	0.1729	0.5476	68.4%	0.0356	0.1629	-0.0356	0.4666	0.1539	0.2469
68	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
69	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
70	0.1760	0.5561	68.3%	0.0362	0.1664	-0.0362	0.4189	0.1113	0.2001
71	0.1812	0.5617	67.7%	0.0374	0.1695	-0.0374	0.3375	0.0816	0.1329
72	0.1827	0.5646	67.6%	0.0380	0.1713	-0.0380	0.3096	0.0678	0.1116
73	0.1881	0.5677	66.9%	0.0393	0.1731	-0.0393	0.2819	0.0571	0.0915
74	0.1946	0.5713	65.9%	0.0405	0.1749	-0.0405	0.2585	0.0465	0.0756
75	0.1946	0.5750	66.2%	0.0405	0.1761	-0.0405	0.1969	0.0233	0.0384
76	0.2009	0.5794	65.3%	0.0416	0.1778	-0.0416	0.1385	0.0044	0.0092
77	0.2116	0.5871	64.0%	0.0434	0.1808	-0.0434	0.0831	-0.0226	-0.0142
78	0.2180	0.5916	63.1%	0.0447	0.1826	-0.0447	0.0485	-0.0374	-0.0277
79	0.2214	0.5952	62.8%	0.0452	0.1837	-0.0452	0.0225	-0.0507	-0.0367
80	0.2250	0.5994	62.5%	0.0458	0.1849	-0.0458	0.0064	-0.0649	-0.0418
81	0.2344	0.6058	61.3%	0.0470	0.1872	-0.0470	0.0000	-0.0779	-0.0419
82	0.2398	0.6079	60.6%	0.0476	0.1879	-0.0476	0.0000	-0.0782	-0.0411
83	0.2492	0.6117	59.3%	0.0488	0.1890	-0.0488	0.0000	-0.0785	-0.0399
84	0.2542	0.6137	58.6%	0.0493	0.1896	-0.0493	0.0000	-0.0786	-0.0391
85	0.2597	0.6160	57.8%	0.0499	0.1901	-0.0499	0.0000	-0.0786	-0.0383
86	0.2661	0.6187	57.0%	0.0505	0.1907	-0.0505	0.0000	-0.0785	-0.0374
87	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
88	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
89	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
90	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373

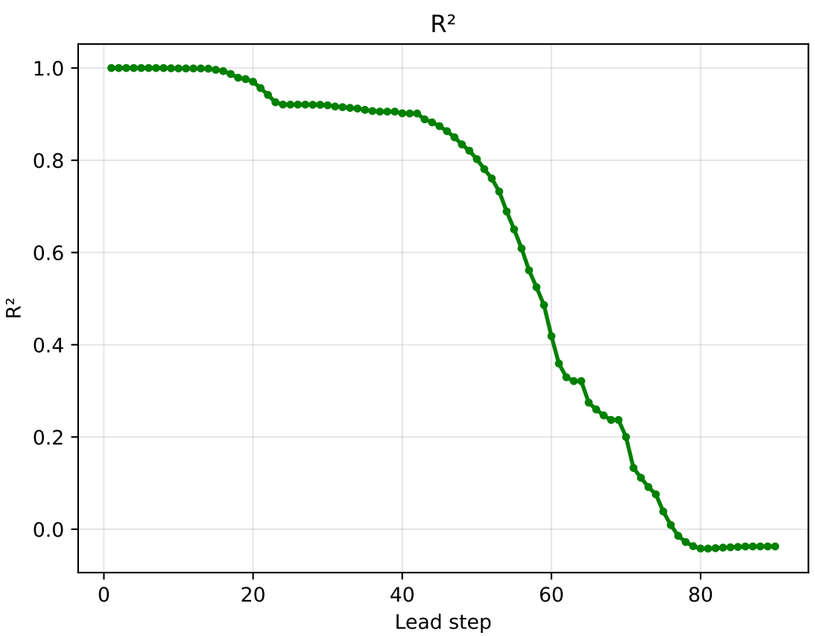
DHW Forecast Skill vs Lead Step - Val - GBR

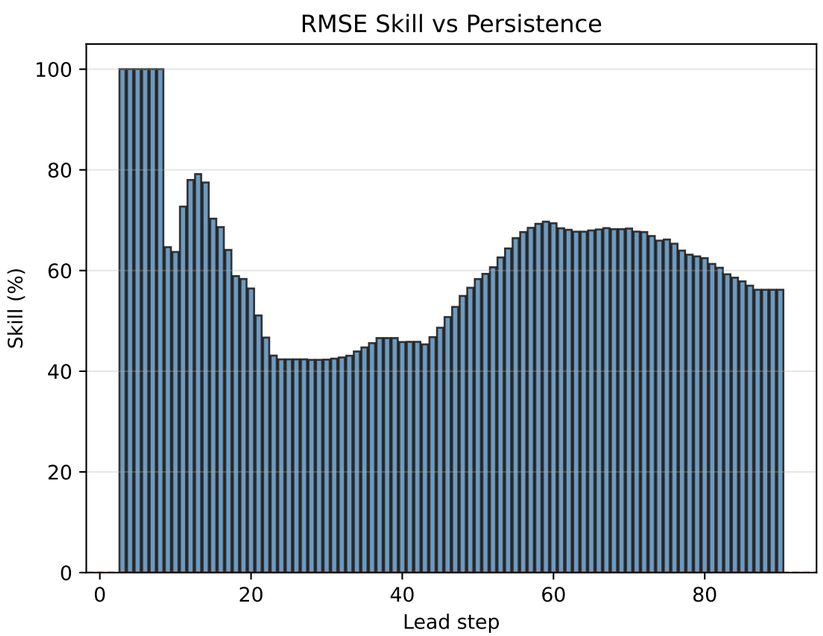


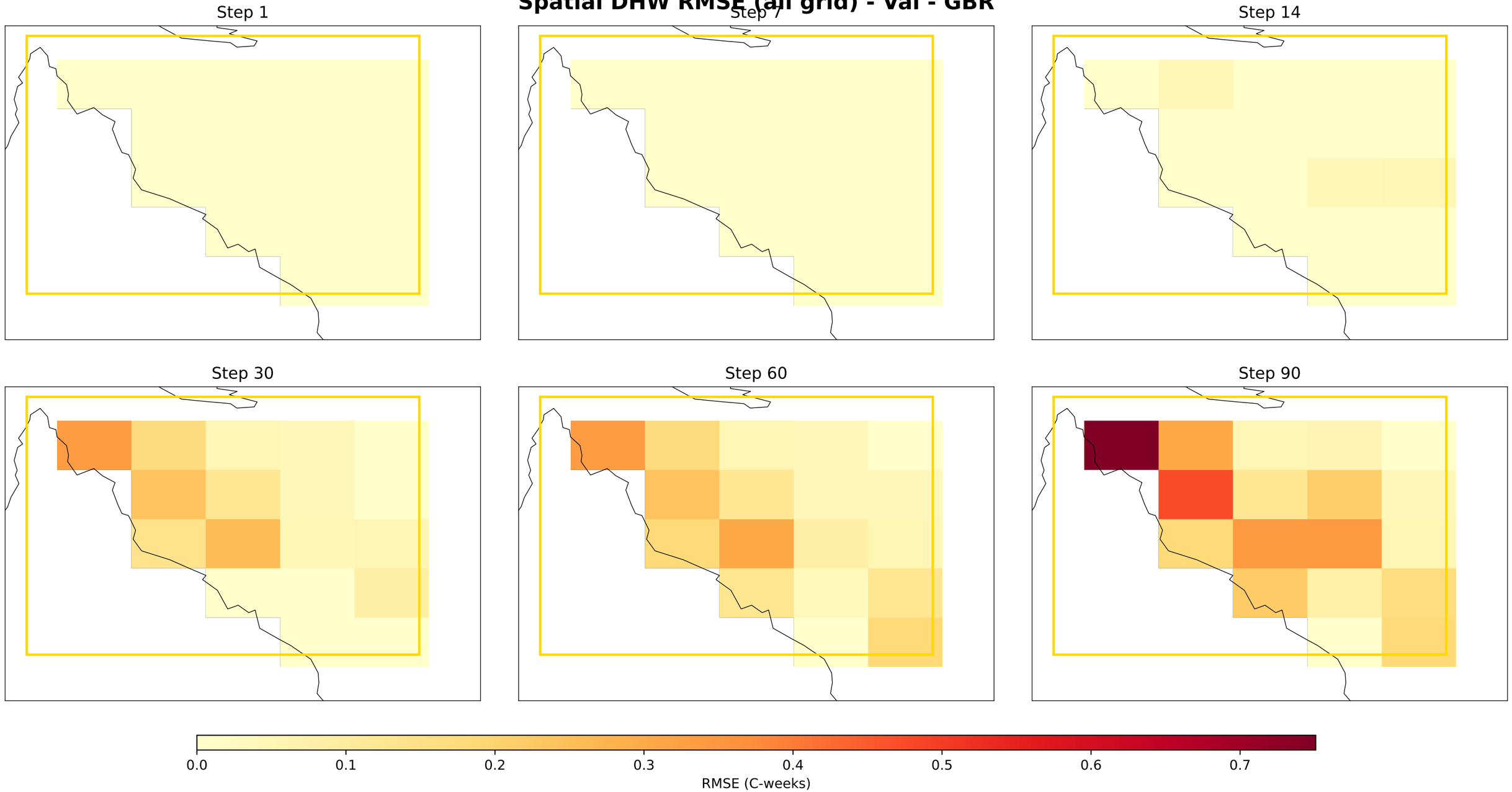


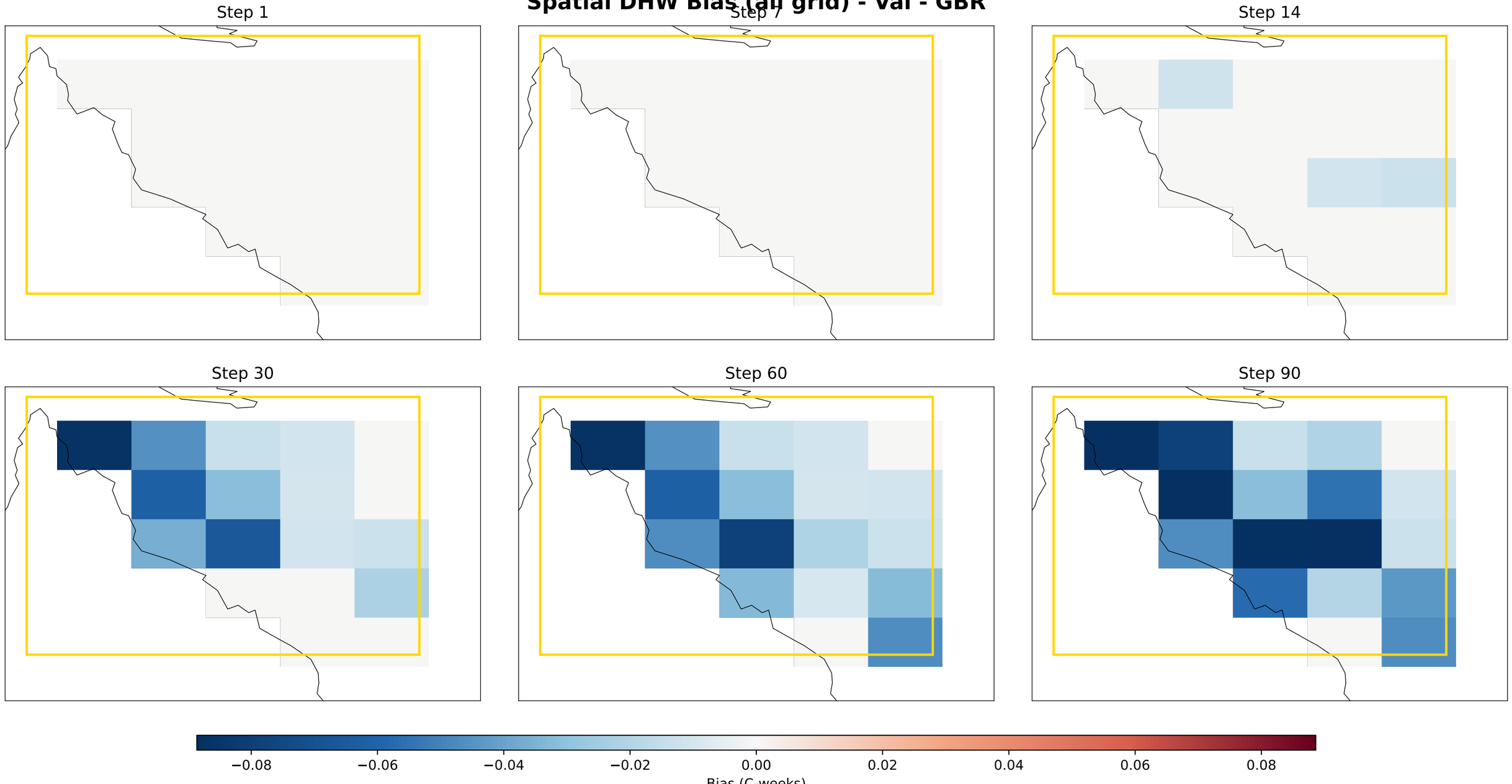


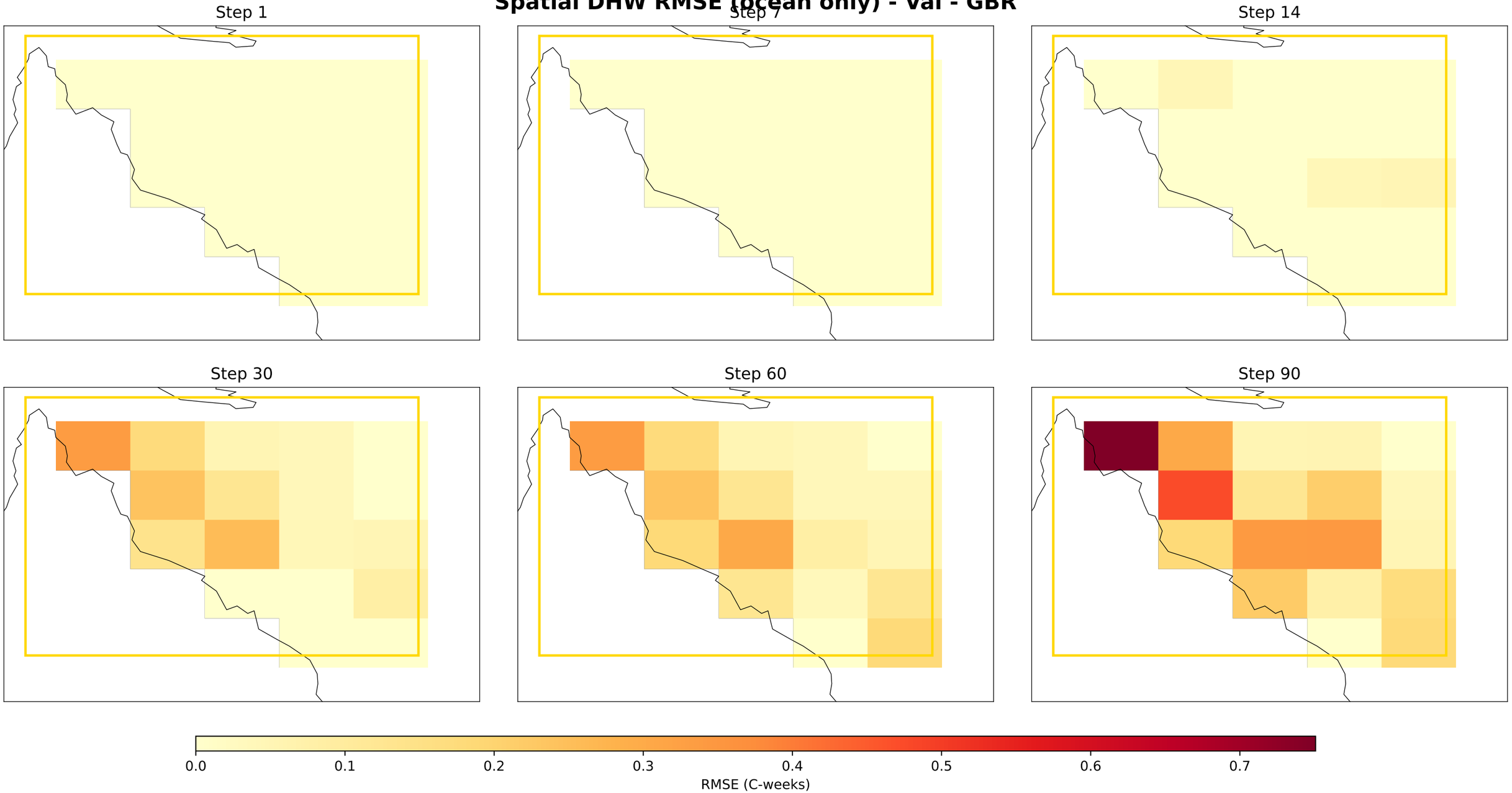


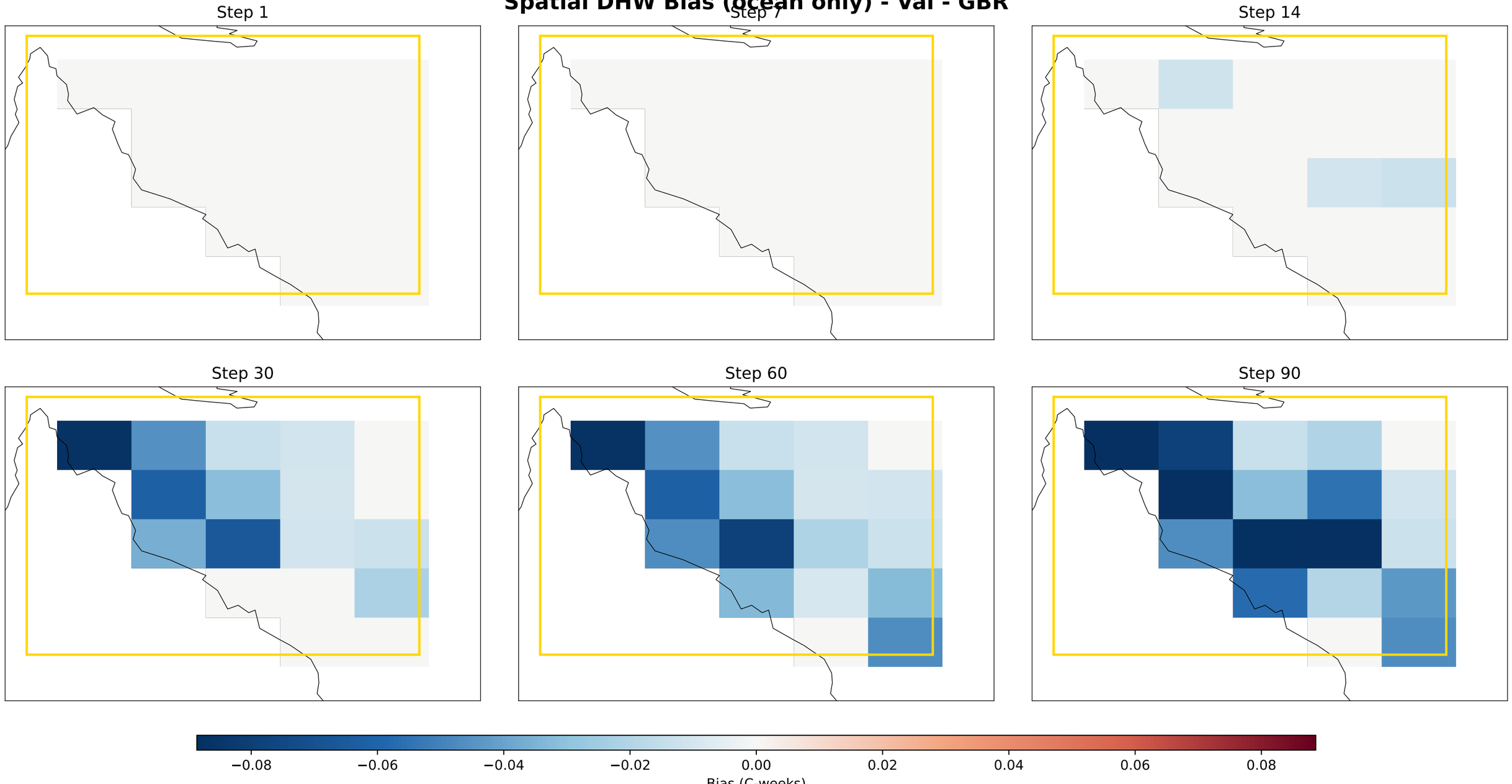





Spatial DHW RMSE (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

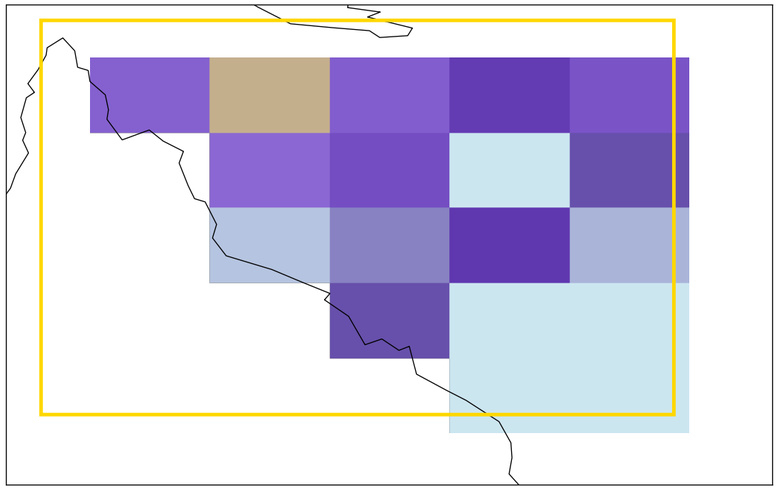

Spatial DHW Bias (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

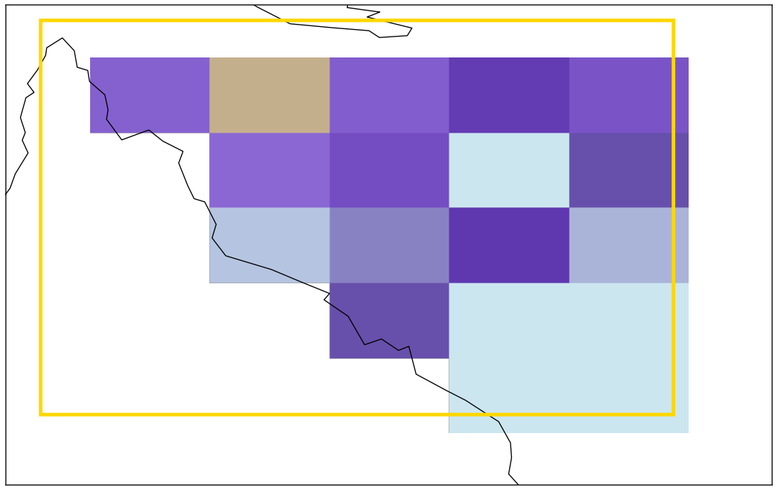

Spatial DHW Bias (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val - GBR (C-weeks)

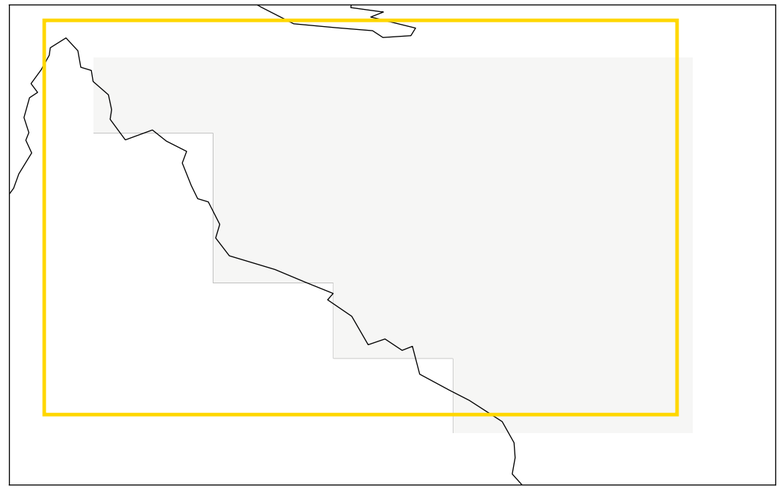
Actual DHW - Step 1



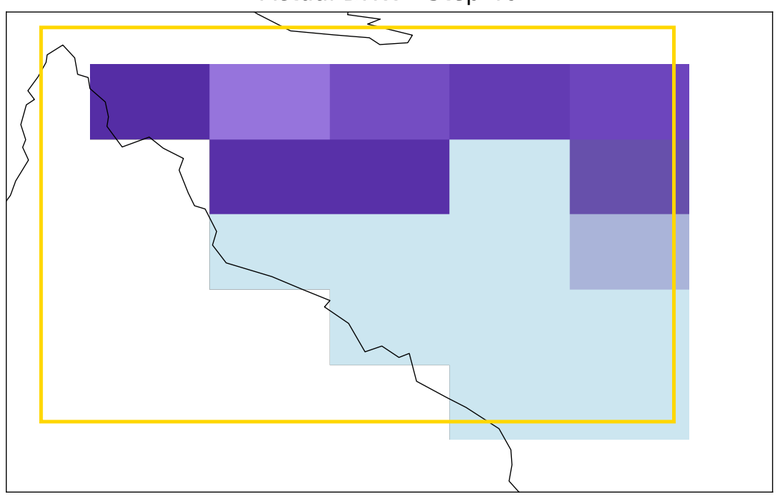
Predicted DHW - Step 1



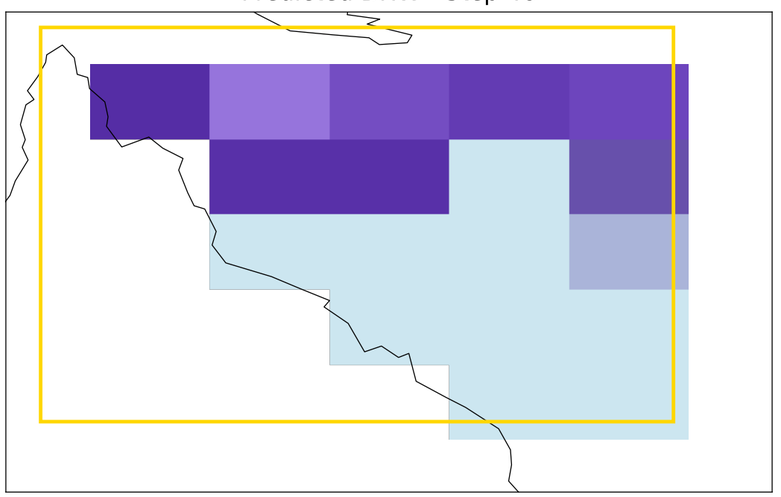
Error (Pred - Actual) - Step 1 [scale ±8]



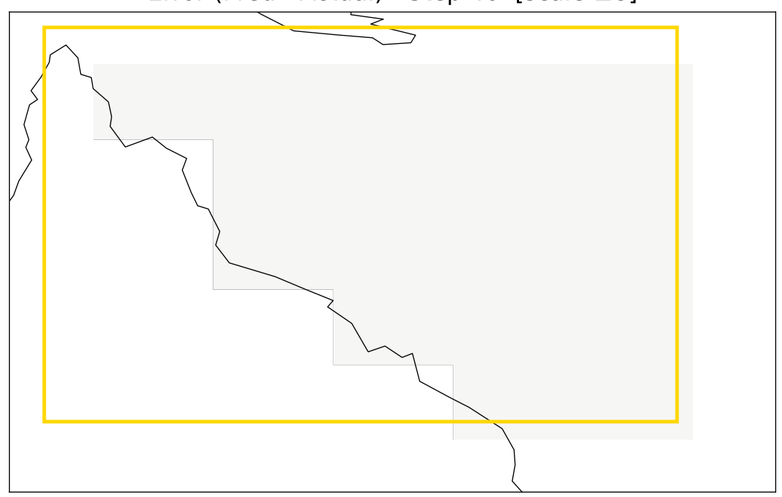
Actual DHW - Step 46



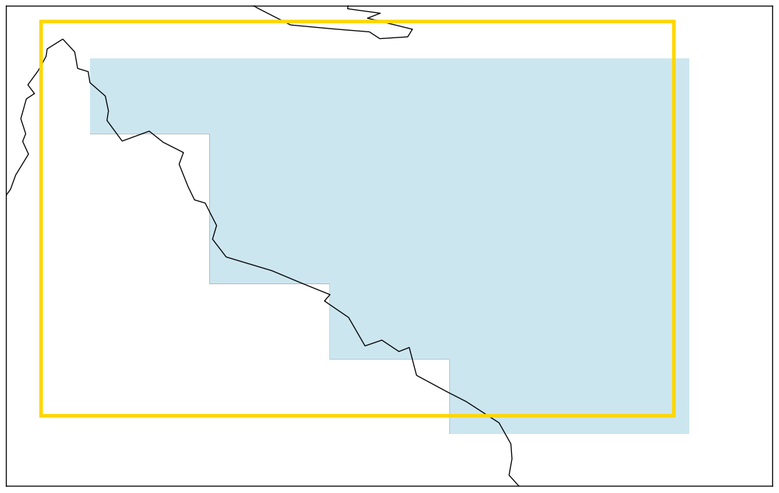
Predicted DHW - Step 46



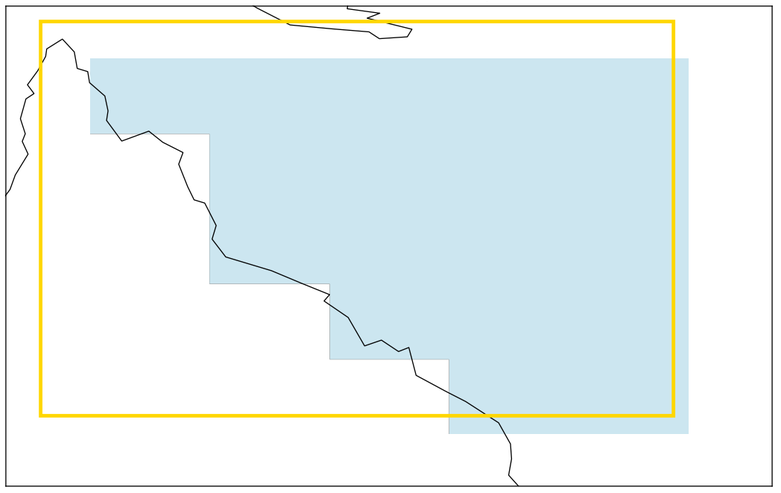
Error (Pred - Actual) - Step 46 [scale ±8]



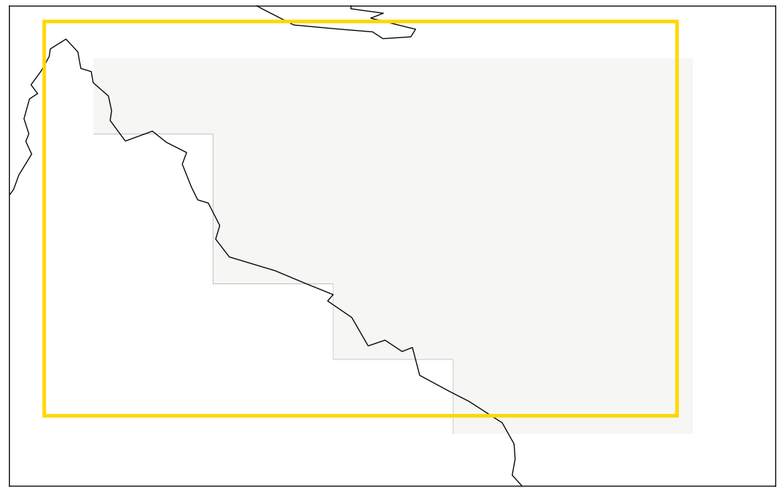
Actual DHW - Step 90

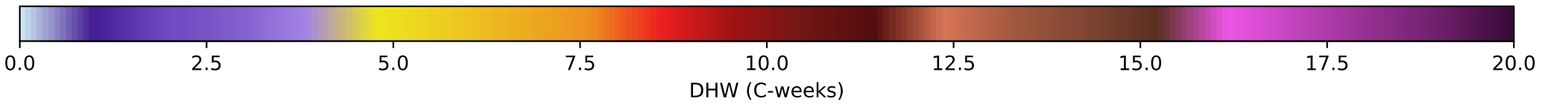


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

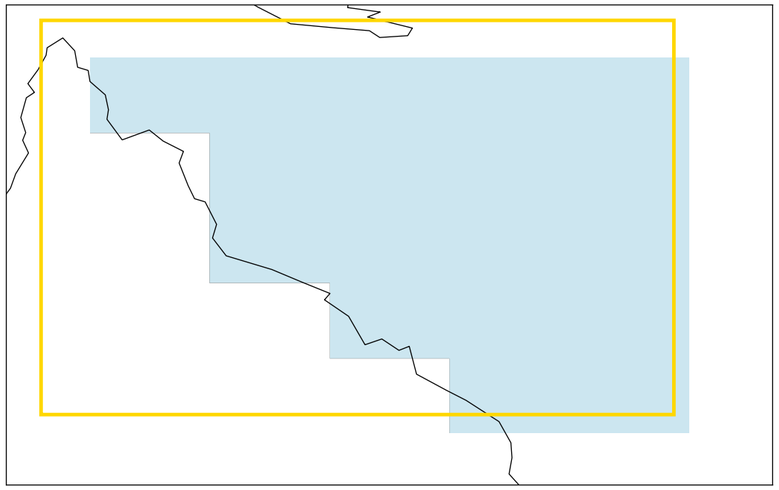




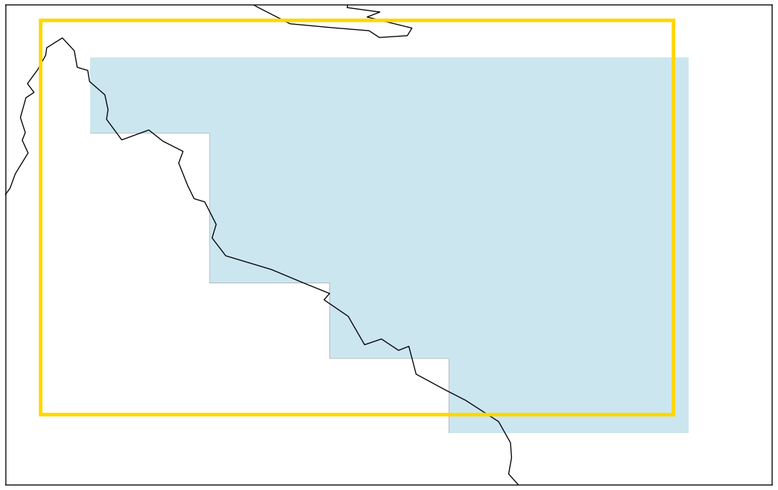


DHW Chain init: 2016-10-07 - Val - GBR (C-weeks)

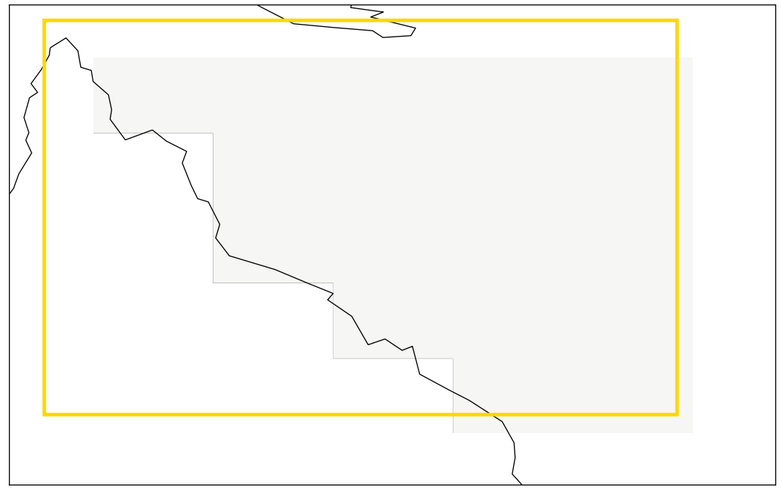
Actual DHW - Step 1



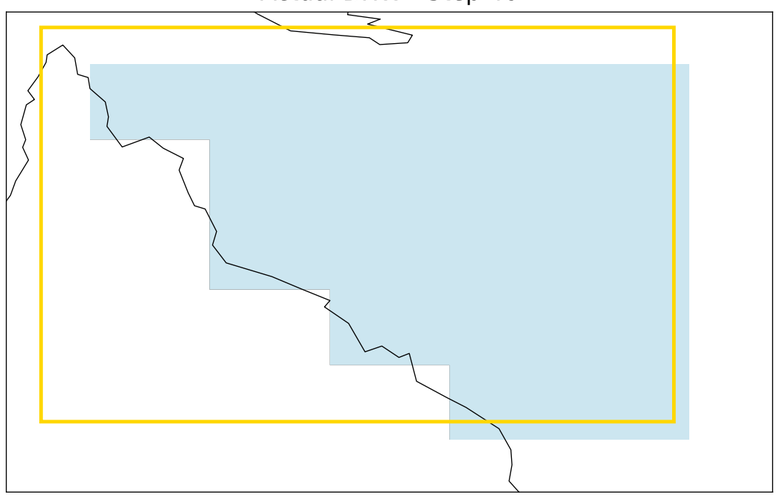
Predicted DHW - Step 1



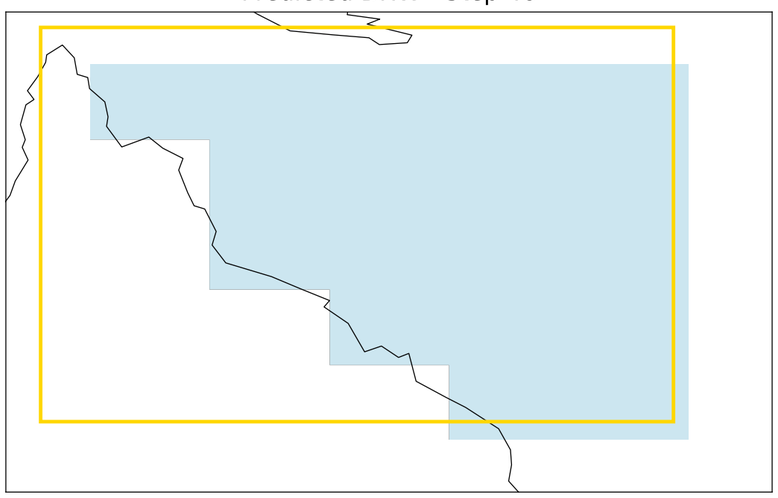
Error (Pred - Actual) - Step 1 [scale ±8]



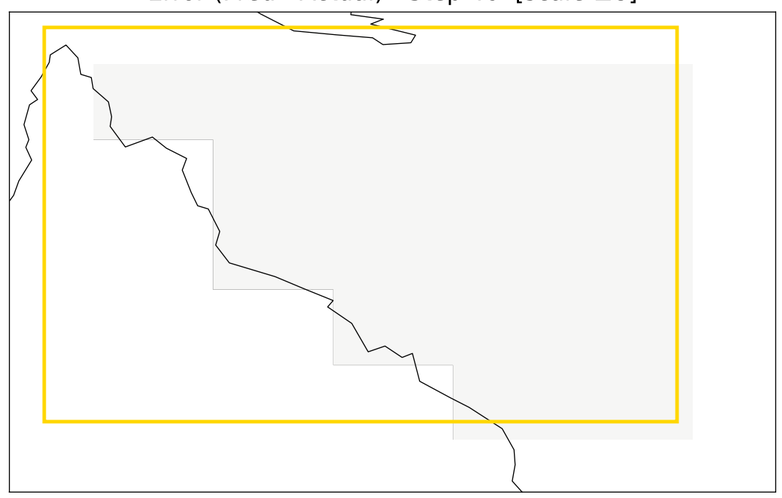
Actual DHW - Step 46



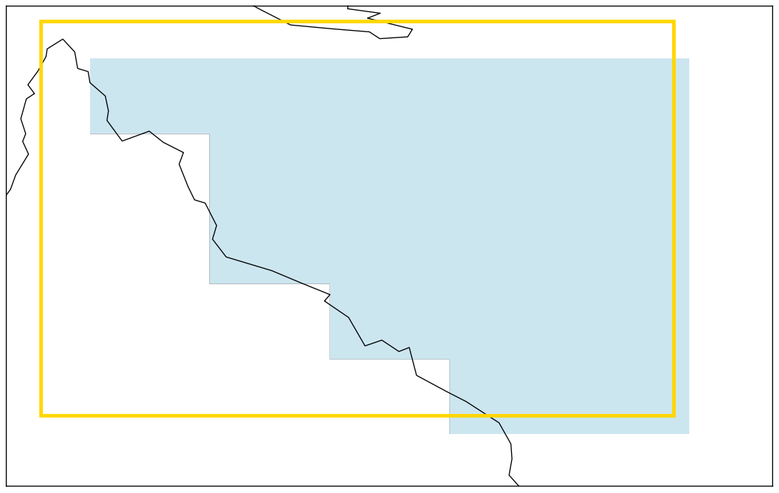
Predicted DHW - Step 46



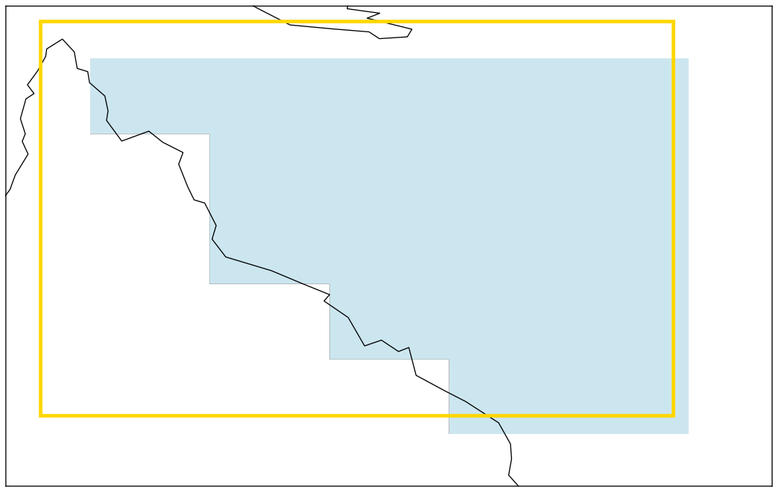
Error (Pred - Actual) - Step 46 [scale ±8]



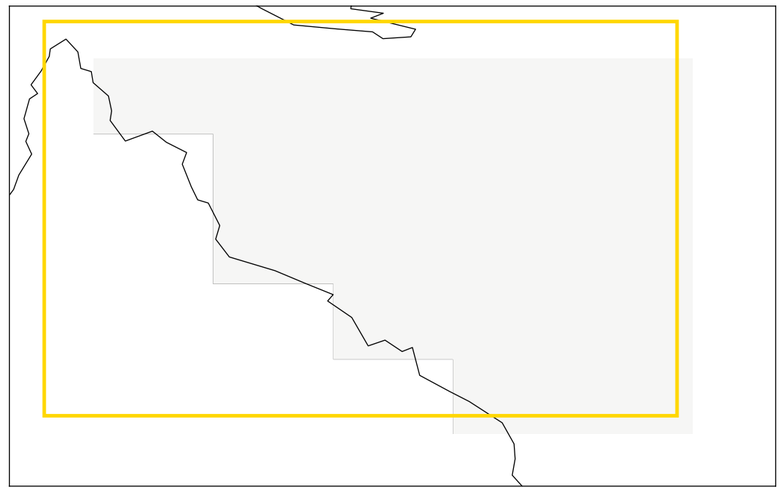
Actual DHW - Step 90

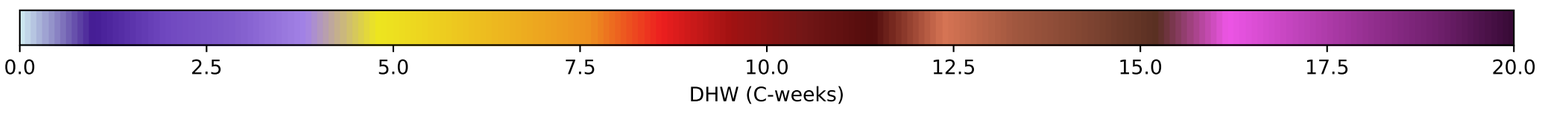


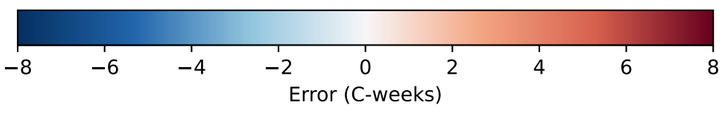
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

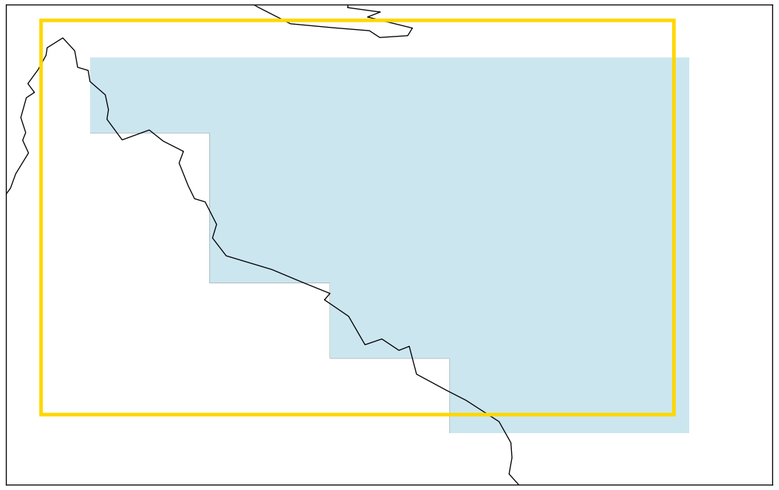




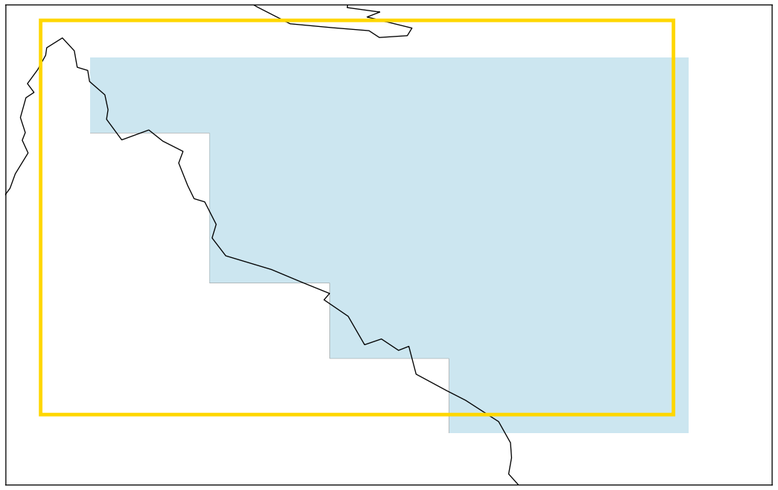


DHW Chain init: 2017-01-05 - Val - GBR (C-weeks)

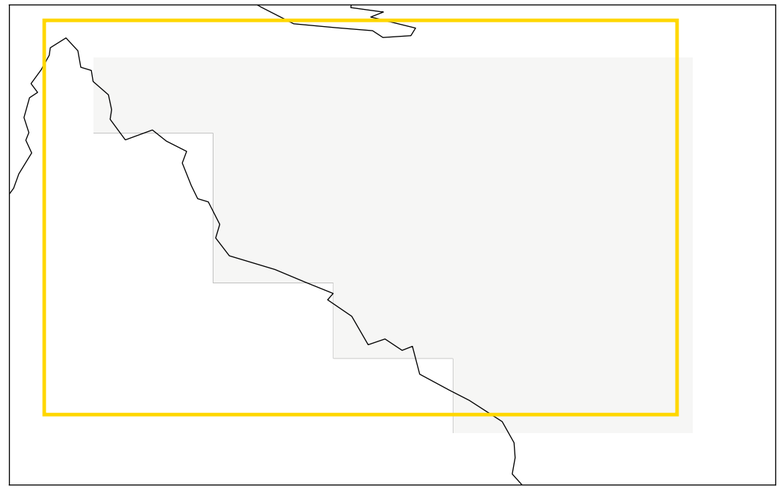
Actual DHW - Step 1



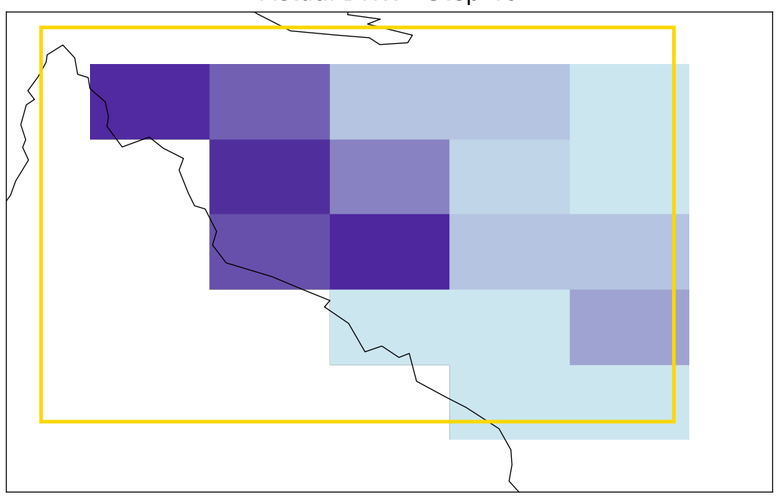
Predicted DHW - Step 1



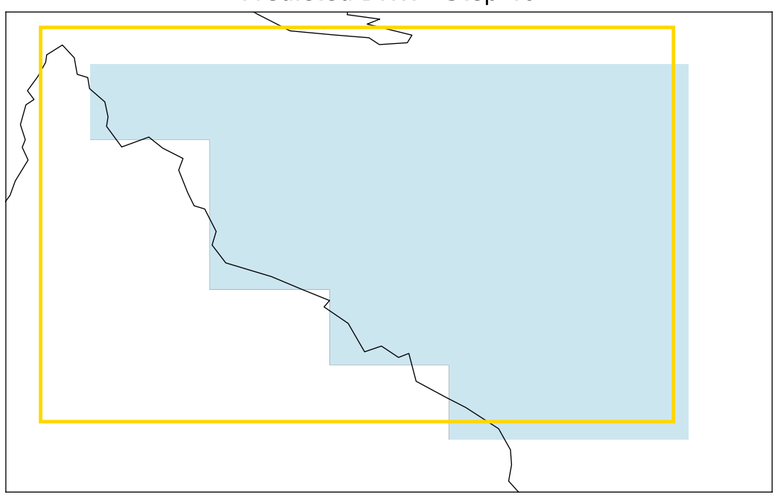
Error (Pred - Actual) - Step 1 [scale ±8]



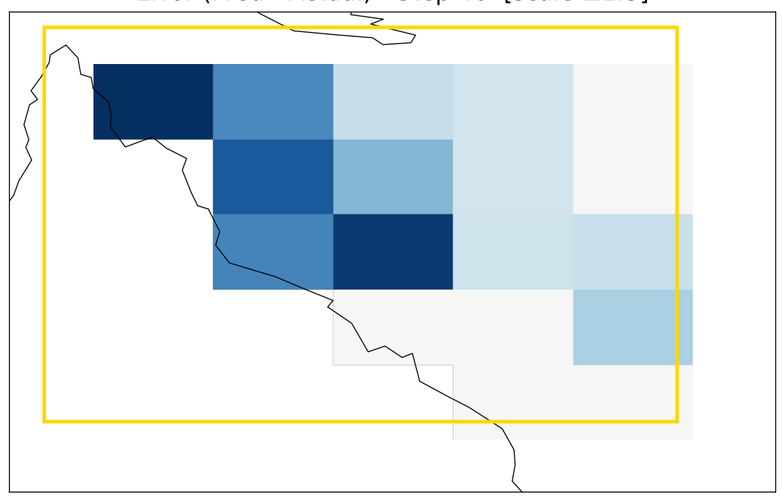
Actual DHW - Step 46



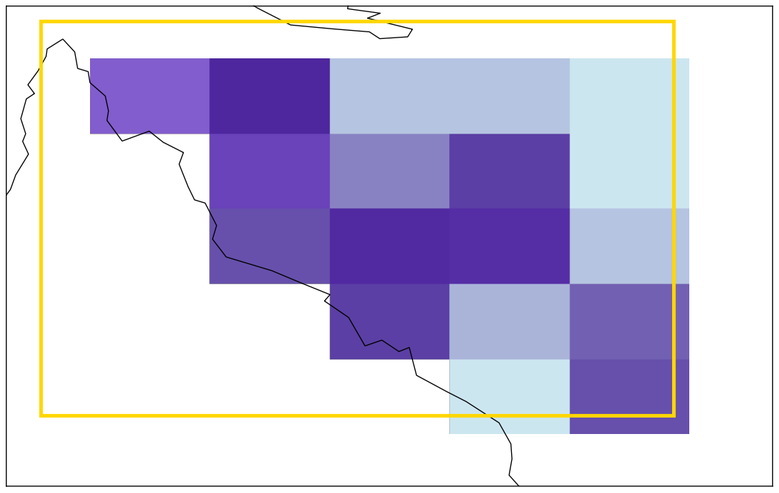
Predicted DHW - Step 46



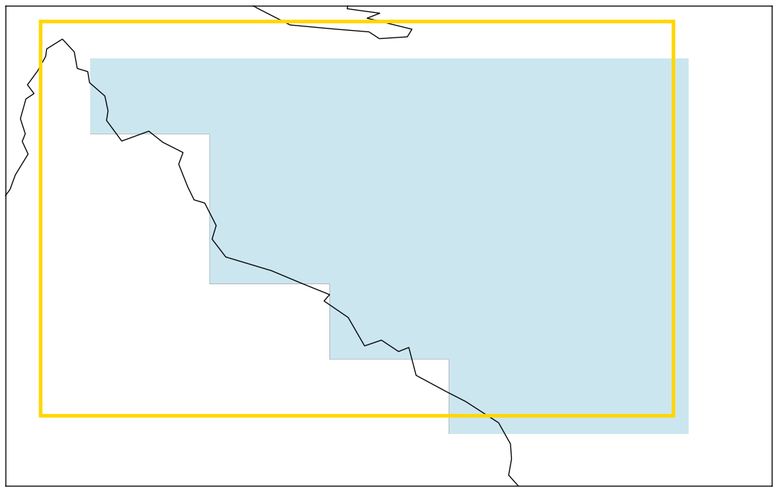
Error (Pred - Actual) - Step 46 [scale ±1.3]



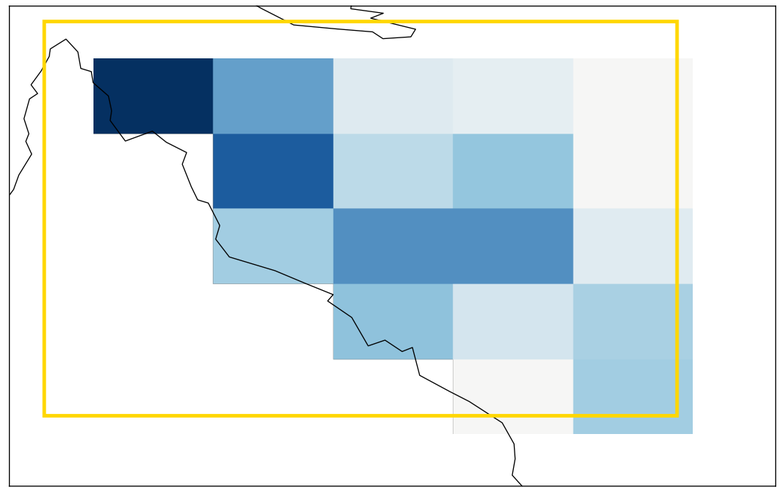
Actual DHW - Step 90

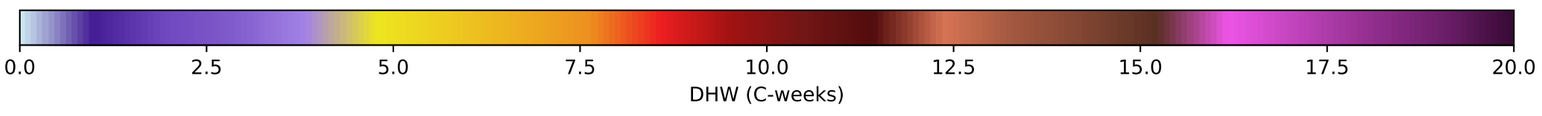


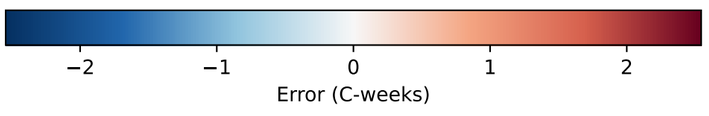
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.5]

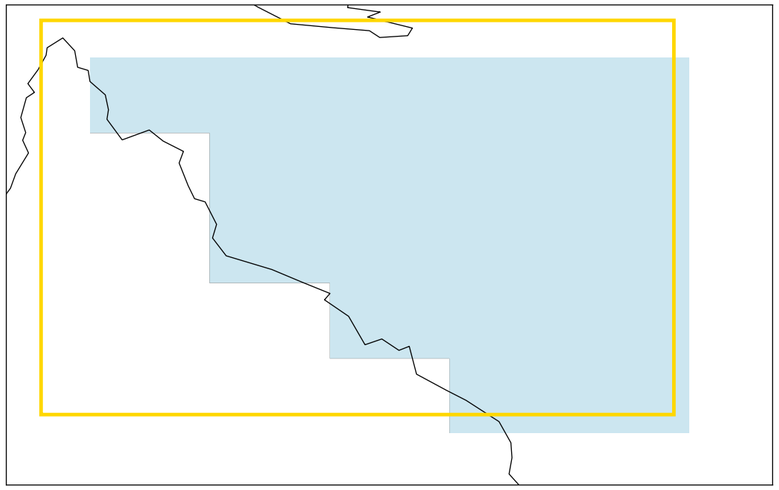




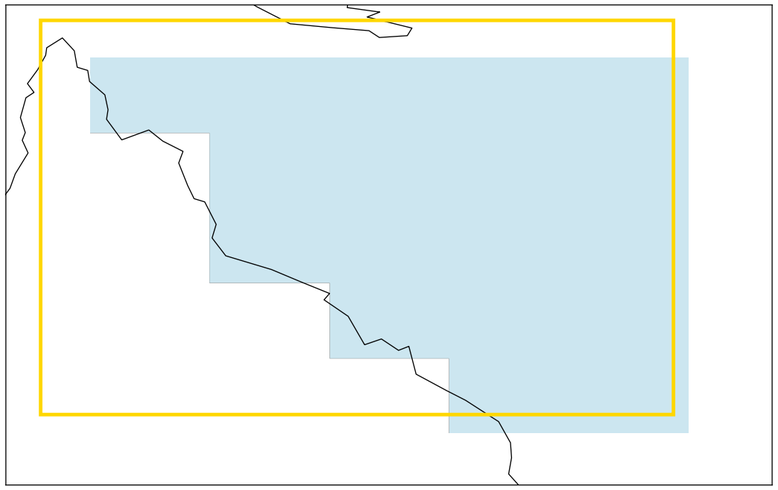


DHW Chain init: 2017-07-04 - Val - GBR (C-weeks)

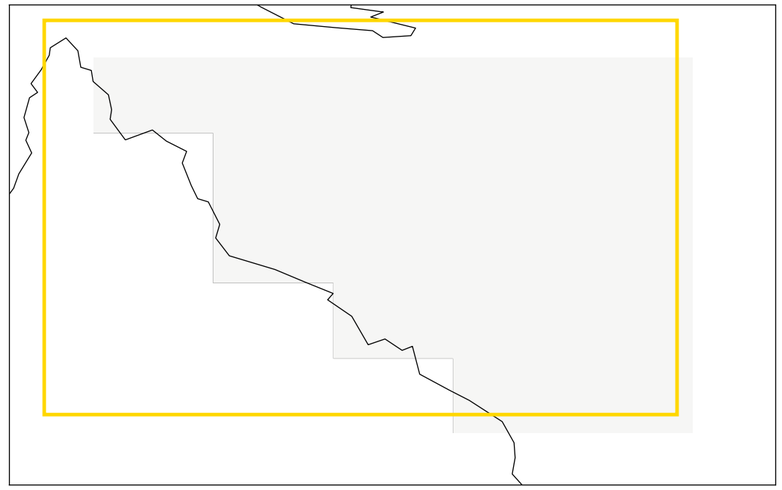
Actual DHW - Step 1



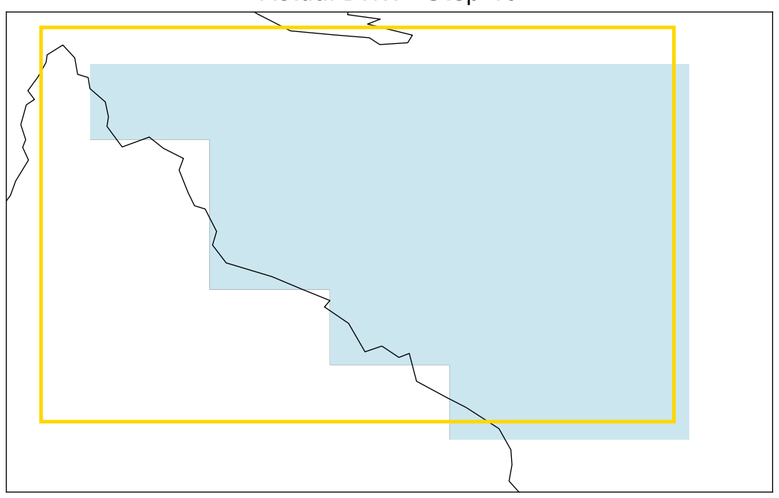
Predicted DHW - Step 1



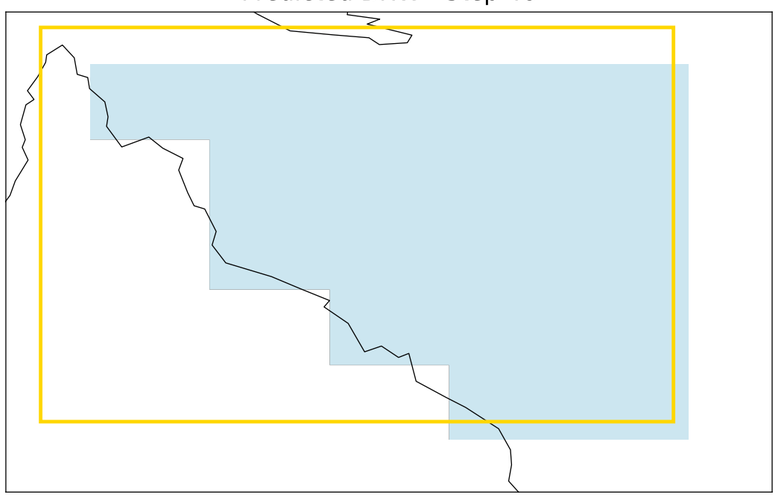
Error (Pred - Actual) - Step 1 [scale ±8]



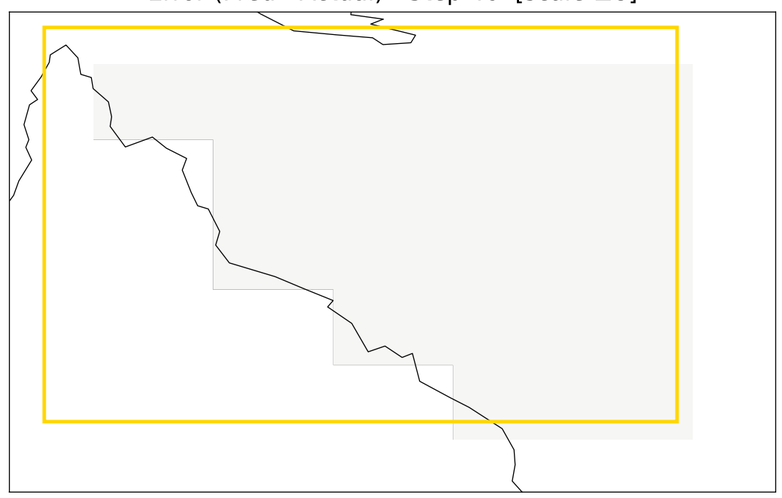
Actual DHW - Step 46



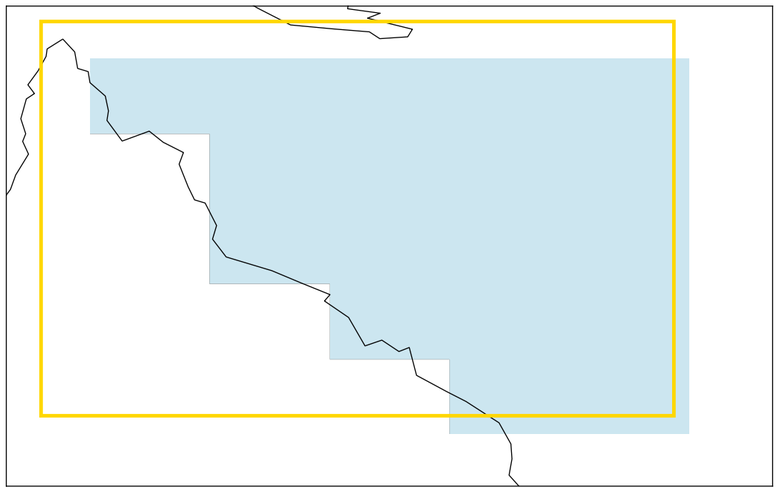
Predicted DHW - Step 46



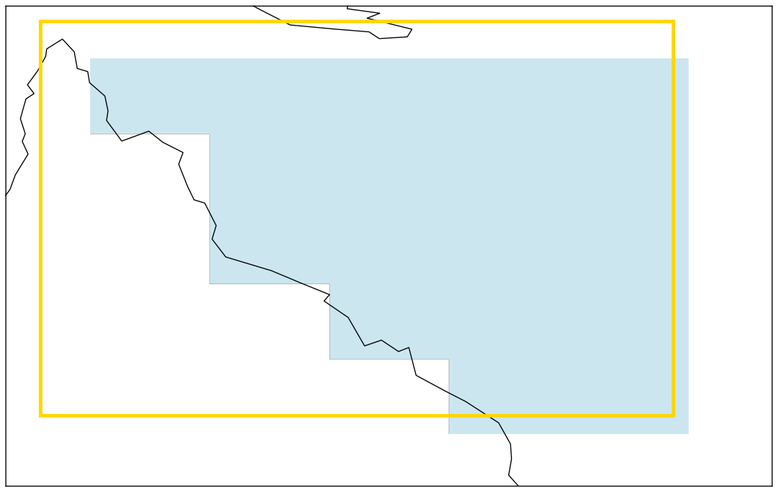
Error (Pred - Actual) - Step 46 [scale ±8]



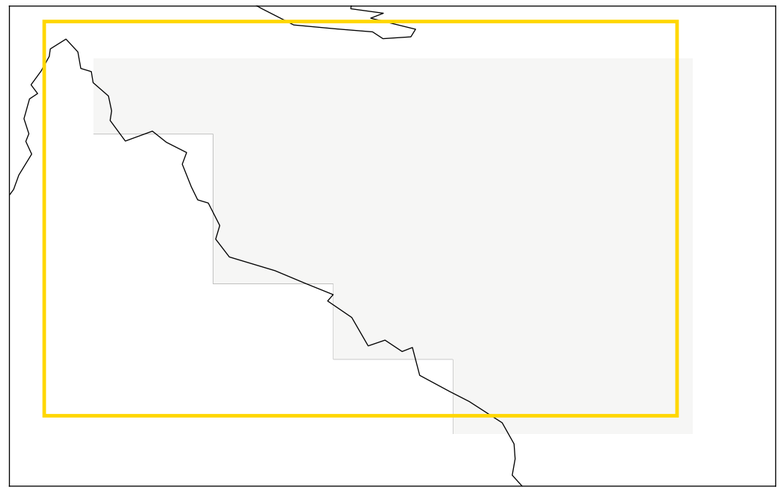
Actual DHW - Step 90

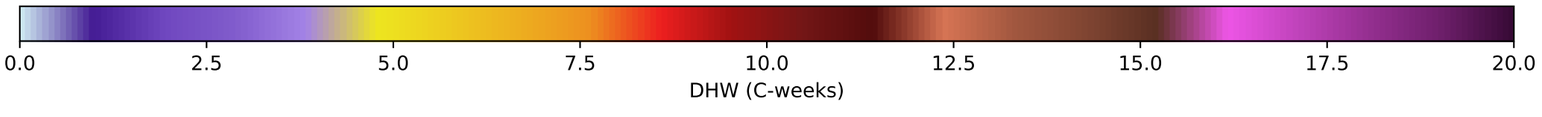


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

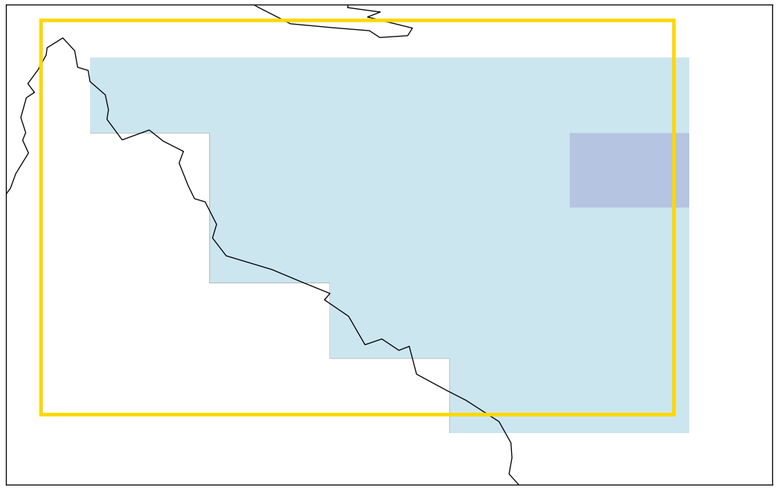




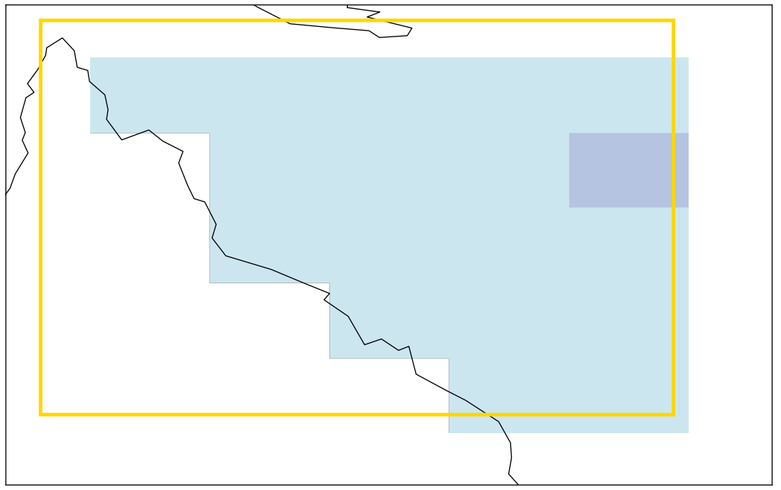


DHW Chain init: 2018-03-31 - Val - GBR (C-weeks)

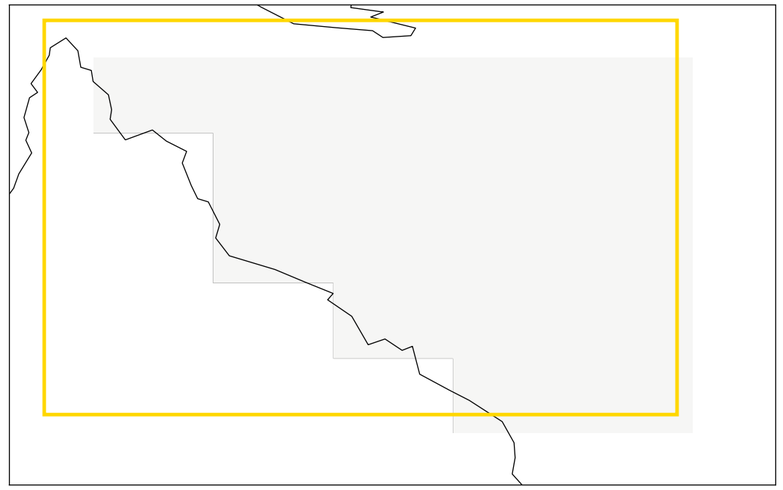
Actual DHW - Step 1



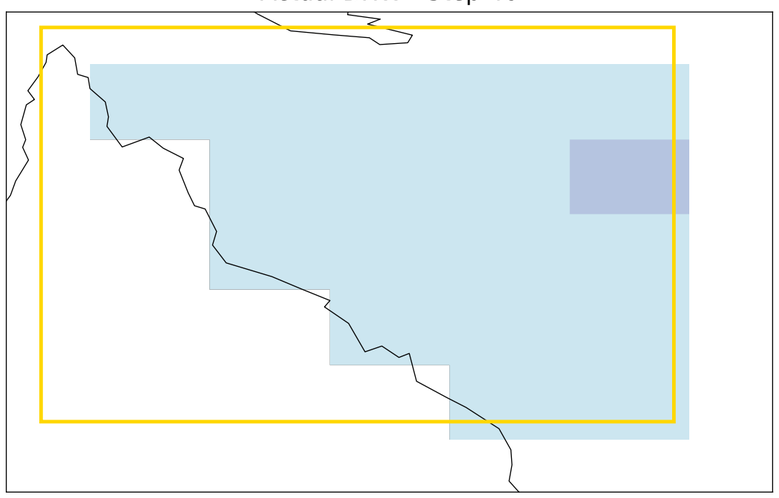
Predicted DHW - Step 1



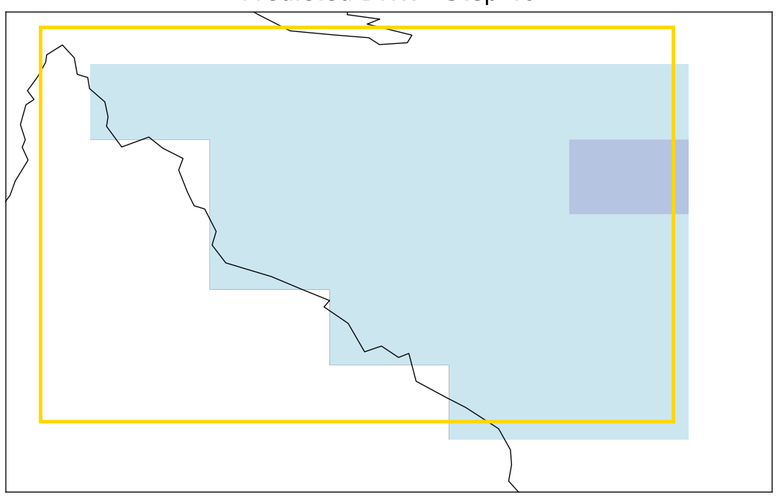
Error (Pred - Actual) - Step 1 [scale ±8]



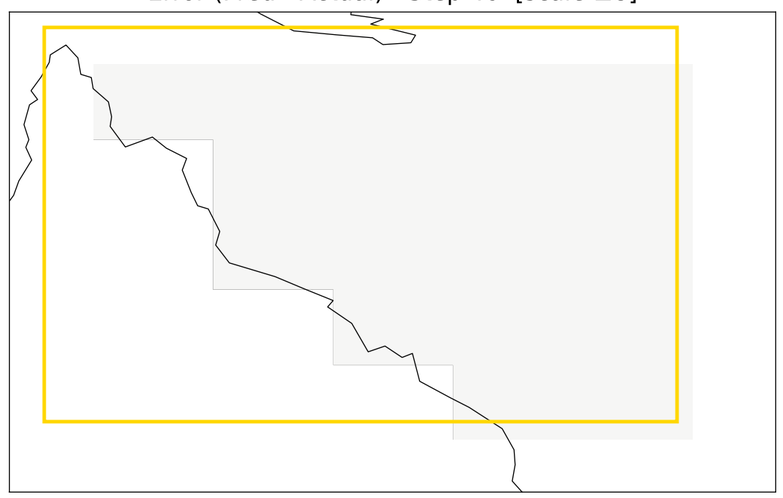
Actual DHW - Step 46



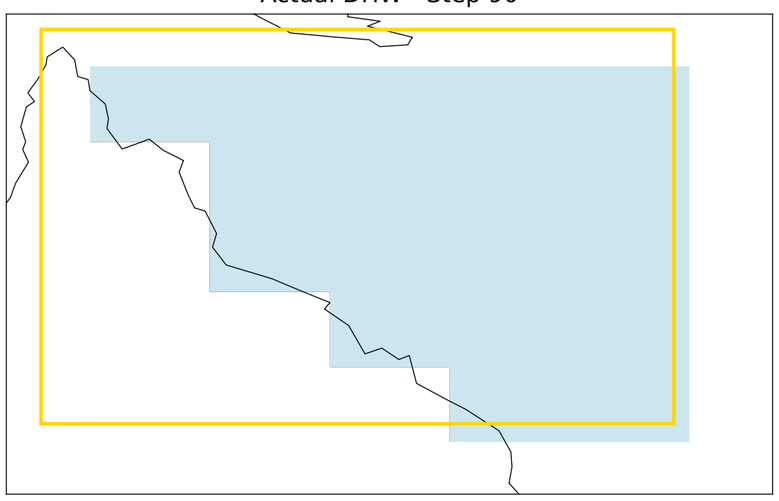
Predicted DHW - Step 46



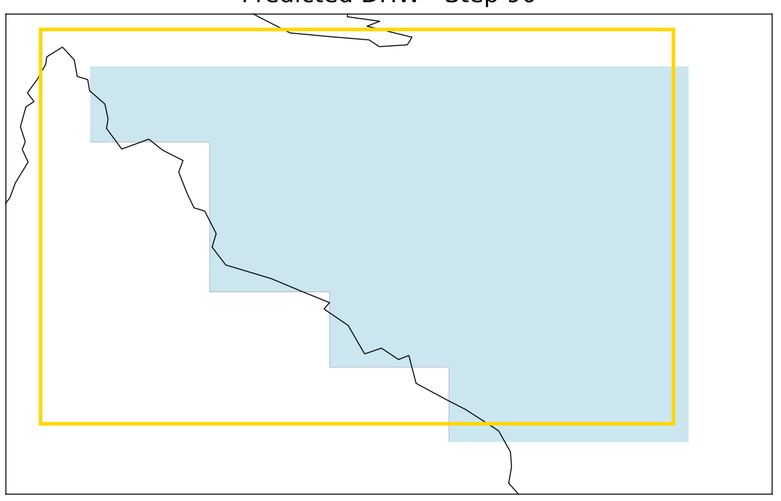
Error (Pred - Actual) - Step 46 [scale ±8]



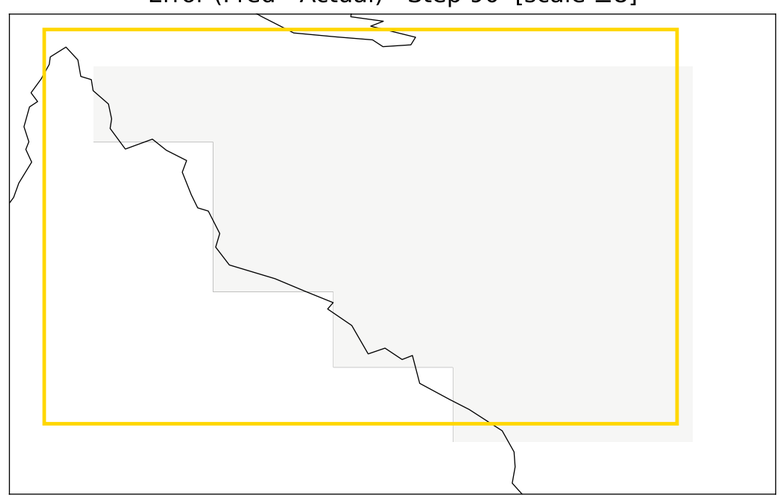
Actual DHW - Step 90

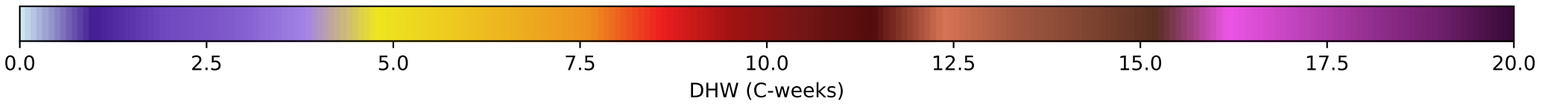


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

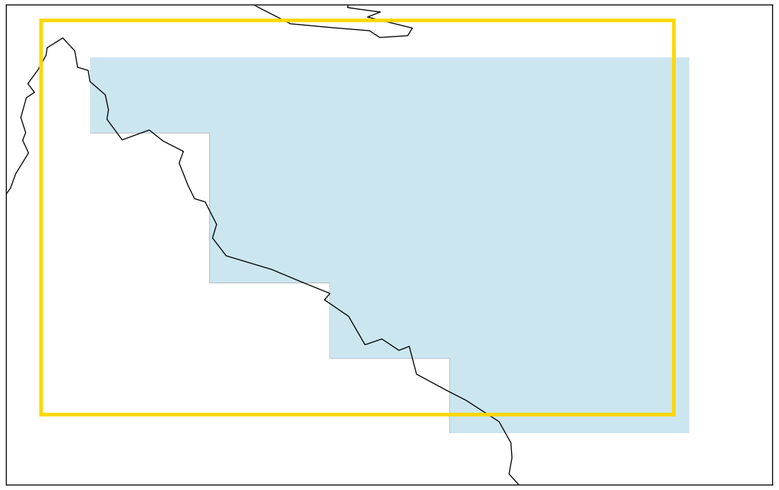




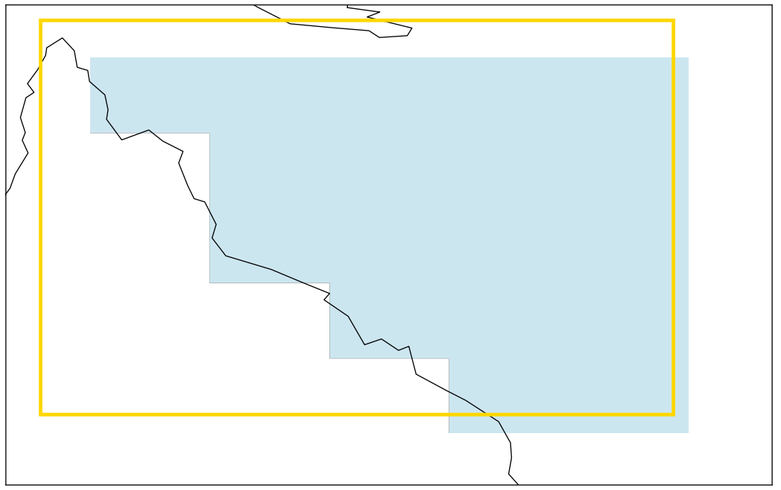


DHW Chain init: 2018-06-29 - Val - GBR (C-weeks)

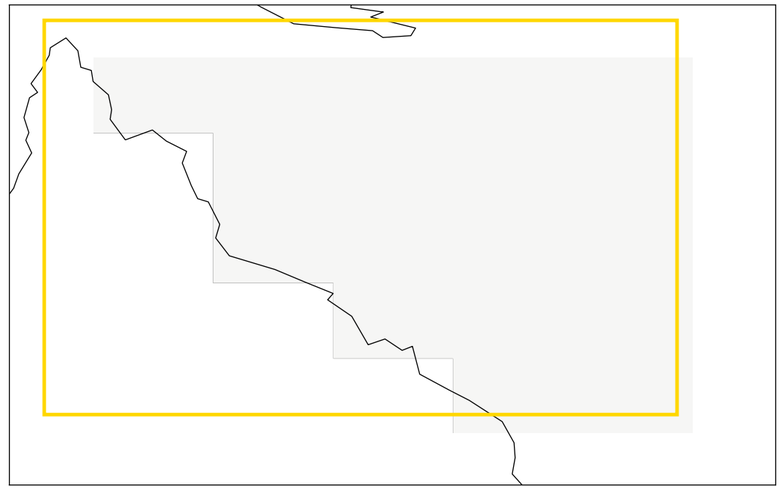
Actual DHW - Step 1



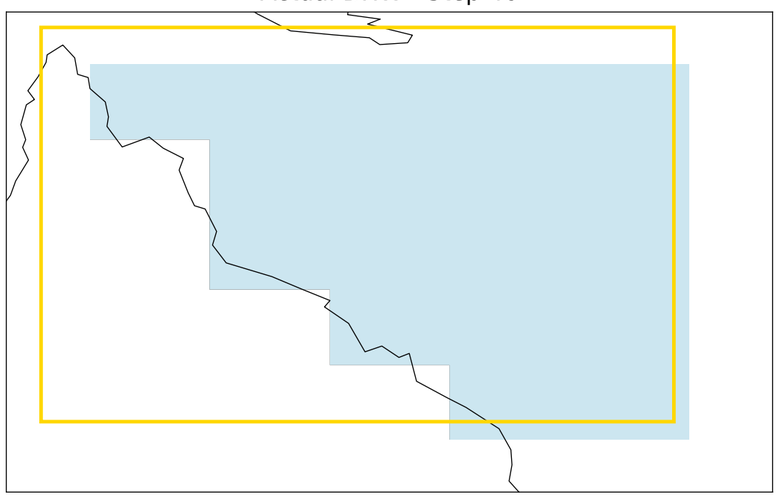
Predicted DHW - Step 1



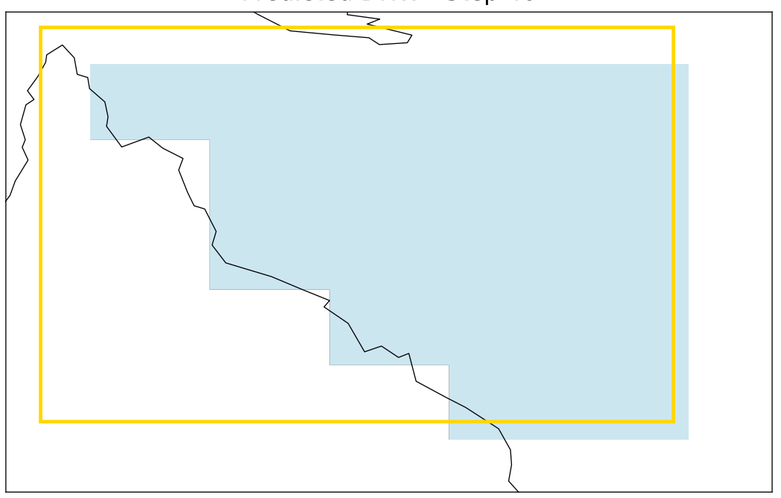
Error (Pred - Actual) - Step 1 [scale ±8]



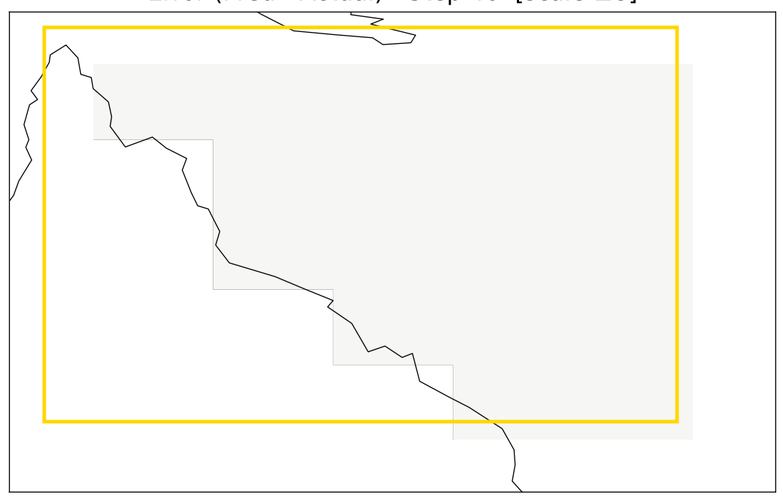
Actual DHW - Step 46



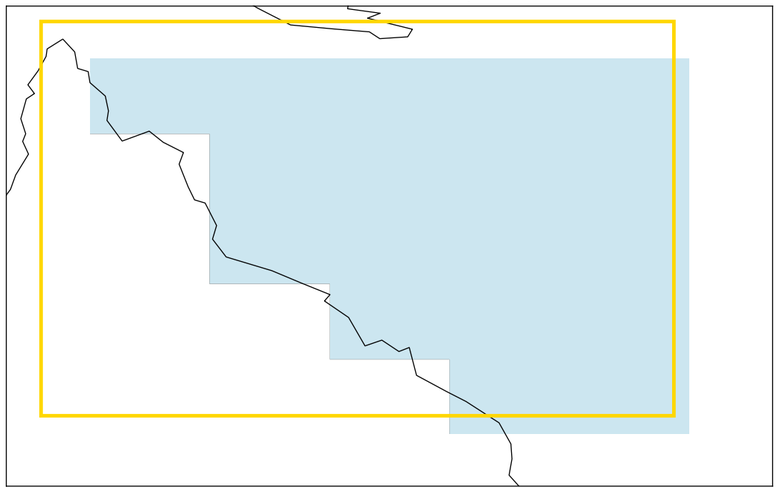
Predicted DHW - Step 46



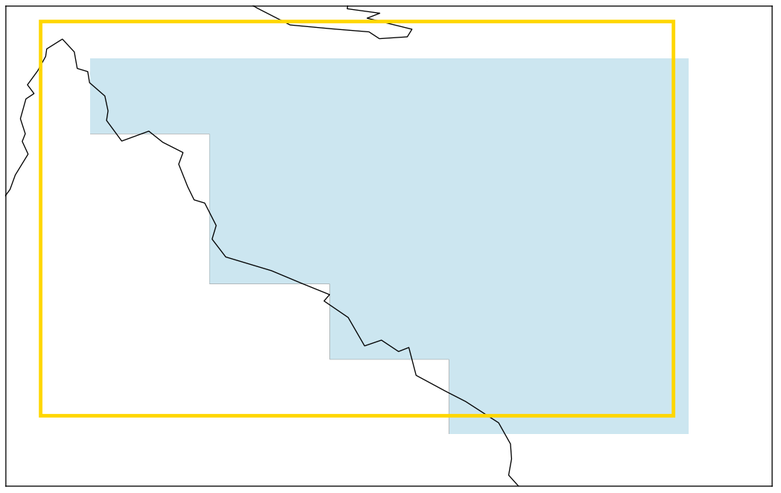
Error (Pred - Actual) - Step 46 [scale ±8]



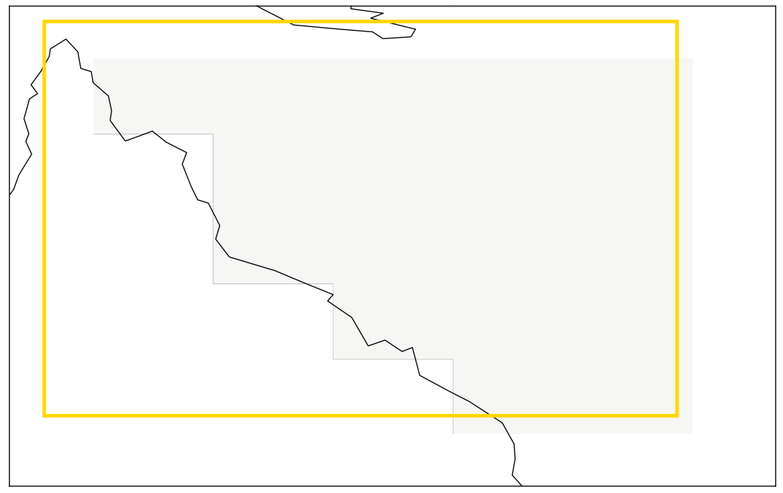
Actual DHW - Step 90

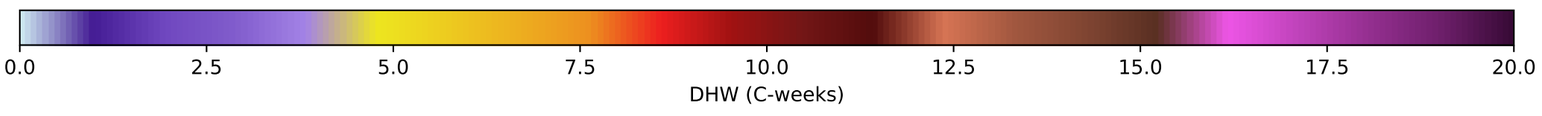


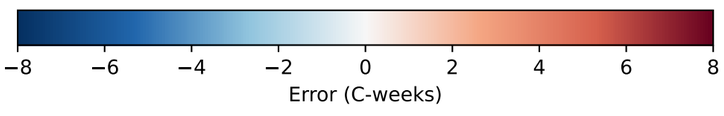
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

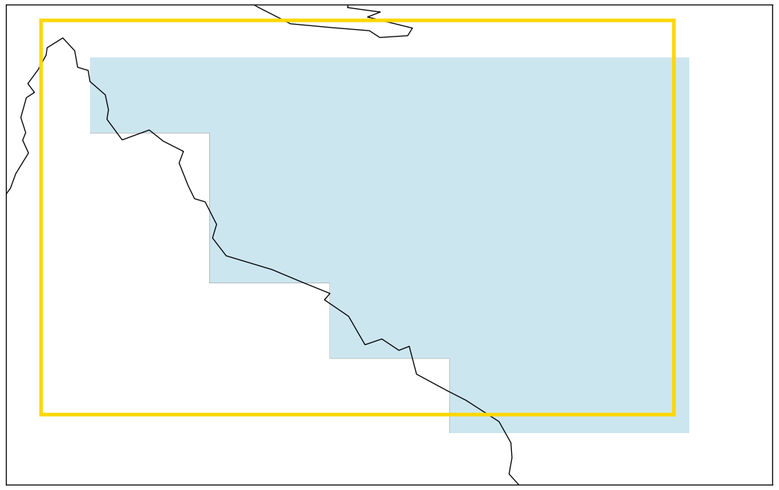




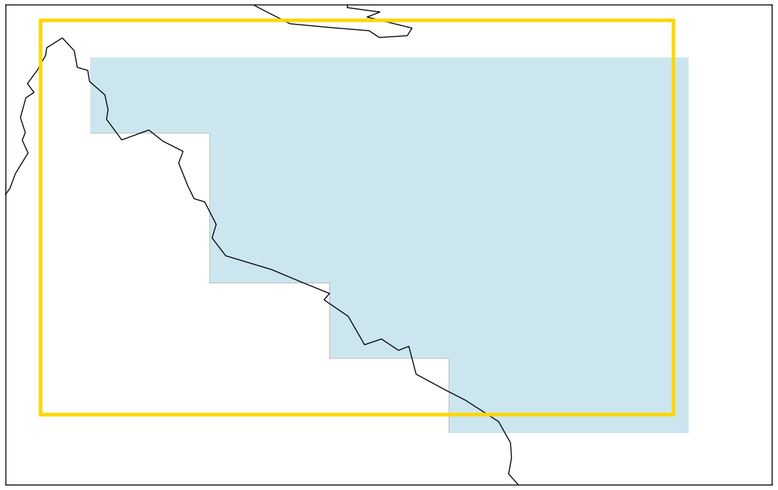


DHW Chain init: 2019-09-22 - Val - GBR (C-weeks)

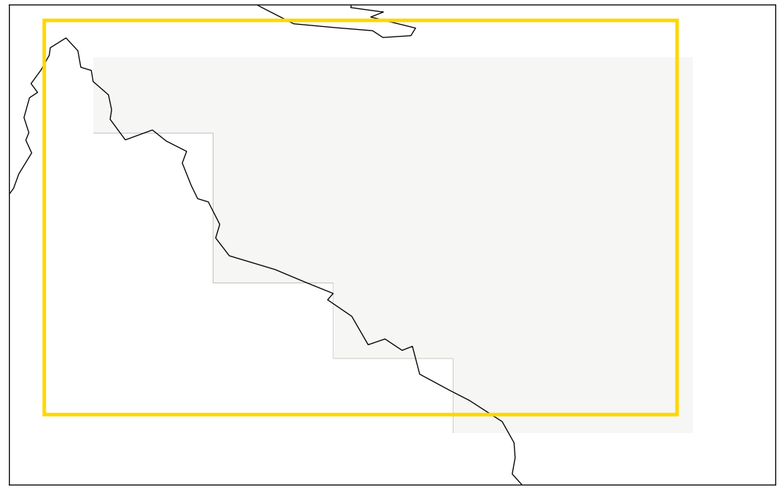
Actual DHW - Step 1



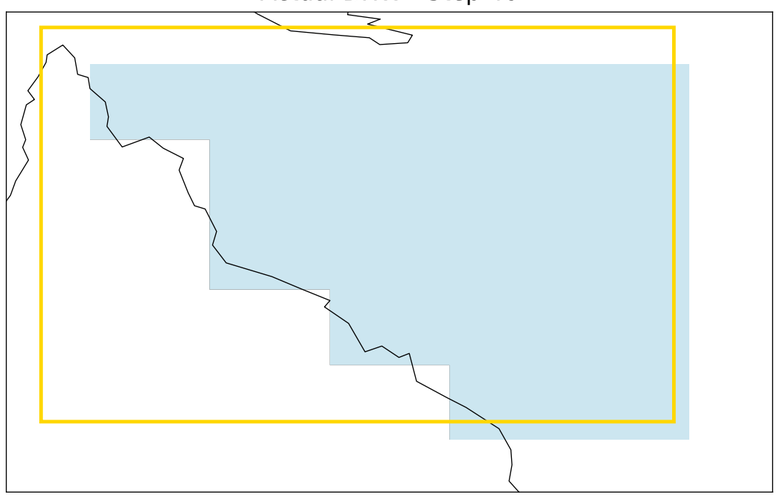
Predicted DHW - Step 1



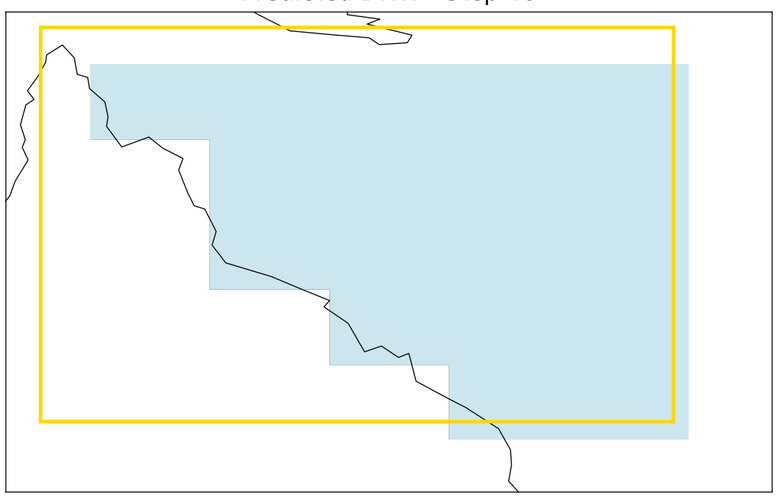
Error (Pred - Actual) - Step 1 [scale ±8]



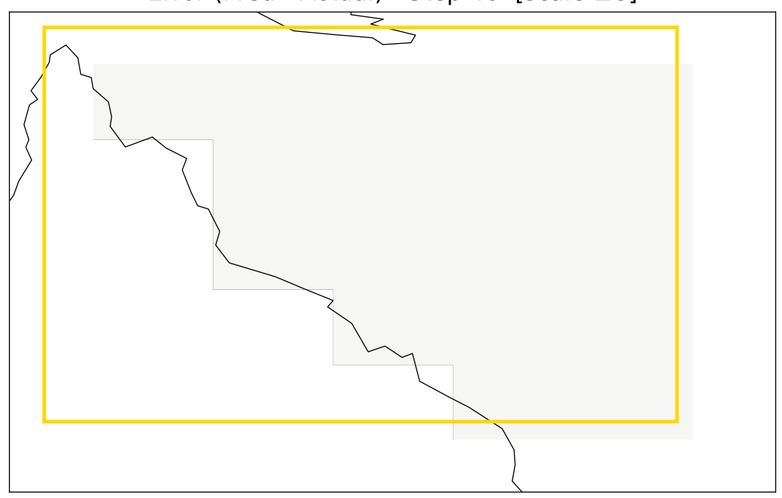
Actual DHW - Step 46



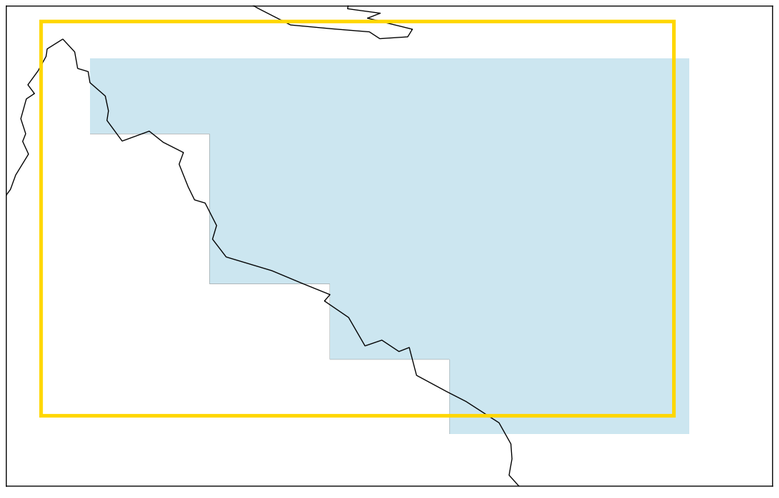
Predicted DHW - Step 46



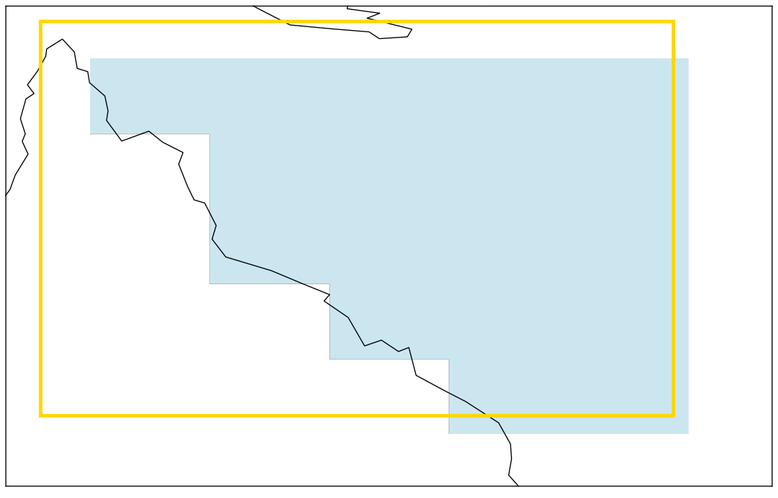
Error (Pred - Actual) - Step 46 [scale ±8]



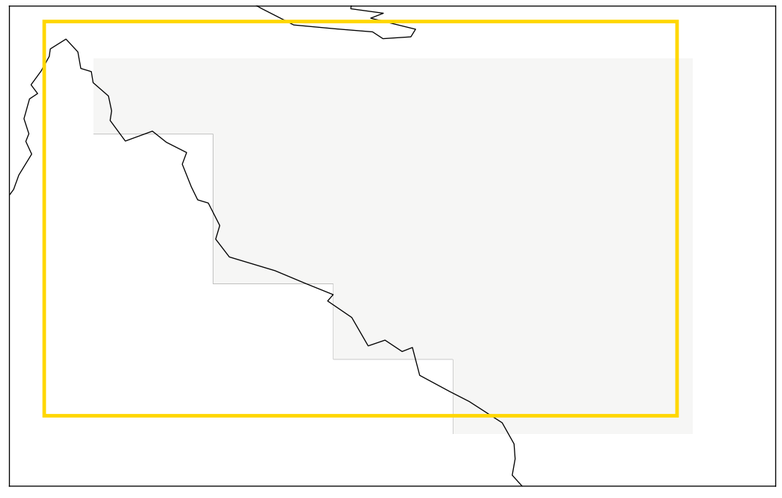
Actual DHW - Step 90

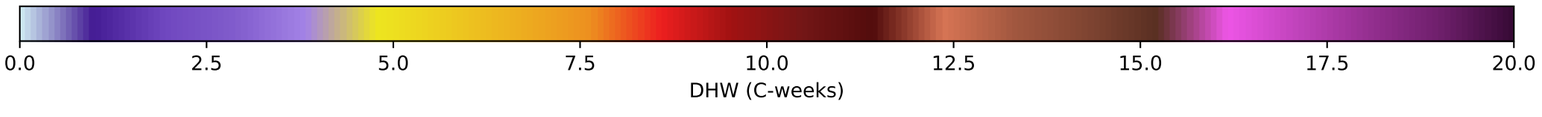


Predicted DHW - Step 90



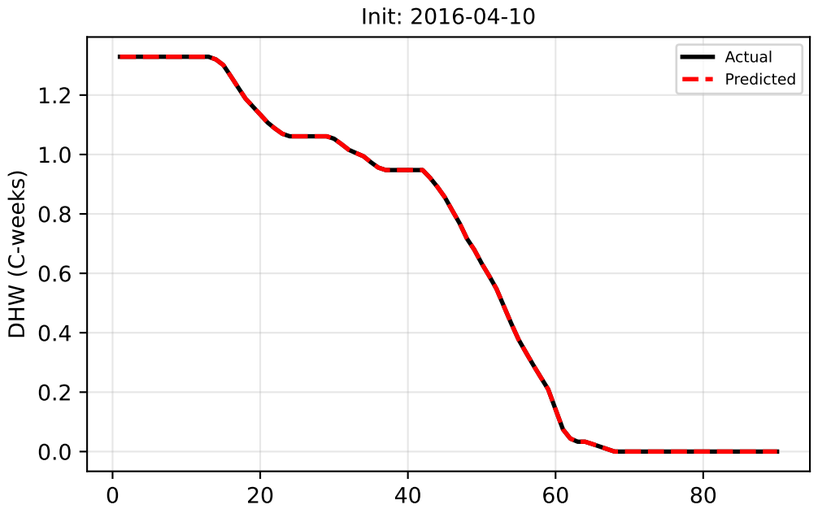
Error (Pred - Actual) - Step 90 [scale ±8]

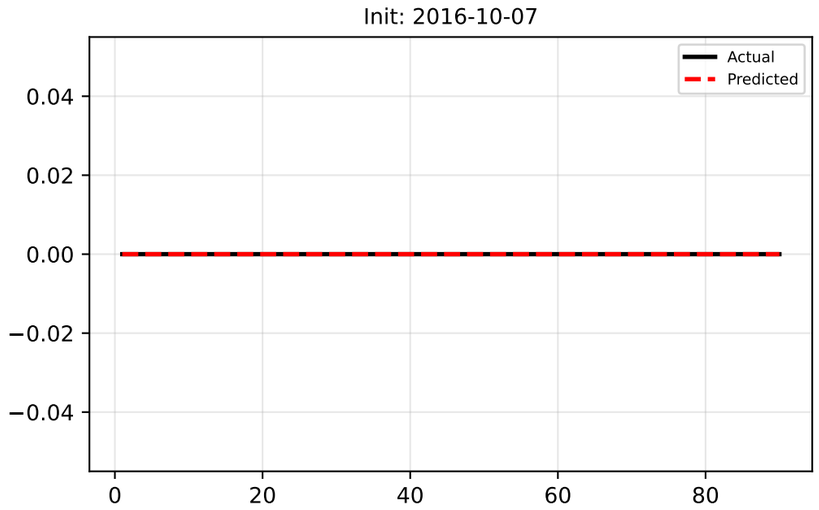


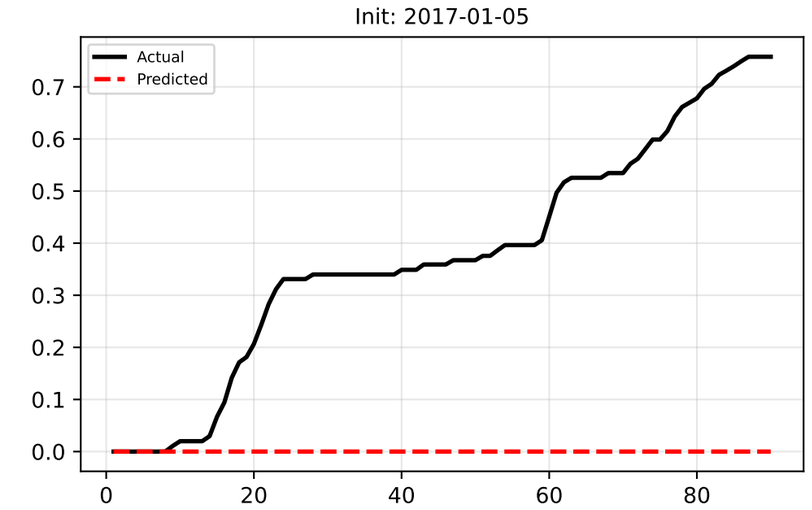


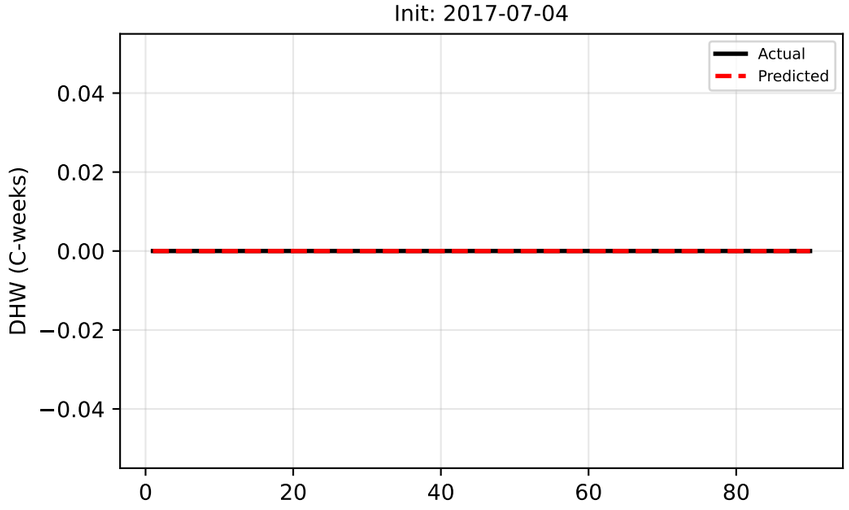


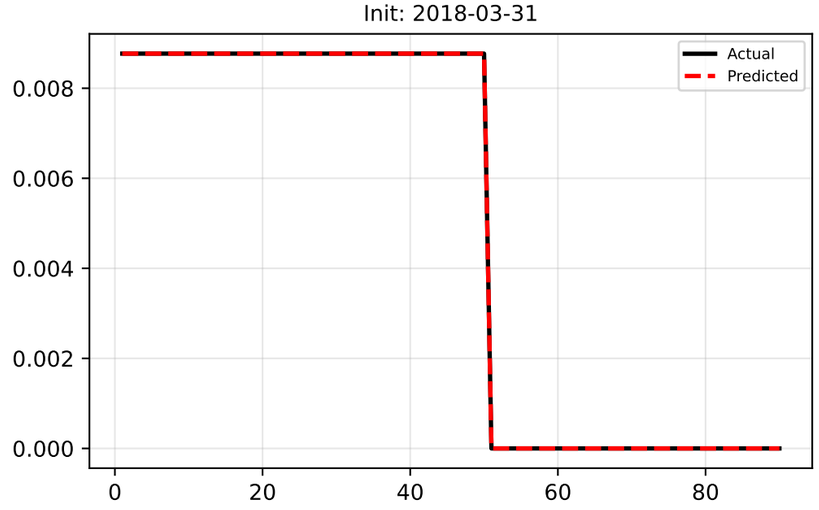
Global-Mean DHW - Val - GBR (C-weeks)

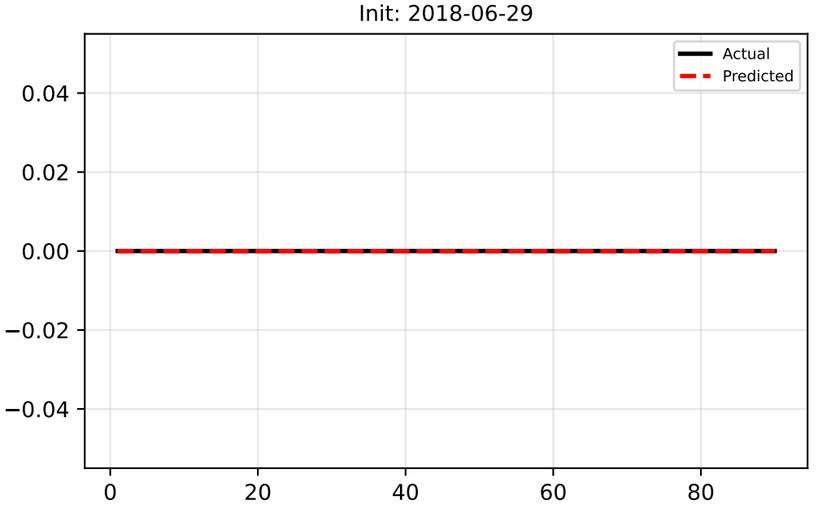


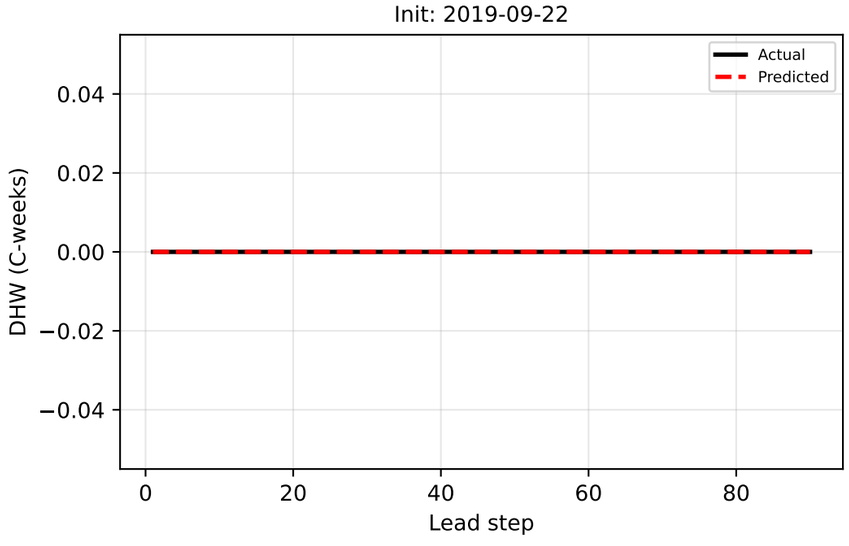


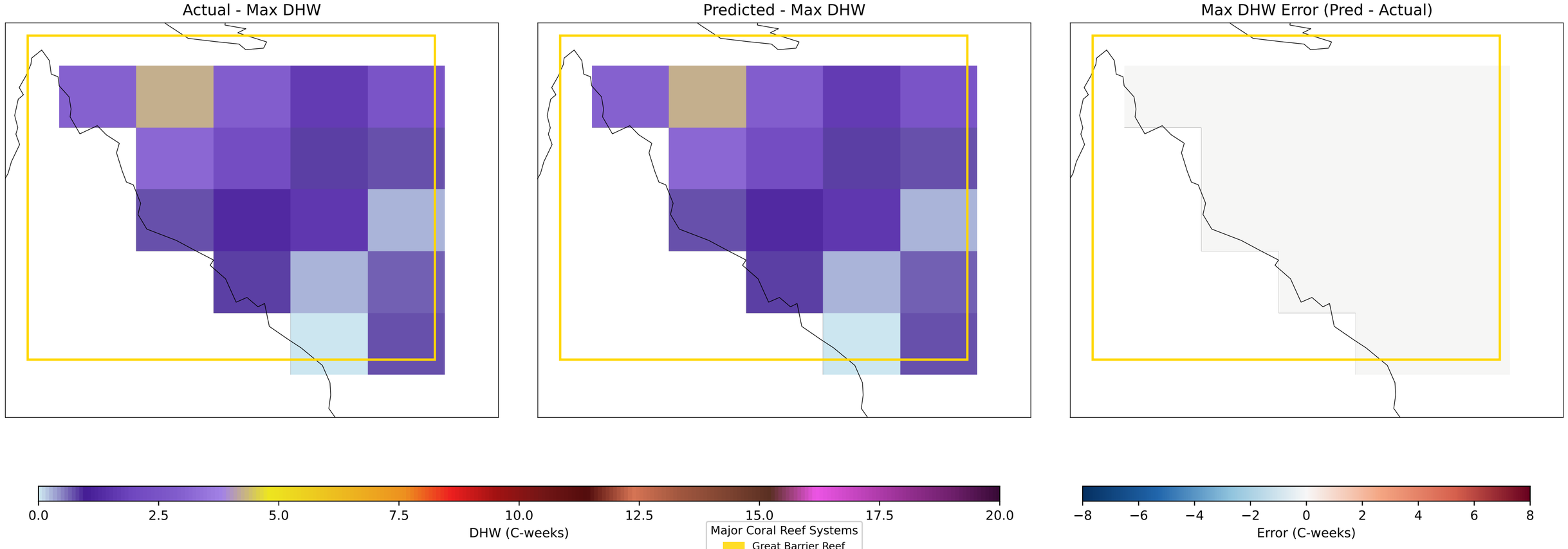


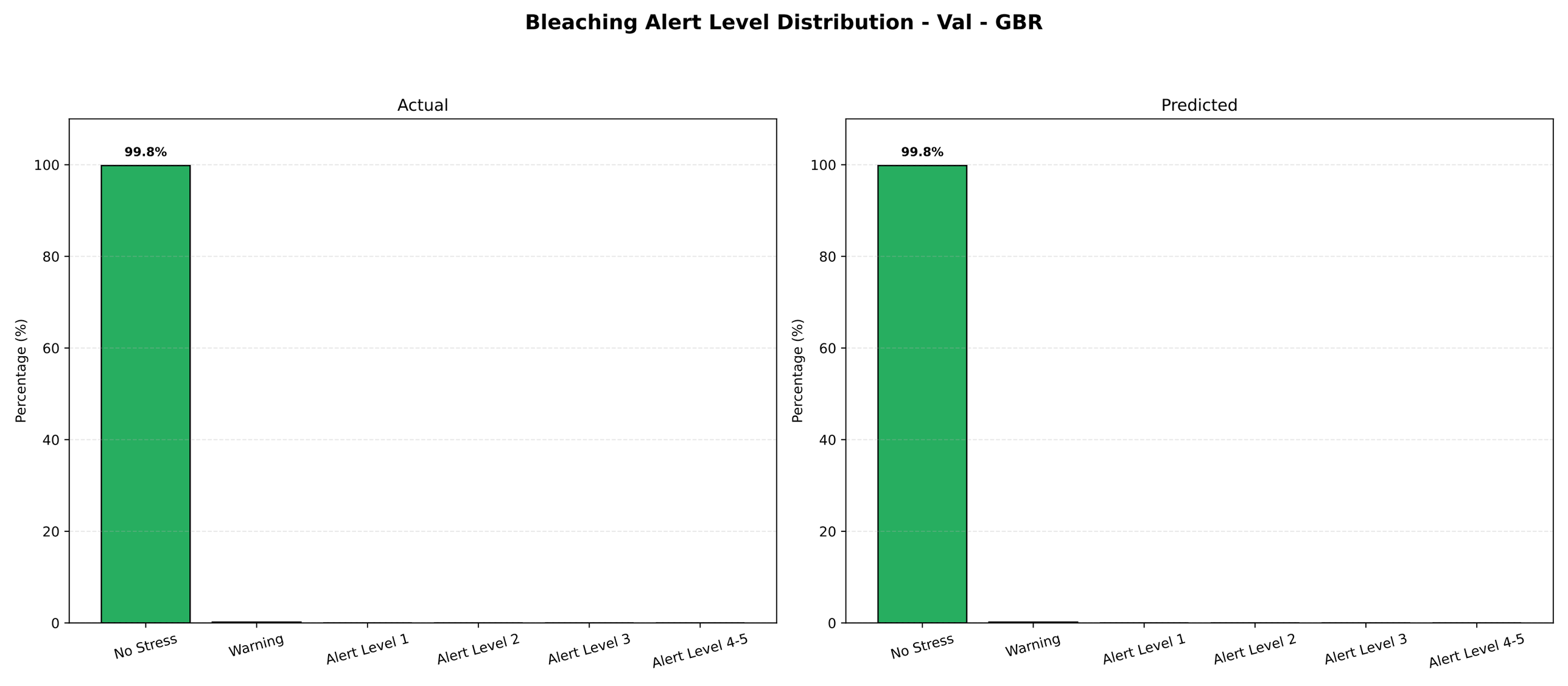






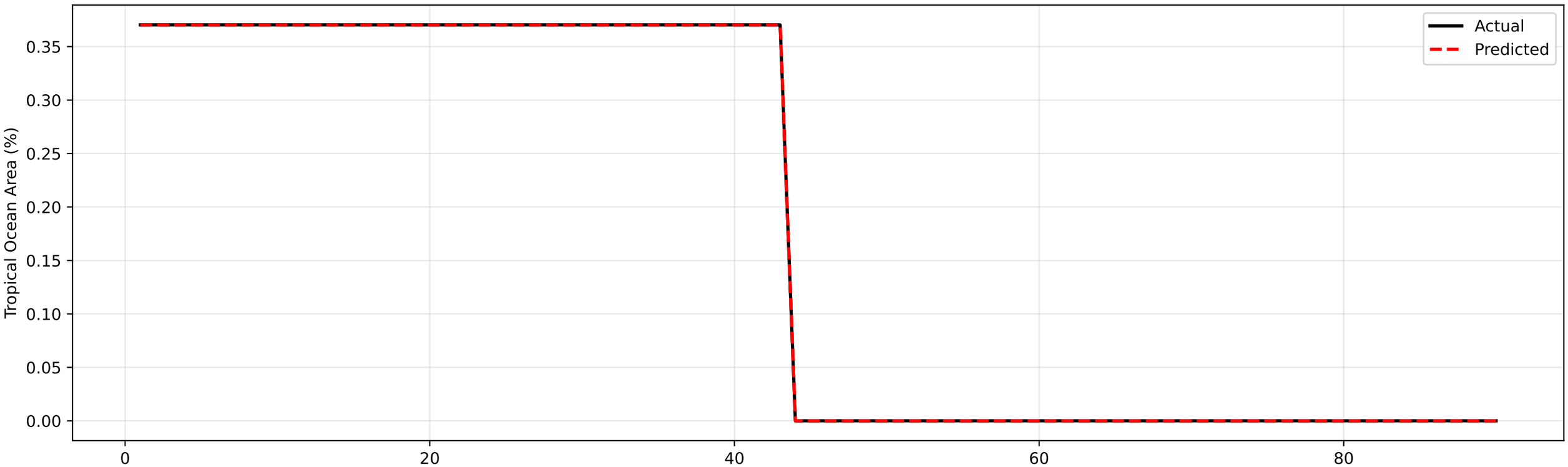



Peak DHW (max over all chains & leads) - Val - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−8
−6
−4
−2
0
2
4
6
8
Error (C-weeks)


Bleaching Alert Level Distribution - Val - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
99.8%
99.8%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

